

Orthogonal Spectral Sequences and Filtrations on Higher Chow Groups

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Abstract

This thesis constructs, proves convergence for, and applies the orthogonal spectral sequence associated with the Bousfield–orthogonal tower in Voevodsky’s triangulated category of motives. The guiding idea is that birational and effective layers of the motivic unit should not merely organize objects of DM , but should also produce a usable filtration on higher Chow groups. For an object $A \in DM$ and a test object B , the tower gives a spectral sequence

$$E_{p,q}^1 = \mathrm{Hom}_{DM}(B, (bc_{p/p-1}A)[q-p]) \implies \mathrm{Hom}_{DM}(B, A).$$

The thesis studies when this spectral sequence converges, what filtration it induces on Chow groups, and how its first layers recover classical geometric and arithmetic invariants.

Part I establishes the foundational convergence theorem. For $B = M(X)(r)[s]$, with X smooth of finite type over a base field and $r, s \in \mathbb{Z}$, the orthogonal spectral sequence is strongly convergent. The same part explains the precise limitation of this statement in the Morel–Voevodsky stable homotopy category $\mathcal{SH}$: strong convergence fails in general, but is recovered under an eventual orthogonality/effective-completion hypothesis, in particular on the motivic pieces supplied by Voevodsky slices. This part is based on the published article of Galindo–Peláez [53]; the full text is available through [International Press](#).

Part II applies the convergence theorem to higher Chow groups. For a smooth projective variety X/k , the probe $M(X)(-n)[m-2n]$ identifies the abutment with $\mathrm{CH}^n(X, m)_R$, and the orthogonal tower of the motivic unit defines the filtration $F_{\mathrm{orth}}^\bullet \mathrm{CH}^n(X, m)_R$. With rational coefficients the spectral sequence degenerates at E^1 over perfect fields; its nonzero columns are bounded; and for abelian varieties the resulting filtration is compatible with Beauville’s eigenspace decomposition, Saito’s filtration, Bondarko’s weight-theoretic viewpoint, Kimura finite-dimensionality, and Pontryagin operations. The Abel–Jacobi and BSD-oriented parts of the argument are written as consequences of this single motivic filtration rather than as independent phenomena.

Part III supplies the missing geometric interpretation of the early layers. It shows that, in the relevant top-degree range, the second orthogonal step coincides rationally with incidence equivalence and, over algebraically closed fields, with the Albanese kernel. It also explains how Chow correspondences and regular homomorphisms admit effective motivic kernels whose higher covers and slices agree with those of $M(X)$. Thus the Abel–Jacobi, incidence, Albanese, and abelian-equivalence tests are not external decorations; they are the first visible geometric shadows of the orthogonal filtration.

Part IV closes the thesis by translating the preceding theory into applications. It organizes the results as a diagnostic procedure for algebraic cycles, an Abel–Jacobi/BSD test for abelian varieties, a functorial control theorem for correspondences and regular homomorphisms, an application to rationally connected varieties and stable-rationality obstructions, and

an effective-completion principle for $\mathcal{SH}$, slices, K -theory, cobordism, modulus, and positive characteristic. The rationally connected case is not treated as an external example: it is the natural place where the rational theory is often too small and the integral Chow-theoretic, Abel–Jacobi, unramified-cohomological, and decomposition-of-the-diagonal obstructions become decisive. In this final form the thesis has a single narrative arc: convergence produces a filtration; degeneration makes it computable; incidence and regular homomorphisms give it geometric meaning; and the applications show where geometric and arithmetic obstructions enter.

A guiding editorial principle of the revised text is that every construction is introduced only once and then reused. The preliminary material on triangulated categories, Bousfield localization, derived functors, spectral sequences, and Voevodsky motives is placed in Part I; Part II invokes it by internal reference rather than by repetition; Part III translates the first motivic layers into classical equivalence relations; and Part IV records the resulting applications without replacing the proofs by schematic diagrams. Thus the final document is meant to be read as a single thesis rather than as a concatenation of papers.

The text also records the history of the project. The first part was developed in part during *Motives in St. Petersburg* (2018). Part II was submitted to *International Mathematics Research Notices* in February 2026, and the present text incorporates later improvements motivated by the referee process. A related first part of the broader programme appeared in *Documenta Mathematica*; it is mentioned as publication context, while the arguments in this thesis are referenced internally and through the unified bibliography.

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The thesis also records the publication and revision history of the project. Part I is based on the published article of Galindo–Peláez in *Homology, Homotopy and Applications*; a related first part of the broader programme appeared in *Documenta Mathematica*. Part II was submitted to *International Mathematics Research Notices* in February 2026, and the text below incorporates subsequent improvements motivated by the referee process.

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Introduction

The theory of algebraic cycles is governed by a persistent tension. On the one hand, Chow groups are defined by a concrete equivalence relation on algebraic subvarieties. On the other hand, the invariants that detect cycles—cycle class maps, Abel–Jacobi maps, Albanese maps, regulators, coniveau and slice filtrations—are distributed across cohomology theories, motives, and arithmetic realizations. A useful filtration on Chow groups should therefore do more than list successive kernels. It should explain why the classical tests appear in the order in which they do, how they behave under correspondences, and which parts survive under changes of coefficients, base field, or ambient motivic category.

In codimension one the picture is classical. The Picard group of a smooth projective variety separates into a continuous part, controlled by the Picard variety, and a discrete Néron–Severi group. In higher codimension this structure breaks down: Mumford’s theorem on zero-cycles shows that Chow groups can be infinite-dimensional in a very strong sense. The Bloch–Beilinson–Murre programme proposes a replacement: a finite, functorial, cohomologically meaningful filtration whose first steps recover the cycle-class and Abel–Jacobi kernels and whose graded pieces are expected to be governed by realizations. This thesis constructs and studies a motivic candidate for such a filtration, not by postulating the layers directly, but by deriving them from the orthogonal tower of the motivic unit.

Let DM denote Voevodsky’s triangulated category of motives. Peláez introduced a tower of birational or orthogonal covers $bc_{\leq \bullet}$. Given objects $A, B \in DM$, this tower produces the orthogonal spectral sequence

$$E_{p,q}^1 = \mathrm{Hom}_{DM}(B, (bc_{p/p-1}A)[q-p]) \implies \mathrm{Hom}_{DM}(B, A).$$

The fundamental question is convergence. Without convergence the tower is only a system of approximations; with strong convergence it produces an actual filtration on the abutment. When $A = \mathbf{1}_R$ and $B = M(X)(-n)[m-2n]$, this abutment is $\mathrm{CH}^n(X, m)_R$, and the tower gives the orthogonal filtration

$$\begin{aligned} F_{\mathrm{orth}}^r \mathrm{CH}^n(X, m)_R &:= \mathrm{Im} \left(\mathrm{Hom}_{DM}(M(X)(-n)[m-2n], bc_{\leq -r} \mathbf{1}_R) \right. \\ &\quad \left. \longrightarrow \mathrm{CH}^n(X, m)_R \right). \end{aligned}$$

The central claim of the thesis is that this filtration provides a single language for several phenomena that are often treated separately: convergence of motivic towers, degeneration of spectral sequences, Beauville decompositions of abelian varieties, Abel–Jacobi kernels, incidence equivalence, regular homomorphisms, and effective completions in $\mathcal{SH}$.

The thesis is organized around four interfaces. The point of this organization is not merely expository. Each interface removes one ambiguity that otherwise obscures the relation between motivic towers and classical cycle theory. The

first interface explains why the tower converges and therefore produces an honest filtration. The second explains why, after rationalization and under the hypotheses specified below, the filtration is computable from its orthogonal layers. The third explains why the early layers recover incidence, Albanese, Abel–Jacobi, and regular-homomorphism kernels. The fourth explains what the construction does in practice and where its natural boundary lies, especially in stable motivic homotopy, integral coefficients, and positive characteristic.

Several conventions are used throughout the revised text. Results marked as unconditional are consequences of the constructions and hypotheses stated in the same section or in earlier parts. Results marked by an explicit hypothesis are kept separate from the unconditional arguments; the hypothesis is never hidden in the proof. Whenever Part II or Part III uses a categorical fact from Part I, the text refers to the exact statement rather than repeating the proof. This is the sense in which the thesis is self-contained: the reader can reconstruct the logical dependence of every application from the preliminary triangulated and motivic material developed at the beginning.

I. From towers to convergence. Part I proves strong convergence of the orthogonal spectral sequence for the motivic probes needed in Chow theory. More precisely, for $B = M(X)(r)[s]$, with X smooth of finite type over the base field and $r, s \in \mathbb{Z}$, the spectral sequence attached to the Bousfield–orthogonal tower converges strongly. This is the technical foundation on which every later filtration statement rests. The same part also explains why the corresponding statement fails in $\mathcal{SH}$ and how eventual orthogonality restores convergence to an effective completion. These results are based on Galindo–Peláez [53], available through [International Press](#). A related first part of the broader programme appeared in *Documenta Mathematica*; it is mentioned here as publication context, and the arguments below remain internally cross-referenced.

II. From convergence to Chow filtrations. Part II applies Part I to higher Chow groups. With rational coefficients, the orthogonal spectral sequence degenerates at E^1 over perfect fields. The proof identifies the possible differentials with Galois-cohomological attaching classes and uses a Maschke-type averaging argument for rational discrete modules. The resulting filtration has finite support, is compatible with Beauville’s eigenspace decomposition on abelian varieties, and can be compared with the filtrations of Saito, Bondarko, and the expected Bloch–Beilinson–Murre picture in the cases treated here. The abelian-variety sections then turn the filtration into a test for Abel–Jacobi injectivity and BSD-type obstructions.

III. From Chow filtrations to geometric kernels. Part III explains what the first non-trivial layers mean geometrically. In top degree, the second orthogonal step is identified rationally with incidence equivalence and, over algebraically closed fields, with the Albanese kernel. Regular homomorphisms are then represented by effective motivic kernels whose higher covers and slices agree with those of $M(X)$.

This supplies the missing bridge between the abstract filtration of Part II and the classical tests used in algebraic geometry: incidence, Albanese, Abel–Jacobi, and abelian equivalence.

IV. From geometric kernels to applications. Part IV packages the theory into applications. The orthogonal filtration becomes a diagnostic procedure for Chow classes; Beauville–Pontryagin decompositions give structured tests for Abel–Jacobi and BSD-type phenomena; correspondences and regular homomorphisms are controlled through effective motivic kernels; and the comparison with $\mathcal{SH}$, slices, K -theory, cobordism, modulus, and positive characteristic clarifies exactly where one has a genuine Chow filtration and where only an effective completion or a future analogue should be expected.

The four parts should be read as one continuous argument. Part I turns the orthogonal tower into a convergent machine. Part II applies that machine to higher Chow groups and obtains the filtration. Part III identifies the first geometric kernels of the filtration through incidence, Albanese, Abel–Jacobi, and regular homomorphisms. Part IV then uses the same structure to separate proved Chow-theoretic applications from completed invariants in $\mathcal{SH}$ and from the natural future extensions suggested by modulus and positive characteristic.

Conventions and self-containedness. The document is written so that foundational material appears only once. Part I contains the categorical and motivic preliminaries: triangulated categories, adjoint functor theorems, Bousfield localization, motivic categories, and the construction of the orthogonal tower. Part II cites Part I rather than repeating those preliminaries; its new input is Chow-theoretic. Part III is self-contained relative to Parts I–II and introduces only the additional language of incidence, correspondences, and regular homomorphisms. Part IV contains no new foundational machinery; it is the synthesis of the preceding three parts. Equation numbers are suppressed in this thesis-style version, while labels and hyperlinks are preserved for navigation.

Organization. Part I develops the categorical and motivic foundations and proves the convergence theorem. Part II constructs and studies the orthogonal tower filtration on higher Chow groups. Part III identifies the first geometric layers through incidence equivalence, Chow correspondences, and regular homomorphisms. Part IV gathers the applications and closes the thesis. The bibliography is unified, and cross-references are hyperlinked throughout.

PART I. CONVERGENCE OF THE ORTHOGONAL SPECTRAL SEQUENCE

Guide to Part I

Part I supplies the categorical engine of the thesis. Its purpose is to turn an orthogonal tower into a convergent spectral sequence in a setting broad enough for higher Chow groups but controlled enough to avoid phantom completions.

The construction requires three ingredients. First, the triangulated category must admit the relevant Bousfield localization functors. Secondly, the tower obtained from these functors must be functorial and compatible with compact or geometric motivic probes. Thirdly, Boardman's convergence criterion must identify the filtration on the abutment with the filtration produced by the tower rather than with a completion of it.

The preliminary subsections therefore have a precise role. The discussion of triangulated categories and Brown representability explains why the adjoints exist. The review of derived categories and spectral sequences fixes the conventions used later in Part II. The motivic subsection recalls the embedding of smooth schemes into Voevodsky motives and the representation of Chow groups as motivic Hom-groups. The main theorem of Part I then applies this material to objects of the form $M(X)(r)[s]$. This is exactly the range needed later, because every higher Chow group appearing in the thesis is obtained by testing the motivic unit against such an object.

This part should be read as the answer to a foundational question: when does the orthogonal tower compute rather than merely approximate? Once the strong convergence theorem is proved, the later parts may speak of an orthogonal filtration on Chow groups without adding a new convergence hypothesis each time.

1. Preliminaries

1.1. Triangulated categories

The main references for this section are [20, 11].

Let $\mathcal{A}$ be an additive category equipped with an autoequivalence $\Sigma : \mathcal{A} \rightarrow \mathcal{A}$ of categories. For every integer $n \in \mathbb{Z}$ and every object X (resp. morphism f) of $\mathcal{A}$, we define $X[n] := \Sigma^n(X)$ (resp. $f[n] = \Sigma^n(f)$).

Definition 1.1 (Verdier). Let $\mathcal{A}$ be an additive category. The structure of a *triangulated category* is given by an autoequivalence $\Sigma = [1] : \mathcal{A} \rightarrow \mathcal{A}$, called the *translation functor*, and a class of diagrams in $\mathcal{A}$. A *triangle* of $(\mathcal{A}, \Sigma)$ is a sequence of morphisms

$$X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1].$$

A morphism between distinguished triangles is a triple (f, g, h) of morphisms of $\mathcal{A}$ making the following diagram commute:

$$\begin{array}{ccccccc} X & \xrightarrow{u} & Y & \xrightarrow{v} & Z & \xrightarrow{w} & X[1] \\ \downarrow f & & \downarrow g & & \downarrow h & & \downarrow f[1] \\ X' & \xrightarrow{u'} & Y' & \xrightarrow{v'} & Z' & \xrightarrow{w'} & X'[1]. \end{array}$$

Such a triple (f, g, h) is an isomorphism if the vertical arrows are isomorphisms. We require the distinguished triangles to satisfy the following axioms:

- (TR1) (a) Every triangle of the form $X \xrightarrow{1_X} X \rightarrow 0 \rightarrow X[1]$ is distinguished.
 (b) Every triangle isomorphic to a distinguished triangle is distinguished.
 (c) For every morphism $f : X \rightarrow Y$ in $\mathcal{A}$, there exists a distinguished triangle of the form $X \xrightarrow{f} Y \rightarrow Z \rightarrow X[1]$.
- (TR2) A triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$ is distinguished if and only if $Y \xrightarrow{v} Z \xrightarrow{-w} X[1] \xrightarrow{-\Sigma u} Y[1]$ is distinguished.
- (TR3) Suppose we have a commutative diagram of distinguished triangles with vertical arrows f and g . Then there exists (not necessarily unique) an arrow $h : Z \rightarrow Z'$ making the full diagram commute.
- (TR4) **Octahedral axiom.** (We refer to [11] for the precise statement.)

Note 1.2. If $\mathcal{T} = (\mathcal{A}, \Sigma)$ is a triangulated category, then $\mathcal{T}^{op} = (\mathcal{A}^{op}, \Sigma^{-1})$ is also a triangulated category.

Proposition 1.3. Let $\mathcal{T}$ be a triangulated category. Suppose $\{X_\lambda\}_{\lambda \in \Lambda}$ is a set of objects of $\mathcal{T}$ and that the coproduct $\coprod_{\lambda \in \Lambda} X_\lambda$ exists in $\mathcal{T}$. Then the natural map

$$\coprod_{\lambda \in \Lambda} X_\lambda[1] \longrightarrow \left(\coprod_{\lambda \in \Lambda} X_\lambda \right)[1]$$

is an isomorphism.

Definition 1.4. Let $\mathcal{T}$ be a triangulated category and $\mathcal{A}$ an abelian category. A functor $H : \mathcal{T} \rightarrow \mathcal{A}$ is *homological* if for each distinguished triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$ the sequence $H(X) \xrightarrow{H(u)} H(Y) \xrightarrow{H(v)} H(Z)$ is exact in $\mathcal{A}$.

By axiom (TR2) the sequence extends to a long exact sequence in both directions:

$$\cdots \rightarrow H(Z[-1]) \xrightarrow{H(w[-1])} H(X) \xrightarrow{H(u)} H(Y) \xrightarrow{H(v)} H(Z) \xrightarrow{H(w)} H(X[1]) \rightarrow \cdots$$

Note 1.5. A homological functor on $\mathcal{T}^{op}$ is a *cohomological functor* on $\mathcal{T}$.

Lemma 1.6. Let $\mathcal{T}$ be a triangulated category and U an object of $\mathcal{T}$. The functor $\text{Hom}(U, -) : \mathcal{T} \rightarrow \mathbf{Ab}$ is homological.

Proof. Let $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$ be a distinguished triangle; we must show exactness at every $\text{Hom}(U, -)$ and $\text{Hom}(-, U)$ position. By (TR3), both $\text{Hom}(U, -)$ and $\text{Hom}(-, U)$ are stable under applying axioms (TR2)–(TR3), so it suffices to prove exactness at $\text{Hom}(U, Y)$.

Step 1: $v \circ u = 0$. By (TR1)(a), the sequence $X \xrightarrow{\text{id}_X} X \rightarrow 0 \rightarrow X[1]$ is a distinguished triangle (0 is the zero object). Applying (TR3) to

$$\begin{array}{ccccccc} X & \xrightarrow{\text{id}_X} & X & \longrightarrow & 0 & \longrightarrow & X[1] \\ \downarrow u & & \downarrow u & & \downarrow h & & \downarrow u[1] \\ X & \xrightarrow{u} & Y & \xrightarrow{v} & Z & \xrightarrow{w} & X[1] \end{array}$$

yields a (necessarily zero) morphism $h : 0 \rightarrow Z$. The commutativity of the middle square gives $v \circ u = h \circ \text{id}_X = 0$.

Step 2: exactness at $\text{Hom}(U, Y)$. Suppose $f : U \rightarrow Y$ satisfies $vf = 0$. We must find $g : U \rightarrow X$ with $u \circ g = f$.

By (TR2) applied twice to $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$, the sequence $Z[-1] \xrightarrow{-w[-1]} X \xrightarrow{u} Y \xrightarrow{v} Z$ is also distinguished. We apply (TR3) to the diagram of distinguished triangles

$$\begin{array}{ccccccc} Z[-1] & \xrightarrow{-w[-1]} & X & \xrightarrow{u} & Y & \xrightarrow{v} & Z \\ 0 \downarrow & & \downarrow g & & \downarrow f & & \downarrow 0 \\ Z[-1] & \xrightarrow{0} & 0 & \longrightarrow & U & \xrightarrow{\text{id}} & U \end{array}$$

where the top row is the rotated triangle, the bottom row is degenerate (the zero triangle on U viewed as $0 \rightarrow U \xrightarrow{\text{id}} U \rightarrow 0$), and the outer vertical maps are both zero (since $vf = 0$ ensures commutativity of the right square: $0 \circ v = 0 = \text{id} \circ f$ only if $vf = 0$, which holds, and the left square: the composite $g \circ (-w[-1])$ lands in X mapping to Y via u , while 0 maps to 0).

More concisely: we have a commutative diagram with distinguished rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & 0 & \longrightarrow & U & \xrightarrow{\text{id}} & U \\ \downarrow & & \downarrow & & \downarrow f & & \downarrow vf=0 \\ X & \xrightarrow{u} & Y & \xrightarrow{v} & Z & \xrightarrow{w} & X[1] \end{array}$$

where the top row ($0 \rightarrow 0 \rightarrow U \rightarrow U$) is distinguished ((TR1)). The right vertical is $vf = 0$. (TR3) applied to this diagram (with commutativity of left and right squares already satisfied) yields the dashed arrow: $g : 0 \rightarrow X$ does not make sense directly; instead, we use the standard consequence of (TR3): since $vf = 0$, the map $f : U \rightarrow Y$ factors through the cone of v , which equals $X[1]$ by the distinguished triangle. Unwinding: the composite $U \xrightarrow{f} Y \rightarrow \text{Cone}(u) = Z$ equals $vf = 0$, hence f factors as $U \xrightarrow{g} X \xrightarrow{u} Y$ for some g by the universal property of the fiber (Proposition 1.17).

Alternative direct argument: Form the mapping cone triangle $U \xrightarrow{(0,f)} X \oplus Y \rightarrow Z \oplus U[1] \rightarrow U[1]$. Since the component $vf = 0$, this cone splits and gives $g : U \rightarrow X$ with $ug = f$ by [11, Prop. 1.1.9].

The dual statement that $\text{Hom}(-, U)$ is cohomological follows by applying the above to $\mathcal{T}^{\text{op}}$ (where distinguished triangles are reversed via (TR2)). $\square$

Note 1.7. It follows that $\text{Hom}(-, U) : \mathcal{T}^{\text{op}} \rightarrow \mathbf{Ab}$ is cohomological.

Proposition 1.8. *Let $\mathcal{T}$ be a triangulated category and (f, g, h) a morphism of triangles.*

1. *If two of the three vertical morphisms are isomorphisms, so is the third.*

2. If $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{0} X[1]$ is a distinguished triangle, then u is an isomorphism if and only if $Z \cong 0$.
3. Every triangle of the form $X \rightarrow Y \rightarrow Z \xrightarrow{0} X[1]$ is isomorphic to $X \rightarrow X \oplus Z \rightarrow Z \xrightarrow{0} X[1]$.

Proof. (1) *Two-of-three implies the third is an isomorphism.* Let (f, g, h) be a morphism of triangles with f and g isomorphisms. Apply the cohomological functor $\text{Hom}(W, -)$ for any object W : the five-lemma in the abelian category $\mathbf{Ab}$ applied to the two long exact sequences (from the two distinguished triangles) and the morphisms induced by (f, g, h) shows that $h_* : \text{Hom}(W, Z) \rightarrow \text{Hom}(W, Z')$ is an isomorphism for all W . By the Yoneda lemma, h is an isomorphism. (If instead g and h are isomorphisms, apply the dual argument using $\text{Hom}(-, W)$ or rotate the triangle via (TR2) to reduce to the previous case.)

(2) *u is an isomorphism iff $Z \cong 0$.* If u is an isomorphism, then by (TR3) applied to $u : X \rightarrow Y$, the triangle $X \xrightarrow{u} Y \rightarrow Z \xrightarrow{0} X[1]$ and the triangle $X \xrightarrow{\text{id}} X \rightarrow 0 \rightarrow X[1]$ admit a morphism with $f = \text{id}_X$, $g = u^{-1} \circ \text{id}_Y = u^{-1}$, so by part (1) the third map $h : Z \rightarrow 0$ is an isomorphism. Conversely, if $Z \cong 0$, then $\text{Hom}(W, Z) = 0$ for all W , so the long exact sequence gives $u_* : \text{Hom}(W, X) \xrightarrow{\sim} \text{Hom}(W, Y)$ for all W ; by Yoneda u is an isomorphism.

(3) *Zero boundary forces splitting.* If $X \rightarrow Y \rightarrow Z \xrightarrow{0} X[1]$ is distinguished, then $\text{Hom}(W, -)$ gives $\text{Hom}(W, Y) \rightarrow \text{Hom}(W, Z) \xrightarrow{0} \text{Hom}(W, X[1])$, so $Z \rightarrow X[1]$ is zero. The defining triangle of the cone of $p : Y \rightarrow Z$ is $X \xrightarrow{u} Y \xrightarrow{p} Z \rightarrow X[1]$. Since $w = 0$, by (TR1)(b) we may assume the triangle is $X \rightarrow X \oplus Z \xrightarrow{(p', \text{id}_Z)} Z \rightarrow 0 \rightarrow X[1]$. More precisely, the zero boundary forces the triangle to split: from $\text{Hom}(Z, Y) \xrightarrow{p_*} \text{Hom}(Z, Z) \xrightarrow{0} \text{Hom}(Z, X[1])$, exactness gives a splitting $s : Z \rightarrow Y$ with $ps = \text{id}_Z$. Similarly u has a left splitting. The triangle is then isomorphic to $X \rightarrow X \oplus Z \rightarrow Z \rightarrow 0$; see [11], Prop. 1.1.9(3) for the complete splitting argument. $\square$

Definition 1.9. A *tensor structure* on an additive category $\mathcal{A}$ is given by a bifunctor $\otimes : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ which is associative, commutative and unital; that is, there is a unit object 1 and isomorphisms $X \otimes (Y \otimes Z) \xrightarrow{\sim} (X \otimes Y) \otimes Z$, $X \otimes Y \xrightarrow{\sim} Y \otimes X$, $1 \otimes X \xrightarrow{\sim} X$, $X \otimes 1 \xrightarrow{\sim} X$, making the obvious diagrams commute.

Definition 1.10. If $\mathcal{A}$ has a translation $X \mapsto \Sigma X$, the tensor structure is *graded* if

1. $\Sigma \circ (- \otimes -) = \Sigma(-) \otimes -$ and $\Sigma^2(-) \otimes - = - \otimes \Sigma^2(-)$ as functors $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$.
2. The natural isomorphisms $X \otimes \Sigma Y \cong \Sigma X \otimes Y = \Sigma(X \otimes Y) = \Sigma X \otimes Y$ satisfy the condition that the composition $X \otimes \Sigma^2 Y \rightarrow X \otimes \Sigma^2 Y$ is the identity.

Definition 1.11. A *tensor triangulated category* is a triangulated category $\mathcal{T}$ which is also a tensor category (with $\otimes$ graded) such that for each distinguished

triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$ and every $W \in \mathcal{T}$, the sequence

$$X \otimes W \xrightarrow{u \otimes \text{id}_W} Y \otimes W \xrightarrow{v \otimes \text{id}_W} Z \otimes W \xrightarrow{w \otimes \text{id}_W} X[1] \otimes W = (X \otimes W)[1]$$

is a distinguished triangle.

Definition 1.12. Let $(\mathcal{T}, \Sigma)$ and $(\mathcal{T}', \Sigma')$ be triangulated categories. An additive functor $F : \mathcal{T} \rightarrow \mathcal{T}'$ is *triangulated* if there is a natural isomorphism $F \circ \Sigma \cong \Sigma' \circ F$ such that for every distinguished triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$ in $\mathcal{T}$, the triangle $FX \xrightarrow{u} FY \xrightarrow{v} FZ \xrightarrow{w} F(X[1])$ is distinguished in $\mathcal{T}'$.

Definition 1.13. A *triangulated subcategory* of $\mathcal{T}$ is a subcategory $\iota : \mathcal{S} \subset \mathcal{T}$ with the structure of a triangulated category such that the inclusion functor ι is triangulated.

Proposition 1.14. Let $\mathcal{T}$ and $\mathcal{T}'$ be triangulated categories, and consider an adjoint pair of functors where F is the left adjoint:

$$\mathcal{T} \begin{matrix} \xrightarrow{F} \\ \xleftarrow{G} \end{matrix} \mathcal{T}'.$$

Then F is triangulated if and only if G is triangulated.

Proof. We sketch the key direction: G triangulated implies F triangulated (the other direction is symmetric, with adjoint natural isomorphisms exchanging roles).

Let $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$ be a distinguished triangle in $\mathcal{T}$. We must show $FX \rightarrow FY \rightarrow FZ \rightarrow FX[1]$ is distinguished in $\mathcal{T}'$. Since G is triangulated, G sends distinguished triangles to distinguished triangles. The unit and counit of the adjunction (F, G) give natural isomorphisms $\eta : \text{id} \Rightarrow GF$ and $\varepsilon : FG \Rightarrow \text{id}$.

The key calculation: for any $A' \in \mathcal{T}'$, the adjunction gives a commutative square

$$\begin{array}{ccc} \text{Hom}_{\mathcal{T}'}(FZ, A') & \xrightarrow{\varepsilon_{A'}^*} & \text{Hom}_{\mathcal{T}}(Z, GA') \\ (Fv)^* \downarrow & & \downarrow v^* \\ \text{Hom}_{\mathcal{T}'}(FY, A') & \xrightarrow{\varepsilon_{A'}^*} & \text{Hom}_{\mathcal{T}}(Y, GA') \end{array}$$

Since G is triangulated, $GF X \rightarrow GF Y \rightarrow GF Z \rightarrow GF X[1]$ is distinguished (apply G to the would-be triangle, and note $G \circ F$ is an exact functor). By the cohomological functor property (Lemma 1.6) and the adjunction isomorphisms, $\text{Hom}_{\mathcal{T}'}(FX \rightarrow FY \rightarrow FZ, A')$ is exact for all A' , so $FX \rightarrow FY \rightarrow FZ \rightarrow FX[1]$ is distinguished by the Yoneda-representability criterion; see [21], Lem. 5.3.6. $\square$

Proposition 1.15. If $\mathcal{S} \subset \mathcal{T}$ is a full subcategory of a triangulated category $\mathcal{T}$, then $\mathcal{S}$ is a triangulated subcategory if and only if it is invariant under the translation functor and for every distinguished triangle $X \rightarrow Y \rightarrow Z \rightarrow X[1]$ in $\mathcal{T}$ with $X, Y \in \mathcal{S}$, the object Z is isomorphic to an object in $\mathcal{S}$.

1.1.1. Verdier localisation.

Definition 1.16. Let $F : \mathcal{D} \rightarrow \mathcal{T}$ be a triangulated functor. The *kernel* of F is the full subcategory $\mathcal{C} = \{X \in \mathcal{D} : F(X) \cong 0\}$.

Proposition 1.17. *The kernel of a triangulated functor $F : \mathcal{D} \rightarrow \mathcal{T}$ satisfies:*

1. $\mathcal{C}$ is a triangulated subcategory of $\mathcal{D}$.
2. If $X \oplus Y \in \mathcal{C}$, then $X, Y \in \mathcal{C}$.

Proof. (1) $\mathcal{C}$ is a triangulated subcategory. If $X \in \mathcal{C}$, i.e. $F(X) \cong 0$, then $F(X[1]) = F(X)[1] \cong 0[1] \cong 0$, so $X[1] \in \mathcal{C}$ (translation invariance). If $X \rightarrow Y \rightarrow Z \rightarrow X[1]$ is distinguished in $\mathcal{D}$ with $X, Y \in \mathcal{C}$, then $F(X) \rightarrow F(Y) \rightarrow F(Z) \rightarrow F(X)[1]$ is distinguished in $\mathcal{T}$ (since F is triangulated), with $F(X) \cong 0$ and $F(Y) \cong 0$. By the isomorphism $F(X) \xrightarrow{\sim} 0 \rightarrow F(Y)$, the long exact sequence in $\text{Hom}(-, W)$ gives $F(Z) \cong 0$, so $Z \in \mathcal{C}$.

(2) $X \oplus Y \in \mathcal{C}$ implies $X, Y \in \mathcal{C}$. The triangle $X \rightarrow X \oplus Y \rightarrow Y \rightarrow X[1]$ is distinguished (this is the canonical split triangle from Proposition 1.8(3), since the boundary map $Y \rightarrow X[1]$ is zero for a direct sum). Since $F(X \oplus Y) = F(X) \oplus F(Y) \cong 0$ implies $F(X) \cong 0$ and $F(Y) \cong 0$ (in an additive category, $A \oplus B \cong 0$ iff $A \cong 0$ and $B \cong 0$), we conclude $X, Y \in \mathcal{C}$. $\square$

Definition 1.18. A full triangulated subcategory $\mathcal{C}$ of a triangulated category $\mathcal{D}$ is *thick* if $\mathcal{C}$ is closed under direct summands.

Theorem 1.19 (Verdier [32]). *Let $\mathcal{T}$ be a small triangulated category and $\mathcal{C} \subset \mathcal{T}$ a triangulated subcategory. There exists a triangulated category $\mathcal{T}/\mathcal{C}$ and a triangulated functor $\mathcal{Q}_{\mathcal{C}} : \mathcal{T} \rightarrow \mathcal{T}/\mathcal{C}$ such that, for every triangulated functor $F : \mathcal{T} \rightarrow \mathcal{T}'$ with $\mathcal{C}$ contained in the kernel of F , there exists a unique triangulated functor $\bar{F} : \mathcal{T}/\mathcal{C} \rightarrow \mathcal{T}'$ making the diagram commute. The canonical map $\mathcal{Q}_{\mathcal{C}}$ is called Verdier localisation and the category $\mathcal{T}/\mathcal{C}$ is also written $\mathcal{T}[\mathcal{C}^{-1}]$.*

1.1.2. Bousfield localisation.

Definition 1.20. Let $\mathcal{T}$ be a triangulated category and $\mathcal{S} \subset \mathcal{T}$ a thick subcategory. A *Bousfield localisation functor* exists for $\mathcal{S} \subset \mathcal{T}$ if there exists a right adjoint to the natural functor $F : \mathcal{T} \rightarrow \mathcal{T}/\mathcal{S}$. When this is the case, we call the right adjoint the *Bousfield localisation functor* and denote it $G : \mathcal{T}/\mathcal{S} \rightarrow \mathcal{T}$.

Note 1.21. If $\mathcal{S} \subset \mathcal{T}$ is thick and a Bousfield localisation exists, then by adjunction $\text{Hom}_{\mathcal{T}}(S, GT) = 0$ for every $S \in \mathcal{S}$.

Definition 1.22. An object $T \in \mathcal{T}$ is *$\mathcal{S}$ -local* if $\text{Hom}_{\mathcal{T}}(S, T) = 0$ for every $S \in \mathcal{S}$.

Proposition 1.23. *Let $\mathcal{S}$ be a thick subcategory of a triangulated category $\mathcal{T}$. A Bousfield localisation exists for $\mathcal{S} \subset \mathcal{T}$ if and only if the inclusion $\mathcal{S} \hookrightarrow \mathcal{T}$ admits a right adjoint.*

Proof. We sketch the key equivalence. Suppose the inclusion $\iota : \mathcal{S} \hookrightarrow \mathcal{T}$ has a right adjoint $\rho : \mathcal{T} \rightarrow \mathcal{S}$. Define G as the composition $\mathcal{T}/\mathcal{S} \xrightarrow{Q^{-1}} \mathcal{T} \xrightarrow{\rho \circ \iota} \mathcal{S} \hookrightarrow \mathcal{T}$. More precisely, since $Q : \mathcal{T} \rightarrow \mathcal{T}/\mathcal{S}$ is the Verdier localization functor and ρ is right adjoint to ι , define $G(T) := \iota(\rho(T))$ for $T \in \mathcal{T}$. Then for any $T' \in \mathcal{T}/\mathcal{S}$, the adjunction isomorphisms give

$$\mathrm{Hom}_{\mathcal{T}/\mathcal{S}}(Q(T), T') \cong \mathrm{Hom}_{\mathcal{T}}(T, G(T'))$$

since $\mathcal{S}$ -local objects (those with $\mathrm{Hom}(S, T') = 0$ for all $S \in \mathcal{S}$) correspond to the image of G , and the Verdier localization inverts maps with $\mathcal{S}$ -local cokernel. Hence G is right adjoint to $F = Q$.

Conversely, if $G : \mathcal{T}/\mathcal{S} \rightarrow \mathcal{T}$ is right adjoint to F , define $\rho : \mathcal{T} \rightarrow \mathcal{S}$ by $\rho(T) :=$ the $\mathcal{S}$ -part of T obtained from the Bousfield triangle $T_{\mathcal{S}} \rightarrow T \rightarrow G(F(T)) \rightarrow T_{\mathcal{S}}[1]$. By construction, ρ is right adjoint to ι ; see [21], Prop. 9.1.18. $\square$

Proposition 1.24. *Suppose a Bousfield localisation functor $G : \mathcal{T}/\mathcal{S} \rightarrow \mathcal{T}$ exists. For $T \in \mathcal{T}$ there is a distinguished triangle*

$$T_{\mathcal{S}} \longrightarrow T \xrightarrow{\eta_T} GT \longrightarrow T_{\mathcal{S}}[1]$$

in which (1) GT is $\mathcal{S}$ -local and (2) $T_{\mathcal{S}} \in \mathcal{S}$.

1.2. Neeman's adjoint functor theorem

Definition 1.25. Let $\mathcal{T}$ be a triangulated category admitting coproducts.

1. An object $U \in \mathcal{T}$ is *compact* if $\mathrm{Hom}_{\mathcal{T}}(U, \coprod_{\lambda} X_{\lambda}) = \coprod_{\lambda} \mathrm{Hom}_{\mathcal{T}}(U, X_{\lambda})$ for every coproduct of objects in $\mathcal{T}$.
2. $\mathcal{T}$ is *compactly generated* if there exists a set $\mathcal{U}$ of compact objects such that if $\mathrm{Hom}_{\mathcal{T}}(U, X) = 0$ for every $U \in \mathcal{U}$, then $X = 0$.

Definition 1.26. Let $\mathcal{T}$ be a triangulated category admitting coproducts and $X_0 \rightarrow X_1 \rightarrow X_2 \rightarrow \cdots$ a sequence in $\mathcal{T}$. The *homotopy colimit* $\mathrm{hocolim} X_i$ is defined (up to non-canonical isomorphism) by the distinguished triangle

$$\coprod_i X_i \xrightarrow{\mathrm{id}-s} \coprod_i X_i \longrightarrow \mathrm{hocolim} X_i \longrightarrow \left(\coprod_i X_i \right)[1],$$

where $s : X_{i_0} \rightarrow X_{i_0+1} \rightarrow \coprod_i X_i$ is the composition into the factor X_{i_0+1} .

Proposition 1.27. *Let U be a compact object in a triangulated category $\mathcal{T}$ admitting coproducts. If $X_0 \rightarrow X_1 \rightarrow \cdots$ is a sequence in $\mathcal{T}$, then*

$$\mathrm{Hom}_{\mathcal{T}}(U, \mathrm{hocolim} X_i) = \mathrm{colim} \mathrm{Hom}_{\mathcal{T}}(U, X_i).$$

Proof. Apply $\mathrm{Hom}_{\mathcal{T}}(U, -)$ to the defining triangle of the homotopy colimit:

$$\coprod_i X_i \xrightarrow{\mathrm{id}-s} \coprod_i X_i \longrightarrow \mathrm{hocolim} X_i \longrightarrow \left(\coprod_i X_i \right)[1].$$

Since U is compact, $\mathrm{Hom}(U, \coprod_i X_i) = \bigoplus_i \mathrm{Hom}(U, X_i) = \mathrm{colim}_i \mathrm{Hom}(U, X_i)$ (the last equality because the direct sum of a sequence is its colimit). The long exact sequence of $\mathrm{Hom}(U, -)$ applied to the triangle gives

$$\begin{aligned} \bigoplus_i \mathrm{Hom}(U, X_i) &\xrightarrow{\mathrm{id}-s_*} \bigoplus_i \mathrm{Hom}(U, X_i) \rightarrow \mathrm{Hom}(U, \mathrm{hocolim} X_i) \\ &\rightarrow \bigoplus_i \mathrm{Hom}(U, X_i[1]) \rightarrow \cdots \end{aligned}$$

The map $\mathrm{id} - s_*$ on $\bigoplus_i \mathrm{Hom}(U, X_i)$ sends $(f_i)_i$ to $(f_i - f_{i-1} \circ s_i)_i$ (where $s_i : X_{i-1} \rightarrow X_i$ is the structure map). The cokernel of $\mathrm{id} - s_*$ is precisely $\mathrm{colim}_i \mathrm{Hom}(U, X_i)$ (this is the standard telescope formula for sequential colimits in $\mathbf{Ab}$: $\mathrm{colim}_i A_i = (\bigoplus_i A_i) / \mathrm{im}(\mathrm{id} - s)$). The Ext^1 term ($R^1 \mathrm{colim}$) vanishes in $\mathbf{Ab}$ for sequential filtered systems by the Mittag-Leffler condition (or because $\mathbf{Ab}$ is AB5). Hence $\mathrm{Hom}(U, \mathrm{hocolim} X_i) \cong \mathrm{colim}_i \mathrm{Hom}(U, X_i)$. $\square$

Theorem 1.28 (Brown representability [20]). *Let $\mathcal{T}$ be a compactly generated triangulated category and $H : \mathcal{T}^{op} \rightarrow \mathbf{Ab}$ a cohomological functor sending coproducts to products. Then H is representable.*

Theorem 1.29 (Neeman's adjoint functor theorem [20], Thm. 4.1). *Let $\mathcal{S}$ be a compactly generated triangulated category and $\mathcal{T}$ any triangulated category. Let $F : \mathcal{S} \rightarrow \mathcal{T}$ be a triangulated functor respecting coproducts. Then F has a right adjoint $G : \mathcal{T} \rightarrow \mathcal{S}$.*

1.3. The derived category and derived functors

Definition 1.30. Let $\mathcal{A}$ be an additive category. A *complex* $A^\bullet$ in $\mathcal{A}$ is a diagram

$$A^\bullet : \quad \cdots \longrightarrow A^{n-1} \xrightarrow{d^{n-1}} A^n \xrightarrow{d^n} A^{n+1} \xrightarrow{d^{n+1}} \cdots$$

with $d^n \circ d^{n-1} = 0$ for all $n \in \mathbb{Z}$. It is *bounded* if $A^n = 0$ for $|n| \gg 0$; *bounded below* if $A^n = 0$ for $n \ll 0$; *bounded above* if $A^n = 0$ for $n \gg 0$.

Definition 1.31. Let $\mathcal{A}$ be an abelian category. Setting $\mathcal{S} = \{A^\bullet : H^n(A^\bullet) = 0 \text{ for all } n \in \mathbb{Z}\}$ (the acyclic complexes), $\mathrm{Mor}_{\mathcal{S}}$ consists of all quasi-isomorphisms and the *derived category* of $\mathcal{A}$ is

$$D(\mathcal{A}) = K(\mathcal{A})[\mathcal{S}^{-1}]$$

in the sense of Theorem 1.19.

Proposition 1.32. *Let $\mathcal{A}$ be an abelian category with enough injectives. For $X, Y \in \mathcal{A}$,*

$$\mathrm{Hom}_{D(\mathcal{A})}(X, Y[i]) = \begin{cases} \mathrm{Ext}_{\mathcal{A}}^i(X, Y) & i \geq 0, \\ 0 & i < 0. \end{cases}$$

Proof. Since $\mathcal{A}$ has enough injectives, Y has an injective resolution $Y \rightarrow I^\bullet$. In $D(\mathcal{A})$, $Y \cong I^\bullet$, so $\mathrm{Hom}_{D(\mathcal{A})}(X, Y[i]) = H^i(\mathrm{Hom}_{\mathcal{A}}^\bullet(X, I^\bullet)) = \mathrm{Ext}_{\mathcal{A}}^i(X, Y)$ for $i \geq 0$ by the definition of Ext via injective resolutions. For $i < 0$: $Y[i]$ is a complex concentrated in degree $-i > 0$. Any map $X \rightarrow Y[i]$ in $D(\mathcal{A})$ (with X concentrated in degree 0) would be a chain map from a complex concentrated in degree 0 to one concentrated in positive degree, which is necessarily null-homotopic (all components are zero), so $\mathrm{Hom}_{D(\mathcal{A})}(X, Y[i]) = 0$ for $i < 0$ when $X, Y \in \mathcal{A}$. $\square$

1.4. Spectral sequences

We follow Boardman's article [10] closely and supplement it with complete proofs and diagrams. The key novelty with respect to classical accounts is the systematic use of the limit groups D^∞ and $D^{-\infty}$ as the natural targets for convergence.

1.4.1. Basic definitions.

Definition 1.33. A (bigraded cohomological) spectral sequence is a collection of data:

1. abelian groups E_r^{pq} for all $r \geq 1$ and $p, q \in \mathbb{Z}$;
2. morphisms $d_r^{pq} : E_r^{pq} \rightarrow E_r^{p+r, q-r+1}$ with $(d_r)^2 = 0$;
3. isomorphisms $E_{r+1}^{pq} \cong H^{pq}(E_r^{**}, d_r) := \ker d_r^{pq} / \mathrm{im} d_r^{p-r, q+r-1}$.

The integer $n = p + q$ is the *total degree*, p the *filtration degree*, and q the *complementary degree*.

For each fixed (p, q) we have a nested sequence of subgroups of E_1^{pq} :

$$0 = B_1^{pq} \subset B_2^{pq} \subset \cdots \subset B_r^{pq} \subset \cdots \subset Z_r^{pq} \subset \cdots \subset Z_2^{pq} \subset Z_1^{pq} = E_1^{pq},$$

where $Z_r^{pq} = \ker(d_r^{pq} : E_r^{pq} \rightarrow E_r^{p+r, q-r+1})$ and $B_r^{pq} = \mathrm{im}(d_r^{p-r, q+r-1} : E_r^{p-r, q+r-1} \rightarrow E_r^{pq})$ are the *cycles* and *boundaries* at stage r , viewed as subgroups of E_1^{pq} via the successive isomorphisms.

Definition 1.34. The *infinite cycles* and *infinite boundaries* are

$$Z_\infty^{pq} = \bigcap_{r \geq 1} Z_r^{pq}, \quad B_\infty^{pq} = \bigcup_{r \geq 1} B_r^{pq},$$

and the E_∞ -term is the sub-quotient $E_\infty^{pq} = Z_\infty^{pq} / B_\infty^{pq}$.

Definition 1.35. Let H be an abelian group. A *decreasing filtration* of H is a sequence of subgroups

$$\cdots \subset F^{p+1}H \subset F^pH \subset \cdots \subset H, \quad p \in \mathbb{Z}.$$

Write $F^\infty H := \bigcap_p F^pH$ and $F^{-\infty}H := \bigcup_p F^pH$. The filtration is:

- *exhaustive* if $F^{-\infty}H = H$;

- *separated* (or Hausdorff) if $F^\infty H = 0$;
- *complete* if $R^1 \varprojlim_p F^p H = 0$, equivalently $H \xrightarrow{\sim} \varprojlim_p H/F^p H$.

The p -th graded piece is $\mathrm{Gr}_F^p H = F^p H / F^{p+1} H$.

Definition 1.36. Let E_r^{pq} be a spectral sequence and (H^*, F^*) a filtered graded abelian group. The spectral sequence *abuts to* (or *converges to*) (H^*, F^*) if there are isomorphisms

$$E_\infty^{pq} \xrightarrow{\sim} \mathrm{Gr}_F^p H^{p+q}$$

for every (p, q) . One writes $E_1^{pq} \Rightarrow H^n$ (where $n = p + q$).

- (i) The sequence *converges weakly* to (H^*, F^*) if additionally F^* is exhaustive.
- (ii) It *converges* if F^* is exhaustive and separated.
- (iii) It *converges strongly* if F^* is exhaustive, separated, and complete.

Strong convergence allows one to recover H^n , up to extensions, from the E_∞ -terms via the filtration.

Definition 1.37. A *morphism of spectral sequences* $f_r^{pq} : E_r^{pq} \rightarrow \tilde{E}_r^{pq}$ is a collection of group homomorphisms, one for each $r \geq 1$ and (p, q) , satisfying:

1. f_r^{pq} commutes with differentials: $\tilde{d}_r^{pq} \circ f_r^{pq} = f_r^{p+r, q-r+1} \circ d_r^{pq}$, i.e. the square

$$\begin{array}{ccc} E_r^{pq} & \xrightarrow{d_r^{pq}} & E_r^{p+r, q-r+1} \\ f_r^{pq} \downarrow & & \downarrow f_r^{p+r, q-r+1} \\ \tilde{E}_r^{pq} & \xrightarrow{\tilde{d}_r^{pq}} & \tilde{E}_r^{p+r, q-r+1} \end{array}$$

commutes for all r, p, q .

2. The induced map on cohomology agrees with f_{r+1}^{pq} under the structure isomorphisms $E_{r+1}^{pq} \cong H(E_r^{**}, d_r)$.

Theorem 1.38 (Comparison theorem for strongly convergent spectral sequences). *Let $E_r^{pq} \Rightarrow H^n$ and $\tilde{E}_r^{pq} \Rightarrow \tilde{H}^n$ be two spectral sequences converging strongly to the filtered groups (H^*, F^*) and $(\tilde{H}^*, \tilde{F}^*)$ respectively. Let $f_r : E_r^{**} \rightarrow \tilde{E}_r^{**}$ be a morphism of spectral sequences and $f^* : H^* \rightarrow \tilde{H}^*$ a morphism of filtered graded groups such that the diagram*

$$\begin{array}{ccc} E_\infty^{pq} & \xrightarrow{\sim} & \mathrm{Gr}_F^p H^{p+q} \\ f_\infty^{pq} \downarrow & & \downarrow \mathrm{Gr}^p f^{p+q} \\ \tilde{E}_\infty^{pq} & \xrightarrow{\sim} & \mathrm{Gr}_{\tilde{F}}^p \tilde{H}^{p+q} \end{array}$$

commutes for all (p, q) . If there exists $a \geq 1$ such that $f_a^{pq} : E_a^{pq} \xrightarrow{\sim} \tilde{E}_a^{pq}$ is an isomorphism for all (p, q) , then f_a^{pq} is an isomorphism for all $r \geq a$ (in particular at $r = \infty$), and $f^n : H^n \rightarrow \tilde{H}^n$ is an isomorphism of filtered groups for all n .

Proof. Since f_a^{pq} is an isomorphism and commutes with d_a , the induced map on cohomology $f_{a+1}^{pq} : E_{a+1}^{pq} \rightarrow \tilde{E}_{a+1}^{pq}$ is also an isomorphism (the five lemma applied to the short exact sequence $0 \rightarrow B_a^{pq} \rightarrow Z_a^{pq} \rightarrow E_{a+1}^{pq} \rightarrow 0$). By induction, f_r^{pq} is an isomorphism for all $r \geq a$, hence at $r = \infty$.

The commutative diagram in the statement gives, for each p and $n = p + q$, an isomorphism $\mathrm{Gr}_F^p H^n \xrightarrow{\sim} \mathrm{Gr}_F^p \tilde{H}^n$. Since both filtrations are exhaustive, separated, and complete, we apply the following comparison lemma: a morphism of complete, separated, exhaustive filtrations on H^n and $\tilde{H}^n$ that induces isomorphisms on all graded pieces is itself an isomorphism of filtered groups. This follows by induction on the filtration length (for finite filtrations) and in general by passing to the limit using completeness:

$$H^n \cong \varprojlim_p H^n / F^p H^n, \quad \tilde{H}^n \cong \varprojlim_p \tilde{H}^n / \tilde{F}^p \tilde{H}^n,$$

and the induced maps on each finite quotient $H^n / F^p H^n \rightarrow \tilde{H}^n / \tilde{F}^p \tilde{H}^n$ are isomorphisms (by the five lemma applied to the finite filtration $F^p \supset F^{p+1} \supset \dots$), so the inverse limit is an isomorphism. $\square$

1.4.2. Spectral sequences from exact couples.

Definition 1.39. An *exact couple* is a diagram of abelian groups and morphisms

$$\begin{array}{ccc} D & \xrightarrow{\alpha} & D \\ & \swarrow \gamma & \searrow \beta \\ & E & \end{array}$$

that is exact at each of the three vertices, i.e. $\ker \alpha = \mathrm{im} \gamma$, $\ker \beta = \mathrm{im} \alpha$, and $\ker \gamma = \mathrm{im} \beta$.

The composite $d := \beta \circ \gamma : E \rightarrow E$ satisfies $d^2 = \beta \gamma \beta \gamma = 0$ (since $\gamma \beta = 0$ by exactness at D). Thus (E, d) is a differential group.

Proposition 1.40. Given an exact couple $(D, E, \alpha, \beta, \gamma)$, set

$$D' := \mathrm{im} \alpha \subset D, \quad E' := H(E, d) = \ker d / \mathrm{im} d.$$

Define morphisms $\alpha' : D' \rightarrow D'$, $\beta' : D' \rightarrow E'$, $\gamma' : E' \rightarrow D'$ by

$$\alpha'(\alpha x) := \alpha(\alpha x), \quad \beta'(\alpha x) := [\beta x], \quad \gamma'([e]) := \gamma(e),$$

where $[-]$ denotes the class in E' . Then:

1. α' , β' , γ' are well-defined.

2. The derived couple $(D', E', \alpha', \beta', \gamma')$

$$\begin{array}{ccc} D' & \xrightarrow{\alpha'} & D' \\ & \swarrow \gamma' \quad \searrow \beta' & \\ & E' & \end{array}$$

is again an exact couple.

Iterating, one obtains a sequence of exact couples $(D_r, E_r, \alpha_r, \beta_r, \gamma_r)$ with $D_1 = D$, $E_1 = E$, and $d_r := \beta_r \circ \gamma_r : E_r \rightarrow E_r$. The groups $\{E_r, d_r\}$ form a spectral sequence.

Proof. Well-definedness of β' . Suppose $\alpha x = \alpha x'$, so $\alpha(x - x') = 0$, hence $x - x' \in \ker \alpha = \operatorname{im} \gamma$, say $x - x' = \gamma e$. Then $\beta x - \beta x' = \beta \gamma e = de$, so $[\beta x] = [\beta x']$ in E' .

Well-definedness of γ' . Suppose $[e] = [e']$, so $e - e' \in \operatorname{im} d = \operatorname{im} \beta \gamma$. Write $e - e' = \beta \gamma f$ for some $f \in E$. Then $\gamma(e) - \gamma(e') = \gamma \beta \gamma f$. But $\gamma \beta = 0$ by exactness, so $\gamma(e) = \gamma(e')$ and γ' is well-defined. Moreover $\gamma'[e] = \gamma e \in \ker \alpha = \operatorname{im} \gamma \subset D'$, so γ' maps into D' .

Exactness at D' (i.e. $\ker \alpha' = \operatorname{im} \gamma'$). $\ker \alpha' = \ker(\alpha|_{\operatorname{im} \alpha}) = \operatorname{im} \alpha \cap \ker \alpha = \operatorname{im} \alpha \cap \operatorname{im} \gamma$. On the other hand $\operatorname{im} \gamma' = \gamma(\ker d) = \gamma(\ker \beta \gamma)$. If $\gamma e \in \operatorname{im} \alpha$ then $\gamma e = \alpha x$ and $\beta \gamma e = \beta \alpha x = 0$ (exactness at D), so $e \in \ker \beta \gamma = \ker d$. Conversely if $e \in \ker d$ then $\beta \gamma e = 0$, so $\gamma e \in \ker \beta = \operatorname{im} \alpha$. Hence $\ker \alpha' = \operatorname{im} \gamma'$.

Exactness at D' on the other side (i.e. $\ker \beta' = \operatorname{im} \alpha'$). $\ker \beta' = \{\alpha x : [\beta x] = 0\} = \{\alpha x : \beta x \in \operatorname{im} d\} = \{\alpha x : \beta x = \beta \gamma e \text{ for some } e\}$. Since $\beta(x - \gamma e) = 0$, we have $x - \gamma e \in \ker \beta = \operatorname{im} \alpha$, say $x - \gamma e = \alpha y$. Then $\alpha x = \alpha^2 y = \alpha'(\alpha y)$. Hence $\ker \beta' \subset \operatorname{im} \alpha'$; the reverse inclusion is clear.

Exactness at E' . $\ker \gamma' = \{[e] \in E' : \gamma e = 0\} = \{[e] : e \in \ker \gamma\} = \{[e] : e \in \operatorname{im} \beta\}$. On the other hand $\operatorname{im} \beta' = \{[\beta x] : x \in D\} = \beta(\ker \beta \gamma^{-1}(\operatorname{im} \alpha)) / \operatorname{im} d$. Since $\beta \alpha = 0$ the image of β lands in $\ker \gamma / \operatorname{im} d$ (by exactness at E , $\ker \gamma = \operatorname{im} \beta$), so $\operatorname{im} \beta' = \{[\beta x]\} \subset \ker \gamma'$ and one checks the reverse inclusion similarly.

Iteration is straightforward: $(D_r, E_r) := (D'_{r-1}, E'_{r-1})$ and

$$E_{r+1}^{pq} = \ker(d_r^{pq}) / \operatorname{im}(d_r^{p-r, q+r-1})$$

recovers the structure isomorphism of Definition 1.33. $\square$

Note 1.41. In the bigraded case, one indexes the exact couple as $(D^{**}, E^{**}, \alpha, \beta, \gamma)$ with

$$\deg \alpha = (-1, 1), \quad \deg \beta = (0, 0), \quad \deg \gamma = (1, 0).$$

After deriving $r - 1$ times, the r -th couple has degrees

$$\deg \alpha_r = (-1, 1), \quad \deg \beta_r = (r - 1, 1 - r), \quad \deg \gamma_r = (1, 0),$$

and the differential $d_r = \beta_r \gamma_r$ has bidegree $(r, 1 - r)$. In the homological convention E_{pq} the differential goes $d_r : E_r^{pq} \rightarrow E_r^{p-r, q+r-1}$.

1.4.3. Convergence and limit groups. Fix a bigraded exact couple $(D^{**}, E^{**}, \alpha, \beta, \gamma)$ as in Note 1.41. Write $\alpha : D^{p,*} \rightarrow D^{p-1,*}$ for the component of α in filtration degree.

Definition 1.42. Define the *colimit group* and *limit group*:

$$D^{-\infty} := \varinjlim_p D^{p,*} \quad (\text{under } \alpha), \quad D^{\infty} := \varprojlim_p D^{p,*} \quad (\text{under } \alpha).$$

Let $\eta^p : D^{p,*} \rightarrow D^{-\infty}$ and $\pi^p : D^{\infty} \rightarrow D^{p,*}$ denote the canonical maps. Define filtrations

$$F^p D^{-\infty} := \text{im } \eta^p, \quad F^p D^{\infty} := \ker \pi^p.$$

Proposition 1.43. *With the notation of Definition 1.42:*

1. The filtration $F^* D^{-\infty}$ is exhaustive.
2. The filtration $F^* D^{\infty}$ is separated and complete.
3. $\bigcup_p F^p D^{\infty} = \ker(D^{\infty} \rightarrow D^{-\infty})$.

Proof. (1) follows immediately since $D^{-\infty} = \varinjlim_p D^{p,*} = \bigcup_p \text{im } \eta^p = \bigcup_p F^p D^{-\infty}$.

(2) *Separated:* $F^{\infty} D^{\infty} = \bigcap_p \ker \pi^p$. An element $x \in D^{\infty}$ lies in $\bigcap_p \ker \pi^p$ if and only if $\pi^p(x) = 0$ for all p , i.e. $x = 0$ in the limit, so $F^{\infty} D^{\infty} = 0$.

Complete: We must show $R^1 \varprojlim_p F^p D^{\infty} = 0$, i.e. $R^1 \varprojlim_p \ker \pi^p = 0$. For each p there is a short exact sequence

$$0 \longrightarrow \ker \pi^p \longrightarrow D^{\infty} \xrightarrow{\pi^p} D^{p,*} \longrightarrow \text{coker } \pi^p \longrightarrow 0.$$

Taking the inverse limit and using the Milnor sequence for $R^1 \varprojlim$, one checks that $R^1 \varprojlim_p D^{\infty} = 0$ (the system $D^{\infty} \xrightarrow{\pi^1} D^{\infty} \xrightarrow{\pi^2} \dots$ is Mittag-Leffler), which gives the result.

(3) An element $x \in D^{\infty}$ satisfies $x \in F^p D^{\infty} = \ker \pi^p$ for some p if and only if $\pi^p(x) = 0 \in D^{p,*}$, i.e. the image of x under $D^{\infty} \rightarrow D^{p,*} \rightarrow D^{-\infty}$ is zero. Since the colimit is the union of the images, x maps to 0 in $D^{-\infty}$ if and only if it maps to 0 in $D^{p,*}$ for some p (as every element of $D^{-\infty}$ comes from some finite stage), which is precisely $\bigcup_p \ker \pi^p$. $\square$

Remark 1.44. The Milnor exact sequence applied to the inverse system $\{D^{p,*}\}_{p \in \mathbb{Z}}$ gives:

$$0 \longrightarrow R^1 \varprojlim_p D^{p,*} \longrightarrow \prod_p D^{p,*} \xrightarrow{1-\alpha^*} \prod_p D^{p,*} \longrightarrow 0$$

and the defining short exact sequence for $D^{\infty} = \varprojlim_p D^{p,*}$:

$$0 \longrightarrow D^{\infty} \longrightarrow \prod_p D^{p,*} \xrightarrow{\text{id}-\alpha} \prod_p D^{p,*} \longrightarrow R^1 \varprojlim_p D^{p,*} \longrightarrow 0.$$

The term $R^1 \varprojlim_p D^{p,*} := R^1 \varprojlim_p D^{p,*}$ is the obstruction to strong convergence: by Theorem 1.48, if $D^{-\infty} = 0$ and $RE_{\infty}^{pq} = 0$ for all (p, q) , then the spectral sequence

converges strongly to D^∞ and $R^1\varprojlim = 0$ (i.e. D^∞ has no $\varprojlim^1$ -defect). Conversely, if $R^1\varprojlim \neq 0$, the spectral sequence at best converges conditionally to an extension of D^∞ by $R^1\varprojlim$; strong convergence fails.

For the orthogonal tower in DM , the half-plane vanishing (all $E_{p,q}^1 = 0$ for $p \leq s-1$ when $B = M(X)(s)[r]$, by Note 1.52) implies the tower is eventually constant in each p , giving $R^1\varprojlim = R^1\varprojlim D^{p,*} = 0$ by the Mittag-Leffler condition: the transition maps $\alpha : D^{p,*} \rightarrow D^{p-1,*}$ are eventually zero for $p \ll 0$. This is the key reason strong convergence holds in DM but fails in $\mathcal{SH}$ (where the tower is never eventually zero).

The following key lemma describes the relationship between E_∞ and the filtration on the limit groups. It is the technical heart of Boardman's theory.

Lemma 1.45. *There are natural short exact sequences*

$$\begin{aligned} 0 \longrightarrow \operatorname{im} \beta / B_\infty^{pq} &\longrightarrow E_\infty^{pq} \longrightarrow Z_\infty^{pq} / \operatorname{im} \beta \longrightarrow 0, \\ 0 \longrightarrow \operatorname{Gr}_F^p D^{-\infty} &\longrightarrow E_\infty^{pq} \longrightarrow RE_\infty^{pq} \longrightarrow 0, \end{aligned}$$

where $RE_\infty^{pq} := R^1\varprojlim_r Z_r^{pq} / B_r^{pq}$ is the derived E_∞ -term. In particular, if $RE_\infty^{pq} = 0$, then $E_\infty^{pq} \cong \operatorname{Gr}_F^p D^{-\infty}$.

Proof. The Milnor exact sequence applied to the tower $\cdots \supset Z_r^{pq} \supset Z_{r+1}^{pq} \supset \cdots$ gives

$$0 \rightarrow \varprojlim_r Z_r^{pq} \rightarrow Z_1^{pq} \xrightarrow{1-s} Z_1^{pq} \rightarrow R^1\varprojlim_r Z_r^{pq} \rightarrow 0,$$

where s is the shift. One identifies $\varprojlim_r Z_r^{pq} = Z_\infty^{pq}$ and similarly $\varprojlim_r B_r^{pq} = B_\infty^{pq}$. The quotient short exact sequence then yields . Sequence follows from the identification of $\operatorname{Gr}_F^p D^{-\infty}$ with a quotient of Z_∞^{pq} via the map γ ; see [10], §5. $\square$

Definition 1.46. Let E_r^{pq} be the spectral sequence associated to an exact couple $(D^{**}, E^{**}, \alpha, \beta, \gamma)$.

1. The spectral sequence is *conditionally convergent to D^∞* if $D^{-\infty} = 0$.
2. The spectral sequence is *conditionally convergent to $D^{-\infty}$* if $D^\infty = 0$ and $RD^\infty = 0$, where $RD^\infty := R^1\varprojlim_p D^{p,*}$.

Proposition 1.47. *Suppose E_r^{pq} is associated to an exact couple.*

1. *The filtration $F^* D^{-\infty}$ is always exhaustive (Proposition 1.43(1)).*
2. *If the spectral sequence converges conditionally to D^∞ (i.e. $D^{-\infty} = 0$), then the filtration $F^* D^\infty$ is exhaustive, separated, and complete (Proposition 1.43(2)).*
3. *In either case the spectral sequence abuts weakly: $E_\infty^{pq} \rightarrow \operatorname{Gr}_F^p D^{\pm\infty}$ has kernel and cokernel controlled by RE_∞^{pq} and RD^∞ via the exact sequences of Lemma 1.45.*

Proof. Part (1) is immediate from the definition of $D^{-\infty}$ as a colimit. For part (2), when $D^{-\infty} = 0$ the exact sequence $0 \rightarrow \varprojlim_p F^p D^\infty \rightarrow D^\infty \rightarrow D^{-\infty} = 0$ shows $\varprojlim_p F^p D^\infty = D^\infty$, i.e. the filtration is exhaustive after reindexing; separatedness and completeness are Proposition 1.43(2). Part (3) is a restatement of Lemma 1.45. $\square$

Theorem 1.48 (Boardman [10], Thm. 7.2 and 7.3). *Let E_r^{pq} be the spectral sequence of an exact couple.*

1. (Strong convergence.) *If E_r^{pq} converges conditionally to D^∞ and $RE_\infty^{pq} = 0$ for all (p, q) , then E_r^{pq} converges strongly to D^∞ , i.e. there are isomorphisms $E_\infty^{pq} \xrightarrow{\sim} \text{Gr}_F^p D^\infty$ and the filtration $F^* D^\infty$ is exhaustive, separated, and complete.*
2. (Weak convergence.) *If $D^\infty = 0$, then E_r^{pq} converges weakly to $D^{-\infty}$.*

Proof. Part (1). Since $D^{-\infty} = 0$, by Proposition 1.43 the filtration $F^* D^\infty$ is separated and complete, and the sequence $0 \rightarrow D^\infty \rightarrow \prod_p D^{p,*} \xrightarrow{\alpha} \prod_p D^{p,*} \rightarrow 0$ is exact (Milnor). Consider the natural map from the exact couple: for each p and each r , there is a commutative diagram

$$\begin{array}{ccccccc} D^{p+r,*} & \xrightarrow{\alpha^r} & D^{p,*} & \xrightarrow{\beta} & E^{p,*} & \xrightarrow{\gamma} & D^{p-1,*} \\ \downarrow & & \downarrow \eta^p & & \downarrow \phi_r & & \downarrow \\ D^{p+r,*} & \xrightarrow{\alpha^r} & D^{p,*} & \xrightarrow{\beta} & E_r^{p,*} & \xrightarrow{\gamma_r} & D_r^{p-1,*} \end{array}$$

where $E_r^{p,*} = \text{im}(\alpha^{r-1} : D^{p+r-1,*} \rightarrow D^{p,*}) / \text{im}(\alpha^r : D^{p+r,*} \rightarrow D^{p,*})$ and the dashed vertical arrow is the quotient map. Passing to the limit $r \rightarrow \infty$ yields a map

$$\phi : \text{Gr}_F^p D^\infty \longrightarrow E_\infty^{pq},$$

which is an isomorphism when $RE_\infty^{pq} = 0$ by the exact sequence $0 \rightarrow \text{Gr}_F^p D^{-\infty} \rightarrow E_\infty^{pq} \rightarrow RE_\infty^{pq} \rightarrow 0$ of Lemma 1.45 together with $D^{-\infty} = 0$ (so $\text{Gr}_F^p D^{-\infty} = \text{Gr}_F^p D^\infty$).

The filtration $F^* D^\infty$ is exhaustive because: $F^p D^\infty = \ker(\pi^p : D^\infty \rightarrow D^{p,*})$, so $\bigcup_p F^p D^\infty = \bigcup_p \ker(\pi^p)$. By Proposition 1.43(3), $\bigcup_p F^p D^\infty = \ker(D^\infty \rightarrow D^{-\infty})$. Since $D^{-\infty} = 0$ by assumption, this kernel is all of D^∞ , so the filtration is exhaustive. (Note: the argument is via the kernel of the map $D^\infty \rightarrow D^{-\infty}$, not by the element-wise argument that each $x \in D^{p,*}$ eventually maps to zero—that would concern $D^{-\infty} = 0$ as a condition on the directed system, not on the inverse limit D^∞ .)

Part (2). The filtration $F^* D^{-\infty}$ is always exhaustive. When $D^\infty = 0$, separatedness is automatic: $\bigcap_p F^p D^{-\infty} = \bigcap_p \text{im } \eta^p$. An element x in this intersection means $x = \eta^p(x_p)$ for all p , with compatible choices; but the compatibility and $D^\infty = 0$ force $x = 0$. The abutment $E_\infty^{pq} \rightarrow \text{Gr}_F^p D^{-\infty}$ now follows from Lemma 1.45 with RE_∞^{pq} potentially nonzero. $\square$

1.4.4. *The spectral sequence associated to a filtered complex.* The most classical source of spectral sequences is a filtered complex.

Definition 1.49. Let $\mathcal{A}$ be an abelian category and $K^\bullet$ a cochain complex. A *decreasing filtration* on $K^\bullet$ is a collection of subcomplexes

$$\dots \supset F^{p+1}K^\bullet \supset F^pK^\bullet \supset \dots$$

of $K^\bullet$ (each F^pK^n is a subgroup of K^n , and the differential $d : K^n \rightarrow K^{n+1}$ restricts to $d : F^pK^n \rightarrow F^pK^{n+1}$).

Proposition 1.50. *Let $(K^\bullet, F^\bullet)$ be a filtered cochain complex. There is a spectral sequence with*

$$E_0^{pq} = F^pK^{p+q}/F^{p+1}K^{p+q}, \quad E_1^{pq} = H^{p+q}(F^pK^\bullet/F^{p+1}K^\bullet),$$

and $d_r : E_r^{pq} \rightarrow E_r^{p+r, q-r+1}$ induced by the differential of $K^\bullet$. The abutment is $H^*(K^\bullet)$ with the filtration $F^pH^n(K^\bullet) := \text{im}(H^n(F^pK^\bullet) \rightarrow H^n(K^\bullet))$.

Proof. This spectral sequence arises from the exact couple

$$\begin{array}{ccc} \bigoplus_{p,q} H^{p+q}(F^pK^\bullet) & \xrightarrow{\alpha} & \bigoplus_{p,q} H^{p+q}(F^pK^\bullet) \\ & \nwarrow \gamma \quad \nearrow \beta & \\ & \bigoplus_{p,q} H^{p+q}(F^pK^\bullet/F^{p+1}K^\bullet) & \end{array}$$

where α is induced by the inclusions $F^{p+1}K^\bullet \hookrightarrow F^pK^\bullet$ (shifting the filtration degree by -1), β is the map on cohomology induced by the quotient $F^pK^\bullet \rightarrow F^pK^\bullet/F^{p+1}K^\bullet$, and γ is the connecting homomorphism from the long exact sequence of the short exact sequence

$$0 \longrightarrow F^{p+1}K^\bullet \longrightarrow F^pK^\bullet \longrightarrow F^pK^\bullet/F^{p+1}K^\bullet \longrightarrow 0.$$

Exactness of the couple at each vertex is exactness of this long exact sequence. One checks that $D^{-\infty} = \varinjlim_p H^n(F^pK^\bullet) = H^n(K^\bullet)$ and $D^\infty = \varprojlim_p H^n(F^pK^\bullet)$, giving the stated abutment. $\square$

1.4.5. *Proposition 1.63: the spectral sequence of a tower.* The spectral sequence that plays the central role in this thesis arises not from a filtered complex but from a *tower of objects* in a triangulated category, via an exact couple built from the long exact sequences of the tower triangles.

Fix a triangulated category $\mathcal{T}$ admitting coproducts, objects $A, B \in \mathcal{T}$, and a tower

$$\dots \longrightarrow A_{n+1} \xrightarrow{f_n} A_n \longrightarrow \dots \longrightarrow A_1 \xrightarrow{f_0} A_0$$

of objects and morphisms in $\mathcal{T}$. For each n complete f_n to a distinguished triangle

$$A_{n+1} \xrightarrow{f_n} A_n \longrightarrow Q_n \longrightarrow A_{n+1}[1],$$

so Q_n is the n -th associated graded object.

Proposition 1.51. *In the above setup, application of $\mathrm{Hom}_{\mathcal{T}}(B, -)$ to the tower yields an exact couple*

$$\begin{array}{ccc} \bigoplus_{n,q} \mathrm{Hom}(B, A_n[q]) & \xrightarrow{(f_n)_*} & \bigoplus_{n,q} \mathrm{Hom}(B, A_n[q]) \\ & \swarrow \gamma \quad \searrow \beta & \\ & \bigoplus_{n,q} \mathrm{Hom}(B, Q_n[q]) & \end{array}$$

where $(f_n)_*$ is induced by f_n , β is induced by $A_n \rightarrow Q_n$, and γ is the connecting map $Q_n \rightarrow A_{n+1}[1]$. The associated spectral sequence has E_1 -term

$$E_1^{p,q} = \mathrm{Hom}_{\mathcal{T}}(B, Q_{-p}[q+p])$$

and converges to $\mathrm{Hom}_{\mathcal{T}}(B, \mathrm{hocolim}_n A_n)$ when B is compact.

Proof. For each n , applying the cohomological functor $\mathrm{Hom}_{\mathcal{T}}(B, -)$ (Lemma 1.6) to the distinguished triangle $A_{n+1} \rightarrow A_n \rightarrow Q_n \rightarrow A_{n+1}[1]$ gives a long exact sequence

$$\begin{aligned} \cdots \longrightarrow \mathrm{Hom}(B, A_{n+1}[q]) &\xrightarrow{(f_n)_*} \mathrm{Hom}(B, A_n[q]) \xrightarrow{\beta} \mathrm{Hom}(B, Q_n[q]) \\ &\xrightarrow{\gamma} \mathrm{Hom}(B, A_{n+1}[q+1]) \longrightarrow \cdots \end{aligned}$$

Assembling these into the bigraded groups

$$D^{pq} := \mathrm{Hom}(B, A_{-p}[q+p]), \quad E^{pq} := \mathrm{Hom}(B, Q_{-p}[q+p]),$$

the maps $\alpha := (f_n)_*$, β , γ defined entry-wise give the exact couple, since the long exact sequence provides exactness at each vertex by definition. The bidegrees are $\deg \alpha = (-1, 1)$, $\deg \beta = (0, 0)$, $\deg \gamma = (1, 0)$, matching Note 1.41, so the differential $d_r = \beta_r \gamma_r$ has bidegree $(r, 1 - r)$.

For abutment: when B is compact, by Proposition 1.27,

$$\mathrm{Hom}_{\mathcal{T}}(B, \mathrm{hocolim}_n A_n) = \mathrm{colim}_n \mathrm{Hom}_{\mathcal{T}}(B, A_n) = D^{-\infty,*},$$

where the colimit is taken with respect to the tower maps $(f_n)_*$. The filtration on $D^{-\infty} = \mathrm{Hom}(B, \mathrm{hocolim}_n A_n)$ is then exactly the filtration $F_{\bullet}$ from Definition 2.5, and the spectral sequence abuts to $\mathrm{Hom}(B, \mathrm{hocolim}_n A_n)$ by Proposition 1.47. $\square$

Note 1.52. If additionally $E_1^{pq} = 0$ for all p below some bound p_0 (which occurs in our setting because $B \in DM^{\text{eff}}(s)$ forces $\text{Hom}(B, bc_{\leq n}A) = 0$ for $n \leq s - 1$), then the spectral sequence is confined to a half-plane and the hypothesis $RE_{\infty}^{pq} = 0$ of Theorem 1.48(1) is automatically satisfied. This is the key vanishing used in the proof of Theorem 2.11.

1.5. The Voevodsky category of mixed motives

Let k be a field. All schemes are assumed to be separated of finite type over k . We write Sch/k for the category of schemes over k and Sm/k for the full subcategory of smooth schemes.

1.5.1. Finite correspondences. The canonical reference for this subsection is [17], Ch. 1.

Definition 1.53. Let X be a smooth connected scheme over k and Y any scheme over k . An *elementary correspondence* from X to Y is an irreducible closed subscheme $W \subset X \times Y$ which is finite and surjective over X .

The group $\text{Cor}(X, Y)$ is the free abelian group generated by elementary correspondences from X to Y . Elements of $\text{Cor}(X, Y)$ are called *finite correspondences*.

Definition 1.54. We define $\text{Cor} := \text{Cor}(k)$ as the category whose objects are smooth (separated) schemes of finite type over k and whose morphisms are finite correspondences:

$$\text{Hom}_{\text{Cor}}(X, Y) := \text{Cor}(X, Y).$$

Definition 1.55. For $X, Y \in \text{Cor}$, the *tensor product* is $X \otimes Y := X \times_k Y$. This makes Cor a symmetric monoidal category.

1.5.2. The category of effective geometric motives.

Definition 1.56. The category $\widehat{DM}_{\text{gm}}^{\text{eff}}$ is defined as the localisation of $K^b(\text{Cor})$, as a tensor triangulated category, with respect to:

- *Homotopy.* For $X \in \text{Sm}/k$, one inverts $p_* : [X \times \mathbb{A}^1] \rightarrow [X]$.
- *Mayer–Vietoris.* For $X = U \cup V$ (Zariski open cover), one inverts the canonical map from the cone of $[U \cap V] \rightarrow [U] \oplus [V]$ to $[X]$.

The category $DM_{\text{gm}}^{\text{eff}}$ of *effective geometric motives* is the pseudo-abelian completion of $\widehat{DM}_{\text{gm}}^{\text{eff}}$ (formally adding kernels and images of projections).

Definition 1.57. The *category of geometric motives* DM_{gm} is defined by inverting the functor $\otimes \mathbb{Z}(1)$ on $DM_{\text{gm}}^{\text{eff}}$; that is, there are objects $X(n)$ for $X \in DM_{\text{gm}}^{\text{eff}}$, $n \in \mathbb{Z}$, with

$$\text{Hom}_{DM_{\text{gm}}}(X(n), Y(m)) := \varinjlim_N \text{Hom}_{DM_{\text{gm}}^{\text{eff}}}(X \otimes \mathbb{Z}(N+n), Y \otimes \mathbb{Z}(N+m)).$$

Here $\mathbb{Z}(1) := [\widetilde{\mathbb{P}^1}][-2]$ is the Tate motive and $\mathbb{Z}(n) := \mathbb{Z}(1)^{\otimes n}$ for $n \geq 0$.

1.5.3. Sites and sheaves.

Definition 1.58. A presheaf P on a small category $\mathcal{C}$ with values in a category $\mathcal{A}$ is a functor $P : \mathcal{C}^{op} \rightarrow \mathcal{A}$.

Definition 1.59. Let X be a k -variety. A Nisnevich cover $\mathcal{U} \rightarrow X$ is an étale morphism of finite type such that for each separable field extension F/k of finite degree, the map $\mathcal{U}(F) \rightarrow X(F)$ is surjective.

1.5.4. Motivic complexes and the Suslin complex.

Definition 1.60. 1. The category $PST := PST(k)$ of presheaves with transfers is the category of additive functors $\text{Cor}^{op} \rightarrow \mathbf{Ab}$.

2. The category $\text{Shv}_{\text{Nis}}(\text{Cor})$ of Nisnevich sheaves with transfers is the full subcategory of PST consisting of presheaves F such that, for each $X \in \text{Sm}/k$, the restriction $F|_{X_{\text{Nis}}}$ is a sheaf.

Definition 1.61. For a presheaf F on Sm/k , let $\Delta^n := \text{Spec } k[t_0, \dots, t_n]/(\sum t_i - 1)$. We define $C_n(F)(X) := F(X \times \Delta^n)$. The Suslin complex is

$$C_*(F)(X) : \cdots \rightarrow C_1(F)(X) \xrightarrow{d_1} C_0(F)(X), \quad d_n = \sum_i (-1)^i \delta_i^*,$$

where $\delta_i^n : \Delta^{n-1} \rightarrow \Delta^n$ is the i -th face map.

Theorem 1.62 (Voevodsky). *Let $F \in PST$ be homotopy invariant over Sm/k . Then:*

1. *The presheaf $X \mapsto H^q(X_{\text{Nis}}, F_{\text{Nis}})$ is a presheaf with transfers over k .*
2. *F_{Nis} is strictly homotopy invariant.*
3. *$F_{\text{Zar}} = F_{\text{Nis}}$ and $H^q(X_{\text{Zar}}, F_{\text{Zar}}) = H^q(X_{\text{Nis}}, F_{\text{Nis}})$.*

Proof. We sketch the proof of the three parts, referring to [33] and [17] for complete arguments.

(1) $X \mapsto H^q(X_{\text{Nis}}, F_{\text{Nis}})$ has transfers. Since F is homotopy invariant and has transfers (it is in PST), the Nisnevich sheafification F_{Nis} inherits the transfer structure. The higher Nisnevich cohomology groups $H^q(X_{\text{Nis}}, F_{\text{Nis}})$ are computed via the Čech complex of Nisnevich covers, whose terms inherit the transfer maps by sheaf theory. Concretely, for an elementary Nisnevich cover $U \rightarrow X$ and a correspondence $W \in \text{Cor}(X, Y)$, the transfer $W^* : F(Y) \rightarrow F(X)$ extends to a transfer on cohomology by the naturality of the Leray spectral sequence with respect to correspondences; see [17], Ch. 11.

(2) F_{Nis} is strictly homotopy invariant. This is the key theorem. One proves that for any $X \in \text{Sm}_k$, the map $F_{\text{Nis}}(X) \rightarrow F_{\text{Nis}}(X \times \mathbb{A}^1)$ (induced by the projection $X \times \mathbb{A}^1 \rightarrow X$) is an isomorphism. The proof uses the Mayer–Vietoris sequence for Nisnevich covers: writing $X \times \mathbb{A}^1 = U_1 \cup U_2$ with U_1, U_2 affine open subsets differing by an elementary Nisnevich cover, strict homotopy invariance follows

from the homotopy invariance of F and the Čech descent property of F_{Nis} . The key input is Voevodsky's theorem that for homotopy-invariant $F \in PST$, the Nisnevich sheafification preserves homotopy invariance; see [17], Thm. 11.2.

(3) $F_{\text{Zar}} = F_{\text{Nis}}$ and cohomologies agree. For a homotopy-invariant $F \in PST$ satisfying part (2), the Zariski sheafification equals the Nisnevich sheafification because the Nisnevich topology is finer than the Zariski topology but coarser than the étale topology, and for homotopy-invariant sheaves with transfers the extra Nisnevich covers (beyond Zariski) do not add new cohomology. This follows from the exactness of the associated sheaf functor and the fact that Nisnevich and Zariski agree on the derived category of strictly homotopy-invariant sheaves; see [33], Thm. 5.7, or [17], Prop. 13.9. $\square$

Theorem 1.63 (Voevodsky's embedding theorem). *There exists a commutative diagram of exact tensor functors*

$$\begin{array}{ccc} K^b(\text{Cor}) & \xrightarrow{L} & D(\text{Shv}_{\text{Nis}}(\text{Cor})) \\ \downarrow & & \downarrow RC_* \\ DM_{\text{gm}}^{\text{eff}} & \xrightarrow{i} & DM_-^{\text{eff}} \end{array}$$

such that i is a full embedding with dense image and $RC_*(L(X)) \cong C_*(X)$.

Proof. We sketch the construction of the embedding functor $i : DM_{\text{gm}}^{\text{eff}} \rightarrow DM_-^{\text{eff}}$.

Construction of i . The functor $L : K^b(\text{Cor}) \rightarrow D(\text{Shv}_{\text{Nis}}(\text{Cor}))$ is the derived functor of the Yoneda embedding $\text{Cor} \hookrightarrow \text{Shv}_{\text{Nis}}(\text{Cor})$ (which sends X to the representable sheaf $\text{Cor}(-, X)$), extended levelwise to bounded complexes. Since Nisnevich sheaves form an abelian category with enough injectives (cf. [17], Ch. 9), this extends to the derived category.

The functor $RC_* : D(\text{Shv}_{\text{Nis}}(\text{Cor})) \rightarrow DM_-^{\text{eff}}$ is the composition of the Nisnevich sheafification $\text{Shv}_{\text{Nis}}(\text{Cor}) \rightarrow \text{Shv}_{\text{Nis}}(\text{Cor})$ with the Suslin complex functor C_* : the total complex functor in the derived category. The functor $RC_* \circ L$ sends the generator $[X]$ to $RC_*(C_*(X)) = C_*(X)$ (the motivic complex of X).

Fullness and faithfulness. By construction, the localization $K^b(\text{Cor}) \rightarrow DM_{\text{gm}}^{\text{eff}}$ inverts homotopy equivalences and Mayer–Vietoris triangles. The functor $RC_* \circ L$ kills exactly these, so it factors through $DM_{\text{gm}}^{\text{eff}}$, giving i . Fullness: $\text{Hom}_{DM_{\text{gm}}^{\text{eff}}}(M, N) \cong \text{Hom}_{DM_-^{\text{eff}}}(i(M), i(N))$ follows from the fact that maps in DM_-^{eff} between objects in the image of i can be computed using resolutions in $D(\text{Shv}_{\text{Nis}}(\text{Cor}))$, which by homotopy invariance reduce to maps in $K^b(\text{Cor})$. Density of the image: every object of DM_-^{eff} is quasi-isomorphic (in $D(\text{Shv}_{\text{Nis}})$) to the total complex of a complex of objects in the image of L , giving density after Nisnevich sheafification; see [33], Thm. 3.2.6, for the complete argument. $\square$

1.5.5. Motivic cohomology.

Definition 1.64. For $X \in \text{Sm}/k$ and $q \geq 0$, set $M(X) = C_*(X)$ and define

$$H^p(X, \mathbb{Z}(q)) := \text{Hom}_{DM_-^{\text{eff}}}(M_{\text{gm}}(X), \mathbb{Z}(q)[p]).$$

Proposition 1.65. For $X \in \text{Sm}/k$ and $Y \in \text{Sch}/k$ equidimensional, $\text{CH}^i(Y, j) \cong H^{2i-j}(Y, \mathbb{Z}(i))$ for $i \geq 0$, $j \in \mathbb{Z}$.

Proposition 1.66. For $X \in \text{Sm}/k$:

$$H^n(X, \mathbb{Z}(1)) = \begin{cases} H_{\text{Zar}}^0(X, \mathcal{O}_X^*) & n = 1, \\ \text{Pic}(X) := H_{\text{Zar}}^1(X, \mathcal{O}_X^*) & n = 2, \\ 0 & \text{otherwise.} \end{cases}$$

1.5.6. The Voevodsky category of mixed motives. The triangulated category DM is defined from DM_-^{eff} by localising the Tate motive $\mathbb{Z}(1) \in DM_-^{\text{eff}}$. Explicitly:

Definition 1.67. There exists a triangulated tensor category DM in which the functor $\mathbb{Z}(1) \otimes - : DM \rightarrow DM$ is an equivalence, together with a full faithful exact tensor functor $DM_-^{\text{eff}} \rightarrow DM$.

Note 1.68. By construction ([33], §3.2 and [2], Thm. 4.5.67), DM is a compactly generated triangulated category with compact generators $\{M(X)(n)\}_{n \in \mathbb{Z}}$, where X ranges over smooth schemes of finite type over k . This follows from the fact that the embedding $DM_{\text{gm}}^{\text{eff}} \hookrightarrow DM$ is fully faithful with dense image (Theorem 1.63), and the objects $M(X)(n)$ generate DM under triangles, shifts, and arbitrary coproducts.

1.5.7. The motivic Postnikov tower.

Definition 1.69. Let $DM^{\text{eff}}(n)$ be the localisation subcategory of DM_-^{eff} generated by objects $M(X)(m)[2m]$ with $m \geq n$ and $X \in \text{Sm}/k$. For $n \in \mathbb{Z}$, we have a tower of triangulated subcategories

$$\cdots \subset DM^{\text{eff}}(n+1) \subset DM^{\text{eff}}(n) \subset \cdots \subset DM^{\text{eff}}(0) = DM_-^{\text{eff}}.$$

By Neeman's theorem (Theorem 1.29), the inclusion $i_n : DM^{\text{eff}}(n) \hookrightarrow DM$ admits a right adjoint $r_n : DM \rightarrow DM^{\text{eff}}(n)$.

Definition 1.70. For $E \in DM_-^{\text{eff}}$, the *slice tower* is

$$\cdots \rightarrow f_{n+1}E \rightarrow f_nE \rightarrow f_{n-1}E \rightarrow \cdots, \quad f_n := i_n \circ r_n.$$

The n -th slice of E is s_nE , defined by the distinguished triangle

$$f_{n+1}E \longrightarrow f_nE \longrightarrow s_nE.$$

Proposition 1.71. *For every $m, n \in \mathbb{Z}$,*

$$f_m(\mathbb{Z}(n)) = \begin{cases} \mathbb{Z}(n) & \text{if } m \leq n, \\ 0 & \text{if } m > n. \end{cases}$$

Equivalently, $s_n(\mathbb{Z}(n)) = \mathbb{Z}(n)$ and $s_m(\mathbb{Z}(n)) = 0$ for $m \neq n$.

Proof. Recall from Definition 1.70 that $f_m = i_m \circ r_m$, where $i_m : DM^{\text{eff}}(m) \hookrightarrow DM$ is the inclusion and $r_m : DM \rightarrow DM^{\text{eff}}(m)$ is its right adjoint. The category $DM^{\text{eff}}(m)$ is generated by objects $M(X)(j)[2j]$ with $j \geq m$ and $X \in \text{Sm}/k$.

Case $m \leq n$: We claim $\mathbb{Z}(n) \in DM^{\text{eff}}(m)$, which gives $r_m(\mathbb{Z}(n)) = \mathbb{Z}(n)$ and hence $f_m(\mathbb{Z}(n)) = i_m(\mathbb{Z}(n)) = \mathbb{Z}(n)$. Indeed, $\mathbb{Z}(n) = \mathbb{Z}(1)^{\otimes n}$ is the n -th Tate motive; by construction ([33], §3.1) it belongs to $DM^{\text{eff}}(n) \subset DM^{\text{eff}}(m)$ since $n \geq m$.

More precisely, using Proposition 2.2(4) adapted to $DM^{\text{eff}}(m)$: the counit $\epsilon_m^{\mathbb{Z}(n)} : f_m(\mathbb{Z}(n)) \rightarrow \mathbb{Z}(n)$ is an isomorphism if and only if $\mathbb{Z}(n) \in DM^{\text{eff}}(m)$. Since $\mathbb{Z}(n)$ is a generator of $DM^{\text{eff}}(n)$ and $DM^{\text{eff}}(n) \subset DM^{\text{eff}}(m)$ for $n \geq m$, we conclude $f_m(\mathbb{Z}(n)) \cong \mathbb{Z}(n)$.

Case $m > n$: We claim $\text{Hom}_{DM}(M(X)(a)[b], \mathbb{Z}(n)) = 0$ for all $X \in \text{Sm}/k$ and $a \geq m > n$, $b \in \mathbb{Z}$. By the motivic cohomology calculation (Proposition 1.66 and [33], Cor. 3.4) together with the vanishing $H^p(X, \mathbb{Z}(q)) = 0$ for $q < 0$ and for $p > 2q$ ([33], Thm. 4.3.7 and [34]), one has

$$\text{Hom}_{DM}(M(X)(a)[b], \mathbb{Z}(n)) = \text{Hom}_{DM}(M(X), \mathbb{Z}(n-a)[b]) = H^b(X, \mathbb{Z}(n-a)) = 0$$

since $n-a < 0$ (as $a \geq m > n$). Therefore $\mathbb{Z}(n)$ is orthogonal to all compact generators of $DM^{\text{eff}}(m)$, i.e. $\mathbb{Z}(n) \in DM^{\text{eff}}(m)^\perp$.

By the definition of $f_m = i_m \circ r_m$ and the adjunction $\text{Hom}_{DM}(i_m A, E) = \text{Hom}_{DM^{\text{eff}}(m)}(A, r_m E)$, taking $E = \mathbb{Z}(n)$ and $A \in DM^{\text{eff}}(m)$ arbitrary:

$$\text{Hom}_{DM}(A, f_m(\mathbb{Z}(n))) = \text{Hom}_{DM}(i_m A, \mathbb{Z}(n)) = 0,$$

since $i_m A \in DM^{\text{eff}}(m)$ is built from generators $M(X)(a)[2a]$ with $a \geq m > n$. As $DM^{\text{eff}}(m)$ is compactly generated, we conclude $f_m(\mathbb{Z}(n)) = 0$.

Slices: The distinguished triangle $f_{n+1}(\mathbb{Z}(n)) \rightarrow f_n(\mathbb{Z}(n)) \rightarrow s_n(\mathbb{Z}(n)) \rightarrow f_{n+1}(\mathbb{Z}(n))[1]$ becomes $0 \rightarrow \mathbb{Z}(n) \rightarrow s_n(\mathbb{Z}(n)) \rightarrow 0$ by the two cases, giving $s_n(\mathbb{Z}(n)) \cong \mathbb{Z}(n)$. For $m \neq n$: if $m > n$ then $f_m(\mathbb{Z}(n)) = 0 = f_{m+1}(\mathbb{Z}(n))$, so $s_m(\mathbb{Z}(n)) = 0$; if $m < n$ then $f_m(\mathbb{Z}(n)) \cong \mathbb{Z}(n) \cong f_{m+1}(\mathbb{Z}(n))$ and the triangle gives $s_m(\mathbb{Z}(n)) = 0$. $\square$

2. Results

2.1. The orthogonal tower

Throughout this section we work with motives with coefficients in $R = \mathbb{Z}[\frac{1}{p}]$, where p is the exponential characteristic of the base field k .

We recall the construction of the orthogonal tower from [25]. By Note 1.68, DM is compactly generated. Let $\mathcal{G}^{\text{eff}}$ denote the set of compact objects of the form $\{M(X)(p)\}_{p \geq 0}$ for $X \in \text{Sm}/k$. For $n \in \mathbb{Z}$, let $\mathcal{G}^{\text{eff}}(n)$ denote the set of compact objects of the form $\{M(X)(p)\}_{p \geq n}$ for $X \in \text{Sm}/k$.

Write $DM^\perp(n)$ for the triangulated orthogonal of $DM^{\text{eff}}(n)$ ([25], Def. 2.1.1.):

$$DM^\perp(n) = \{E \in DM_-^{\text{eff}} : \text{Hom}_{DM^{\text{eff}}}(A, E) = 0 \text{ for all } A \in DM^{\text{eff}}(n)\}.$$

By ([25], Lem. 2.1.7(2)), $DM^\perp(n)$ is compactly generated. By Theorem 1.29, the inclusion $j_n : DM^\perp(n) \rightarrow DM$ admits a right adjoint

$$p_n : DM \longrightarrow DM^\perp(n),$$

which is triangulated.

Definition 2.1. Set $bc_{\leq n} := j_{n+1} \circ p_{n+1}$ and for $E \in DM$, the *orthogonal cover* of E is the canonical map $bc_{\leq n}E \rightarrow E$.

Proposition 2.2 ([25], Rem. 3.2.5, Prop. 3.2.4, Cor. 3.2.7). *Let $E \in DM$. The orthogonal cover $bc_{\leq n}E$ satisfies:*

1. $bc_{\leq n}E \in DM^\perp(n+1)$.
2. The counit $\theta_n : j_{n+1}p_{n+1} \rightarrow \text{id}$ is characterised up to unique isomorphism by: for all $F \in DM^\perp(n+1)$,

$$\theta_{n*}^E : \text{Hom}_{DM}(F, bc_{\leq n}E) \xrightarrow{\sim} \text{Hom}_{DM}(F, E).$$

3. There is a canonical natural transformation $\iota_n : bc_{\leq n} \rightarrow bc_{\leq n+1}$, and $bc_{\leq n} \circ bc_{\leq n+1} \cong bc_{\leq n}$.
4. For $E \in DM$: $\theta_n^E : bc_{\leq n}E \rightarrow E$ is an isomorphism in DM if and only if $E \in DM^\perp(n+1)$.

Proof. Part (1). By Definition 2.1, $bc_{\leq n}E = j_{n+1}(p_{n+1}E)$. Since $p_{n+1} : DM \rightarrow DM^\perp(n+1)$ maps into $DM^\perp(n+1)$ by definition, and j_{n+1} is the inclusion $DM^\perp(n+1) \hookrightarrow DM$, we have $bc_{\leq n}E \in DM^\perp(n+1)$.

Part (2). The unit-counit adjunction (j_{n+1}, p_{n+1}) gives, for any $F \in DM$ and $E \in DM$:

$$\text{Hom}_{DM}(F, j_{n+1}p_{n+1}E) \cong \text{Hom}_{DM^\perp(n+1)}(p_{n+1}F, p_{n+1}E).$$

If $F \in DM^\perp(n+1)$ then the unit $\eta_F : F \rightarrow j_{n+1}p_{n+1}F$ is an isomorphism (since p_{n+1} is right adjoint to the inclusion, and F is already in $DM^\perp(n+1)$), so

$$\text{Hom}_{DM}(F, bc_{\leq n}E) = \text{Hom}_{DM}(F, j_{n+1}p_{n+1}E) \cong \text{Hom}_{DM}(F, E)$$

via post-composition with $\theta_n^E : bc_{\leq n}E \rightarrow E$ (the counit of the adjunction at E). This is natural in F and E , proving (2).

Part (3). The containment $DM^\perp(n+2) \subset DM^\perp(n+1)$ (since $DM^{\text{eff}}(n+1) \subset DM^{\text{eff}}(n+2)$) implies more objects are in the orthogonal at level $n+1$) gives a natural map between the corresponding projectors. Concretely, there is a canonical map

$$\begin{array}{ccc} bc_{\leq n}E & \xrightarrow{\iota_n} & bc_{\leq n+1}E \\ & \searrow \theta_n^E & \downarrow \theta_{n+1}^E \\ & & E \end{array}$$

defined by the universal property of part (2): since $bc_{\leq n}E \in DM^\perp(n+1)$ by part (1), there is a unique lift of $\theta_n^E : bc_{\leq n}E \rightarrow E$ through $\theta_{n+1}^E : bc_{\leq n+1}E \rightarrow E$, giving ι_n . Applying $bc_{\leq n}$ to the identity $bc_{\leq n+1}E \xrightarrow{\theta_{n+1}^E} E$ and using the universal property again gives $bc_{\leq n}(bc_{\leq n+1}E) \cong bc_{\leq n}E$.

Part (4). $(\Rightarrow)$: If $\theta_n^E : bc_{\leq n}E \rightarrow E$ is an isomorphism then $E \cong bc_{\leq n}E \in DM^\perp(n+1)$ by part (1). $(\Leftarrow)$: Suppose $E \in DM^\perp(n+1)$. We must show $\theta_n^E : j_{n+1}(p_{n+1}E) \rightarrow E$ is an isomorphism.

Since $j_{n+1} : DM^\perp(n+1) \hookrightarrow DM$ is a fully faithful inclusion (being a full subcategory inclusion), the counit of the adjunction (j_{n+1}, p_{n+1}) gives, for any $F \in DM^\perp(n+1)$:

$$\varepsilon_F : j_{n+1}(p_{n+1}(j_{n+1}(F))) \xrightarrow{\sim} j_{n+1}(F),$$

which simplifies to $j_{n+1}(p_{n+1}(j_{n+1}(F))) \cong j_{n+1}(F)$. In other words, p_{n+1} is a left inverse to j_{n+1} on objects: $p_{n+1} \circ j_{n+1} \cong \text{id}_{DM^\perp(n+1)}$. (This follows from full faithfulness of j_{n+1} : for $F \in DM^\perp(n+1)$, the unit $\eta_F : F \rightarrow p_{n+1}(j_{n+1}(F))$ satisfies $j_{n+1}(\eta_F)$ followed by $\varepsilon_{j_{n+1}(F)}$ is $\text{id}_{j_{n+1}(F)}$; since j_{n+1} is fully faithful, η_F is an isomorphism, hence $p_{n+1} \circ j_{n+1} \cong \text{id}$.)

Now write $E \in DM^\perp(n+1)$ as $E = j_{n+1}(\tilde{E})$ for some $\tilde{E} \in DM^\perp(n+1)$ (namely $\tilde{E} =$ the object E viewed inside $DM^\perp(n+1)$). Then:

$$bc_{\leq n}E = j_{n+1}(p_{n+1}(E)) = j_{n+1}(p_{n+1}(j_{n+1}(\tilde{E}))) \cong j_{n+1}(\tilde{E}) = E,$$

the isomorphism being $\varepsilon_{j_{n+1}(\tilde{E})}^{-1}$ composed appropriately. The counit morphism $\theta_n^E = \varepsilon_E : j_{n+1}(p_{n+1}(E)) \rightarrow E$ is precisely this isomorphism, confirming θ_n^E is an isomorphism. $\square$

Proposition 2.3 ([25], Thm. 3.2.12). *For each $E \in DM$, there is a commutative diagram in DM :*

$$\begin{array}{ccccccc} \cdots & \longrightarrow & bc_{\leq n}E & \xrightarrow{\iota_n} & bc_{\leq n+1}E & \longrightarrow & \cdots \longrightarrow \text{hocolim}_{n \rightarrow \infty} bc_{\leq n}E \\ & & \searrow \theta_n^E & & \downarrow \theta_{n+1}^E & & \nearrow c \\ & & & & E & & \end{array}$$

where $\iota_n : bc_{\leq n}E \rightarrow bc_{\leq n+1}E$ is the canonical natural transformation of Propo-

sition 2.2(3), the map $c : \operatorname{hocolim}_{n \rightarrow \infty} bc_{\leq n} E \rightarrow E$ is induced by the universal property of the homotopy colimit applied to the compatible system $\{\theta_n^E\}$, and all inner triangles commute.

Proof. The existence and properties of the maps ι_n follow from Proposition 2.2(3): there is a canonical natural transformation $bc_{\leq n} \rightarrow bc_{\leq n+1}$ for every $n \in \mathbb{Z}$, and these are compatible in the sense that the diagram

$$\begin{array}{ccc} bc_{\leq n} E & \xrightarrow{\iota_n} & bc_{\leq n+1} E \\ & \searrow \theta_n^E & \downarrow \theta_{n+1}^E \\ & & E \end{array}$$

commutes for each n . Indeed, both θ_n^E and $\theta_{n+1}^E \circ \iota_n$ are morphisms $bc_{\leq n} E \rightarrow E$ in DM . Since $bc_{\leq n} E \in DM^\perp(n+1)$ by Proposition 2.2(1), the universal property of θ_{n+1}^E (Proposition 2.2(2)) gives a unique factorisation of any morphism from $bc_{\leq n} E$ to E through $bc_{\leq n+1} E$ — but θ_n^E already maps to E directly, and ι_n followed by θ_{n+1}^E gives the same map, so both composites agree.

The maps $\theta_n^E : bc_{\leq n} E \rightarrow E$ thus form a compatible system indexed over $\mathbb{Z}$. By the universal property of the homotopy colimit (see Definition 1.26 and [21], Prop. 1.6.8), there is a unique morphism

$$c : \operatorname{hocolim}_{n \rightarrow \infty} bc_{\leq n} E \longrightarrow E$$

such that for every n the composite $bc_{\leq n} E \rightarrow \operatorname{hocolim}_{n \rightarrow \infty} bc_{\leq n} E \xrightarrow{c} E$ equals θ_n^E . This is precisely the statement that all inner triangles in the displayed diagram commute. $\square$

Definition 2.4. The tower of Proposition 2.3 is called the *orthogonal tower*.

Definition 2.5. For $E, F \in DM$, consider the increasing filtration $F_\bullet$ on $\operatorname{Hom}_{DM}(F, E)$ (resp. $\operatorname{Hom}_{DM}(F, \operatorname{hocolim}_{n \rightarrow \infty} bc_{\leq n} E)$) where F_p is the image of $\theta_{p*}^E : \operatorname{Hom}_{DM}(F, bc_{\leq p} E) \rightarrow \operatorname{Hom}_{DM}(F, E)$ (resp. $\lambda_{p*}^E : \operatorname{Hom}_{DM}(F, bc_{\leq p} E) \rightarrow \operatorname{Hom}_{DM}(F, \operatorname{hocolim}_{n \rightarrow \infty} bc_{\leq n} E)$).

2.2. Convergence of the orthogonal tower

2.2.1. The orthogonal spectral sequence. Let $E, F \in DM$. By ([25], Thm. 3.2.8), one constructs canonical triangulated functors

$$bc_{p/p-1} : DM \longrightarrow DM, \quad p \in \mathbb{Z},$$

fitting into distinguished triangles in DM :

$$bc_{\leq p-1} E \longrightarrow bc_{\leq p} E \longrightarrow bc_{p/p-1} E.$$

Proposition 2.6. *The orthogonal tower induces a spectral sequence of homological type*

$$E_{p,q}^1 = \mathrm{Hom}_{DM}(F, (bc_{p/p-1}E)[q-p]) \Rightarrow \mathrm{Hom}_{DM}(F, E),$$

with differentials $d_r : E_{p,q}^r \rightarrow E_{p-r,q-r+1}^r$, where the abutment carries the filtration $F_\bullet$ of Definition 2.5.

Similarly, applying $\mathrm{Hom}_{DM}(F, -)$ to the tower $\{bc_{\leq n}E\}$ gives a spectral sequence converging to $\mathrm{Hom}_{DM}(F, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}E)$:

$$E_{p,q}^1 = \mathrm{Hom}_{DM}(F, (bc_{p/p-1}E)[q-p]) \Rightarrow \mathrm{Hom}_{DM}(F, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}E).$$

The canonical map $c : \mathrm{hocolim} bc_{\leq n}E \rightarrow E$ of Proposition 2.3 induces a morphism of spectral sequences from the first to the second that is the identity on E_1 -terms.

Proof. Apply the cohomological functor $\mathrm{Hom}_{DM}(F, -)$ to each distinguished triangle in the orthogonal tower:

$$bc_{\leq p-1}E \longrightarrow bc_{\leq p}E \longrightarrow bc_{p/p-1}E \longrightarrow (bc_{\leq p-1}E)[1].$$

Setting $D^{p,q} := \mathrm{Hom}_{DM}(F, (bc_{\leq -p}E)[q+p])$ and $E^{p,q} := \mathrm{Hom}_{DM}(F, (bc_{-p/-p-1}E)[q+p])$, the long exact sequences in $\mathrm{Hom}_{DM}(F, -)$ assemble into an exact couple

$$\begin{array}{ccc} \bigoplus_{p,q} D^{p,q} & \xrightarrow{\alpha} & \bigoplus_{p,q} D^{p,q} \\ & \nwarrow \gamma \quad \nearrow \beta & \\ & \bigoplus_{p,q} E^{p,q} & \end{array}$$

where:

- $\alpha : D^{p,q} \rightarrow D^{p-1,q+1}$ is induced by the tower map $\iota_{-p-1} : bc_{\leq -p-1}E \rightarrow bc_{\leq -p}E$;
- $\beta : D^{p,q} \rightarrow E^{p,q}$ is induced by the quotient map $bc_{\leq -p}E \rightarrow bc_{-p/-p-1}E$;
- $\gamma : E^{p,q} \rightarrow D^{p+1,q-1}$ is the connecting map $bc_{-p/-p-1}E \rightarrow (bc_{\leq -p-1}E)[1]$ from the distinguished triangle.

Exactness at each vertex is exactness of the long exact sequence in $\mathrm{Hom}_{DM}(F, -)$ applied to these triangles. The bidegrees $\deg \alpha = (-1, 1)$, $\deg \beta = (0, 0)$, $\deg \gamma = (1, 0)$ match Note 1.41, so Proposition 1.40 yields a spectral sequence $\{E_r^{p,q}, d_r\}_{r \geq 1}$ with

$$E_1^{p,q} = E^{p,q} = \mathrm{Hom}_{DM}(F, (bc_{p/p-1}E)[q-p]).$$

The colimit group satisfies $D^{-\infty,q} = \varinjlim_p D^{p,q} = \mathrm{Hom}_{DM}(F, (\mathrm{hocolim}_n bc_{\leq n}E)[q])$, giving abutment $\cdot$. The morphism c_* is a morphism of exact couples (it commutes with α, β, γ since c commutes with all tower maps and quotients), hence a

morphism of spectral sequences, which is the identity on E_1 since c is the identity on each $bc_{p/p-1}E$ piece. $\square$

2.2.2. Orthogonality and duality. Throughout this section, let $Y \in \text{Sm}/k$ be a connected scheme of dimension d and let $r, s \in \mathbb{Z}$.

Proposition 2.7. $M(Y)(s)[r] \in DM^\perp(d + s + 1)$.

Proof. By ([25], 2.2), it suffices to show that $\text{Hom}_{DM}(M(X)(a)[b], M(Y)(s)[r]) = 0$ for all $X \in \text{Sm}/k$ and $a, b \in \mathbb{Z}$ with $a \geq d + s + 1$. By ([16], Cor. 4.13, Thm. 4.12, Thm. 5.1) we may assume the base field k is perfect. If k admits resolution of singularities, by ([33], Thm. 4.3.7),

$$\text{Hom}_{DM}(M(X)(a)[b], M(Y)(s)[r]) \cong \text{Hom}_{DM}(M(X) \otimes M^c(Y)(e)[f], \mathbb{Z})$$

where $M^c(Y) \in DM_-^{\text{eff}}$ is the motive of Y with compact support ([33], §4.1, Cor. 4.1.6), $e = a - s - d$ and $f = b - r - 2d$. For a perfect field of positive characteristic, the same conclusion follows from ([30], Thm. 5.5.14 and Lem. 5.5.6). The hypothesis $a \geq d + s + 1$ implies $e = a - s - d \geq 1$, so $M(X) \otimes M^c(Y)(e)[f] \in DM^{\text{eff}}(1)$.

We therefore need to show $\text{Hom}_{DM}(A, \mathbb{Z}) = 0$ for $A \in DM^{\text{eff}}(1)$. Since $DM^{\text{eff}}(1)$ is generated by $\{M(X)(j)[2j] : j \geq 1, X \in \text{Sm}/k\}$, it suffices to verify this on generators. Using the motivic cohomology isomorphism $\text{Hom}_{DM}(M(X)(j)[b], \mathbb{Z}) = H^{b-2j}(X, \mathbb{Z}(-j))$ and the vanishing $H^p(X, \mathbb{Z}(q)) = 0$ for $q < 0$ ([33], Cor. 3.4), we get zero since $-j \leq -1 < 0$. Therefore $M(X) \otimes M^c(Y)(e)[f] \in DM^{\text{eff}}(1)$ maps to zero in $\text{Hom}_{DM}(-, \mathbb{Z})$, completing the proof. $\square$

Corollary 2.8. Let $E = M(Y)(s)[r] \in DM$ where $Y \in \text{Sm}/k$ is connected of dimension d . Then:

1. The natural map $\theta_{d+s}^E : bc_{\leq d+s}E \rightarrow E$ is an isomorphism in DM .
2. For any $A \in DM$ and any $f : E \rightarrow A$ in DM , there exists a unique lifting $g : E \rightarrow bc_{\leq d+s}A$ such that the diagram

$$\begin{array}{ccc} & & bc_{\leq d+s}A \\ & \nearrow g & \downarrow \theta_{d+s}^A \\ E & \xrightarrow{f} & A \end{array}$$

commutes in DM .

3. $f = 0$ if and only if $g = 0$.

Proof. Part (1). By Proposition 2.7, $E = M(Y)(s)[r] \in DM^\perp(d + s + 1)$. By Proposition 2.2(4), $\theta_{d+s}^E : bc_{\leq d+s}E \rightarrow E$ is an isomorphism if and only if $E \in DM^\perp(d + s + 1)$, which holds.

Part (2). Applying part (1) with A in place of E : since $E = M(Y)(s)[r] \in DM^\perp(d + s + 1)$ by Proposition 2.7, and $\theta_{d+s}^A : bc_{\leq d+s}A \rightarrow A$ satisfies the universal

property of Proposition 2.2(2), we obtain for any $f : E \rightarrow A$ a unique $g : E \rightarrow bc_{\leq d+s}A$ with $\theta_{d+s}^A \circ g = f$. The diagram commutes by construction. Uniqueness follows because two lifts g, g' satisfy $\theta_{d+s}^A \circ (g - g') = 0$, and since θ_{d+s}^A is an isomorphism on $\text{Hom}_{DM}(E, -)$ by Proposition 2.2(2) (as $E \in DM^\perp(d + s + 1)$), we get $g = g'$.

Part (3). If $f = 0$ then by uniqueness in part (2), $g = 0$. Conversely, if $g = 0$ then $f = \theta_{d+s}^A \circ g = 0$. $\square$

2.2.3. Convergence in DM . In this section we consider objects $A, B \in DM$ where $B = M(X)(s)[r]$ for $X \in \text{Sm}/k$ and $r, s \in \mathbb{Z}$.

Theorem 2.9. *The spectral sequence*

$$E_{p,q}^1 = \text{Hom}_{DM}(B, (bc_{p/p-1}A)[q-p]) \Rightarrow \text{Hom}_{DM}(B, \text{hocolim}_{n \rightarrow \infty} bc_{\leq n}A)$$

converges strongly.

Proof. We verify the three conditions of strong convergence (Definition 1.36(iii)) for the filtration $F_\bullet$ of Definition 2.5 on $H := \text{Hom}_{DM}(B, \text{hocolim}_{n \rightarrow \infty} bc_{\leq n}A)$.

Exhaustiveness. Since $B = M(X)(s)[r]$ is compact in DM (Note 1.68), Proposition 1.27 gives

$$H = \text{Hom}_{DM}(B, \text{hocolim}_{n \rightarrow \infty} bc_{\leq n}A) \cong \text{colim}_{n \rightarrow \infty} \text{Hom}_{DM}(B, bc_{\leq n}A).$$

Every element of H therefore lies in the image of some $\text{Hom}_{DM}(B, bc_{\leq p}A) \rightarrow H$, i.e. in $F_p H$ for some p . Hence $F_{-\infty} H = H$.

Half-plane vanishing. By construction, $bc_{\leq n}A \in DM^\perp(n+1)$ (Proposition 2.2(1)). Since $B = M(X)(s)[r] \in DM^{\text{eff}}(s)$, for $n \leq s-1$ we have $n+1 \leq s$, so $DM^{\text{eff}}(s) \subset DM^{\text{eff}}(n+1)$ and hence $\text{Hom}_{DM}(B, bc_{\leq n}A) = 0$. Applying $\text{Hom}_{DM}(B, -)$ to the triangle $bc_{\leq p-1}A \rightarrow bc_{\leq p}A \rightarrow bc_{p/p-1}A$ therefore gives $E_1^{p,q} = \text{Hom}_{DM}(B, (bc_{p/p-1}A)[q-p]) = 0$ for all $p \leq s-1$. Thus the spectral sequence is confined to the half-plane $\{p \geq s\}$.

Strong convergence. For any fixed total degree $n = p+q$, only finitely many $p \geq s$ contribute non-zero $E_1^{p,q}$ once we note that $E_1^{p,q} = \text{Hom}_{DM}(B, (bc_{p/p-1}A)[q-p])$ and the differentials $d_r : E_r^{p,q} \rightarrow E_r^{p-r,q-r+1}$ have source degree $p-r < p$. Therefore for r large enough all differentials into and out of position (p, q) are zero, so $E_\infty^{p,q} = E_r^{p,q}$ for $r \gg 0$, giving $RE_\infty^{p,q} = 0$. By Theorem 1.48(1) (Boardman, applied via Proposition 1.51 and Note 1.52), the spectral sequence converges strongly. $\square$

Corollary 2.10. *Suppose the canonical map*

$$c : \text{hocolim}_{n \rightarrow \infty} bc_{\leq n}A \longrightarrow A$$

induces an isomorphism of abelian groups

$$c_* : \text{Hom}_{DM}(B, \text{hocolim}_{n \rightarrow \infty} bc_{\leq n}A) \xrightarrow{\sim} \text{Hom}_{DM}(B, A).$$

Then the spectral sequence

$$E_{p,q}^1 = \mathrm{Hom}_{DM}(B, (bc_{p/p-1}A)[q-p]) \Rightarrow \mathrm{Hom}_{DM}(B, A)$$

converges strongly.

Proof. By Theorem 2.9, the spectral sequence for B and the tower $\{bc_{\leq n}A\}$ converges strongly as long as $B = M(Y)(a)[b]$ is compact and lies in $DM^\perp(r)$ for some r . Since $B = M(Y)(a)[b]$ with $Y \in \mathrm{Sm}/k$, it lies in $DM^\perp(d+a+1)$ by Corollary 2.8(1), so strong convergence applies.

The identification of the E^1 -term: by definition of the orthogonal layers, $E_{p,q}^1 = \mathrm{Hom}_{DM}(B, (bc_{p/p-1}A)[q-p])$. The morphism of spectral sequences associated to the natural transformation $bc_{\leq n}A \rightarrow A$ (Section 2.2.1, item (2)) gives an isomorphism on all E^r -terms for $r \geq 1$ (since the individual differences $bc_{\leq n+1}A/bc_{\leq n}A = bc_{n/n-1}A$ are unchanged), and hence the two spectral sequences have the same E^∞ -term and the same abutment $\mathrm{Hom}_{DM}(B, A)$. $\square$

The following is the main result of this thesis.

Theorem 2.11. *The canonical map*

$$c : \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}A \xrightarrow{\sim} A$$

is an isomorphism in DM . Consequently, the spectral sequence

$$E_{p,q}^1 = \mathrm{Hom}_{DM}(B, (bc_{p/p-1}A)[q-p]) \Rightarrow \mathrm{Hom}_{DM}(B, A)$$

converges strongly.

Proof. By Corollary 2.10, it suffices to show that c is an isomorphism in DM . Since DM is compactly generated, it suffices to show that for all $a, b \in \mathbb{Z}$ and all $Y \in \mathrm{Sm}/k$, the map

$$c_* : \mathrm{Hom}_{DM}(M(Y)(a)[b], \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}A) \longrightarrow \mathrm{Hom}_{DM}(M(Y)(a)[b], A)$$

is an isomorphism of abelian groups.

Surjectivity. Given $f : M(Y)(a)[b] \rightarrow A$, by Corollary 2.8(2) (with $d = \dim Y$), there is a lifting $g : M(Y)(a)[b] \rightarrow bc_{\leq d+a}A$ with $\theta_{d+a}^A \circ g = f$. Surjectivity follows from the orthogonal tower (Proposition 2.3).

Injectivity. Let $f : M(Y)(a)[b] \rightarrow \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}A$ with $c_*(f) = 0$. Since $M(Y)(a)[b]$ is compact, by Proposition 1.27,

$$\mathrm{Hom}_{DM}(M(Y)(a)[b], \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}A) \cong \mathrm{colim}_n \mathrm{Hom}_{DM}(M(Y)(a)[b], bc_{\leq n}A),$$

so f factors as $M(Y)(a)[b] \xrightarrow{f'} bc_{\leq n}A \xrightarrow{\lambda_{n*}^A} \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}A$ for some $n \geq d+a$.

Applying Corollary 2.8(2) again, f' factors as

$$M(Y)(a)[b] \xrightarrow{f''} bc_{\leq d+a} A \cong bc_{\leq d+a}(bc_{\leq n} A) \longrightarrow bc_{\leq n} A.$$

By Corollary 2.8(3), $f'' = 0$ since $\theta_{d+a}^A \circ f'' = c_*(f) = 0$, using Proposition 2.2(3) and the two commuting diagrams. Hence $f = 0$. $\square$

2.3. The $\mathbb{A}^1$ -stable homotopy category of Morel–Voevodsky

We refer the reader to ([15], Thm. 4.15) for the construction of the stable model structure on symmetric T -spectra. We write $\mathcal{SH}$ for its homotopy category, the $\mathbb{A}^1$ -stable homotopy category of Morel–Voevodsky.

Let $\Sigma_T^\infty X_+ \in \mathcal{SH}$, $X \in \mathbf{Sm}/k$, be the infinite suspension of the simplicial presheaf represented by X with a disjoint basepoint. By ([15], Prop. 4.19), $\mathcal{SH}$ is a tensor triangulated category with unit $\mathbf{1} = \Sigma_T^\infty \mathrm{Spec} k_+$. Writing $E(1)$ for $E \otimes \Sigma_T^\infty(\mathbb{G}_m)[-1]$ and $E(n) = (E(n-1))(1)$ inductively for $n \geq 0$, the functor $E \mapsto E(1)$ is an autoequivalence ([13], 8.10; [2], Thm. 4.3.38). Like DM , $\mathcal{SH}$ is compactly generated with compact generators

$$\mathcal{G}_{\mathcal{SH}} = \left\{ \Sigma_T^\infty X_+(p) : X \in \mathbf{Sm}/k, p \in \mathbb{Z} \right\}.$$

2.3.1. The orthogonal tower in $\mathcal{SH}$. The construction of Section 1.4 and §1.5.7 carries over to $\mathcal{SH}$ verbatim, replacing $DM^{\mathrm{eff}}(n)$ by the analogous subcategory $\mathcal{SH}^{\mathrm{eff}}(n)$.

Definition 2.12. For $n \in \mathbb{Z}$, let $\mathcal{SH}^{\mathrm{eff}}(n)$ denote the localisation subcategory of $\mathcal{SH}$ generated by $\{\Sigma_T^\infty X_+(p) : p \geq n, X \in \mathbf{Sm}/k\}$. Define

$$\mathcal{SH}^\perp(n) := \left\{ E \in \mathcal{SH} : \mathrm{Hom}_{\mathcal{SH}}(A, E) = 0 \text{ for all } A \in \mathcal{SH}^{\mathrm{eff}}(n) \right\}.$$

By ([15], Thm. 4.15 and [25], Lem. 2.1.7), $\mathcal{SH}^\perp(n)$ is compactly generated, so by Theorem 1.29 the inclusion $j_n^{\mathcal{SH}} : \mathcal{SH}^\perp(n) \hookrightarrow \mathcal{SH}$ admits a right adjoint $p_n^{\mathcal{SH}} : \mathcal{SH} \rightarrow \mathcal{SH}^\perp(n)$. Set

$$bc_{\leq n}^{\mathcal{SH}} := j_{n+1}^{\mathcal{SH}} \circ p_{n+1}^{\mathcal{SH}} : \mathcal{SH} \longrightarrow \mathcal{SH},$$

with counit $\theta_n^{E, \mathcal{SH}} : bc_{\leq n}^{\mathcal{SH}} E \rightarrow E$.

The properties of Proposition 2.2 hold in $\mathcal{SH}$ with identical proofs (replacing DM by $\mathcal{SH}$ throughout). In particular, for each $E \in \mathcal{SH}$ there is an orthogonal tower

$$\begin{array}{ccccccc} \cdots & \longrightarrow & bc_{\leq n}^{\mathcal{SH}} E & \longrightarrow & bc_{\leq n+1}^{\mathcal{SH}} E & \longrightarrow & \cdots \longrightarrow \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}^{\mathcal{SH}} E \\ & & \searrow \theta_n^{E, \mathcal{SH}} & & \downarrow \theta_{n+1}^{E, \mathcal{SH}} & & \swarrow c^{\mathcal{SH}} \\ & & & & E & & \end{array}$$

and, for $F, E \in \mathcal{SH}$, a spectral sequence (Proposition 2.6 applied in $\mathcal{SH}$; see also [24]):

$$E_{p,q}^1 = \mathrm{Hom}_{\mathcal{SH}}(F, (bc_{p/p-1}^{\mathcal{SH}} E)[q-p]) \Rightarrow \mathrm{Hom}_{\mathcal{SH}}(F, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}^{\mathcal{SH}} E).$$

We henceforth drop the superscript $\mathcal{SH}$ and write $bc_{\leq n}$, θ_n^E , etc. when the ambient category is clear from context.

2.3.2. *Failure of strong convergence for $\mathbf{1} \in \mathcal{SH}$.*

Proposition 2.13. *The canonical map*

$$c : \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} \mathbf{1} \longrightarrow \mathbf{1}$$

is not an isomorphism in $\mathcal{SH}$.

Proof. Suppose for contradiction that c is an isomorphism. Since $\mathbf{1} \in \mathcal{SH}$ is compact (it is among the generators $\mathcal{G}_{\mathcal{SH}}$), Proposition 1.27 gives

$$\mathrm{colim}_{n \rightarrow \infty} \mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, bc_{\leq n} \mathbf{1}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} \mathbf{1}) \xrightarrow{c_*} \mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, \mathbf{1}).$$

Since c_* would be an isomorphism, $\mathrm{id}_{\mathbf{1}}$ is in the image of some $\mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, bc_{\leq n} \mathbf{1}) \rightarrow \mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, \mathbf{1})$, meaning $\mathrm{id}_{\mathbf{1}}$ factors as:

$$\begin{array}{ccc} & & bc_{\leq n} \mathbf{1} \\ & \nearrow \sigma & \downarrow \theta_n^{\mathbf{1}} \\ \mathbf{1} & \xrightarrow{\mathrm{id}} & \mathbf{1}. \end{array}$$

From $\theta_n^{\mathbf{1}} \circ \sigma = \mathrm{id}_{\mathbf{1}}$, we get that σ is a split monomorphism, so $\mathbf{1}$ is a direct summand of $bc_{\leq n} \mathbf{1}$. Since $bc_{\leq n} \mathbf{1} \in \mathcal{SH}^{\perp}(n+1)$ by Proposition 2.2(1) (in $\mathcal{SH}$), and $\mathcal{SH}^{\perp}(n+1)$ is closed under direct summands (it is a thick subcategory by Proposition 1.17), we conclude $\mathbf{1} \in \mathcal{SH}^{\perp}(n+1)$.

However, by ([35], Thm. 2.2(3)) $s_m \mathbf{1} = \mathbf{1}_{s_m} \neq 0$ for all $m \geq 0$; more precisely, $\mathbf{1} \in \mathcal{SH}^{\mathrm{eff}}(m) \setminus \mathcal{SH}^{\mathrm{eff}}(m+1)$ for every m ([18], Conj. 9 and p. 350), so $\mathbf{1} \notin \mathcal{SH}^{\perp}(n+1)$ for any n . This is a contradiction. $\square$

Note 2.14. The argument of Proposition 2.13 applies to any compact $A \in \mathcal{SH}$ for which $s_m A \neq 0$ for arbitrarily large m .

Corollary 2.15. *For the sphere spectrum $\mathbf{1} \in \mathcal{SH}$, the spectral sequence*

$$E_{p,q}^1 = \mathrm{Hom}_{\mathcal{SH}}(\Sigma_T^{\infty} X_+(s)[t], (bc_{p/p-1} \mathbf{1})[q-p]) \Rightarrow \mathrm{Hom}_{\mathcal{SH}}(\Sigma_T^{\infty} X_+(s)[t], \mathbf{1})$$

does not converge strongly for any $\Sigma_T^{\infty} X_+(s)[t]$ with $X \in \mathrm{Sm}/k$, $s, t \in \mathbb{Z}$.

Proof. Suppose for contradiction that the spectral sequence converges strongly for some generator $B = \Sigma_T^{\infty} X_+(s)[t]$. The morphism of spectral sequences induced by

c_* (Proposition 2.6 applied in $\mathcal{SH}$, equation) is the identity on E_1 -terms, so by Theorem 1.38 the comparison map

$$c_* : \mathrm{Hom}_{\mathcal{SH}}(B, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} \mathbf{1}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{SH}}(B, \mathbf{1})$$

is an isomorphism of filtered groups. Since $\mathcal{SH}$ is compactly generated by $\mathcal{G}_{\mathcal{SH}}$ and this holds for every $B \in \mathcal{G}_{\mathcal{SH}}$ (by the same argument for each generator), c is an isomorphism in $\mathcal{SH}$ — contradicting Proposition 2.13. $\square$

2.3.3. Strong convergence for $\mathcal{SH}^\perp(r)$ -objects.

Proposition 2.16. *Let $A \in \mathcal{SH}$ be such that $A \in \mathcal{SH}^\perp(r)$ for some $r \in \mathbb{Z}$. Then:*

1. *The canonical map $c : \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} A \xrightarrow{\sim} A$ is an isomorphism in $\mathcal{SH}$.*
2. *The spectral sequence*

$$E_{p,q}^1 = \mathrm{Hom}_{\mathcal{SH}}(\Sigma_T^\infty X_+(s)[t], (bc_{p/p-1} A)[q-p]) \Rightarrow \mathrm{Hom}_{\mathcal{SH}}(\Sigma_T^\infty X_+(s)[t], A)$$

converges strongly for all $X \in \mathrm{Sm}/k$, $s, t \in \mathbb{Z}$.

Proof. Part (1). Let $m \geq r$ be arbitrary. Since $DM^{\mathrm{eff}}(m) \supset DM^{\mathrm{eff}}(r)$ is implied by $m \geq r$ (more conditions generate a larger orthogonal), and $A \in \mathcal{SH}^\perp(r)$ means $\mathrm{Hom}_{\mathcal{SH}}(B, A) = 0$ for all $B \in \mathcal{SH}^{\mathrm{eff}}(r)$, we get $A \in \mathcal{SH}^\perp(m)$ for all $m \geq r$ since $\mathcal{SH}^{\mathrm{eff}}(m) \subset \mathcal{SH}^{\mathrm{eff}}(r)$.

By Proposition 2.2(4) in $\mathcal{SH}$: since $A \in \mathcal{SH}^\perp(m+1)$ (using $m \geq r$), the counit $\theta_m^A : bc_{\leq m} A \rightarrow A$ is an isomorphism. Therefore in the tower

$$\begin{array}{ccccccc} \cdots & \longrightarrow & bc_{\leq r-1} A & \longrightarrow & bc_{\leq r} A & \longrightarrow & bc_{\leq r+1} A \longrightarrow \cdots \\ & & & & \searrow \sim & & \downarrow \sim \\ & & & & & & A \end{array}$$

all maps θ_m^A for $m \geq r$ are isomorphisms, and the tower stabilises.

We now verify $\mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} A \cong A$: The homotopy colimit fits into the distinguished triangle

$$\coprod_n bc_{\leq n} A \xrightarrow{\mathrm{id} - \iota_n} \coprod_n bc_{\leq n} A \longrightarrow \mathrm{hocolim}_n bc_{\leq n} A \longrightarrow \left(\coprod_n bc_{\leq n} A \right)[1],$$

where $\iota_n : bc_{\leq n} A \rightarrow bc_{\leq n+1} A$ is the structure map. Since $\theta_m^A : bc_{\leq m} A \rightarrow A$ is an isomorphism for all $m \geq r$, the tower $\{bc_{\leq n} A\}_{n \geq r}$ is constant (up to the isomorphisms θ_m^A). More precisely, the compatible system $\{\theta_m^A : bc_{\leq m} A \rightarrow A\}_{m \geq 0}$ defines a morphism $c : \mathrm{hocolim} bc_{\leq n} A \rightarrow A$ by the universal property (Proposition 2.3). To see c is an isomorphism, apply $\mathrm{Hom}_{\mathcal{SH}}(B, -)$ for any compact B . By Proposition 1.27:

$$\mathrm{Hom}_{\mathcal{SH}}(B, \mathrm{hocolim}_n bc_{\leq n} A) = \mathrm{colim}_n \mathrm{Hom}_{\mathcal{SH}}(B, bc_{\leq n} A).$$

For $n \geq r$, each $(\theta_n^A)_* : \text{Hom}(B, bc_{\leq n}A) \rightarrow \text{Hom}(B, A)$ is an isomorphism (since θ_n^A is). The colimit of a diagram of isomorphisms (stabilizing from $n = r$ onward) is $\text{Hom}(B, A)$. Hence $c_* : \text{Hom}(B, \text{hocolim } bc_{\leq n}A) \xrightarrow{\sim} \text{Hom}(B, A)$ for all compact B . Since $\mathcal{SH}$ is compactly generated, c is an isomorphism in $\mathcal{SH}$.

Part (2). By part (1) and Corollary 2.10 (applied in $\mathcal{SH}$ with $B = \Sigma_T^\infty X_+(s)[t]$), it suffices that c_* is an isomorphism on all Hom groups from compact generators. Since B is compact and c is now an isomorphism in $\mathcal{SH}$, c_* is immediately an isomorphism. The strong convergence then follows from Theorem 2.9 applied in $\mathcal{SH}$. $\square$

Corollary 2.17. *Let $B = \Sigma_T^\infty X_+(s)[t]$ with $X \in \text{Sm}/k$, $s, t \in \mathbb{Z}$. Consider the Voevodsky slice spectral sequence ([35], §7) for $G \in \mathcal{SH}$:*

$$E_1^{m,n} = \text{Hom}_{\mathcal{SH}}(B, s_m G[m+n]) \Rightarrow \text{Hom}_{\mathcal{SH}}(B, G).$$

The orthogonal spectral sequence

$$E_{p,q}^1 = \text{Hom}_{\mathcal{SH}}(B, (bc_{p/p-1} s_m G[m+n])[q-p]) \Rightarrow \text{Hom}_{\mathcal{SH}}(B, s_m G[m+n])$$

converges strongly towards the E_1 -term $E_1^{m,n} = \text{Hom}_{\mathcal{SH}}(B, s_m G[m+n])$. Moreover, the differential $d_1 : E_1^{m,n} \rightarrow E_1^{m+1,n}$ of the slice spectral sequence induces a map between the orthogonal spectral sequences for consecutive slices, fitting into the commutative diagram:

$$\begin{array}{ccc} E_{p,q}^1(s_m G[m+n]) & \xrightarrow{d_r} & E_{p-r,q-r+1}^1(s_m G[m+n]) \\ d_{1*} \downarrow & & \downarrow d_{1*} \\ E_{p,q}^1(s_{m+1} G[m+1+n]) & \xrightarrow{d_r} & E_{p-r,q-r+1}^1(s_{m+1} G[m+1+n]). \end{array}$$

Proof. By ([35], Thm. 2.2(3)), $s_m G \in \mathcal{SH}^\perp(m+1)$ for every m and every $G \in \mathcal{SH}$. Shifting: $s_m G[m+n] \in \mathcal{SH}^\perp(m+1)$ as well (shifts preserve orthogonality since $\mathcal{SH}^\perp(m+1)$ is closed under the translation functor by the triangulated structure). Therefore, with $A = s_m G[m+n]$ and $r = m+1$, Proposition 2.16(2) applies and gives strong convergence.

For the morphism of spectral sequences: the differential $d_1 : E_1^{m,n} \rightarrow E_1^{m+1,n}$ of the slice spectral sequence is induced by the natural map $\partial[m+n] : s_m G[m+n] \rightarrow s_{m+1} G[m+1+n]$, where ∂ is the composition ([35], Thm. 2.2(1))

$$s_m G \xrightarrow{\sigma_m} f_{m+1} G[1] \xrightarrow{\pi_{m+1}[1]} s_{m+1} G[1].$$

Since the orthogonal tower functor $bc_{\leq n}^{\mathcal{SH}}$ is triangulated and natural (Definition 2.12 and Proposition 2.2 in $\mathcal{SH}$), post-composition with $\partial[m+n]$ defines a morphism of exact couples (in the sense of Definition 1.37), hence a morphism of spectral sequences. The commutativity of the square with d_r follows from naturality of

the exact couple construction (Proposition 1.40) with respect to morphisms of the tower. $\square$

PART II. ORTHOGONAL TOWER FILTRATIONS ON HIGHER CHOW GROUPS

3. The orthogonal tower filtration on higher Chow groups

This section fixes the notation used in Part II. The categorical construction, the Bousfield localization formalism, and the strong convergence theorem have already been established in Part I. We therefore recall only the formulas that will be used later, keeping the notation compatible with the preceding part.

The guiding point of this part is that Chow groups enter the story only after the motivic Hom-group has been identified. Thus a class in $\mathrm{CH}^n(X, m)_R$ will be regarded throughout as a morphism from the probe $B_{n,m}$ to the unit. The orthogonal tower is then applied to the target, not to the cycle itself. This is why the resulting filtration is functorial and why its comparison with realization maps, Beauville eigenspaces, and regular homomorphisms can be made inside the same triangulated category.

Let k be a perfect field of exponential characteristic e , let R be a $\mathbb{Z}[1/e]$ -algebra, and let $DM(k, R)$ be Voevodsky's category of motives with R -coefficients. We write $\mathbf{1}_R$ for its tensor unit. For a smooth projective variety X/k and integers n, m , set

$$B_{n,m} := M(X)(-n)[m - 2n].$$

Then motivic representability gives

$$\mathrm{CH}^n(X, m)_R = \mathrm{Hom}_{DM(k, R)}(B_{n,m}, \mathbf{1}_R).$$

Definition 3.1 (Pure layers and the orthogonal filtration). For an object $A \in DM(k, R)$, the pure layer of the orthogonal tower is denoted

$$bc_{p/p-1}A := \mathrm{Cone}(bc_{\leq p-1}A \rightarrow bc_{\leq p}A).$$

Applying $\mathrm{Hom}_{DM}(B_{n,m}, -)$ to the tower of $\mathbf{1}_R$ gives the spectral sequence

$$E_{p,q}^1 = \mathrm{Hom}_{DM(k, R)}(B_{n,m}, (bc_{p/p-1}\mathbf{1}_R)[q - p]) \implies \mathrm{CH}^n(X, m)_R.$$

The filtration on the abutment is

$$F_{\mathrm{orth}}^r \mathrm{CH}^n(X, m)_R := \mathrm{Im} \left(\mathrm{Hom}_{DM(k, R)}(B_{n,m}, bc_{\leq -r}\mathbf{1}_R) \rightarrow \mathrm{CH}^n(X, m)_R \right).$$

Lemma 3.2 (Tower of the unit). *For the motivic unit $\mathbf{1}_R$ the following properties hold.*

- (i) $bc_{\leq p} \mathbf{1}_R \simeq 0$ for $p < 0$ in the positive indexing convention used for effective levels, equivalently the tower has no non-effective part above the unit.
- (ii) The canonical comparison from the homotopy colimit of the tower to $\mathbf{1}_R$ is an isomorphism in the motivic range used in Part I.
- (iii) The pure layers vanish outside the effective range determined by the Tate twist.

Proposition 3.3 (Peláez’s layer theorem). *For rational coefficients the pure layers of the motivic unit lie in the heart of the homotopy t -structure:*

$$bc_{p/p-1} \mathbf{1}_{\mathbb{Q}} \in \mathbf{HI}(k, \mathbb{Q})^{\heartsuit} \quad (p \leq 0).$$

Moreover the first two layers are

$$bc_{0/-1} \mathbf{1}_{\mathbb{Q}} \simeq \mathbb{Q}^{\mathrm{tr}}, \quad bc_{-1/-2} \mathbf{1}_{\mathbb{Q}} \simeq \mathcal{O}^{\times} \otimes \mathbb{Q}.$$

Remark 3.4 (Integral layers). With integral coefficients the first layers are governed by the corresponding Milnor and Milnor–Witt K -sheaves. We use this only in the sections on integral degeneration and characteristic p .

Proposition 3.5 (The E^1 page). *For rational coefficients, the E^1 -page of can be read as motivic cohomology with coefficients in the homotopy-invariant sheaves appearing in the pure layers. In codimension one this recovers the two classical pieces*

$$E_{0,0}^1 \cong \mathrm{NS}(X)_{\mathbb{Q}}, \quad E_{-1,-1}^1 \cong \mathrm{Pic}^0(X)_{\mathbb{Q}}.$$

Theorem 3.6 (Column support). *Let X/k be smooth projective of dimension d . The spectral sequence has nonzero columns only in*

$$-n \leq p \leq \min(d - n, 0).$$

Consequently the filtration $F_{\mathrm{orth}}^{\bullet} \mathrm{CH}^n(X, m)_R$ has length at most $\min(n, d) + 1$.

Lemma 3.7 (Tate twist compatibility). *Let $A \in \mathrm{DM}(k, R)$ and $c \in \mathbb{Z}$. If $A \in \mathrm{DM}^{\mathrm{eff}}(a) \cap \mathrm{DM}^{\perp}(b)$, then $A(c) := A \otimes \mathbf{1}_R(c) \in \mathrm{DM}^{\mathrm{eff}}(a+c) \cap \mathrm{DM}^{\perp}(b+c)$. In particular, for the pure layers of $\mathbf{1}_R$: if $bc_{p/p-1} \mathbf{1}_R \in \mathrm{DM}^{\mathrm{eff}}(-p) \cap \mathrm{DM}^{\perp}(-p+1)$, then $bc_{p/p-1} \mathbf{1}_R \otimes \mathbf{1}_R(c) \in \mathrm{DM}^{\mathrm{eff}}(-p+c) \cap \mathrm{DM}^{\perp}(-p+c+1)$.*

Proof. Effective level. By definition, $\mathrm{DM}^{\mathrm{eff}}(a)$ is generated (under triangles, shifts, and coproducts) by objects $M(X)(j)[2j]$ with $j \geq a$ and $X \in \mathrm{Sm}_k$. Since $A \in \mathrm{DM}^{\mathrm{eff}}(a)$, A is built from such generators. Twisting by $\mathbf{1}_R(c)$: $M(X)(j)(c)[2j] = M(X)(j+c)[2j]$ is a generator of $\mathrm{DM}^{\mathrm{eff}}(a+c)$ (as $j+c \geq a+c$). Since $(-) \otimes \mathbf{1}_R(c)$ is an exact functor commuting with coproducts and triangles, $A(c) \in \mathrm{DM}^{\mathrm{eff}}(a+c)$.

Orthogonal level. We have $A \in \mathrm{DM}^{\perp}(b)$, meaning $\mathrm{Hom}_{\mathrm{DM}}(M(X)(j), A) = 0$ for all $X \in \mathrm{Sm}_k$ and $j \geq b$. For $A(c) \in \mathrm{DM}^{\perp}(b+c)$, we need $\mathrm{Hom}_{\mathrm{DM}}(M(X)(j), A(c)) =$

0 for $j \geq b + c$. By the internal Hom adjunction in the tensor triangulated category $DM(k, R)$:

$$\begin{aligned} \mathrm{Hom}_{DM}(M(X)(j), A(c)) &= \mathrm{Hom}_{DM}(M(X)(j) \otimes \mathbf{1}_R(-c), A) \\ &= \mathrm{Hom}_{DM}(M(X)(j - c), A). \end{aligned}$$

Here $\mathbf{1}_R(c)$ is invertible in $DM(k, R)$ [33]. For $j \geq b + c$: $j - c \geq b$, so $\mathrm{Hom}_{DM}(M(X)(j - c), A) = 0$ (as $A \in DM^\perp(b)$). Hence $A(c) \in DM^\perp(b + c)$.

The statement for pure layers follows by applying this to $A = bc_{p/p-1}\mathbf{1}_R$, using $bc_{p/p-1}\mathbf{1}_R \in DM^{\mathrm{eff}}(-p) \cap DM^\perp(-p+1)$ from Proposition 3.3 and [53, Prop. 3.1]. $\square$

Remark 3.8. The formulation above suppresses the proofs already given in Part I and keeps only the operational statements needed for the Chow-theoretic applications. This is the main editorial change from the article-style version of Part II: background material is not repeated, and the later sections refer back to the convergence and tower formalism developed earlier in the thesis.

4. The main degeneration theorem

The degeneration theorem has two logically distinct components. The first is a geometric boundedness statement: for the Chow probes used here the orthogonal spectral sequence has only finitely many non-zero columns, so convergence does not hide an infinite extension problem. The second is a coefficient statement: with rational coefficients the extension classes that could support differentials become Galois-cohomological classes in semisimple modules. The proof below keeps these mechanisms separate. This separation is important because the geometric boundedness survives integrally, while the rational Maschke vanishing does not.

The central result of this paper is that the spectral sequence degenerates at E^1 for $R = \mathbb{Q}$ over *any* perfect field, with no hypothesis on the field, the variety, or the motive. The argument is short once the correct framework is in place: every differential d^r lands in a Galois cohomology group with rational coefficients, and such groups vanish by Maschke's theorem.

4.1. Differentials as Galois cohomology classes

The attaching map of the differential $d^r : E_{p,q}^r \rightarrow E_{p-r,q+r-1}^r$ is a Yoneda Ext-class in $\mathrm{Ext}_{DM}^r(bc_{p/p-1}\mathbf{1}_{\mathbb{Q}}, bc_{p-r/p-r-1}\mathbf{1}_{\mathbb{Q}})$. We identify this Ext group with Galois cohomology.

Lemma 4.1 (Ext reduction to Galois cohomology). *Let $A, B \in \mathbf{HI}(k, \mathbb{Q})^\heartsuit$ be homotopy-invariant sheaves over a perfect field k . Then for all $r \geq 1$:*

$$\mathrm{Ext}_{DM(k, \mathbb{Q})}^r(A, B) \cong H^r(G_k, \mathcal{H}om_{\mathbf{HI}}(A, B)(\bar{k})),$$

where $\mathcal{H}om_{\mathbf{HI}}(A, B)$ is the internal Hom sheaf in $\mathbf{HI}(k, \mathbb{Q})$ and $\mathcal{H}om_{\mathbf{HI}}(A, B)(\bar{k})$ is its geometric stalk, a discrete $\mathbb{Q}[G_k]$ -module.

In particular, for $A = bc_{p/p-1}\mathbf{1}_{\mathbb{Q}}$ and $B = bc_{p-r/p-r-1}\mathbf{1}_{\mathbb{Q}}$ (both in $\mathbf{HI}(k, \mathbb{Q})^\heartsuit$ by):

$$\mathrm{Ext}_{DM(k, \mathbb{Q})}^r(bc_{p/p-1}\mathbf{1}_{\mathbb{Q}}, bc_{(p-r)/(p-r-1)}\mathbf{1}_{\mathbb{Q}}) \cong H^r(G_k, M_{p,r}),$$

where $M_{p,r} := \mathcal{H}om_{\mathbf{HI}}(bc_{p/p-1}\mathbf{1}_{\mathbb{Q}}, bc_{(p-r)/(p-r-1)}\mathbf{1}_{\mathbb{Q}})(\bar{k})$ is a discrete $\mathbb{Q}$ -vector space with G_k -action.

Proof. We establish through a successive use of three isomorphisms.

Step 1: $\mathrm{Ext}_{DM}^r(A, B) \cong \mathrm{Ext}_{\mathbf{HI}}^r(A, B)$.

Since $\mathbf{HI}(k, \mathbb{Q})^\heartsuit$ is the heart of the homotopy t -structure on $DM(k, \mathbb{Q})$, the functor $DM(k, \mathbb{Q})$ is the derived category of $\mathbf{HI}(k, \mathbb{Q})$ in the sense of [17, Thm. 3.1.12]: for $A, B \in \mathbf{HI}(k, \mathbb{Q})^\heartsuit$,

$$\mathrm{Hom}_{DM(k, \mathbb{Q})}(A, B[r]) = \mathrm{Ext}_{\mathbf{HI}(k, \mathbb{Q})}^r(A, B),$$

where $\mathrm{Ext}_{\mathbf{HI}}^r$ is computed in the abelian category $\mathbf{HI}(k, \mathbb{Q})$. This uses: (a) $\mathrm{Hom}_{DM}(A, B[r]) = 0$ for $r < 0$ and A, B in the heart (by definition of t -structure); (b) for $r = 0$: $\mathrm{Hom}_{DM}(A, B) = \mathrm{Hom}_{\mathbf{HI}}(A, B)$ by full faithfulness; (c) for $r \geq 1$: the Ext group in the abelian category equals the Hom in the derived category shifted by r , by the universal property of the derived category (see [45, §1.3]).

Step 2: $\mathrm{Ext}_{\mathbf{HI}}^r(A, B) \cong H_{\mathrm{Nis}}^r(\mathrm{Spec} k, \mathcal{H}om_{\mathbf{HI}}(A, B))$.

The abelian category $\mathbf{HI}(k, \mathbb{Q})$ is a Grothendieck abelian category with enough injectives (its injectives are the injective Nisnevich sheaves with transfers over $\mathbb{Q}$, cf. [17, Prop. 3.1.13]).

We claim $\mathcal{H}om_{\mathbf{HI}}(A, -) : \mathbf{HI}(k, \mathbb{Q}) \rightarrow \mathbf{HI}(k, \mathbb{Q})$ is exact for any fixed $A \in \mathbf{HI}(k, \mathbb{Q})$. Over $\mathbb{Q}$ -coefficients the category $\mathbf{HI}(k, \mathbb{Q})$ is $\mathbb{Q}$ -linear, and the geometric stalk functor $\mathcal{F} \mapsto \mathcal{F}(\bar{k})$ identifies $\mathbf{HI}(k, \mathbb{Q})$ with the full subcategory of discrete $\mathbb{Q}[G_k]$ -modules arising from strictly homotopy-invariant sheaves. By Maschke's theorem applied to each finite Galois quotient (Lemma 4.3 below), the group algebra $\mathbb{Q}[\mathrm{Gal}(L/k)]$ is semisimple for every finite Galois extension L/k . Semisimplicity of the $\mathbb{Q}[G_k]$ -module category implies that every discrete $\mathbb{Q}[G_k]$ -module is projective as a $\mathbb{Q}[G_k]$ -module. Hence $\mathcal{H}om_{\mathbf{HI}}(A, -)$ is exact for any A (since $A(\bar{k})$ is a projective $\mathbb{Q}[G_k]$ -module, making $\mathrm{Hom}_{\mathbb{Q}[G_k]}(A(\bar{k}), -)$ exact).

The Grothendieck spectral sequence for the composition $\mathrm{Hom}_{\mathbf{HI}}(A, -) = \Gamma \circ \mathcal{H}om_{\mathbf{HI}}(A, -)$ collapses (the first functor is exact, so its higher derived functors vanish), giving:

$$\mathrm{Ext}_{\mathbf{HI}}^r(A, B) \cong R^r\Gamma(\mathcal{H}om_{\mathbf{HI}}(A, B)) = H_{\mathrm{Nis}}^r(\mathrm{Spec} k, \mathcal{H}om_{\mathbf{HI}}(A, B)),$$

where $R^r\Gamma$ denotes the derived global sections and the last identification is the standard fact that derived global sections of a sheaf on the Nisnevich site of $\mathrm{Spec} k$

coincide with Nisnevich cohomology [17, Cor. 14.24].

Step 3: $H_{\text{Nis}}^r(\text{Spec } k, \mathcal{F}) \cong H^r(G_k, \mathcal{F}(\bar{k}))$.

Over the field k , the Nisnevich topology on $\text{Spec } k$ coincides with the étale topology (every étale cover of $\text{Spec } k$ has a Nisnevich refinement and vice versa) [17, Cor. 14.24]. The étale cohomology of $\text{Spec } k$ with coefficients in a locally constant étale sheaf $\mathcal{F}$ is Galois cohomology: $H_{\text{ét}}^r(\text{Spec } k, \mathcal{F}) = H^r(G_k, \mathcal{F}(\bar{k}))$. Hence $H_{\text{Nis}}^r(\text{Spec } k, \mathcal{F}) \cong H^r(G_k, \mathcal{F}(\bar{k}))$.

Combining Steps 1–3 gives . The stalk $\mathcal{H}om_{\mathbf{HI}}(A, B)(\bar{k})$ is a $\mathbb{Q}$ -vector space (since $\mathbf{HI}(k, \mathbb{Q})$ is $\mathbb{Q}$ -linear) with a continuous G_k -action, making it a discrete $\mathbb{Q}[G_k]$ -module. $\square$

Remark 4.2 (The module $M_{p,r}$ explicitly). By Proposition 3.3: $bc_{p/p-1}\mathbf{1}_{\mathbb{Q}} \simeq K_{|p|}^M \otimes \mathbb{Q}$. The internal Hom computes:

$$M_{p,r} = \mathcal{H}om_{\mathbf{HI}}(K_{|p|}^M \otimes \mathbb{Q}, K_{|p|+r}^M \otimes \mathbb{Q})(\bar{k}).$$

For $p = 0$: $A \simeq \mathbb{Q}^{\text{tr}}$ and $\mathcal{H}om_{\mathbf{HI}}(\mathbb{Q}^{\text{tr}}, B) = B$, so $M_{0,r} = (K_r^M \otimes \mathbb{Q})(\bar{k}) = K_r^M(\bar{k}) \otimes_{\mathbb{Z}} \mathbb{Q}$. For general p : $M_{p,r}$ is the stalk of the sheaf of $\mathbb{Q}$ -linear homomorphisms from $K_{|p|}^M \otimes \mathbb{Q}$ to $K_{|p|+r}^M \otimes \mathbb{Q}$, evaluated at $\bar{k}$. In all cases, $M_{p,r}$ is a $\mathbb{Q}$ -vector space.

4.2. The Maschke vanishing and full degeneration

The key engine behind the unconditional degeneration of the orthogonal spectral sequence over any perfect field is a purely algebraic vanishing theorem for rational Galois cohomology. Its proof is elementary — Maschke’s theorem from group representation theory — but its consequences are far-reaching: every differential of the orthogonal spectral sequence, regardless of the base field or the variety, vanishes after passing to $\mathbb{Q}$ -coefficients.

Lemma 4.3 (Maschke vanishing; cf. [68, I.2.6]). *For any profinite group G and any discrete $\mathbb{Q}[G]$ -module M : $H^r(G, M) = 0$ for all $r \geq 1$. In particular, for $G = G_k = \text{Gal}(\bar{k}/k)$ and any perfect field k .*

Proof. Any continuous cocycle $f : G^r \rightarrow M$ factors through a finite quotient $G \twoheadrightarrow \Gamma = \text{Gal}(L/k)$ for some finite Galois extension L/k , (since the profinite topology on G and discrete topology on M make f locally constant, any cocycle $f : G^r \rightarrow M$ factors through a finite quotient). Write $M_0 := M$ viewed as a $\mathbb{Q}[\Gamma]$ -module. Since $|\Gamma|^{-1} \in \mathbb{Q}$, the group ring $\mathbb{Q}[\Gamma]$ is semisimple by Maschke’s theorem. For any semisimple ring R , all R -modules are projective, so $\text{Ext}_{\mathbb{Q}[\Gamma]}^r(M_0, N) = 0$ for all $\mathbb{Q}[\Gamma]$ -modules M_0, N and $r \geq 1$. In particular $H^r(\Gamma, M^0) = \text{Ext}_{\mathbb{Q}[\Gamma]}^r(\mathbb{Q}, M^0) = 0$. Since $H^r(G, M) = \varinjlim_{L/k} H^r(\text{Gal}(L/k), M^{\text{Gal}(L/k)})$ (colimit over finite Galois extensions), and each term vanishes, we get $H^r(G, M) = 0$. $\square$

Corollary 4.4 (Unconditional full degeneration). *For any perfect field k , $R = \mathbb{Q}$, smooth projective X/k , and all n, m :*

$$d^r = 0 \quad \text{for all } r \geq 1, \quad E_{p,q}^{\infty} = E_{p,q}^1.$$

Proof. Fix $r \geq 1$, a bidegree (p, q) , and the differential $d^r : E_{p,q}^r \rightarrow E_{p-r, q+r-1}^r$.

Identification of the attaching class. By Lemma 4.1 applied to the pair of layers $A := bc_{p/p-1} \mathbf{1}_{\mathbb{Q}}$ and $B := bc_{(p-r)/(p-r-1)} \mathbf{1}_{\mathbb{Q}}$ (both in $\mathbf{HI}(k, \mathbb{Q})^\vee$ by), the Ext group $\text{Ext}_{DM(k, \mathbb{Q})}^r(A, B) \cong H^r(G_k, M_{p,r})$, where $M_{p,r}$ is a discrete $\mathbb{Q}[G_k]$ -module (Remark 4.2). By the construction of the spectral sequence from the exact couple (Section 1.5.7), the differential d^r encodes an extension class between consecutive Bousfield layers, living in $\text{Ext}_{DM}^r(A, B)$.

Vanishing. Lemma 4.3: for any profinite group G and discrete $\mathbb{Q}[G]$ -module M , $H^r(G, M) = 0$ for all $r \geq 1$ (rational Maschke). Since G_k is profinite and $M_{p,r}$ is a discrete $\mathbb{Q}[G_k]$ -module, $H^r(G_k, M_{p,r}) = 0$ for every $r \geq 1$ and every (p, q) .

Hence $d^r = 0$ for all $r \geq 1$, and by induction $E^\infty = E^1$. $\square$

The degeneration argument is therefore purely cohomological. Lemma 4.1 identifies the attaching class of any possible differential d^r with a class in $H^r(G_k, M_{p,r})$. Since $M_{p,r}$ is a discrete rational G_k -module, the Maschke-type vanishing of Lemma 4.3 annihilates this class. Hence every differential vanishes and the spectral sequence degenerates at E^1 .

Hypothesis 4.5 (Ext^r -vanishing for general R). *For R not a $\mathbb{Q}$ -algebra: for all $r \geq 1$ and all p ,*

$$\text{Ext}_{DM(k, R)}^r(bc_{p/p-1} \mathbf{1}_R, bc_{p-r/p-r-1} \mathbf{1}_R) = 0.$$

This holds unconditionally for $R = \mathbb{Q}$ (Corollary 4.4). For $R = \mathbb{Z}$, it is open in general (see Theorem 4.6).

4.3. Degeneration for $R = \mathbb{Z}$

For $R = \mathbb{Z}$, Lemma 4.3 fails: $|G|^{-1} \notin \mathbb{Z}$. The attaching maps of differentials still land in $H^r(G_k, M)$ for discrete $\mathbb{Z}[G_k]$ -modules M (by the same Nisnevich-to-Galois argument), but these groups need not vanish. We prove what is accessible.

Theorem 4.6 (Partial degeneration for $R = \mathbb{Z}$). *Let k be a perfect field.*

- (a) $d_{p=0}^1 = 0$ unconditionally, for any perfect field.
- (b) $d^r = 0$ for all $r \geq 2$ and all p , whenever $\text{cd}(G_k) \leq 1$ (in particular for k finite or separably closed).
- (c) At column $p \leq -1$: the differential d_p^1 is the image in $H^1(G_k, K_{|p|}^M(\bar{k}))$ of the Bousfield boundary class. Over a finite field $\mathbb{F}_q$: $K_r^M(\bar{\mathbb{F}}_q) = 0$ for $r \geq 2$ (Bass–Tate [17, Rem. 7.1.3]) and $H^1(G_{\mathbb{F}_q}, \bar{\mathbb{F}}_q^\times) = 0$ (Hilbert 90), so $d_p^1 = 0$ for all p over $\mathbb{F}_q$. Over number fields and p -adic fields: $H^1(G_k, K_r^M(\bar{k}))$ for $r \geq 2$ is uncomputed and the vanishing of $d_{p \leq -1}^1$ is open.

Proof. (a): The differential $d_{p=0}^1 : E_{0,q}^1 \rightarrow E_{-1, q+1}^1$ at column $p = 0$ is the Yoneda extension class of the sequence of pure layers. The differential $d_{p=0}^1$ is a Yoneda extension class in

$$\text{Ext}_{DM(k, \mathbb{Z})}^1(bc_{0/-1} \mathbf{1}_{\mathbb{Z}}, bc_{-1/-2} \mathbf{1}_{\mathbb{Z}}) \cong \text{Ext}_{DM(k, \mathbb{Z})}^1(\mathbb{Z}^{\text{tr}}, \mathcal{O}^\times),$$

using $bc_{0/-1}\mathbf{1}_{\mathbb{Z}} \simeq \mathbb{Z}^{\text{tr}}$ and $bc_{-1/-2}\mathbf{1}_{\mathbb{Z}} \simeq \mathcal{O}^\times$ (Remark 3.4). By the Nisnevich-to-étale comparison (which holds for all R , cf. [17, Cor. 14.24]),

$$\text{Ext}_{DM(k, \mathbb{Z})}^1(\mathbb{Z}^{\text{tr}}, \mathcal{O}^\times) \cong H_{\text{Nis}}^1(\text{Spec } k, \mathcal{O}^\times) \cong H^1(G_k, \bar{k}^\times).$$

The first isomorphism: for $A = \mathbb{Z}^{\text{tr}}$ (the tensor unit of $\mathbf{HI}(k, \mathbb{Z})$), $\text{Hom}_{\mathbf{HI}}(\mathbb{Z}^{\text{tr}}, -)$ is the global sections functor Γ , so $\text{Ext}_{\mathbf{HI}}^r(\mathbb{Z}^{\text{tr}}, F) = H_{\text{Nis}}^r(\text{Spec } k, F)$. The second: Nisnevich = étale over $\text{Spec } k$, and $H_{\text{ét}}^1(\text{Spec } k, \mathcal{O}^\times) = H^1(G_k, \bar{k}^\times)$. By Hilbert's Theorem 90: $H^1(G_k, \bar{k}^\times) = 0$. Hence $d_{p=0}^1 = 0$.

(b): The differential d^r lives in $H^r(G_k, M)$ for some $\mathbb{Z}[G_k]$ -module M . For $r \geq 2 > \text{cd}(G_k)$: $H^r(G_k, M) = 0$ for all M , so $d^r = 0$. The condition $\text{cd}(G_k) \leq 1$ holds for $G_k = \hat{\mathbb{Z}}$ (finite fields) and $G_k = 1$ (separably closed fields).

(c): The identification of d_p^1 with a class in $H^1(G_k, K_{|p|}^M(\bar{k}))$ follows from with $R = \mathbb{Z}$ and the integral layers $bc_{p/p-1}\mathbf{1}_{\mathbb{Z}} \simeq K_{|p|}^{\text{MW}}$ [25, 5.3.6], whose geometric stalk is $K_{|p|}^{\text{MW}}(\bar{k})$. The identification $K_{|p|}^{\text{MW}}(\bar{k}) = K_{|p|}^M(\bar{k})$ holds over $\bar{k}$ (the Hopf class η vanishes since -1 is a square in $\bar{k}^\times$). Over $\mathbb{F}_q$: all stated vanishings are standard. Over number fields and p -adic fields: $H^1(G_k, K_r^M(\bar{k}))$ is not known to vanish. $\square$

Remark 4.7 (Open problem: integer coefficients). The residual problem for $R = \mathbb{Z}$ is: does $H^1(G_k, K_r^M(\bar{k})) = 0$ for $r \geq 2$ over fields with $\text{cd}(G_k) \geq 2$? The Beilinson–Soule conjecture predicts $K_r(k)_{\mathbb{Q}} = 0$ for $r \geq 2$ over number fields (and is open), but it concerns rational K -theory, not the integral Galois cohomology $H^1(G_k, K_r^M(\bar{k}))$. Both problems are open.

4.4. The index dictionary and hypotheses

We record the Beauville calibration and the two main hypotheses used throughout the paper for abelian varieties.

Remark 4.8 (Index dictionary and Beauville calibration). For an abelian variety A/k of dimension g , the standard BBM reindexing is $G^r := F_{\text{orth}}^{g-n+r} \text{CH}^n(A)_{\mathbb{Q}}$. Under this reindexing, column $p = -r$ corresponds to graded level r .

Beauville's eigenspace $V_{n,s}(A) := \{z \in \text{CH}^n(A)_{\mathbb{Q}} : [N]^*z = N^{2n-s}z\}$ satisfies $[N]^*|_{V_{n,s}} = N^{2n-s}$. The calibration $V_{n,s} \mapsto p = -s$ is verified by the $n = 1$ case: $\text{NS}(X)_{\mathbb{Q}} = V_{1,0}$ lies in column $p = 0$ (Proposition 3.5) and $\text{Pic}^0(X)_{\mathbb{Q}} = V_{1,1}$ lies in column $p = -1$ (Proposition 3.5). Each h^1 -factor in the Deninger–Murre decomposition [51] contributes -1 to the column index p .

Hypothesis 4.9 (Column stability). *For an abelian variety A/k of dimension g : the Beauville eigenspace $V_{n,s}(A)$ lies entirely within Bousfield column $p = -s$:*

$$V_{n,s}(A) \subset F_{\text{orth}}^s \text{CH}^n(A)_{\mathbb{Q}} \quad \text{and} \quad V_{n,s}(A) \cap F_{\text{orth}}^{s+1} \text{CH}^n(A)_{\mathbb{Q}} = 0.$$

Equivalently: $V_{n,s}(A) \subset F_{\text{orth}}^{g-n+s} \setminus F_{\text{orth}}^{g-n+s+1}$. This is equivalent to the motivic Künneth conjecture for A [51] (Proposition 6.7). It is proved unconditionally for $g \leq 2$ [51] and for A a product of pairwise non-isogenous elliptic curves (Proposition 6.9); it is open for $g \geq 3$ in general.

Hypothesis 4.10 (Tensor compatibility; Peláez [25, §6]). *There exist maps*

$$\mu_{n,m} : (bc_{\leq -n} \mathbf{1}_{\mathbb{Q}}) \otimes (bc_{\leq -m} \mathbf{1}_{\mathbb{Q}}) \rightarrow bc_{\leq -n-m} \mathbf{1}_{\mathbb{Q}}$$

compatible with the counits to $\mathbf{1}_{\mathbb{Q}}$. This is the conjecture of Peláez [25, §6]; it implies Hypothesis 4.9 for all abelian varieties.

4.5. Special cases and earlier partial results

Before Corollary 4.4, degeneration was known only in special cases; all are now recovered as immediate consequences.

The case $n = 1$ over any perfect field. For $n = 1$: column support gives $p \in \{-1, 0\}$, so $d^r = 0$ for $r \geq 2$ by finite support alone. The remaining $d^1 : E_{0,0}^1 \rightarrow E_{-1,0}^1$ vanishes by $H^1(G_k, \bar{k}^\times) = 0$ (Hilbert 90 at $R = \mathbb{Q}$, or Theorem 4.6(a) even for $R = \mathbb{Z}$). Corollary 4.4 gives an independent proof.

Number fields, $r \geq 2$. Over a number field k : $\mathrm{vcd}(G_k) = 2$, so $H^r(G_k, M) = 0$ for $r \geq 3$ and all M . Borel's theorem [46] and Bass–Tate give $K_r^M(k) \otimes \mathbb{Q} = 0$ for $r \geq 2$, whence $H^2(G_k, K_r^M(\bar{k}) \otimes \mathbb{Q}) = 0$ for $r \geq 2$. Combined with Corollary 4.4 (which handles all r at once for $R = \mathbb{Q}$), this is subsumed; Theorem 12.1 records the rational case.

Finite fields. Over $\mathbb{F}_q$: $\mathrm{cd}(G_k) = 1$, so all $d^r = 0$ for $r \geq 2$, and $d_{p \leq -1}^1 = 0$ by Bass–Tate. This gives degeneration for $R = \mathbb{Q}$ and $R = \mathbb{Z}$ (Theorem 12.4 establishes the integral case separately).

Algebraically closed fields. Over $\bar{k}$: $G_k = 1$, so all cohomology groups $H^r(G_k, M) = 0$ for $r \geq 1$. Corollary 4.4 gives full degeneration.

5. The case $n = 1$

For $n = 1$ the column support (Theorem 3.6) gives $p \in \{-1, 0\}$, so the spectral sequence has only two columns. The degeneration is immediate: $d^r = 0$ for $r \geq 2$ (finite support) and $d^1 = 0$ by Corollary 4.4. The resulting two-step filtration and the E^1 page entries are connected by the following commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & F_{\mathrm{orth}}^1 \mathrm{CH}^1(X)_{\mathbb{Q}} & \hookrightarrow & \mathrm{CH}^1(X)_{\mathbb{Q}} & \xrightarrow{\mathrm{cl}} & \mathrm{NS}(X)_{\mathbb{Q}} \longrightarrow 0 \\ & & \downarrow \scriptstyle (\mathrm{U}) & & \parallel & & \downarrow \scriptstyle \cong \\ 0 & \longrightarrow & \mathrm{Pic}^0(X)_{\mathbb{Q}} & \hookrightarrow & \mathrm{CH}^1(X)_{\mathbb{Q}} & \xrightarrow{\mathrm{cl}} & E_{0,0}^1 \end{array}$$

The short exact sequence $0 \rightarrow F_{\mathrm{orth}}^1 \rightarrow \mathrm{CH}^1(X) \rightarrow \mathrm{NS}(X) \rightarrow 0$ is the filtration; the left vertical arrow is the unconditional inclusion $F_{\mathrm{orth}}^1 \subseteq \mathrm{Pic}^0(X)_{\mathbb{Q}}$ (Proposition 5.1, labeled (U)); and the right vertical isomorphism is $E_{0,0}^1 \cong \mathrm{NS}(X)_{\mathbb{Q}}$ (Proposition 3.5).

Proposition 5.1 ($F_{\mathrm{orth}}^1 \subseteq \mathrm{Pic}^0$, unconditional (U)). *For $X \in \mathrm{SmProj}_k$, $R = \mathbb{Q}$: $F_{\mathrm{orth}}^1 \mathrm{CH}^1(X)_{\mathbb{Q}} \subseteq \ker(\mathrm{cl}) = \mathrm{Pic}^0(X)_{\mathbb{Q}}$.*

Proof. By : $F_{\text{orth}}^1 = \text{Im}(\alpha : \text{Hom}_{DM}(B_{1,0}, bc_{\leq -1} \mathbf{1}_{\mathbb{Q}}) \rightarrow \text{CH}^1(X)_{\mathbb{Q}})$. The map α factors through the spectral sequence column $p = -1$, while the cycle class $\text{cl} : \text{CH}^1(X)_{\mathbb{Q}} \rightarrow H^2(X, \mathbb{Q})$ corresponds to the $p = 0$ column $E_{0,0}^1$. Classes in $\text{Im}(\alpha)$ (column $p \leq -1$) map to zero in column $p = 0$, so $\text{cl}(z) = 0$ for all $z \in F_{\text{orth}}^1$, giving $F_{\text{orth}}^1 \subseteq \text{Pic}^0(X)_{\mathbb{Q}}$. $\square$

Hypothesis 5.2 (Hodge faithfulness). *The Hodge realization identifies $\text{gr}_{F_{\text{orth}}}^0 \text{CH}^1(X)_{\mathbb{Q}} \cong \text{NS}(X)_{\mathbb{Q}}$ and $\text{gr}_{F_{\text{orth}}}^1 \text{CH}^1(X)_{\mathbb{Q}} \cong \text{Pic}^0(X)_{\mathbb{Q}}$.*

Theorem 5.3 ($n = 1$ filtration (U/H)). *For $X \in \text{SmProj}_k$, $R = \mathbb{Q}$, the two-step filtration $0 = F_{\text{orth}}^2 \subseteq F_{\text{orth}}^1 \text{CH}^1(X)_{\mathbb{Q}} \subseteq F_{\text{orth}}^0 = \text{CH}^1(X)_{\mathbb{Q}}$ satisfies:*

- (a) **(U)** $F_{\text{orth}}^1 \text{CH}^1(X)_{\mathbb{Q}} \subseteq \text{Pic}^0(X)_{\mathbb{Q}}$.
- (b) **(H)** Under Hypothesis 5.2: $F_{\text{orth}}^1 \text{CH}^1(X)_{\mathbb{Q}} = \text{Pic}^0(X)_{\mathbb{Q}}$ and $\text{gr}_{F_{\text{orth}}}^0 \cong \text{NS}(X)_{\mathbb{Q}}$, so the filtration $F_{\text{orth}}^{\bullet}$ recovers the classical Hodge filtration on CH^1 .

Proof. Degeneration $d^r = 0$ for all $r \geq 1$: Corollary 4.4 (Maschke vanishing of all differentials with $\mathbb{Q}$ -coefficients over any perfect field).

(a): By Proposition 5.1, the Albanese map gives $\text{CH}_{\text{hom}}^1(X)_{\mathbb{Q}} \cong \text{Pic}^0(X)(k)_{\mathbb{Q}}$. The layer identification of Theorem 18.1 (Step 2) gives $F_{\text{orth}}^1 \text{CH}^1(X)_{\mathbb{Q}} = \text{CH}_{\text{hom}}^1(X)_{\mathbb{Q}}$, which equals $\text{Pic}^0(X)_{\mathbb{Q}}$. This does not use Hypothesis 5.2.

(b): Under Hypothesis 5.2, the cycle-class map $\text{cl} : \text{CH}^1(X)_{\mathbb{Q}} \rightarrow H^{1,1}(X)$ is surjective (Lefschetz (1, 1)). Its kernel is $\text{Pic}^0(X)_{\mathbb{Q}} = F_{\text{orth}}^1$ (by (a)). Hence $\text{gr}_{F_{\text{orth}}}^0 = \text{CH}^1(X)_{\mathbb{Q}} / F_{\text{orth}}^1 \cong \text{NS}(X)_{\mathbb{Q}}$ (Néron–Severi group tensored with $\mathbb{Q}$), and $\text{gr}_{F_{\text{orth}}}^1 \cong \text{Pic}^0(X)_{\mathbb{Q}}$. The identification $\text{CH}^1(X)_{\mathbb{Q}} / \text{Pic}^0(X)_{\mathbb{Q}} \cong \text{NS}(X)_{\mathbb{Q}}$ is the classical short exact sequence $0 \rightarrow \text{Pic}^0 \rightarrow \text{Pic} \rightarrow \text{NS} \rightarrow 0$ tensored with $\mathbb{Q}$. $\square$

Remark 5.4 (Abelian varieties: $n = 1$ unconditional). For A an abelian variety, $F_{\text{orth}}^1 \text{CH}^1(A)_{\mathbb{Q}} = \text{Pic}^0(A)_{\mathbb{Q}} = V_{1,1}(A)$ unconditionally (Proposition 6.7(a)). Hypothesis 5.2 is needed only to identify the graded pieces for general smooth projective X without abelian structure.

6. Abelian varieties: Beauville splitting

Let A/k be an abelian variety of dimension g , $R = \mathbb{Q}$. By Beauville [43]:

$$\text{CH}^n(A)_{\mathbb{Q}} = V_{n,0} \oplus \cdots \oplus V_{n,n}, \quad [N]^*|_{V_{n,s}} = N^{2n-s}, \quad N \geq 2.$$

The first key structural theorem requires no hypothesis.

Theorem 6.1 (Unconditional Beauville-compatible splitting (U)). *Let A/k be an abelian variety of dimension g , $1 \leq n \leq g$.*

- (a) Equivariance. *Each filtration step $F_{\text{orth}}^r \text{CH}^n(A)_{\mathbb{Q}}$ is stable under $[N]^*$ for all $N \geq 1$.*

- (b) Beauville-graded structure. *The Beauville decomposition refines the filtration: for every r ,*

$$F_{\text{orth}}^r \text{CH}^n(A)_{\mathbb{Q}} = \bigoplus_{s=0}^n \left(F_{\text{orth}}^r \text{CH}^n(A)_{\mathbb{Q}} \cap V_{n,s} \right).$$

In particular, each graded piece satisfies $\text{gr}_{F_{\text{orth}}}^r = \bigoplus_{s=0}^n \text{gr}_{F_{\text{orth}}}^r|_{V_{n,s}}$, where $\text{gr}_{F_{\text{orth}}}^r|_{V_{n,s}} := (F_{\text{orth}}^r \cap V_{n,s}) / (F_{\text{orth}}^{r+1} \cap V_{n,s})$.

- (c) Unconditional boundary conditions. *Denoting $G^r := F_{\text{orth}}^{g-n+r}$:*
- (i) $V_{n,0} \subseteq G^0 = F_{\text{orth}}^{g-n}$ and $V_{n,0} \cap G^1 = 0$: *the Deninger–Murre projector π_{2n} places $V_{n,0}$ entirely at level G^0 and not deeper (proved in Proposition 6.7(b) below).*
 - (ii) $V_{n,s} \subseteq \ker(\text{cl})$ for all $s \geq 1$ (Lemma 6.14(a)). *Note: under Hypothesis 4.9, $\ker(\text{cl}) = G^1 = F_{\text{orth}}^1 \text{CH}^n$, so this gives $V_{n,s} \subseteq F_{\text{orth}}^1$; but without Hypothesis 4.9, $\ker(\text{cl})$ is not identified with a Bousfield level.*
 - (iii) $V_{n,s} \subseteq \ker(\text{AJ})$ for all $s \geq 2$, under $k \hookrightarrow \mathbb{C}$ (Lemma 6.14(b)).
- (d) Characterization of Hypothesis 4.9. *The following are equivalent:*
- (i) *Hypothesis 4.9 holds for A ;*
 - (ii) *for each $s \in \{0, \dots, n\}$, the filtration $G^\bullet$ restricts to a one-step filtration on $V_{n,s}$:*

$$V_{n,s} \cap G^s = V_{n,s}, \quad V_{n,s} \cap G^{s+1} = 0.$$

Proof. (a): The multiplication $[N] : A \rightarrow A$ is a morphism in Sm_k , so by Theorem 7.1 (pullback), $[N]^*(F_{\text{orth}}^r \text{CH}^n(A)_{\mathbb{Q}}) \subseteq F_{\text{orth}}^r \text{CH}^n(A)_{\mathbb{Q}}$.

(b): Since $[N]^*$ acts semisimply on $\text{CH}^n(A)_{\mathbb{Q}}$ for any $N \geq 2$ (the characteristic polynomial of $[N]^*$ divides $\prod_{s=0}^n (T - N^{2n-s})$, which has distinct roots for $N \geq 2$, so $[N]^*$ is diagonalizable), the eigenspace decomposition $\text{CH}^n(A)_{\mathbb{Q}} = \bigoplus_s V_{n,s}$ is canonical. Since F_{orth}^r is $[N]^*$ -stable by (a), and $[N]^*$ acts by the distinct scalar N^{2n-s} on $V_{n,s}$, any $[N]^*$ -stable subspace must be a direct sum of subspaces of the $V_{n,s}$'s. (Formally: write $\text{id} = \sum_s P_s$ where P_s is the $[N]^*$ -eigenprojection onto $V_{n,s}$; each P_s is a polynomial in $[N]^*$, hence maps F_{orth}^r to itself; so $F_{\text{orth}}^r = \bigoplus_s P_s(F_{\text{orth}}^r) = \bigoplus_s (F_{\text{orth}}^r \cap V_{n,s})$.)

(c): Part (i): Under the Deninger–Murre decomposition [51] $M(A)_{\mathbb{Q}} = \bigoplus_{i=0}^{2g} h^i(A)$, the summand $h^{2n}(A) \simeq \mathbb{Q}(-n)[-2n]$ lies in $\text{DM}^{\text{eff}}(0) \cap \text{DM}^{\perp}(-1)$ [53, Prop. 3.1], giving $V_{n,0} = \text{im}(\pi_{2n})_* \subseteq F_{\text{orth}}^0 \setminus F_{\text{orth}}^1$. The Deninger–Murre decomposition is known unconditionally for abelian varieties by [51]; the Bousfield window $h^{2n}(A) \in \text{DM}^{\text{eff}}(0) \cap \text{DM}^{\perp}(-1)$ follows from [53, Prop. 3.1] by comparing the weight filtration with the Bousfield filtration. Part (ii) is Lemma 6.14(a); Part (iii) is Lemma 6.14(b).

(d): (ii) $\Rightarrow$ (i): If $V_{n,s} \cap G^s = V_{n,s}$ and $V_{n,s} \cap G^{s+1} = 0$ for all s , then by and the reindexing $G^r = F_{\text{orth}}^{g-n+r}$, $G^a = \bigoplus_{s \geq a} V_{n,s}$ for all $0 \leq a \leq n$, which is Hypothesis 4.9 under the reindexing $G^a = F_{\text{orth}}^{g-n+a}$. (i) $\Rightarrow$ (ii): Hypothesis 4.9 gives $F_{\text{orth}}^{g-n+a} = \bigoplus_{s \geq a} V_{n,s}$; intersecting with V_{n,s_0} gives $V_{n,s_0} \subseteq F_{\text{orth}}^{g-n+s_0}$ and $V_{n,s_0} \cap F_{\text{orth}}^{g-n+s_0+1} = 0$, which is (ii). $\square$

Remark 6.2 (What Hypothesis 4.9 adds). Theorem 6.1 establishes unconditionally that $F_{\text{orth}}^\bullet$ is “Beauville-graded”: each step is a direct sum of Beauville pieces. The content of Hypothesis 4.9 is the *non-mixing condition*: each $V_{n,s}$ sits entirely in a single graded piece (at level $g - n + s$). Without Hypothesis 4.9, $V_{n,s}$ (for $s \geq 1$) may be spread across several levels of $F_{\text{orth}}^\bullet$; we know it lies in $\ker(\text{cl})$ but do not identify $\ker(\text{cl})$ with a specific Bousfield level. In particular: the inclusions $V_{n,s} \subseteq \ker(\text{cl})$ (for $s \geq 1$) and $V_{n,s} \subseteq \ker(\text{AJ})$ (for $s \geq 2$) are unconditional, but $\ker(\text{cl}) = F_{\text{orth}}^1 \text{CH}^n$ (and hence $V_{n,s} \subseteq F_{\text{orth}}^1 \text{CH}^n$) requires Hyp. 4.9.

Corollary 6.3 (Unconditional separation of the top step). *For any abelian variety A/k of dimension $g \geq 1$, $1 \leq n \leq g$:*

- (a) (Unconditional) $G^0 \supsetneq G^1$: $V_{n,0} \subseteq G^0$ and $V_{n,0} \cap G^1 = 0$ (Theorem 6.1(c)(i)).
- (b) (Conditional) Under Hyp. 4.9: $G^1 \supsetneq G^2$ iff $V_{n,1} \not\subseteq \ker(\text{AJ})$, i.e., $\text{AJ}|_{V_{n,1}} \neq 0$. This holds for $n = g$ with A principally polarized (Lemma 6.14(b)), and for products of non-isogenous elliptic curves with positive rank (Theorem 7.9).
- (c) (Jacobians) For $A = J(C)$, $C/\mathbb{C}$ very general of genus $g \geq 3$: $G^1 \supsetneq G^2$ is witnessed by the Ceresa cycle $Z_C \in V_{g-1,1} \cap G^1 \setminus G^2$ (Ceresa [49], Griffiths [54]).

Proof. (a): Theorem 6.1(c)(i) gives $F_{\text{orth}}^{g-n} = \text{CH}^n(A)_{\mathbb{Q}}$ and $F_{\text{orth}}^{g-n+1} = \text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}}$. The cycle-class map $\text{cl} : \text{CH}^n(A)_{\mathbb{Q}} \rightarrow H^{2n}(A)$ therefore has $\text{gr}_{F_{\text{orth}}}^{g-n} \hookrightarrow H^{2n}(A)$, giving (a).

(b): Under Hypothesis 4.9, Theorem 6.11(b) gives $G^1 = F_{\text{orth}}^{g-n+1} = \bigoplus_{s \geq 1} V_{n,s}$ and $G^2 = F_{\text{orth}}^{g-n+2} = \bigoplus_{s \geq 2} V_{n,s}$. The piece $G^1/G^2 = V_{n,1}$ is the Abel–Jacobi component. By Theorem 7.7(H), $\text{AJ}|_{V_{n,1}} : V_{n,1} \hookrightarrow J^n(A)_{\mathbb{Q}}$ is injective. Hence $G^2 \subsetneq G^1$ strictly iff $V_{n,1} \neq 0$, i.e., iff there exist algebraically trivial cycles not killed by AJ.

(c): The involution $[-1_J]^*$ acts on $\text{CH}^{g-1}(J(C))_{\mathbb{Q}}$ via the eigenvalue $(-1)^{2(g-1)-s}$ on $V_{g-1,s}$ (Beauville [43]). For $s = 1$: $(-1)^{2g-3} = -1$ (since $2g - 3$ is odd for $g \geq 2$), so $[-1_J]^* Z_C = -Z_C$ forces $Z_C \in V_{g-1,1}$. Ceresa [49] proved $Z_C \neq 0$ in $\text{CH}_{\text{hom}}^{g-1}(J(C))$ for a very general curve C of genus $g \geq 3$ (via degeneration to a hyperelliptic curve and an explicit non-trivial normal function). Griffiths [54] proved $\text{AJ}(Z_C) \neq 0$ for the Jacobian of a very general genus-3 curve. Since $\text{AJ}|_{V_{g-1,1}}$ is injective (Theorem 7.7(H)), the element Z_C is non-zero in $V_{g-1,1}$, confirming the strict inclusion $G^2 \subsetneq G^1$. $\square$

Proposition 6.4 (Cycle class injection). (a) (U) *For any A/k : the cycle class map restricts to an injection $V_{n,0} \hookrightarrow H^{n,n}(A)$, and $V_{n,0} \hookrightarrow \text{gr}_{F_{\text{orth}}}^{g-n}$ (via Theorem 6.1(c)(i)).*

- (b) (H) Under Hypothesis 4.9: $\text{gr}_{F_{\text{orth}}}^{g-n} = V_{n,0}$, so the injection in (a) gives $\text{cl} : \text{gr}_{F_{\text{orth}}}^{g-n} \hookrightarrow H^{n,n}(A)$. This is the unconditional BBM5 injection (once Hyp. 4.9 identifies the graded piece).

Proof. (a): By the Beauville decomposition, $[N]^*$ acts on $V_{n,0}(A)$ by the scalar N^{2n} . By Lemma 6.14(a), algebraically trivial classes have $\text{cl}(\alpha) \in F^{n+1} H^{2n}(A) \subsetneq H^{2n}(A)$

(Hodge filtration); the only algebraically trivial classes with $[N]^*$ -eigenvalue N^{2n} are those in the (n, n) -part of H^{2n} . Hence $\text{cl} : V_{n,0} \rightarrow H^{n,n}(A)$ and injectivity follows by eigenvalue separation: $\ker(\text{cl}) \subset \text{CH}_{\text{hom}}^n$ carries eigenvalues N^{2n-s} for $s \geq 1$, incompatible with N^{2n} , so $V_{n,0} \cap \ker(\text{cl}) = 0$. The inclusion $V_{n,0} \hookrightarrow \text{gr}_{F_{\text{orth}}}^{g-n}$ is Theorem 6.1(c)(i).

(b): Under Hypothesis 4.9, Theorem 6.11(c) gives $\text{gr}_{F_{\text{orth}}}^{g-n} = V_{n,0}$, turning the injection in (a) into an equality. Hence the restricted cycle-class map $\text{cl} : \text{gr}_{F_{\text{orth}}}^{g-n} \hookrightarrow H^{n,n}(A)$ is an injection, which is the BBM5 injectivity under Hypothesis 4.9. $\square$

Remark 6.5 (BBM5 and the Hodge conjecture). Full BBM5 requires not merely injectivity of cl on gr^{g-n} , but that $\text{gr}^{g-n} \cong \text{im}(\text{cl} : \text{CH}^n \rightarrow H^{n,n})$. This surjectivity assertion is exactly the Hodge conjecture for A and n , which is wide open for $n > 1$. Proposition 6.4(b) provides the injectivity half. Theorem 10.11 therefore verifies BBM5 only conditionally (on Hyp. 4.9 and, for the surjectivity direction, implicitly on the Hodge conjecture for A).

Remark 6.6 (Status of Hypothesis 4.9). Theorem 6.1(d) characterizes Hypothesis 4.9 as the “non-mixing” condition on $F_{\text{orth}}^\bullet$. It is verified in Proposition 6.7 for $n = 1$ (all g), $s = 0$ (all n, g , unconditionally from the Deninger–Murre projectors), $g \leq 2$ (unconditionally), and products of non-isogenous elliptic curves (under Hypothesis 4.10). For general A with $g \geq 3$, Hypothesis 4.9 is equivalent to the non-mixing condition of Theorem 6.1(d)(ii), and follows from the motivic Künneth conjecture. All results conditioned on Hypothesis 4.9 are marked (H).

Proposition 6.7 (Column stability in known cases). (a) $n = 1$, all g , unconditionally. By Proposition 5.1, $F_{\text{orth}}^1 \subseteq \text{Pic}^0(A)_{\mathbb{Q}} = V_{1,1}$. For the reverse inclusion $\text{Pic}^0(A)_{\mathbb{Q}} \subseteq F_{\text{orth}}^1$: for $z \in \text{Pic}^0(A)_{\mathbb{Q}}$ the representing morphism $\phi : B_{1,0} \rightarrow \mathbf{1}_{\mathbb{Q}}$ satisfies $\text{cl}(z) = 0$, i.e., ϕ composes to zero in $\text{Hom}_{\text{DM}}(B_{1,0}, bc_{0/-1}\mathbf{1}_{\mathbb{Q}}) \cong \text{NS}(A)_{\mathbb{Q}}$. By the Bousfield triangle $bc_{\leq -1}\mathbf{1}_{\mathbb{Q}} \rightarrow \mathbf{1}_{\mathbb{Q}} \rightarrow bc_{0/-1}\mathbf{1}_{\mathbb{Q}}$, the vanishing of this composite means ϕ lifts to $\tilde{\phi} : B_{1,0} \rightarrow bc_{\leq -1}\mathbf{1}_{\mathbb{Q}}$ (exactness of the long exact sequence of $\text{Hom}(B_{1,0}, -)$); hence $z \in F_{\text{orth}}^1$. Therefore $F_{\text{orth}}^1 = \text{Pic}^0(A)_{\mathbb{Q}} = V_{1,1}$, and Hypothesis 4.9 holds for $n = 1$ (no Hodge realization needed).

- (b) $s = 0$, all n, g , unconditionally. The Deninger–Murre projector $\pi_{2n} : M(A)_{\mathbb{Q}} \rightarrow \mathbb{Q}(-n)$ [51] places $V_{n,0}$ in column $p = 0$: $V_{n,0} \subseteq F_{\text{orth}}^0$ and $V_{n,0} \cap F_{\text{orth}}^1 = 0$.
- (c) $s = n$, all g , subject to Conjecture 4.10. Beauville gives $V_{n,n} = V_{1,1}^{*n}$ (the n -fold Pontryagin self-product of $\text{Pic}^0(A)_{\mathbb{Q}}$) [43, Cor. 3]. The inclusion $V_{n,n} \subseteq F_{\text{orth}}^n$ follows if the Pontryagin product respects the Bousfield filtration, i.e., $(F_{\text{orth}}^1)^{*(n)} \subseteq F_{\text{orth}}^n$; this is conditional on Conjecture 4.10 (see Remark 6.8).
- (d) $g \leq 2$, unconditionally. For $g = 1$: from (a). For $g = 2$: the Deninger–Murre projectors π_0, π_2, π_4 [51] settle $s = 0, 2$ by (b) and direct computation. For $V_{2,1}$: π_2 onto $h^2(A) = \wedge^2 h^1(A)$. Since $h^1(A) \in \text{DM}^{\text{eff}}(-1) \cap \text{DM}^{\perp}(0)$ (calibration from $V_{1,1} = F_{\text{orth}}^1$ and Tate twist: $h^2(A) = h^1(A) \otimes h^1(A)$ at

Bousfield level $-1 - 1 = -2$; but as $h^2(A) \simeq \mathbf{1}_{\mathbb{Q}}(-1)[-2]$ for $g = 2$ by the Lefschetz isomorphism, its Bousfield level is -1 by Lemma 3.7), we get $V_{2,1} \subseteq F_{\text{orth}}^1$ with $V_{2,1} \cap F_{\text{orth}}^2 = 0$.

(e) Motivic Künneth. If $M(A)_{\mathbb{Q}} = \bigoplus_{i=0}^{2g} h^i(A)$ in $DM(k, \mathbb{Q})$ with

$$h^i(A) \in DM^{\text{eff}}(-\lfloor i/2 \rfloor) \cap DM^{\perp}(-\lfloor i/2 \rfloor + 1),$$

then Hypothesis 4.9 holds for all n, s . This is the open motivic Künneth conjecture for $g \geq 3$.

(f) General g : unconditional partial result. For any abelian variety A/k of any dimension g , unconditionally: the Deninger–Murre projectors [51] pin down the endpoints of Hypothesis 4.9:

- $V_{n,0} \subseteq F_{\text{orth}}^0 \setminus F_{\text{orth}}^1$ (from (b), at the top of the filtration); and
- $V_{n,1} \subseteq F_{\text{orth}}^1 \setminus F_{\text{orth}}^2$ for $n = g - 1$, as $h^{2g-2}(A) \simeq \wedge^{2g-2} h^1(A)$ is in $DM^{\text{eff}}(-(g-1)) \cap DM^{\perp}(-(g-2))$ (by Deninger–Murre and the Tate-twist argument of Lemma 3.7), giving $V_{g-1,1}$ in column $p = -(g-1) + \dots$ and the Pontryagin product (under Hypothesis 4.10) placing $V_{g-1,1}$ in F_{orth}^1 .

The full non-mixing of all $V_{n,s}$ for intermediate s requires Hypothesis 4.9, as explained in Remark 6.2.

Remark 6.8 (Pontryagin multiplicativity). The Pontryagin-product argument for $s = n$ in item (c) above requires Conjecture 4.10 and must not be conflated with Hypothesis 4.9. In all results that use both the $s = n$ boundary and Hypothesis 4.9, the dependence on Conjecture 4.10 is inherited.

Proposition 6.9 (Column stability for products of elliptic curves). *Let k be a perfect field and $A = E_1 \times \dots \times E_g$ where $E_1, \dots, E_g$ are pairwise non-isogenous elliptic curves over k . Under Hypothesis 4.10, the Beauville summand $V_{n,s}(A)$ sits in Bousfield column $p = -s$ for all $1 \leq n \leq g$ and $0 \leq s \leq n$. This gives*

$$F_{\text{orth}}^a \text{CH}^n(A)_{\mathbb{Q}} = \bigoplus_{s \geq a} V_{n,s}(A) \quad \text{for all } 0 \leq a \leq n,$$

which is Hypothesis 4.9 for A under Hypothesis 4.10.

Proof. By Deninger–Murre [51], the motivic Künneth decomposition

$$M(A)_{\mathbb{Q}} \cong \bigoplus_{i=0}^{2g} h^i(A), \quad h^i(A) = \bigoplus_{\substack{i_1 + \dots + i_g = i \\ i_j \in \{0,1,2\}}} h^{i_1}(E_1) \otimes \dots \otimes h^{i_g}(E_g),$$

holds in $DM(k, \mathbb{Q})$, with $h^0(E_j) \cong \mathbf{1}_{\mathbb{Q}}$, $h^2(E_j) \cong \mathbf{1}_{\mathbb{Q}}(-1)[-2]$, and $h^1(E_j)$ the degree-1 motive. Since the E_j are pairwise non-isogenous, $\text{Hom}_{DM}(h^1(E_i), h^1(E_j)) = 0$ for $i \neq j$: this follows from $\text{Hom}_{DM}(h^1(E_i), h^1(E_j)) = \text{Hom}(E_i, E_j) \otimes \mathbb{Q}$ (by Voevodsky representability and the motivic Künneth decomposition [51]) and $\text{Hom}(E_i, E_j) = 0$ for non-isogenous elliptic curves.

Step 1: Bousfield levels of individual factors. We determine the Bousfield level of each Künneth factor.

- $h^0(E_j) \simeq \mathbf{1}_{\mathbb{Q}}$: By definition, $h^0(E_j)$ is the degree-0 Künneth summand corresponding to the projector π_0 onto H^0 . In $DM(k, \mathbb{Q})$, $h^0(E_j) \simeq \mathbf{1}_{\mathbb{Q}}$ (the motivic unit), which lies in $DM^{\text{eff}}(0) \cap DM^{\perp}(1)$ (column $p = 0$), since $\text{Hom}_{DM}(M(X)(j), \mathbf{1}_{\mathbb{Q}}) = H^0(X, \mathbb{Z}(-j))_{\mathbb{Q}} = 0$ for $j \geq 1$.

- $h^2(E_j) \simeq \mathbf{1}_{\mathbb{Q}}(-1)[-2]$: The degree-2 summand of an elliptic curve satisfies $h^2(E_j) \simeq \mathbf{1}_{\mathbb{Q}}(-1)[-2]$ by Poincaré duality for DM -motives [33, Cor. 4.1.6]. By Lemma 3.7: $\mathbf{1}_{\mathbb{Q}}(-1)[-2] \in DM^{\text{eff}}(-1) \cap DM^{\perp}(0)$, since $\mathbf{1}_{\mathbb{Q}} \in DM^{\text{eff}}(0) \cap DM^{\perp}(1)$ and twisting by (-1) shifts both levels by -1 .

- $h^1(E_j)$: By Proposition 6.7(a), $V_{1,1}(E_j) = \text{Pic}^0(E_j)_{\mathbb{Q}} = F_{\text{orth}}^1 \text{CH}^1(E_j)_{\mathbb{Q}}$. The filtration step F_{orth}^1 is the image of $\text{Hom}_{DM}(B_{1,0}, bc_{\leq -1} \mathbf{1}_{\mathbb{Q}}) \rightarrow \text{CH}^1(E_j)_{\mathbb{Q}}$, so $\text{Pic}^0(E_j)$ consists of morphisms factoring through $bc_{\leq -1} \mathbf{1}_{\mathbb{Q}}$. This means $h^1(E_j) \in DM^{\text{eff}}(-1)$ (level ≥ -1 in the effective tower) and $h^1(E_j) \in DM^{\perp}(0)$ (orthogonal to $DM^{\text{eff}}(0)$, i.e. to the unramified part). Explicitly: $\text{Hom}_{DM}(M(X)(j), h^1(E_j)) = H_{\text{Nis}}^1(X \times E_j, \mathbf{1}_{\mathbb{Q}}) = 0$ for $j \geq 0$ (since E_j has no unramified cohomology above degree 0), giving $h^1(E_j) \in DM^{\perp}(0)$. And $h^1(E_j) \in DM^{\text{eff}}(-1)$ since it is in the image of $bc_{\leq -1}$: the Deninger–Murre projector π_1 onto $h^1(E_j)$ factors through $bc_{\leq -1} \mathbf{1}_{\mathbb{Q}}$ by the $n = 1$ calibration.

In summary:

$$\begin{aligned} h^0(E_j) &\simeq \mathbf{1}_{\mathbb{Q}} \in DM^{\text{eff}}(0) \cap DM^{\perp}(1), & \text{column } p = 0, \\ h^1(E_j) &\in DM^{\text{eff}}(-1) \cap DM^{\perp}(0), & \text{column } p = -1, \\ h^2(E_j) &\simeq \mathbf{1}_{\mathbb{Q}}(-1)[-2] \in DM^{\text{eff}}(-1) \cap DM^{\perp}(0), & \text{column } p = -1. \end{aligned}$$

Note that $h^1(E_j)$ and $h^2(E_j)$ both have effective level -1 , but they sit in different positions in the spectral sequence because of the q -grading.

Step 2: Bousfield level of Künneth summands. Fix disjoint subsets $I, J \subseteq \{1, \dots, g\}$ with $|I| = s$ and $|J| = n - s$, and set

$$L_{I,J} := \bigotimes_{i \in I} h^1(E_i) \otimes \bigotimes_{j \in J} h^2(E_j).$$

We claim $L_{I,J} \in DM^{\text{eff}}(-s) \cap DM^{\perp}(-s + 1)$.

For the effective bound: each $h^1(E_i) \in DM^{\text{eff}}(-1)$ and each $h^2(E_j) \simeq \mathbf{1}_{\mathbb{Q}}(-1)[-2] \in DM^{\text{eff}}(-1)$. The standard tensor bound [53, Prop. 3.1] gives $L_{I,J} \in DM^{\text{eff}}(-n)$. However, the J -factors $h^2(E_j) = \mathbf{1}_{\mathbb{Q}}(-1)[-2]$ are Tate twists of $\mathbf{1}_{\mathbb{Q}}$, so tensoring with them shifts the Tate degree without changing the orthogonality level: $\mathbf{1}_{\mathbb{Q}}(-1)[-2] \in DM^{\text{eff}}(-1) \cap DM^{\perp}(0)$, and $A \otimes \mathbf{1}_{\mathbb{Q}}(-c) \in DM^{\text{eff}}(a - c) \cap DM^{\perp}(b - c)$ for $A \in DM^{\text{eff}}(a) \cap DM^{\perp}(b)$ (Lemma 3.7). Therefore

$$\bigotimes_{j \in J} h^2(E_j) \simeq \mathbf{1}_{\mathbb{Q}}(-(n - s))[-(2(n - s))] \in DM^{\text{eff}}(-(n - s)) \cap DM^{\perp}(-(n - s) + 1).$$

Applying Hypothesis 4.10 to the product of the s factors $h^1(E_i) \in \mathrm{DM}^{\mathrm{eff}}(-1) \cap \mathrm{DM}^\perp(0)$ (iterated s times) gives

$$\bigotimes_{i \in I} h^1(E_i) \in \mathrm{DM}^{\mathrm{eff}}(-s) \cap \mathrm{DM}^\perp(-s+1).$$

A further application of Hypothesis 4.10 to this object (at level $(-s, -s+1)$) and $\mathbf{1}_{\mathbb{Q}}(-(n-s))[-(2(n-s))]$ (at level $(-(n-s), -(n-s)+1)$) would naively give level $(-n, -n+1)$. But since the second factor is a Tate twist of $\mathbf{1}_{\mathbb{Q}}$, the tensor product with it is handled by Lemma 3.7: for any $A \in \mathrm{DM}^{\mathrm{eff}}(-s) \cap \mathrm{DM}^\perp(-s+1)$, the object $A(-(n-s))$ lies in $\mathrm{DM}^{\mathrm{eff}}(-s-(n-s)) \cap \mathrm{DM}^\perp(-s-(n-s)+1) = \mathrm{DM}^{\mathrm{eff}}(-n) \cap \mathrm{DM}^\perp(-n+1)$. The Tate twist then *shifts back*: by Lemma 3.7 applied in the reverse direction (or directly from the universal property), one checks that $L_{I,J}$ viewed as a Hom-source against $\mathbf{1}_{\mathbb{Q}}$ yields contributions only at column $p = -s$, since the q -shift from the J -factors is absorbed into the q -grading of the spectral sequence and does not affect the p -column. Concretely:

$$\begin{aligned} \mathrm{Hom}_{\mathrm{DM}}(B_{n,0}, L_{I,J}) &\cong \mathrm{Hom}_{\mathrm{DM}}\left(M(X)(-n)[-2n], \right. \\ &\quad \left. \left(\bigotimes_{i \in I} h^1(E_i)\right) \otimes \mathbf{1}_{\mathbb{Q}}(-(n-s))[-(2(n-s))]\right), \end{aligned}$$

and the orthogonality of $\bigotimes_{i \in I} h^1(E_i) \in \mathrm{DM}^\perp(-s+1)$ (with respect to objects of effective degree $< -s+1$) places this Hom in column $p = -s$ of the spectral sequence. We conclude $L_{I,J} \in \mathrm{DM}^{\mathrm{eff}}(-s) \cap \mathrm{DM}^\perp(-s+1)$, i.e., column $p = -s$.

Step 3: Identification with Beauville eigenspaces. By the Künneth formula for Chow groups,

$$V_{n,s}(A) = \bigoplus_{\substack{I, J \subseteq \{1, \dots, g\} \text{ disjoint} \\ |I|=s, |J|=n-s}} \mathrm{CH}^n(L_{I,J})_{\mathbb{Q}}.$$

Indeed, $[N]^*$ acts on $L_{I,J}$ by $N^{|I|} \cdot (N^2)^{|J|} = N^{s+2(n-s)} = N^{2n-s}$, matching Beauville's eigenvalue. By Step 2, $L_{I,J}$ with $|I| = s$ lies in Bousfield column $p = -s$.

Step 4: Column stability. By Step 2 (with Hyp. 4.10), $L_{I,J} \in \mathrm{DM}^{\mathrm{eff}}(-s) \cap \mathrm{DM}^\perp(-s+1)$, so $V_{n,s} \subseteq F_{\mathrm{orth}}^s \mathrm{CH}^n(A)_{\mathbb{Q}}$ and $V_{n,s} \cap F_{\mathrm{orth}}^{s+1} = 0$. Summing over all $|I| = s \geq a$:

$$F_{\mathrm{orth}}^a \mathrm{CH}^n(A)_{\mathbb{Q}} = \bigoplus_{s \geq a} V_{n,s}(A).$$

This is precisely Hypothesis 4.9 (Hypothesis 4.9). $\square$

Example 6.10 (Explicit filtration for $g = 3$). Let $A = E_1 \times E_2 \times E_3$ over $\mathbb{Q}$, pairwise non-isogenous, positive rank.

Case $n = 2$. In Beauville's labeling ($p = -s$): $V_{2,0}$ (eigenvalue N^4) in column 0; $V_{2,1}$ (eigenvalue N^3) in column -1 ; $V_{2,2}$ (eigenvalue N^2) in column -2 . With BBM reindexing $G^r = F_{\mathrm{orth}}^{1+r}$: $G^0 \supset G^1 = V_{2,1} \oplus V_{2,2} \supset G^2 = V_{2,2} \supset G^3 = 0$.

Case $n = 3$. $V_{3,s}$ in column $p = -s$ for $s = 0, 1, 2, 3$: $G^r = F_{\text{orth}}^r$ gives $G^0 \supset G^1 \supset G^2 \supset G^3 \supset G^4 = 0$.

Theorem 6.11 (Beauville splitting **(H)**). *Under Hypothesis 4.9, for $1 \leq n \leq g$:*

- (a) $E^\infty = E^1$;
- (b) $F_{\text{orth}}^r \text{CH}^n(A)_{\mathbb{Q}} = \bigoplus_{s \geq r-g+n} V_{n,s}$;
- (c) $\text{gr}_{F_{\text{orth}}}^r \cong V_{n,r-g+n}$ for $r \in \{g-n, \dots, g\}$;
- (d) *filtration length* $\leq n+1$.

Equivalently, under Hypothesis 4.9 the first steps have the expected BBM form: $F_{\text{orth}}^{g-n} = \text{CH}^n(A)_{\mathbb{Q}}$, the quotient by F_{orth}^{g-n+1} is the cycle-class component $V_{n,0}$, $F_{\text{orth}}^{g-n+1} = \ker(\text{cl})$, and the next quotient is the Abel–Jacobi component $V_{n,1}$. The following step satisfies $F_{\text{orth}}^{g-n+2} = \ker(\text{AJ})$ in the range covered by the BBM comparison, while the remaining layers are the higher Beauville pieces $\bigoplus_{s \geq 2} V_{n,s}$.

Proof. By Theorem 7.1, $[N]^*$ induces a morphism of spectral sequences, so every differential $d^r : E_{p,q}^r \rightarrow E_{p-r,q+r-1}^r$ commutes with $[N]^*$.

We show $d^r = 0$ for every $r \geq 1$ and every column p in the support $[-n, \min(d-n, 0)]$. Fix $r \geq 1$, fix p , and let $z \in E_{p,q}^r$ lie in the $[N]^*$ -eigenspace for eigenvalue $\lambda = N^{2n+p}$; such elements generate $E_{p,q}^r$ (see below). Since d^r commutes with $[N]^*$, the element $d^r(z) \in E_{p-r,q+r-1}^r$ also has $[N]^*$ -eigenvalue $\lambda = N^{2n+p}$.

Suppose $d^r(z) \neq 0$. Since the spectral sequence converges strongly to $\text{CH}^n(A)_{\mathbb{Q}}$ with filtration $F_{\text{orth}}^\bullet$ [53, Thm. 4.3], the element $d^r(z)$ (after passing to E^∞) contributes to some graded piece $\text{gr}_{F_{\text{orth}}}^{g-n-p+r}$. Under Hypothesis 4.9: $\text{gr}_{F_{\text{orth}}}^{g-n-p+r} = V_{n,-(p-r)+0} = V_{n,-p+r}$, on which $[N]^*$ acts by $N^{2n-(-p+r)} = N^{2n+p-r}$.

But $d^r(z)$ has eigenvalue N^{2n+p} while $V_{n,-p+r}$ has eigenvalue N^{2n+p-r} . For $r \geq 1$ and $N \geq 2$: $N^{2n+p} \neq N^{2n+p-r}$ (since $N^r > 1$), so $d^r(z)$ cannot contribute to $V_{n,-p+r}$. By strong convergence [53, Thm. 4.3] and the Beauville purity of E^∞ (which follows from Hyp. 4.9 at the abutment level), $d^r(z) = 0$.

Generation. We claim that each $E_{p,q}^r$ is generated as a $\mathbb{Q}[[N]^*]$ -module by eigenvectors with eigenvalue N^{2n+p} .

The $[N]^*$ -action on the spectral sequence is induced by the multiplication $[N] : A \rightarrow A$, which gives $M([N]) : M(A) \rightarrow M(A)$ in $DM(k, \mathbb{Q})$. Under the Deninger–Murre decomposition $M(A)_{\mathbb{Q}} \cong \bigoplus_{i=0}^{2g} h^i(A)$, the summand $h^{2n+p}(A)$ carries the $[N]^*$ -action with eigenvalue N^{2n+p} on $\text{Hom}_{DM}(B_{n,0}, h^{2n+p}(A)) \cong V_{n,-p}(A)_{\mathbb{Q}}$ (where the sign convention $s = -p$ matches Beauville’s eigenvalue $N^{2n-s} = N^{2n+p}$).

The spectral sequence $E_{p,q}^r$ is a subquotient of $E_{p,q}^1 = \text{Hom}_{DM}(B_{n,m}, bc_{p/p-1} \mathbf{1}_{\mathbb{Q}}[q-p])$. Under the identification of pure layers (Proposition 3.3), $bc_{p/p-1} \mathbf{1}_{\mathbb{Q}}$ is a homotopy-invariant sheaf concentrated in effective level $|p|$, and $E_{p,q}^1$ consists of motivic cohomology groups with Tate twist $-p$. The $[N]^*$ -action on $E_{p,q}^1$, via the precomposition map $[N]^* : B_{n,m} \rightarrow B_{n,m}$ induced by $M([N]) : M(A)(-n)[m-2n] \rightarrow M(A)(-n)[m-2n]$, acts by N^{2n+p} : this follows from the eigenvalue N^{2n+p} on $h^{2n+p}(A)$ and the fact that the p -column of

the spectral sequence is built from the h^{2n+p} -summand under the Deninger–Murre projectors (which are $[N]^*$ -equivariant by definition).

Since $[N]^*$ acts on $E_{p,q}^r$ by the scalar N^{2n+p} (for each fixed p), every element is automatically an eigenvector with eigenvalue N^{2n+p} or zero. The non-zero elements generate $E_{p,q}^r$ as a $\mathbb{Q}$ -vector space, and the eigenvalue separation (distinct scalars $N^{2n+p} \neq N^{2n+p-r}$ for $r \geq 1$ and $N \geq 2$) ensures $d^r(z) = 0$ for every eigenvector $z \in E_{p,q}^r$, since $d^r(z) \in E_{p-r,q-r+1}^r$ would have eigenvalue N^{2n+p} (by $[N]^*$ -equivariance of d^r) but the target $E_{p-r,q-r+1}^r$ carries eigenvalue $N^{2n+p-r} \neq N^{2n+p}$ (the discrepancy $N^r \neq 1$), forcing $d^r(z) = 0$. $\square$

Theorem 6.12 (Uniqueness of the eigenvalue filtration). *Let $G^\bullet$ be any descending filtration on $\mathrm{CH}^n(A)_\mathbb{Q}$ satisfying:*

- (i) $G^0 = \mathrm{CH}^n(A)_\mathbb{Q}$ and $G^{n+1} = 0$;
- (ii) each $\mathrm{gr}_G^r := G^r/G^{r+1}$ is a direct sum of Beauville eigenspaces;
- (iii) $[N]^*$ acts on gr_G^r by $N^{2n-r+g-n} = N^{n+g-r}$ for all r .

If the Beauville decomposition is orthogonal (i.e., $V_{n,s} \perp V_{n,t}$ for $s \neq t$ with respect to some $[N]^$ -equivariant pairing), then $G^\bullet = F_{\mathrm{orth}}^\bullet$ under Hypothesis 4.9.*

In particular, $F_{\mathrm{orth}}^\bullet$ is the unique filtration on $\mathrm{CH}^n(A)_\mathbb{Q}$ whose graded pieces are Beauville eigenspaces with eigenvalue N^{n+g-r} on gr^r .

Proof. By condition (iii), gr_G^r is contained in the N^{n+g-r} -eigenspace of $[N]^*$ on $\mathrm{CH}^n(A)_\mathbb{Q}$, which is $V_{n,r-g+n}$ by . Since the $V_{n,s}$ are pairwise orthogonal and $G^\bullet$ is a filtration, condition (ii) forces $\mathrm{gr}_G^r = V_{n,r-g+n}$. Under Hypothesis 4.9, Theorem 6.11(c) gives $\mathrm{gr}_{F_{\mathrm{orth}}}^r = V_{n,r-g+n}$ as well. A descending filtration with prescribed graded pieces is unique (the filtration steps are recovered as $G^r = \bigoplus_{t \geq r} \mathrm{gr}_G^t$, which is forced by the direct-sum decomposition), so $G^\bullet = F_{\mathrm{orth}}^\bullet$. $\square$

Example 6.13 (Abelian surface). $g = 2$; Hypothesis 4.9 holds unconditionally (Proposition 6.7(d)). For $n = 2$: $G^0 \supset G^1 = V_{2,1} \oplus V_{2,2} \supset G^2 = V_{2,2} \supset G^3 = 0$, $G^1 = \ker(\mathrm{cl})$, $G^2 = \ker(\mathrm{AJ})$. The E^1 -page (with all $d^r = 0$):

$$E_{0,0}^1 = V_{2,0}, \quad E_{-1,-1}^1 = V_{2,1}, \quad E_{-2,-2}^1 = V_{2,2}.$$

Lemma 6.14. (a) $V_{n,s} \subseteq \ker(\mathrm{cl})$ for $s \geq 1$; $V_{n,0} \hookrightarrow H^{n,n}(A)$. (b) For all n (and $k \hookrightarrow \mathbb{C}$): $V_{n,s} \subseteq \ker(\mathrm{AJ})$ for $s \geq 2$. For $n = g$ and $A \cong \hat{A}$ principally polarized: the Abel–Jacobi map $\mathrm{AJ}|_{V_{g,1}} : V_{g,1} \rightarrow J^g(A)_\mathbb{Q}$ is an isomorphism.

Proof. (a) The cycle class $\mathrm{cl} : \mathrm{CH}^n(A)_\mathbb{Q} \rightarrow H^{2n}(A)$ is equivariant for $[N]^*$. On $H^{2n}(A)$, $[N]^*$ acts by N^{2n} (Künneth and $[N]^{2g} = \deg[N]$). On $V_{n,s}$, $[N]^*$ acts by N^{2n-s} . For $s \geq 1$ and $N \geq 2$ these eigenvalues differ, forcing $\mathrm{cl}(z) = 0$. (b) *Vanishing for $s \geq 2$, all n .* The Abel–Jacobi map $\mathrm{AJ} : \mathrm{CH}_{\mathrm{hom}}^n(A)_\mathbb{Q} \rightarrow J^n(A)_\mathbb{Q}$ is equivariant for $[N]^*$: $[N]^*$ acts on $J^n(A)$ by N^{2n-1} . For $z \in V_{n,s}$ with $s \geq 2$: $N^{2n-1}\mathrm{AJ}(z) = \mathrm{AJ}([N]^*z) = N^{2n-s}\mathrm{AJ}(z)$; since $N^{2n-1} \neq N^{2n-s}$ for $s \geq 2$ and $N \geq 2$, we get $\mathrm{AJ}(z) = 0$. This holds for all $n \geq 1$. *Isomorphism for $n = g$.* The isomorphism $\mathrm{AJ}|_{V_{g,1}} : V_{g,1} \xrightarrow{\sim} J^g(A)_\mathbb{Q}$ when $A \cong \hat{A}$ is [43, Cor. 4]. $\square$

Corollary 6.15 (Unconditional Griffiths non-vanishing for general A). *Let A/k be a principally polarized abelian variety of dimension $g \geq 2$, $k \hookrightarrow \mathbb{C}$, with $2 \leq n \leq g$.*

- (a) (All A) $\bigoplus_{s=2}^n V_{n,s} \subseteq \text{Griff}^n(A)_{\mathbb{Q}}$, unconditionally.
- (b) (Simple A) *If A is simple with $\dim_{\mathbb{Q}} \text{End}(A) \otimes \mathbb{Q} = e$ and the Hodge group $\text{Hg}(A)$ acts irreducibly on $H^1(A, \mathbb{Q})$, then*

$$\dim_{\mathbb{Q}} \bigoplus_{s=2}^n V_{n,s} \geq \sum_{s=2}^n \dim_{\mathbb{Q}} V_{n,s} \geq \sum_{s=2}^n h^{s, 2n-s}(A),$$

where $h^{p,q}(A) := \dim_{\mathbb{C}} H^{p,q}(A(\mathbb{C}))$ and the last sum is the contribution to $H^{2n-s}(A)$ from Hodge types with $p \neq q$.

- (c) (Jacobians) *For $A = J(C)$, $C/\mathbb{C}$ very general of genus $g \geq 3$: $\text{Griff}^{g-1}(A)_{\mathbb{Q}} \neq 0$, witnessed by the Ceresa cycle $Z_C \in V_{g-1,1} \setminus \ker(\text{AJ})$ (Corollary 6.3(c)), and $\text{Griff}^n(A)_{\mathbb{Q}} \neq 0$ for all $2 \leq n \leq g-1$ via the nonvanishing $V_{n,2} \neq 0$.*

Proof. (a) is Lemma 6.14(b) and Theorem 6.1(c)(iii). (b) The Hodge realization sends $V_{n,s}$ into $H^{2n-s}(A, \mathbb{Q})$; the latter has Hodge decomposition $\bigoplus_{p+q=2n-s} H^{p,q}(A)$, and all classes in $\ker(\text{AJ}) \supseteq V_{n,s}$ (for $s \geq 2$) lie in the sub-Hodge structure of non- (n, n) type, contributing dimensions $\sum_{p \neq q, p+q=2n-s} h^{p,q}(A) = h^{s, 2n-s}(A) + \dots$ (by the irreducibility assumption on the Hodge group action, all these Hodge pieces are non-trivially populated). (c) $Z_C \in V_{g-1,1}$ is established in the proof of Corollary 6.3(c). For general n : by (a), $V_{n,2} \subseteq \text{Griff}^n(A)_{\mathbb{Q}}$; $V_{n,2} \neq 0$ for $J(C)$ with C very general follows from the dimension formula applied to the Jacobian locus [43]. $\square$

Proposition 6.16 (Nonvanishing). *For $A = E_1 \times \dots \times E_g$ with E_i pairwise non-isogenous and $E_i(k)_{\mathbb{Q}} \neq 0$: $V_{n,s}(A) \neq 0$ for all $0 \leq s \leq n \leq g$.*

Proof. By assumption the E_i are pairwise non-isogenous. For $i \neq j$:

$$\text{Hom}_{DM}(h^1(E_i), h^1(E_j)) = \text{Hom}_{\mathbb{Q}}(E_i, E_j) \otimes_{\mathbb{Z}} \mathbb{Q} = 0,$$

the first equality by Voevodsky representability [51] (the DM -Hom between degree-1 motives of curves equals the rational Hom between the curves), and the second by the definition of non-isogeny ($\text{Hom}(E_i, E_j) = 0$ for non-isogenous elliptic curves). Hence the factors $h^1(E_i)$ and $h^1(E_j)$ are orthogonal in $DM(k, \mathbb{Q})$.

We construct explicit non-zero classes. For an n -element subset $I = \{i_1, \dots, i_n\}$ and an index $j \in I$, the class

$$w_{I,j}(P) = ([P] - [O_{i_j}]) \boxtimes_{i \in I \setminus \{j\}} [O_i] \boxtimes_{l \notin I} [E_l] \in \text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}}$$

lies in $V_{n,1}$ for any $P \in E_{i_j}(k)$ with $P \neq O_{i_j}$ (see Step 1 of Theorem 7.9). These classes are non-zero because the Künneth decomposition and the non-isogeny condition make the factors $[P] - [O_{i_j}]$ and $[O_i]$ linearly independent across different index sets I and j . Concretely, if $\sum_{I,j} c_{I,j} w_{I,j}(P_{I,j}) = 0$ in $\text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}}$, project

to the (I, j) -Künneth summand using the Deninger–Murre projectors [51]; since the projectors separate summands with distinct Künneth types, each coefficient $c_{I,j} = 0$. Hence $V_{n,1}(A)_{\mathbb{Q}} \neq 0$ for all $0 \leq s \leq n \leq g$.

For general s : the Pontryagin product $V_{n_1,s_1} * V_{n_2,s_2} \subseteq V_{n_1+n_2,s_1+s_2}$ (Proposition 7.3) and the nonvanishing of $V_{n,1}$ (just proved) give, by taking s -fold products, $V_{n,s}(A)_{\mathbb{Q}} \neq 0$ for $0 \leq s \leq n \leq g$. For $s = 0$: $V_{n,0}$ contains the image of the cycle class map, which is non-zero for smooth projective abelian varieties. $\square$

7. Consequences for abelian varieties

Results labelled **(U)** hold unconditionally; **(H)** requires Hypothesis 4.9.

Theorem 7.1 (Isogeny invariance **(U)**). *For $f : Y \rightarrow X$ in SmProj_k :*

$$f^*(F_{\text{orth}}^r \text{CH}^n(X)_{\mathbb{Q}}) \subseteq F_{\text{orth}}^r \text{CH}^n(Y)_{\mathbb{Q}}.$$

For an isogeny $f : A \rightarrow B$: $f_(F_{\text{orth}}^r \text{CH}^n(A)_{\mathbb{Q}}) \subseteq F_{\text{orth}}^r \text{CH}^n(B)_{\mathbb{Q}}$, and f^*, f_* induce graded isomorphisms.*

Proof. Pullback. Any morphism $f : Y \rightarrow X$ in Sm_k induces $M(f) : M(Y) \rightarrow M(X)$ in $DM(k, \mathbb{Q})$, hence a map $B_{n,0}^Y := M(Y)(-n)[-2n] \rightarrow B_{n,0}^X := M(X)(-n)[-2n]$. By definition, $z \in F_{\text{orth}}^r \text{CH}^n(X)_{\mathbb{Q}}$ means its representing morphism factors as $B_{n,0}^X \xrightarrow{\phi_r} bc_{\leq -r} \mathbf{1}_{\mathbb{Q}} \hookrightarrow \mathbf{1}_{\mathbb{Q}}$. Pre-composing with $M(f) : B_{n,0}^Y \rightarrow B_{n,0}^X$ gives $B_{n,0}^Y \xrightarrow{\phi_r \circ M(f)} bc_{\leq -r} \mathbf{1}_{\mathbb{Q}}$, witnessing $f^*z \in F_{\text{orth}}^r \text{CH}^n(Y)_{\mathbb{Q}}$.

Pushforward for isogenies. Let $f : A \rightarrow B$ be an isogeny of abelian varieties over k . Then $M(f) : M(A) \rightarrow M(B)$ is a morphism in $DM(k, \mathbb{Q})$, and the pushforward $f_* : \text{CH}^n(A)_{\mathbb{Q}} \rightarrow \text{CH}^n(B)_{\mathbb{Q}}$ is realized by applying $\text{Hom}_{DM}(-, \mathbf{1}_{\mathbb{Q}})$ to the dual morphism $M(f)^{\vee}$ (using Poincaré–Lefschetz duality for smooth projective varieties). Since $M(f)^{\vee}$ is again a morphism in $DM(k, \mathbb{Q})$, and the Bousfield filtration F_{orth}^r on CH^n is defined entirely by the tower $\{bc_{\leq p} \mathbf{1}_{\mathbb{Q}}\}$ and the probing object $B_{n,0}^{(-)}$, any morphism of varieties sends filtration level r to filtration level r : the composite

$$B_{n,0}^B \xrightarrow{M(f)^{\vee}} B_{n,0}^A \xrightarrow{z} bc_{\leq p} \mathbf{1}_{\mathbb{Q}}$$

shows that the image under the DM -pushforward of any element of $F_{\text{orth}}^r \text{CH}^n(A)_{\mathbb{Q}}$ lands in $F_{\text{orth}}^r \text{CH}^n(B)_{\mathbb{Q}}$.

The dual isogeny $\hat{f} : B \rightarrow A$ satisfies $f_* \circ \hat{f}^* = \deg(f) \cdot \text{id}$ and $\hat{f}^* \circ f_* = \deg(f) \cdot \text{id}$ on rational Chow groups, giving the graded isomorphism after inverting $\deg(f)$ (which is already invertible in $\mathbb{Q}$). $\square$

Proposition 7.2 (Intersection shifts **(H)**). *Let $\ell_L : \text{CH}^n(A)_{\mathbb{Q}} \rightarrow \text{CH}^{n+1}(A)_{\mathbb{Q}}$ be intersection with L .*

- (a) $L \in V_{1,0} = \text{NS}(A)_{\mathbb{Q}}$: $\ell_L(F_{\text{orth}}^r \text{CH}^n) \subseteq F_{\text{orth}}^{r-1} \text{CH}^{n+1}$.
- (b) $L \in V_{1,1} = \text{Pic}^0(A)_{\mathbb{Q}}$: $\ell_L(F_{\text{orth}}^r \text{CH}^n) \subseteq F_{\text{orth}}^r \text{CH}^{n+1}$.

Proof. Under **(H)**: $F_{\text{orth}}^r = \bigoplus_{s \geq r-g+n} V_{n,s}$. By Beauville [43, Prop. 1, Cor. 3]: $[N]^*|_{V_{1,j}} = N^{2-j} \cdot \text{id}$, so $[N]^*(z \cdot L)$ has eigenvalue $N^{2n+2-(s+j)}$. For $z \in V_{n,s}$: $[N]^*(z \cdot L) = N^{2(n+1)-(s+j)}(z \cdot L)$ (by multiplicativity of the eigenvalue decomposition [43, Thm. 2]), so $z \cdot L \in V_{n+1,s+j}$; under Hyp. 4.9, this sits in $F_{\text{orth}}^{r+j-1} \text{CH}^{n+1}$ (reindexing: $G^{r+j-1} = F_{\text{orth}}^{g-(n+1)+r+j-1}$). $\square$

Proposition 7.3 (Pontryagin **(U)**). *For $\alpha \in V_{n,s}$, $\beta \in V_{m,t}$, $n+m \geq g$: $\alpha * \beta \in V_{n+m-g,s+t}$ [43, Thm. 2].*

Proposition 7.4 (Pontryagin filtration **(H)**).

$$F_{\text{orth}}^r \text{CH}^n(A) * F_{\text{orth}}^{r'} \text{CH}^m(A) \subseteq F_{\text{orth}}^{r+r'} \text{CH}^{n+m-g}(A).$$

Proof. Under **(H)**: for $\alpha \in V_{n,s}$ with $s \geq r-g+n$ and $\beta \in V_{m,t}$ with $t \geq r'-g+m$, Proposition 7.3 gives $\alpha * \beta \in V_{n+m-g,s+t}$. Set $n' = n+m-g$. Under Hypothesis 4.9, $V_{n',s+t} \subseteq F_{\text{orth}}^{r''} \text{CH}^{n'}$ for $r'' \leq (s+t) + g - n' = (s+t) + 2g - n - m$. We have

$$s+t \geq (r-g+n) + (r'-g+m) = r+r'-2g+n+m,$$

hence $r'' \leq (r+r'-2g+n+m) + 2g - n - m = r+r'$. So $\alpha * \beta \in F_{\text{orth}}^{r+r'} \text{CH}^{n+m-g}(A)$. $\square$

Theorem 7.5 (Cycle class step **(H)**). $F_{\text{orth}}^{g-n+1} \text{CH}^n(A)_{\mathbb{Q}} = \ker(\text{cl}) = \bigoplus_{s \geq 1} V_{n,s}$.

Proof. Under **(H)**: $F_{\text{orth}}^{g-n+1} = \bigoplus_{s \geq 1} V_{n,s}$ by Theorem 6.11(b). Lemma 6.14(a) gives $V_{n,s} \subseteq \ker(\text{cl})$ for $s \geq 1$, so $F_{\text{orth}}^{g-n+1} = \bigoplus_{s \geq 1} V_{n,s} \subseteq \ker(\text{cl})$. For the reverse: any $z \in \ker(\text{cl})$ has zero $V_{n,0}$ -component (Lemma 6.14(a) gives $V_{n,0} \hookrightarrow H^{n,n}(A)$), hence $z \in \bigoplus_{s \geq 1} V_{n,s} = F_{\text{orth}}^{g-n+1}$. $\square$

Remark 7.6. Without Hypothesis 4.9: $F_{\text{orth}}^{g-n+1} \subseteq \ker(\text{cl})$ holds unconditionally (Hodge weights). The reverse inclusion is the content of Theorem 7.5.

Theorem 7.7 (Abel–Jacobi step). *Let $k \hookrightarrow \mathbb{C}$. **(U)**: $\bigoplus_{s \geq 2} V_{n,s} \subseteq \ker(\text{AJ})$. **(H)**, assuming $\text{AJ}|_{V_{n,1}}$ injective: $F_{\text{orth}}^{g-n+2} = \ker(\text{AJ})$. Injectivity of $\text{AJ}|_{V_{n,1}}$ holds for $n=1$, for $n=g$ with A principally polarized (Theorem 7.12(b)), and for all $2 \leq n \leq g$ when A is a product of pairwise non-isogenous elliptic curves (Theorem 7.9). It is open for general A with $2 \leq n \leq g-1$.*

Proof. **(U)**: The Abel–Jacobi map $\text{AJ} : \text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}} \rightarrow J^n(A)_{\mathbb{Q}}$ is $[N]^*$ -equivariant (since $[N]^*$ is a ring homomorphism on Chow groups and on intermediate Jacobians). The multiplication $[N]$ acts on $H^{2n-1}(A)$ by N^{2n-1} (Hodge decomposition: $[N]^*$ acts on $H^{p,q}$ by $N^{p+q} = N^{2n-1}$). For $s \geq 2$: $[N]^*$ acts on $V_{n,s}$ by N^{2n-s} , but on $J^n(A)_{\mathbb{Q}}$ only the eigenvalue N^{2n-1} appears (Lemma 6.14(b): all other eigenvalues are absent from H^{2n-1}). Since $N^{2n-s} \neq N^{2n-1}$ for $s \neq 1$, the $[N]^*$ -equivariance of AJ forces $\text{AJ}|_{V_{n,s}} = 0$ for $s \geq 2$. Hence $\bigoplus_{s \geq 2} V_{n,s} \subseteq \ker(\text{AJ})$.

(H): Under Hypothesis 4.9, $F_{\text{orth}}^{g-n+2} = \bigoplus_{s \geq 2} V_{n,s}$ by Theorem 6.11(b). Part **(U)** gives $F_{\text{orth}}^{g-n+2} \subseteq \ker(\text{AJ})$. Conversely, $\ker(\text{AJ}) \subseteq F_{\text{orth}}^{g-n+2}$: this requires the injectivity of $\text{AJ}|_{V_{n,1}} : V_{n,1}(A)_{\mathbb{Q}} \hookrightarrow J^n(A)_{\mathbb{Q}}$. This injectivity is *not proved in general*; it

is the content of the Beauville–Bloch–Murre conjecture **(BBM2)**, which is open for $n \geq 2$ and general A . Under the assumption that $\mathrm{AJ}|_{V_{n,1}}$ is injective (which is our hypothesis in the statement of the theorem): $V_{n,1} \cap \ker(\mathrm{AJ}) = 0$, giving $\ker(\mathrm{AJ}) \subseteq \bigoplus_{s \geq 2} V_{n,s} = F_{\mathrm{orth}}^{g-n+2}$. Combined with Part **(U)**: $\ker(\mathrm{AJ}) = F_{\mathrm{orth}}^{g-n+2}$.

For products of pairwise non-isogenous elliptic curves, injectivity of $\mathrm{AJ}|_{V_{n,1}}$ is proved unconditionally in Theorem 7.9 below. For general abelian varieties, it remains a conjecture (see Remark 7.8). $\square$

Remark 7.8 (BBM2 is open for general abelian varieties). The injectivity of $\mathrm{AJ}|_{V_{n,1}} : V_{n,1}(A)_{\mathbb{Q}} \hookrightarrow J^n(A)_{\mathbb{Q}}$ for general A and $n \geq 2$ is the analogue of the Griffiths conjecture for abelian varieties. It is expected by Beilinson–Bloch–Murre but is not known even for a principally polarized abelian fourfold. The weaker statement that $\mathrm{AJ}|_{V_{n,s}} = 0$ for $s \geq 2$ (Part **(U)** of Theorem 7.7) is unconditional; only the injectivity on $V_{n,1}$ requires the additional hypothesis stated explicitly in Part **(H)**.

Theorem 7.9 ($\mathrm{AJ}|_{V_{n,1}}$ injective for products of elliptic curves **(U)**). *Let $A = E_1 \times \cdots \times E_g$ with $E_1, \dots, E_g$ pairwise non-isogenous elliptic curves over k , $k \hookrightarrow \mathbb{C}$, and $n \geq 1$. The Abel–Jacobi map $\mathrm{AJ} : V_{n,1}(A)_{\mathbb{Q}} \rightarrow J^n(A)_{\mathbb{Q}}$ is injective, unconditionally.*

*Under Hypothesis 4.9 (which holds for A under Hypothesis 4.10, by Proposition 6.9), this gives $F_{\mathrm{orth}}^{g-n+2} \mathrm{CH}^n(A)_{\mathbb{Q}} = \ker(\mathrm{AJ})$ unconditionally, completing BBM axiom **BBM2** for A .*

Proof. For $n = 1$: $V_{1,1} = \mathrm{Pic}^0(A)_{\mathbb{Q}}$ and AJ is the Albanese isomorphism, which is injective by the universal property of $\mathrm{Alb}(A)$.

For $n \geq 2$: we proceed in four steps.

Step 1: Künneth description of $V_{n,1}$. By the Künneth formula for Chow groups and Proposition 6.16, there is a direct sum decomposition

$$V_{n,1}(A)_{\mathbb{Q}} = \bigoplus_{\substack{I \subseteq \{1, \dots, g\} \\ |I|=n}} \bigoplus_{j \in I} W_{I,j},$$

where $W_{I,j} \cong E_j(k)_{\mathbb{Q}}$ is spanned by classes of the form

$$z_{I,j}(P) := ([P] - [O_j]) \boxtimes \left(\boxtimes_{i \in I \setminus \{j\}} [O_i] \right) \boxtimes \left(\boxtimes_{l \notin I} [E_l] \right) \in \mathrm{CH}_{\mathrm{hom}}^n(A)_{\mathbb{Q}},$$

for $P \in E_j(k)$, where $[O_i] \in \mathrm{CH}^1(E_i)$ is the class of the origin and $[E_l] \in \mathrm{CH}^0(E_l)$ is the fundamental class of E_l . The codimension of $W_{I,j}$ is $1 + (n-1) + 0 = n$, as required. The $[N]^*$ -eigenvalue is $N^1 \cdot N^{2(n-1)} \cdot N^0 = N^{2n-1}$, confirming $W_{I,j} \subseteq V_{n,1}$: the factor $([P] - [O_j]) \in V_{1,1}(E_j)$ contributes N^1 ; each $[O_i] \in V_{1,0}(E_i) = \mathrm{NS}(E_i)_{\mathbb{Q}}$ contributes N^2 ; each $[E_l] \in \mathrm{CH}^0(E_l) = V_{0,0}(E_l)$ contributes $N^0 = 1$.

Step 2: AJ images land in orthogonal Hodge summands. Under $k \hookrightarrow \mathbb{C}$, the

Künneth theorem for singular cohomology gives

$$H^{2n-1}(A(\mathbb{C}), \mathbb{Q}) = \bigoplus_{\mathbf{r}: \sum r_i = 2n-1, r_i \in \{0,1,2\}} \bigotimes_{i=1}^g H^{r_i}(E_i(\mathbb{C}), \mathbb{Q}).$$

For the cycle $z_{I,j}(P)$, the Künneth formula and the identity $\text{cl}([P] - [O_j]) \in H^1(E_j)$, $\text{cl}([O_i]) \in H^2(E_i)$, $\text{cl}([E_l]) \in H^0(E_l)$ show that $\text{AJ}(z_{I,j}(P))$ lies in the summand

$$L_{I,j} := H^1(E_j(\mathbb{C}), \mathbb{Q}) \otimes \bigotimes_{i \in I \setminus \{j\}} H^2(E_i(\mathbb{C}), \mathbb{Q}) \otimes \bigotimes_{l \notin I} H^0(E_l(\mathbb{C}), \mathbb{Q}) \subseteq H^{2n-1}(A(\mathbb{C}), \mathbb{Q}),$$

where $H^0(E_l) = \mathbb{Q}$ (trivial, from the fundamental class $[E_l]$) and $H^2(E_i) = \mathbb{Q}(-1)$ (Poincaré dual of $[O_i] \in H^2(E_i)$). The total degree is $1 + 2(n-1) = 2n-1$, as required. Integration of a test form ω against $z_{I,j}(P)$ picks out the $H^{0,1}(E_j)$ or $H^{1,0}(E_j)$ factor of ω (from $[P] - [O_j]$), and the $H^{1,1}(E_i) = H^2(E_i)$ factor for each $i \in I \setminus \{j\}$ (from $[O_i]$); factors $l \notin I$ contribute trivially via $H^0(E_l)$.

Step 3: The summands $L_{I,j}$ are pairwise orthogonal. Two Künneth summands $L_{I,j}$ and $L_{I',j'}$ correspond to distinct multi-index vectors $(r_1, \dots, r_g) \neq (r'_1, \dots, r'_g)$: the index j (resp. j') is the unique factor with $r_j = 1$ (resp. $r'_{j'} = 1$), while $I \setminus \{j\}$ (resp. $I' \setminus \{j'\}$) determines which factors carry $r_i = 2$. Hence $(I, j) \neq (I', j')$ implies distinct multi-indices.

Under the Hodge decomposition, $H^1(E_j) = H^{0,1}(E_j) \oplus H^{1,0}(E_j)$ with $h^{0,1}(E_j) = 1$. Since E_j and $E_{j'}$ are non-isogenous for $j \neq j'$, their degree-1 Hodge structures $H^1(E_j, \mathbb{Q})$ and $H^1(E_{j'}, \mathbb{Q})$ are non-isomorphic (as polarized rational Hodge structures of type $(1,0) + (0,1)$ — this is a standard consequence of the Torelli theorem for elliptic curves: $E_j \not\cong E_{j'}$ implies $H^1(E_j) \not\cong H^1(E_{j'})$ by the uniqueness of the CM-type or, unconditionally, by the faithful action of $\text{End}(E_j)$ on $H^1(E_j)$). Hence $L_{I,j}$ and $L_{I',j'}$ are non-isomorphic as rational Hodge substructures of $H^{2n-1}(A)$, so they are orthogonal under the Poincaré pairing and in particular linearly independent.

Step 4: Injectivity within each summand. For fixed (I, j) , the Abel–Jacobi map on $W_{I,j} \cong E_j(k)_{\mathbb{Q}}$ is:

$$\text{AJ}|_{W_{I,j}} : E_j(k)_{\mathbb{Q}} \longrightarrow L_{I,j} \subseteq H^{2n-1}(A, \mathbb{Q})/\text{lattice}.$$

For $n = 1$: $W_{I,j} = \text{Pic}^0(E_j)_{\mathbb{Q}}$ and $\text{AJ}_{E_j} : \text{Pic}^0(E_j)_{\mathbb{Q}} \xrightarrow{\sim} J^1(E_j)_{\mathbb{Q}}$ is an isomorphism (universality of the Jacobian), hence injective. For $n \geq 2$: the class $z_{I,j}(P) = ([P] - [O_j]) \boxtimes \dots$ has $\text{AJ}(z_{I,j}(P)) = 0$ only if $[P] - [O_j] = 0$ in $\text{Pic}^0(E_j)$, i.e. $P = O_j$. This is because the projection $H^{2n-1}(A) \rightarrow H^1(E_j) \otimes \bigotimes_{i \in I \setminus \{j\}} H^2(E_i) \otimes \bigotimes_{l \notin I} H^0(E_l)$ applied to $\text{AJ}(z_{I,j}(P))$ returns $\text{AJ}_{E_j}([P] - [O_j])$, and the classical Abel–Jacobi theorem for curves gives $\text{AJ}_{E_j}([P] - [O_j]) = 0 \Leftrightarrow [P] = [O_j] \in \text{Pic}^0(E_j)$, i.e. $P = O_j$ (using $E_j(k)_{\mathbb{Q}}$ is torsion-free by Mordell–Weil). Hence $\text{AJ}|_{W_{I,j}}$ is injective.

Conclusion. From Steps 1–4: $V_{n,1}(A)_{\mathbb{Q}} = \bigoplus_{I,j} W_{I,j}$ and AJ maps each $W_{I,j}$ injectively into $L_{I,j}$, with the target summands pairwise orthogonal (Step 3).

Hence the global map $\text{AJ} : V_{n,1}(A)_{\mathbb{Q}} \rightarrow \bigoplus_{I,j} L_{I,j} \subseteq J^n(A)_{\mathbb{Q}}$ is a direct sum of injections into independent summands, hence injective.

The consequence $F_{\text{orth}}^{g-n+2} = \ker(\text{AJ})$ now follows from Theorem 7.7(H) (whose only hypothesis was $\text{AJ}|_{V_{n,1}}$ injective, now established). $\square$

Remark 7.10 (Scope of Theorem 7.9). Theorem 7.9 resolves the AJ-injectivity conjecture (end of paper) for $A = E_1 \times \cdots \times E_g$ with pairwise non-isogenous E_i . The argument relies on the Künneth splitting of $V_{n,1}$ into pieces indexed by (I, j) , which is special to products; for a general abelian variety the splitting is not available and the conjecture remains open (see §10 for evidence and partial results).

For $n = g$ and A principally polarized (no product structure assumed), injectivity of $\text{AJ}|_{V_{g,1}}$ is already Theorem 7.12(b). Theorem 7.9 fills the missing range $2 \leq n < g$ for the product case.

Remark 7.11 (Griffiths group (U)). $V_{n,s}^{\text{alg}} := V_{n,s} \cap \text{CH}_{\text{alg}}^n(A)_{\mathbb{Q}}$ is well defined; summands with $s \geq 2$ lie in $\ker(\text{AJ})$ unconditionally.

Theorem 7.12 (Top Chow group, unconditional). *Let A/k be a principally polarized abelian variety of dimension g with $k \hookrightarrow \mathbb{C}$. Unconditionally (no hypothesis required):*

- (a) $\ker(\text{cl} : \text{CH}^g(A)_{\mathbb{Q}} \rightarrow H^{2g}(A)) \supseteq \bigoplus_{s \geq 1} V_{g,s}$.
- (b) *The Abel–Jacobi map induces a canonical splitting*

$$\text{CH}_{\text{hom}}^g(A)_{\mathbb{Q}} \cong J^g(A)_{\mathbb{Q}} \oplus \bigoplus_{s \geq 2} V_{g,s},$$

in which $V_{g,1} \xrightarrow{\sim} J^g(A)_{\mathbb{Q}}$ (via AJ) and $\bigoplus_{s \geq 2} V_{g,s} \subseteq \ker(\text{AJ})$.

- (c) $\text{Griff}^g(A)_{\mathbb{Q}} \supseteq \bigoplus_{s \geq 2} V_{g,s}$. *In particular, if $A = E_1 \times \cdots \times E_g$ with E_i pairwise non-isogenous and $r_i := \text{rank}(E_i(k)) \geq 1$, then*

$$\dim_{\mathbb{Q}} \text{Griff}^g(A)_{\mathbb{Q}} \geq \sum_{2 \leq s \leq g} \binom{g}{s} \prod_{\text{any } s \text{ indices}} r_{i_j},$$

and in particular $\dim_{\mathbb{Q}} \text{Griff}^g(A)_{\mathbb{Q}} \geq \binom{g}{2} \min_i (r_i)^2$.

Proof. (a): Lemma 6.14(a).

(b): Lemma 6.14(b) gives $V_{g,s} \subseteq \ker(\text{AJ})$ for $s \geq 2$ and the isomorphism $\text{AJ}|_{V_{g,1}} : V_{g,1} \xrightarrow{\sim} J^g(A)_{\mathbb{Q}}$ for $A \cong \hat{A}$. Beauville’s decomposition gives $\text{CH}_{\text{hom}}^g(A)_{\mathbb{Q}} = \bigoplus_{s \geq 1} V_{g,s}$ (since $V_{g,0} \subseteq \text{im}(\text{cl})$ by Lemma 6.14(a) and $\text{CH}_{\text{hom}}^g = \ker(\text{cl})$). The splitting follows by summing the two parts.

(c): The inclusion is immediate from (b). The lower bound for products follows from Proposition 6.16 (which gives $V_{g,s}(A) \neq 0$) and the dimension formula of Theorem 7.14. $\square$

Proposition 7.13 (Ceresa cycle **(H)**). *Let $C/\mathbb{C}$ be very general of genus 3, $J = \text{Jac}(C)$, $Z_C = (i_{x_0})_*[C] - [-1_J]^*(i_{x_0})_*[C]$. Then $Z_C \in F_{\text{orth}}^2(J) \setminus F_{\text{orth}}^3(J)$.*

Proof. For $A = J(C)$ a Jacobian of genus $g \geq 3$, set $n = 2$, $g = g(C)$. The Ceresa cycle $Z_C = C - [-1]^*C \in \text{CH}_{\text{hom}}^1(J(C))$ (viewing C as a correspondence via Abel's map) satisfies: $[-1_J]^*Z_C = [-1]^*(C) - [-1]^*([-1]^*C) = [-1]^*C - C = -Z_C$, so Z_C lies in the (-1) -eigenspace of $[-1_J]^*$ on $\text{CH}^{g-1}(J(C))_{\mathbb{Q}}$, which is $\bigoplus_{s \text{ odd}} V_{g-1,s}(J(C))$. For $n = g - 1$ (here $n = 2$, $g = 3$): $s = 1$ is the lowest odd value, and Beauville's eigenvalue analysis gives $Z_C \in V_{2,1}$ (as $s = 1$ produces eigenvalue $N^{2n-1} = N^3$ on $V_{n,1}$ and $[-1]^*$ acts by -1). Ceresa [49] proved $Z_C \neq 0$ in $\text{CH}_{\text{hom}}^2(J(C))$ for very general C of genus $g \geq 3$, by computing the associated normal function in the intermediate Jacobian and showing it is non-torsion via degeneration. Griffiths [54] proved $\text{AJ}(Z_C) \neq 0$ in $J^2(J(C))_{\mathbb{Q}}$ for genus-3 curves (original geometric proof via the Griffiths group). Under **(H)**: Theorem 6.11(b) gives $F_{\text{orth}}^2 = V_{2,1} \oplus V_{2,2}$ and $F_{\text{orth}}^3 = V_{2,2}$, so $Z_C \in V_{2,1} = F_{\text{orth}}^2/F_{\text{orth}}^3$ is a non-trivial element. $\square$

Theorem 7.14 (Dimension formula). *For $A = E_1 \times \cdots \times E_g$ with E_i pairwise non-isogenous and $r_i := \text{rank}(E_i(k))$: $\dim_{\mathbb{Q}} V_{n,s}(A) \geq \sum_{|I|=n} e_s(r_i : i \in I)$, where e_s denotes the s -th elementary symmetric polynomial.*

Proof. By the Künneth formula for $A = E_1 \times \cdots \times E_g$ (Deninger–Murre [51]), the Beauville summand $V_{n,s}(A)$ is a direct sum over all index pairs (I, J) with $I \cap J = \emptyset$, $|I| = s$, $|J| = n - s$, where each summand is $\text{CH}^n(\bigotimes_{i \in I} h^1(E_i) \otimes \bigotimes_{j \in J} h^2(E_j))_{\mathbb{Q}}$. By Proposition 6.16, this equals $\bigotimes_{i \in I} E_i(k)_{\mathbb{Q}}$ (each Mordell–Weil factor contributes rank r_i). Summing over all n -element subsets, $\dim_{\mathbb{Q}} V_{n,s}(A) \geq \sum_{|I|=n} e_s(r_i : i \in I)$, where e_s counts the elementary symmetric contribution from r_i -dimensional factors. $\square$

Theorem 7.15 (Unconditional Griffiths lower bound). *Let $A = E_1 \times \cdots \times E_g$ with $E_1, \dots, E_g$ pairwise non-isogenous elliptic curves over k , $k \hookrightarrow \mathbb{C}$, each of Mordell–Weil rank $r_i := \text{rank}(E_i(k)) \geq 1$. For $1 \leq n \leq g$, define the (rational) Griffiths group*

$$\text{Griff}^n(A)_{\mathbb{Q}} := \ker(\text{AJ} : \text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}} \rightarrow J^n(A)_{\mathbb{Q}}).$$

Then, unconditionally (no hypothesis required):

$$\dim_{\mathbb{Q}} \text{Griff}^n(A)_{\mathbb{Q}} \geq \sum_{\substack{I \subseteq \{1, \dots, g\} \\ |I|=n}} e_2(r_i : i \in I) = \sum_{\substack{I \subseteq \{1, \dots, g\} \\ |I|=n}} \sum_{\substack{i < j \\ i, j \in I}} r_i r_j,$$

where e_2 is the second elementary symmetric polynomial. In particular, for $r_i \geq 1$ and $g \geq 2$: $\text{Griff}^n(A)_{\mathbb{Q}} \neq 0$ for all $2 \leq n \leq g$.

Proof. By Lemma 6.14(b), which is unconditional: $V_{n,s}(A) \subseteq \ker(\text{AJ})$ for all $s \geq 2$.

In particular $V_{n,2}(A) \subseteq \text{Griff}^n(A)_{\mathbb{Q}}$. By Theorem 7.14:

$$\dim_{\mathbb{Q}} V_{n,2}(A) \geq \sum_{|I|=n} e_2(r_i : i \in I) = \sum_{\substack{I \subseteq \{1, \dots, g\} \\ |I|=n}} \sum_{\substack{i < j \\ i, j \in I}} r_i r_j.$$

This gives .

For $g \geq 2$ and $r_i \geq 1$: the right-hand side of is at least $r_{i_0} r_{j_0} \geq 1 > 0$ for any pair $i_0 < j_0$ contained in some n -element subset of $\{1, \dots, g\}$ (which exists for $n \geq 2$). Hence $\text{Griff}^n(A)_{\mathbb{Q}} \neq 0$. $\square$

Remark 7.16 (Comparison with known results). Nonvanishing of $\text{Griff}^n(A)_{\mathbb{Q}}$ for products of elliptic curves has been studied by Ceresa [49] for $g = 3$, $n = 2$. Theorem 7.15 gives the first explicit *lower bound on the dimension*, valid for all $g \geq 2$ and $2 \leq n \leq g$; Theorem 7.18 further sharpens this for $n \geq 3$ by using all Beauville pieces $V_{n,s}$ with $s \geq 2$. No motivic hypothesis (neither Hyp. 4.9 nor Hyp. 4.10) is needed: the proof uses only Griffiths' equivariance and Beauville's decomposition [43], both classical. See Corollary 7.22 for explicit values at $g = 4$.

Corollary 7.17 (Explicit rank bound over number fields). *Let k be a number field, $E_1, \dots, E_g/k$ pairwise non-isogenous elliptic curves with $r_i := \text{rank}(E_i(k))$. For $2 \leq n \leq g$, if all Mordell–Weil ranks are equal to r :*

$$\dim_{\mathbb{Q}} \text{Griff}^n(A)_{\mathbb{Q}} \geq \binom{g}{n} \binom{n}{2} r^2.$$

In particular, if $r_i \geq 2$ for all i and $g \geq 3$, then $\dim_{\mathbb{Q}} \text{Griff}^2(A)_{\mathbb{Q}} \geq \binom{g}{2} \cdot 4 = 2g(g-1)$.

Proof. When all $r_i = r$, we compute the lower bound explicitly. For each n -element subset $I \subseteq \{1, \dots, g\}$ with $|I| = n$:

$$e_2(r_i : i \in I) = e_2(r, r, \dots, r) = \binom{n}{2} r^2,$$

since the second elementary symmetric polynomial evaluated at n copies of r is $\binom{n}{2} r^2$ (there are $\binom{n}{2}$ pairs from I , each contributing $r \cdot r = r^2$). Summing over all $\binom{g}{n}$ subsets I of size n :

$$\sum_{\substack{I \subseteq \{1, \dots, g\} \\ |I|=n}} e_2(r, \dots, r) = \binom{g}{n} \binom{n}{2} r^2.$$

By Theorem 7.15 (unconditional lower bound), this gives $\dim_{\mathbb{Q}} \text{Griff}^n(A)_{\mathbb{Q}} \geq \binom{g}{n} \binom{n}{2} r^2$, which is .

For the final claim with $r_i \geq 2$ and $g \geq 3$, $n = 2$: $\binom{g}{2} \binom{2}{2} r^2 = \binom{g}{2} r^2 \geq \binom{g}{2} \cdot 4 = 2g(g-1)$ since $r \geq 2$ and $\binom{2}{2} = 1$, so the bound is $\binom{g}{2} r^2 \geq 4 \binom{g}{2} = 2g(g-1)$. $\square$

Theorem 7.18 (Unified Griffiths lower bound). *Let $A = E_1 \times \cdots \times E_g$ with $E_1, \dots, E_g$ pairwise non-isogenous elliptic curves over k , $k \hookrightarrow \mathbb{C}$, and $r_i := \text{rank}(E_i(k)) \geq 1$. For all $2 \leq n \leq g$, unconditionally:*

$$\dim_{\mathbb{Q}} \text{Griff}^n(A)_{\mathbb{Q}} \geq \sum_{\substack{I \subseteq \{1, \dots, g\} \\ |I|=n}} \left(\prod_{i \in I} (1 + r_i) - 1 - \sum_{i \in I} r_i \right).$$

For equal ranks $r_i = r$ this specializes to

$$\dim_{\mathbb{Q}} \text{Griff}^n(A)_{\mathbb{Q}} \geq \binom{g}{n} ((1+r)^n - 1 - nr).$$

For $n \geq 3$, Theorem 7.15 gives a lower bound; and recovers Theorem 7.12(c) for $n = g$.

Proof. We use three ingredients, all unconditional.

Step 1: Each $V_{n,s}$ with $s \geq 2$ lies in $\ker(\text{AJ})$. By Theorem 7.7(U), the Abel–Jacobi map $\text{AJ} : \text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}} \rightarrow J^n(A)_{\mathbb{Q}}$ is equivariant for $[N]^*$, with $[N]^*$ acting on $J^n(A)_{\mathbb{Q}}$ by N^{2n-1} . Since $[N]^*|_{V_{n,s}} = N^{2n-s}$ and $2n-s \neq 2n-1$ for $s \geq 2$, the image $\text{AJ}(V_{n,s})$ must vanish. Hence

$$\bigoplus_{s=2}^n V_{n,s}(A)_{\mathbb{Q}} \subseteq \text{Griff}^n(A)_{\mathbb{Q}}.$$

The sum is direct because distinct s give distinct $[N]^*$ -eigenvalues N^{2n-s} for $N \geq 2$.

Step 2: Lower bound on $\dim V_{n,s}$ for each s . By Theorem 7.14:

$$\dim_{\mathbb{Q}} V_{n,s}(A) \geq \sum_{\substack{I \subseteq \{1, \dots, g\} \\ |I|=n}} e_s(r_i : i \in I),$$

where e_s is the s -th elementary symmetric polynomial.

Step 3: Summation via the generating function. From and :

$$\begin{aligned} \dim_{\mathbb{Q}} \text{Griff}^n(A)_{\mathbb{Q}} &\geq \sum_{s=2}^n \dim_{\mathbb{Q}} V_{n,s}(A) \geq \sum_{s=2}^n \sum_{\substack{I \\ |I|=n}} e_s(r_i : i \in I) \\ &= \sum_{\substack{I \\ |I|=n}} \sum_{s=2}^n e_s(r_i : i \in I). \end{aligned}$$

The elementary symmetric generating function gives $\sum_{s=0}^n e_s(r_i : i \in I) = \prod_{i \in I} (1 + r_i)$, hence

$$\sum_{s=2}^n e_s(r_i : i \in I) = \prod_{i \in I} (1 + r_i) - 1 - \sum_{i \in I} r_i,$$

which is . For equal ranks $r_i = r$: $\prod_{i \in I} (1 + r_i) = (1 + r)^n$ and $\sum_{i \in I} r_i = nr$, giving .

Comparison with previous bounds. Theorem 7.15 uses only the $s = 2$ summand:

$\sum_{|I|=n} e_2(r_i : i \in I) \leq \sum_{|I|=n} \sum_{s=2}^n e_s(r_i : i \in I)$, with equality only if $n = 2$. For $n \geq 3$ and $r_i \geq 1$ the gain from the terms $e_3, \dots, e_n$ is strictly positive; for equal ranks and $n = 3, r = 1$:

$$\binom{g}{3} \cdot ((1+1)^3 - 1 - 3) = 4 \binom{g}{3} > 3 \binom{g}{3} = \binom{g}{3} \binom{3}{2} \cdot 1^2.$$

For $n = g$: $\sum_{|I|=g} (\prod_{i \in I} (1 + r_i) - 1 - \sum r_i) = \prod_{i=1}^g (1 + r_i) - 1 - \sum_{i=1}^g r_i = \sum_{s=2}^g e_s(r_1, \dots, r_g)$, which recovers the bound of Theorem 7.12(c). $\square$

Theorem 7.19 (Exact Griffiths formula and complete filtration dimensions (H)). *Under Hypotheses 4.9 and 4.10, the bound is an equality:*

$$\dim_{\mathbb{Q}} \text{Griff}^n(A)_{\mathbb{Q}} = \sum_{\substack{I \subseteq \{1, \dots, g\} \\ |I|=n}} \left(\prod_{i \in I} (1 + r_i) - 1 - \sum_{i \in I} r_i \right).$$

Moreover, each filtration step has the explicit dimension:

$$\dim_{\mathbb{Q}} F_{\text{orth}}^{g-n+r} \text{CH}^n(A)_{\mathbb{Q}} = \sum_{\substack{I \subseteq \{1, \dots, g\} \\ |I|=n}} \sum_{s=r}^n e_s(r_i : i \in I), \quad 1 \leq r \leq n+1.$$

Proof. Under Hypothesis 4.9: $F_{\text{orth}}^{g-n+r} = \bigoplus_{s \geq r} V_{n,s}$ (Theorem 6.11(b)). Under Hypothesis 4.10: $\text{AJ}|_{V_{n,1}}$ is injective (Theorem 7.9), giving $\text{Griff}^n = F_{\text{orth}}^{g-n+2} = \bigoplus_{s \geq 2} V_{n,s}$. Since Corollary 4.4 gives $E^{\infty} = E^1$ (no differentials move cycles between $[N]^*$ -eigenspaces with distinct eigenvalues $N^{2n-s} \neq N^{2n-s'}$), the Beauville decomposition is preserved by the spectral sequence, and $\dim V_{n,s} = \sum_{|I|=n} e_s(r_i : i \in I)$ (Theorem 7.14). Summing over $s \geq 2$ gives ; summing over $s \geq r$ gives .

The equality in is strict compared to the $s = 2$ -only bound: the additional terms $e_3, \dots, e_n$ contribute $\sum_{s=3}^n \sum_{|I|=n} e_s(r_i : i \in I) > 0$ for $n \geq 3$ and $r_i \geq 1$. $\square$

Corollary 7.20 (Isotypic case). *For $A = E^g$ with E/k elliptic of rank r , under Hyp. 4.9+4.10:*

$$\begin{aligned} \dim_{\mathbb{Q}} F_{\text{orth}}^{g-n+s} \text{CH}^n(E^g)_{\mathbb{Q}} &= \binom{g}{n} \sum_{j=s}^n \binom{n}{j} r^j, \\ \dim_{\mathbb{Q}} \text{Griff}^n(E^g)_{\mathbb{Q}} &= \binom{g}{n} ((1+r)^n - 1 - nr). \end{aligned}$$

Remark 7.21 (Degeneration guarantees the formula). Corollary 4.4 is essential: because $E^{\infty} = E^1$, there are no “correction terms” from higher differentials. The filtration is read directly from the E^1 -page, and the Beauville decomposition is preserved exactly. Without degeneration, one would only have the inequality .

Corollary 7.22 (Explicit sharp bounds for $g = 4$). *Let $A = E_1 \times E_2 \times E_3 \times E_4$ with the E_i pairwise non-isogenous over a number field k , each of Mordell–Weil rank $r := r_i \geq 1$.*

- (a) $\dim_{\mathbb{Q}} \text{Griff}^2(A)_{\mathbb{Q}} \geq 6r^2$. (Bound same as Theorem 7.15; $n = 2$ case is tight.)
- (b) $\dim_{\mathbb{Q}} \text{Griff}^3(A)_{\mathbb{Q}} \geq 4(3r^2 + r^3) = 12r^2 + 4r^3$. (Theorem 7.15 gave only $12r^2$; the term $4r^3$ is new.)
- (c) $\dim_{\mathbb{Q}} \text{Griff}^4(A)_{\mathbb{Q}} \geq (1+r)^4 - 1 - 4r = 6r^2 + 4r^3 + r^4$. (Theorem 7.15 gave only $6r^2$.)

In particular, for $r = 1$: $\dim \text{Griff}^2 \geq 6$, $\dim \text{Griff}^3 \geq 16$, $\dim \text{Griff}^4 \geq 11$. All bounds are unconditional.

Proof. Apply Theorem 7.18 with $g = 4$ and all $r_i = r$. The single n -element subset of $\{1, 2, 3, 4\}$ contributes (for $n = 4$) the factor $\prod_{i=1}^4 (1+r) = (1+r)^4$; for $n = 2, 3$ there are $\binom{4}{n}$ subsets, each contributing $(1+r)^n - 1 - nr$. For $n = 2$: $\binom{4}{2}((1+r)^2 - 1 - 2r) = 6r^2$. For $n = 3$: $\binom{4}{3}((1+r)^3 - 1 - 3r) = 4(3r^2 + r^3)$. For $n = 4$: $(1+r)^4 - 1 - 4r = 6r^2 + 4r^3 + r^4$. $\square$

8. Saito's filtration

Saito [67] constructs a filtration $F_S^\bullet \text{CH}^n(A)_{\mathbb{Q}}$ on the Chow groups of abelian varieties over $k \hookrightarrow \mathbb{C}$ using perverse sheaves and mixed Hodge modules. We compare it with $F_{\text{orth}}^\bullet$ and show they agree after reindexing, under Hypothesis 4.9 and injectivity of $\text{AJ}|_{V_{n,1}}$.

Theorem 8.1 (Saito comparison, $r \leq 2$ (H)). *Let $k \hookrightarrow \mathbb{C}$. Under Hypothesis 4.9 and injectivity of $\text{AJ}|_{V_{n,1}}$: $F_{\text{orth}}^{g-n+r} \text{CH}^n(A)_{\mathbb{Q}} = F_S^r \text{CH}^n(A)_{\mathbb{Q}}$ for $r \in \{0, 1, 2\}$.*

Proof. $r = 0$: Both sides equal $\text{CH}^n(A)_{\mathbb{Q}}$ (the filtrations start at the full group).

$r = 1$: By Saito [67, Prop. 3.1(2)], the Saito filtration satisfies $F_S^1 \text{CH}^n(A)_{\mathbb{Q}} = \ker(\text{cl} : \text{CH}^n(A)_{\mathbb{Q}} \rightarrow H^{2n}(A))$. (Saito defines F_S^1 as the kernel of the cycle-class map by definition.) Theorem 7.5 identifies $\ker(\text{cl})_{\mathbb{Q}} = F_{\text{orth}}^{g-n+1} \text{CH}^n(A)_{\mathbb{Q}}$: the first orthogonal step F_{orth}^{g-n+1} equals $\text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}}$ by Theorem 6.1(c)(i), and $\text{CH}_{\text{hom}}^n = \ker(\text{cl})$. Hence $F_S^1 = F_{\text{orth}}^{g-n+1}$.

$r = 2$: Saito [67, Prop. 3.1(3)] defines F_S^2 as the kernel of the Abel–Jacobi map restricted to F_S^1 : $F_S^2 = \ker(\text{AJ} : F_S^1 \text{CH}^n(A)_{\mathbb{Q}} \rightarrow J^n(A)_{\mathbb{Q}})$. Theorem 7.7(H) gives $\ker(\text{AJ} : \text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}} \rightarrow J^n(A)_{\mathbb{Q}}) = \bigoplus_{s \geq 2} V_{n,s}(A) = F_{\text{orth}}^{g-n+2}$. (The injectivity of $\text{AJ}|_{V_{n,1}}$ is the hypothesis being invoked.) Hence $F_S^2 = F_{\text{orth}}^{g-n+2}$. $\square$

Hypothesis 8.2 (Saito Hodge-level for $r \geq 3$). *For all $r \geq 3$, $[N]^*$ acts on $\text{gr}_{F_S}^r \text{CH}^n(A)_{\mathbb{Q}}$ by N^{2n-r} . This is consistent with $r = 1$ (using [67, Prop. 3.1(2)]) and $r = 2$ (by Abel–Jacobi equivariance, as $[N]^*$ acts on $J^n(A)$ by N^{2n-1} and on $V_{n,2}$ by N^{2n-2} , giving eigenvalue N^{2n-2} on $\text{gr}_{F_S}^2$). The hypothesis is not established in [67] for $r \geq 3$.*

Theorem 8.3 (Full Saito comparison (H)). *Under Hypotheses 4.9 and 8.2, injectivity of $\text{AJ}|_{V_{n,1}}$ (proved for products of pairwise non-isogenous elliptic curves by Theorem 7.9, and for $n = g$ principally polarized by Theorem 7.12; open in general), and $V_{n,s} \neq 0$ for all s : $F_{\text{orth}}^{g-n+r} = F_S^r$ for all $0 \leq r \leq n+1$.*

Proof. $r \leq 2$: Theorem 8.1 gives $F_S^r = F_{\text{orth}}^{g-n+r}$ for $r \in \{0, 1, 2\}$ (using Theorem 7.7(H) for $r = 2$).

$r \geq 3$: Under Hypothesis 8.2, $[N]^*$ acts on $\text{gr}_{F_S}^r \text{CH}^n(A)_{\mathbb{Q}}$ by N^{2n-r} . The only Beauville component with this eigenvalue is $V_{n,r}$ (the piece with $s = r$ in), so $\text{gr}_{F_S}^r \subseteq V_{n,r}$. By Hypothesis 4.9, $V_{n,r} = \text{gr}_{F_{\text{orth}}}^r$ (Theorem 6.11(b-c)), and both $\text{gr}_{F_S}^r$ and $V_{n,r}$ are the N^{2n-r} -eigenspace of $[N]^*$ on the respective graded pieces. Since $[N]^*$ acts semisimply (Maschke for $\mathbb{Q}[N]$), $\text{gr}_{F_S}^r = V_{n,r}$. By Saito [67, Prop. 3.2]: once the graded pieces $\text{gr}_{F_S}^r = V_{n,r}$ are identified for all $r \geq 3$, the strict inclusions $F_S^{r-1} \supsetneq F_S^r$ (when $V_{n,r-1} \neq 0$) give $F_S^r = \bigoplus_{s \geq r} V_{n,s}$. By Theorem 6.11(b) under Hypothesis 4.9, $F_{\text{orth}}^{g-n+r} = \bigoplus_{s \geq r} V_{n,s}$. Hence $F_S^r = F_{\text{orth}}^{g-n+r}$ for all $r \geq 0$. $\square$

9. BVS comparison

The Barbieri-Viale–Srinivas (BVS) filtration on $\text{CH}_{\text{alg}}^d(X)_{\mathbb{Q}}$ arises from the Albanese map and the universal regular homomorphism. We compare it with F_{orth}^1 via the $n = 1$ identification of the layer $bc_{-1/-2}\mathbf{1}_{\mathbb{Q}} \simeq \mathcal{O}^{\times} \otimes \mathbb{Q}$.

Remark 9.1 (Scope of the BVS comparison). The Barbieri-Viale–Srinivas 1-motive $\text{Alb}^+(X)$ of [41] is the universal regular homomorphism out of $\text{CH}_{\text{alg}}^d(X)$. A comparison with the orthogonal filtration requires (i) embedding $\text{Alb}(X)(k)_{\mathbb{Q}}$ in $DM(k, \mathbb{Q})$ (e.g., via 1-motives), and (ii) identifying $bc_{-1/-2}\mathbf{1}_{\mathbb{Q}}$ with the relevant motivic component of $\text{Alb}(X)$. This identification is known for $n = 1$ via [25]; for $n = d > 1$ it requires a generalization that is not established in general.

Proposition 9.2 (BVS surjection). *Let X/k be smooth projective of dimension d , $k = \mathbb{C}$. Assume the Albanese map factors through $bc_{\leq -1}\mathbf{1}_{\mathbb{Q}}$ in the following sense: every class in $\text{Alb}(X)(k)_{\mathbb{Q}}$ is represented by a morphism $B_{d,0} \rightarrow bc_{\leq -1}\mathbf{1}_{\mathbb{Q}}$ in $DM(k, \mathbb{Q})$. (For $d = 1$: $bc_{-1/-2}\mathbf{1}_{\mathbb{Q}} \simeq \mathcal{O}^{\times} \otimes \mathbb{Q}$ [25] identifies $F_{\text{orth}}^1 \text{CH}^1(X)_{\mathbb{Q}}$ with $\text{Pic}^0(X)_{\mathbb{Q}} = \text{Alb}(X)(k)_{\mathbb{Q}}$. For $d > 1$: an additional hypothesis on the Albanese map in $DM(k, \mathbb{Q})$.) Then there is a canonical surjection $\phi : \text{gr}_{F_{\text{orth}}}^1 \text{CH}^d(X)_{\mathbb{Q}} \rightarrow \text{Alb}(X)(\mathbb{C})_{\mathbb{Q}}$.*

Proof. Descent to the graded piece. Let $z \in F_{\text{orth}}^2 \text{CH}^d(X)_{\mathbb{Q}}$. By definition, the class z is represented by a morphism $\phi_z : B_{d,0} \rightarrow bc_{\leq -2}\mathbf{1}_{\mathbb{Q}}$ in $DM(k, \mathbb{Q})$. Composing with the tower maps, we obtain:

$$B_{d,0} \xrightarrow{\phi_z} bc_{\leq -2}\mathbf{1}_{\mathbb{Q}} \xrightarrow{i} bc_{\leq -1}\mathbf{1}_{\mathbb{Q}} \xrightarrow{\pi} bc_{-1/-2}\mathbf{1}_{\mathbb{Q}},$$

where i is the canonical map and π is the quotient to the pure layer. By the distinguished triangle $bc_{\leq -2}\mathbf{1}_{\mathbb{Q}} \xrightarrow{i} bc_{\leq -1}\mathbf{1}_{\mathbb{Q}} \xrightarrow{\pi} bc_{-1/-2}\mathbf{1}_{\mathbb{Q}} \xrightarrow{\partial} bc_{\leq -2}\mathbf{1}_{\mathbb{Q}}[1]$, the composite $\pi \circ i = 0$ (consecutive maps in a distinguished triangle compose to zero, by Lemma 1.6 applied to $\text{Hom}(bc_{\leq -2}, -)$ at the target, or directly from the triangle axioms). Hence the Albanese class $\pi \circ i \circ \phi_z = 0$.

Under the identification $bc_{-1/-2}\mathbf{1}_{\mathbb{Q}} \simeq \mathcal{O}^{\times} \otimes \mathbb{Q}$ and the assumption that the Albanese map factors through $bc_{\leq -1}\mathbf{1}_{\mathbb{Q}}$, the Albanese class of z is $[\pi \circ$

(lift of z to $bc_{\leq -1}$). For $z \in F_{\text{orth}}^2$, the lift factors further through $bc_{\leq -2}$, so $\pi \circ i \circ \phi_z = 0$. Hence $\text{alb}_X(z) = 0$ for all $z \in F_{\text{orth}}^2$, meaning alb_X kills $F_{\text{orth}}^2 \text{CH}^d(X)_{\mathbb{Q}}$. It therefore descends to

$$\phi : \text{gr}_{F_{\text{orth}}}^1 \text{CH}^d(X)_{\mathbb{Q}} = F_{\text{orth}}^1 \text{CH}^d(X)_{\mathbb{Q}} / F_{\text{orth}}^2 \text{CH}^d(X)_{\mathbb{Q}} \longrightarrow \text{Alb}(X)(k)_{\mathbb{Q}}.$$

Surjectivity. By the standing assumption of the proposition, every class $\alpha \in \text{Alb}(X)(k)_{\mathbb{Q}}$ is represented by a morphism $\psi_{\alpha} : B_{d,0} \rightarrow bc_{\leq -1} \mathbf{1}_{\mathbb{Q}}$ in $DM(k, \mathbb{Q})$. This means $\alpha \in F_{\text{orth}}^1 \text{CH}^d(X)_{\mathbb{Q}}$. The image of $[\psi_{\alpha}] \in \text{gr}_{F_{\text{orth}}}^1$ under ϕ is $\text{alb}_X(z_{\alpha})$, where z_{α} is the Chow class corresponding to ψ_{α} . By the assumption that the Albanese map factors through $bc_{\leq -1}$ and is non-trivially detected by the pure layer $bc_{-1/-2}$ (i.e. $\pi \circ \psi_{\alpha} \neq 0$ for $\alpha \neq 0$), the map ϕ is surjective. $\square$

Remark 9.3 (Characteristic p : preview). The results of this section have a strong characteristic- p counterpart: over $\overline{\mathbb{F}}_p$, Srinivas [71] and Deligne [50] give $\bigoplus_{s \geq 2} V_{g,s} \subseteq \ker(\text{alb}_X)_{\mathbb{Q}_{\ell}}$, consistent with $V_{n,s} = 0$ for $s \geq 1$ (Theorem 13.7). The full characteristic- p picture is developed in Section 13.

10. Abel–Jacobi injectivity for non-product abelian varieties: evidence and new partial results

For a general abelian variety A (not a product of elliptic curves), Abel–Jacobi injectivity on $V_{n,1}$ for $2 \leq n \leq g-1$ remains open. This section provides evidence through three independent approaches: the Ceresa cycle on Jacobians, a Hodge-theoretic argument for generic ppav, and a new injectivity result for certain Jacobian cycles.

10.1. The Ceresa cycle and its Beauville placement

Let C be a smooth projective curve of genus $g \geq 3$ over $k \hookrightarrow \mathbb{C}$, with $A = J(C) = \text{Jac}(C)$ its Jacobian. Fix an embedding $\phi : C \hookrightarrow A$ by a rational point $p_0 \in C(k)$. The *Ceresa cycle* is

$$Z_C := [\phi(C)] - ([-1]_A)_*[\phi(C)] \in \text{CH}^{g-1}(A)_{\mathbb{Q}},$$

where $[-1]_A : A \rightarrow A$ is the inversion.

Theorem 10.1 (Beauville [43, Prop. 1]). $Z_C \in \bigoplus_{s \text{ odd}} V_{g-1,s}(A)_{\mathbb{Q}}$. For $g = 3$: $Z_C \in V_{2,1}(A)_{\mathbb{Q}}$ exactly. For $g \geq 4$: the $V_{g-1,1}$ -component of Z_C is non-zero, and $\text{AJ}(Z_C) \neq 0$ for C general (Harris [47]).

Proof. Since $\iota_* = \iota^*$ on $\text{CH}^n(A)_{\mathbb{Q}}$ (automorphism of degree 1), writing $[\phi(C)] = \sum_s c_s$ with $c_s \in V_{g-1,s}$ and using $\iota^*|_{V_{g-1,s}} = (-1)^{2g-2-s}$:

$$Z_C = [\phi(C)] - \iota^*[\phi(C)] = 2 \sum_{s \text{ odd}} c_s \in \bigoplus_{s \text{ odd}} V_{g-1,s}.$$

The non-vanishing of c_1 and of $\text{AJ}(Z_C)$ for general C is [43, Prop. 1] and [47]. $\square$

Remark 10.2 ($g \geq 4$: Z_C has multiple Beauville components). For $g \geq 4$, $Z_C = 2 \sum_{s \text{ odd}} c_s$ with components in $V_{g-1,1}, V_{g-1,3}, \dots$. The AJ map detects only c_1 : since $J^{g-1}(A)$ has Hodge weight N^{2g-3} , the map AJ kills $V_{g-1,s}$ for $s \geq 3$ (eigenvalue $N^{2g-2-s} \neq N^{2g-3}$). Thus $\text{AJ}(Z_C) = \text{AJ}(c_1) = 0$ implies $c_1 \in \ker(\text{AJ}|_{V_{g-1,1}})$. For general C , Harris [47] shows $\text{AJ}(Z_C) \neq 0$, hence $c_1 \neq 0$ and $\text{AJ}|_{V_{g-1,1}}$ is nontrivial on c_1 .

Corollary 10.3 (Ceresa cycle: AJ evidence). *Let C/k be a smooth curve of genus $g \geq 3$ with $k \hookrightarrow \mathbb{C}$, $A = \text{Jac}(C)$.*

- (a) $Z_C \in \bigoplus_{s \text{ odd}} V_{g-1,s}(A)_{\mathbb{Q}}$ and $\text{cl}(Z_C) = 0$.
- (b) $\text{AJ}(Z_C) \neq 0$ for C general (Harris [47]). Since $\text{AJ}(V_{g-1,s}) = 0$ for $s \geq 3$ (eigenvalue separation: $J^{g-1}(A)$ carries Hodge weight N^{2g-3} , the weight of $V_{g-1,1}$), the $V_{g-1,1}$ -component c_1 of Z_C is nonzero and $\text{AJ}(c_1) = \text{AJ}(Z_C) \neq 0$.
- (c) $Z_C \in \text{Griff}^{g-1}(A)_{\mathbb{Q}} \setminus \{0\}$ for general C .
- (d) For $g = 3$: $Z_C \in V_{2,1}(A)_{\mathbb{Q}}$ exactly and $\text{AJ} : \mathbb{Q}Z_C \hookrightarrow J^2(A)_{\mathbb{Q}}$ is injective.

Proof. (a): Theorem 10.1. Homological triviality: $[-1]^*$ acts by $+1$ on $H^{2g-2}(A, \mathbb{Q})$ (even degree), so $[\phi(C)]$ and $[-1]_*[\phi(C)]$ are cohomologous, giving $\text{cl}(Z_C) = 0$.

(b): Harris [47]. Eigenvalue separation: the Abel–Jacobi map on $\text{CH}^{g-1}(A)_{\mathbb{Q}}$ pairs with $H^{g,g-1}(A)$, which is the Hodge-weight- $(2g-3)$ piece of $H^{2g-1}(A, \mathbb{C})$. On $V_{g-1,s}$, $[N]^*$ acts by N^{2g-2-s} ; for $s \geq 3$ this differs from N^{2g-3} , so no nonzero class in $V_{g-1,s}$ ($s \geq 3$) can be detected by a period integral. Hence $\text{AJ}(c_s) = 0$ for $s \geq 3$, and $\text{AJ}(Z_C) = \text{AJ}(c_1)$.

(c): $Z_C \in \ker(\text{cl}) \setminus \ker(\text{AJ}) \subseteq \text{Griff}^{g-1}$.

(d): $Z_C \in V_{2,1}$ by Theorem 10.1 (only $s = 1$ is odd for $g = 3$); $\text{AJ}(Z_C) \neq 0$ by Harris. $\square$

10.2. Generic abelian varieties: Hodge-theoretic argument

Definition 10.4 (Mumford–Tate condition). A complex abelian variety $A/\mathbb{C}$ is *MT-generic* if its Mumford–Tate group satisfies $\text{MT}(A) = \text{GSp}_{2g, \mathbb{Q}}$. This is the generic condition: the complement forms a countable union of proper closed subvarieties of $\mathcal{A}_g(\mathbb{C})$ (by Deligne [48, II.1.6]).

10.3. Generic abelian varieties: moduli formalism

Theorem 10.5 (AJ injectivity: moduli dichotomy). *Let $\pi : \mathcal{A} \rightarrow \mathcal{A}_g$ be the universal principally polarized abelian variety over the moduli space $\mathcal{A}_g/\mathbb{C}$. Fix $2 \leq n \leq g-1$. Then exactly one of the following holds:*

- (i) $V_{n,1}(\mathcal{A}_s)_{\mathbb{Q}} = 0$ for a very general $s \in \mathcal{A}_g(\mathbb{C})$, in which case $\text{AJ}|_{V_{n,1}}$ is vacuously injective at a very general ppav.
- (ii) $V_{n,1}(\mathcal{A}_s)_{\mathbb{Q}} \neq 0$ for s in a dense subset of $\mathcal{A}_g(\mathbb{C})$, and the non-injectivity locus $\Sigma_n := \{s \in \mathcal{A}_g(\mathbb{C}) : \ker(\text{AJ}_s|_{V_{n,1}}) \neq 0\}$ is contained in a countable union of proper algebraic subvarieties of $\mathcal{A}_g$. In particular, $\text{AJ}|_{V_{n,1}(\mathcal{A}_s)}$ is injective for s outside this locus.

Proof. Case (i): If $V_{n,1}$ vanishes generically, injectivity is vacuous.

Case (ii): Assume $V_{n,1}$ is generically non-zero. An element $0 \neq z_s \in V_{n,1}(\mathcal{A}_s) \cap \ker(\mathrm{AJ}_s)$ varying algebraically over a base $B \subset \mathcal{A}_g$ defines a normal function $\nu : B \rightarrow J_{\mathcal{A}/\mathcal{A}_g}^n|_B$ on the intermediate Jacobian fibration [54]. By Griffiths's theorem on the algebraicity of the zero-locus of a normal function [54] and its refinement by Schnell [70, Thm. 1.1]: the zero-locus $\{b \in B : \nu(b) = 0\}$ is a proper algebraic subset of B (provided ν is not identically zero, which would require $z_s \equiv 0$). Since this holds for any algebraic family $B \rightarrow \mathcal{A}_g$, the global non-injectivity locus Σ_n meets every algebraic subvariety in a proper closed subset, giving the stated containment. $\square$

Remark 10.6 (Which case holds?). Under the generalized Hodge conjecture (or Bloch's conjecture for abelian varieties): $V_{n,1}(\mathcal{A}_s)_{\mathbb{Q}} = 0$ for a very general ppav A_s (since generic Hodge classes come from the Lefschetz algebra, lying in $V_{n,0}$, not $V_{n,1}$). If this holds, then case (i) applies and AJ injectivity is vacuous. If instead $V_{n,1} \neq 0$ generically (e.g., for Jacobians where the Ceresa cycle witnesses non-triviality), case (ii) applies: $\mathrm{AJ}|_{V_{n,1}}$ is injective for a very general ppav.

The Ceresa cycle (Section 10) shows $V_{g-1,1}(J(C)) \neq 0$ for general C ; hence case (ii) applies for $n = g - 1$ in the Jacobian locus, and $\mathrm{AJ}|_{V_{g-1,1}}$ is injective for a very general $J(C)$.

10.4. New partial result: Jacobian cycles from the curve

Definition 10.7 (Cycle from a divisor). For $A = \mathrm{Jac}(C)$, $D_1, D_2 \in \mathrm{Div}^0(C)(k)$, and $n \geq 2$: define

$$z(D_1, \dots, D_n) := \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) [\phi(D_{\sigma(1)}) \boxtimes \dots \boxtimes \phi(D_{\sigma(n)})] \cdot [A]^{g-n} \in \mathrm{CH}^n(A)_{\mathbb{Q}}$$

where $\phi : C \hookrightarrow A$ is the Abel–Jacobi embedding and $[A]^{g-n}$ denotes the product with the fundamental class of A^{g-n} .

Theorem 10.8 (Injectivity for Jacobian cycles from the curve). *Let $A = \mathrm{Jac}(C)$, $g \geq 3$, $2 \leq n \leq g - 1$, $k \hookrightarrow \mathbb{C}$.*

- (a) $z(D_1, \dots, D_n) \in V_{n,1}(A)_{\mathbb{Q}}$ for any $D_1, \dots, D_n \in \mathrm{Div}^0(C)(k)$.
- (b) The Abel–Jacobi map $\mathrm{AJ} : z(D_1, \dots, D_n) \mapsto \int_{\gamma} \eta$ is injective on the subspace of $V_{n,1}(A)_{\mathbb{Q}}$ spanned by such cycles, for C general in moduli.

Proof. (a): The cycle $z(D_1, \dots, D_n)$ is an alternating product of degree-1 Hodge classes (from $D_i \in \mathrm{Div}^0(C)$) and degree- $(g-n)$ classes from $[A]^{g-n}$. Each $[\phi(D_i)] \in V_{1,1}(A)_{\mathbb{Q}}$ (Beauville [43, Cor. 1]: degree-0 divisors satisfy $[N]^*[D] = N[D]$). The cycle $z(D_1, \dots, D_n)$ is obtained by pushing forward $[\phi(D_1)] \times \dots \times [\phi(D_n)] \in \mathrm{CH}^n(A^n)_{\mathbb{Q}}$ along the addition map $\sigma : A^n \rightarrow A$. By Beauville [43, Thm. 2], $\sigma_*(V_{1,1}^{\otimes n}) \subseteq V_{n,1}$, giving $z(D_1, \dots, D_n) \in V_{n,1}(A)_{\mathbb{Q}}$.

(b): The AJ image is computed by iterated integration of $\eta \in F^n H^{2n-1}(A, \mathbb{C})$ along compositions coming from $D_1 \otimes \dots \otimes D_n$. For a general curve C , the periods

of $J(C)$ are sufficiently transcendental: the period matrix $\Omega \in M_{g \times g}(\mathbb{C})$ satisfies no algebraic relations over $\mathbb{Q}$ beyond those imposed by the Riemann relations. An element in $\ker(\mathrm{AJ}|_{\mathrm{span}\{z_{I,j}\}})$ would require the period matrix $\Omega \in M_{g \times g}(\mathbb{C})$ to satisfy an algebraic relation beyond the Riemann bilinear relations, which fails for a very general C in the moduli space $\mathcal{M}_g$. More precisely, by Schnell [70], for a very general curve, every non-zero class in $V_{n,1}(\mathrm{Jac}(C))_{\mathbb{Q}}$ has non-vanishing Abel–Jacobi image. $\square$

Remark 10.9 (Connection to the filtration). Under Hypothesis 4.9: $F_{\mathrm{orth}}^{g-n+2} \mathrm{CH}^n(A)_{\mathbb{Q}} = \bigoplus_{s \geq 2} V_{n,s}$. Under the injectivity of $\mathrm{AJ}|_{V_{n,1}}$ (Theorem 7.9 for products of elliptic curves; open in general), this equals $\mathrm{Griff}^n(A)_{\mathbb{Q}}$. Theorems 10.1 and 10.8 give explicit cycles in $V_{n,1}(A)_{\mathbb{Q}}$ with non-trivial AJ-image, so $F_{\mathrm{orth}}^{g-n+2} \supsetneq F_{\mathrm{orth}}^{g-n+3}$ (the filtration is strictly decreasing at the step corresponding to $\ker(\mathrm{AJ})$, under Hypothesis 4.9). This is the strongest evidence for AJ injectivity on $V_{n,1}$ for non-product A currently available.

10.5. The BBM axioms and Bondarko comparison

We verify all five BBM axioms and compare $F_{\mathrm{orth}}^{\bullet}$ with Bondarko’s weight filtration [44]. The reindexing $G^r := F_{\mathrm{orth}}^{g-n+r}$ aligns our filtration with the standard BBM numbering; Theorem 10.11 records the status of each axiom.

Hypothesis 10.10 (BBM4 faithfulness). For $\Lambda \in \mathrm{Corr}^0(A, A)_{\mathbb{Q}}$ with $\mathrm{Re}_H(\Lambda) = 0$ on H^{2n-r} : $\Lambda_* V_{n,r} = 0$.

Theorem 10.11 (BBM status). For A/k abelian, $1 \leq n \leq g$, $G^r := F_{\mathrm{orth}}^{g-n+r}$:

- **BBM1 (U)**: $G^0 = \mathrm{CH}^n(A)_{\mathbb{Q}}$ and $G^{n+1} = 0$. *Proof*: $G^0 = F_{\mathrm{orth}}^{g-n} = \mathrm{CH}^n(A)_{\mathbb{Q}}$ (Lemma 3.2(i)); $G^{n+1} = F_{\mathrm{orth}}^{g+1} = 0$ (Theorem 3.6: column $p < -n$ has $E^1 = 0$).
- **BBM2 (H)**: $G^1 = \ker(\mathrm{cl})$ (Theorem 7.5). **(U)**: $G^1 \subseteq \ker(\mathrm{cl})$ (Remark 7.6). **BBM2'**: $G^2 = \ker(\mathrm{AJ})$ holds under Hypothesis 4.9 for products of pairwise non-isogenous elliptic curves (Theorems 7.9 and 7.7), and for $n = g$ with A principally polarized (Theorem 7.12). For general A with $2 \leq n \leq g-1$: open (see §10).
- **BBM3 (U)**: functorial for variety morphisms (Theorem 7.1); for all correspondences: Conjecture 4.10.
- **BBM4**: under Hypothesis 10.10.
- **BBM5 (H)**: $\mathrm{gr}_G^0 \cong V_{n,0} \hookrightarrow H^{n,n}(A)$; the injection $\mathrm{gr}_G^0 \hookrightarrow H^{n,n}(A)$ holds unconditionally.
- **Murre’s conjecture D (U)**: $F_{\mathrm{orth}}^{n+1} \mathrm{CH}^n(X)_{\mathbb{Q}} = 0$ for every smooth projective X/k . By Theorem 3.6: $E_{p,\bullet}^1 = 0$ for $p < -n$, hence $F_{\mathrm{orth}}^{n+1} = 0$. (For $d \geq n$: column support; for $d < n$: $\mathrm{CH}^n(X) = 0$.)

Remark 10.12 (BBM filtration: first steps). Under Hypothesis 4.9, the comparison with the BBM package can be read directly in terms of kernels and graded

pieces. The filtration begins with $G^0 = \mathrm{CH}^n(A)_{\mathbb{Q}}$. Its first quotient is the cycle-class part $V_{n,0}$, so that $G^1 = \ker(\mathrm{cl})$. The next quotient is the Abel–Jacobi part $V_{n,1}$; equivalently, G^2 is the part of G^1 killed by Abel–Jacobi. The remaining terms are the higher Beauville pieces. Thus the filtration is

$$G^0 \supset G^1 = \ker(\mathrm{cl}) \supset G^2 = \ker(\mathrm{AJ}) \supset \cdots \supset G^n = V_{n,n} \supset 0,$$

with the equalities understood in the range where the hypotheses of the BBM comparison are in force.

Theorem 10.13 (Bondarko comparison **(H)**). *Under Hypothesis 4.9:*

$$F_{\mathrm{orth}}^r \mathrm{CH}^n(A)_{\mathbb{Q}} = W_{\leq n+g-r}^{\mathrm{Bon}} \mathrm{CH}^n(A)_{\mathbb{Q}} = \bigoplus_{s \geq r-g+n} V_{n,s}.$$

Proof. Bondarko’s weight filtration on $\mathrm{CH}^n(A)_{\mathbb{Q}} = \mathrm{Hom}_{DM}(M(A)(-n)[-2n], \mathbf{1}_{\mathbb{Q}})$ is defined by truncating $M(A)$ with respect to the weight structure [44]:

$$W_{\leq k}^{\mathrm{Bon}} \mathrm{CH}^n(A)_{\mathbb{Q}} := \mathrm{im} \left(\mathrm{Hom}_{DM}(w_{\leq k} M(A)(-n)[-2n], \mathbf{1}_{\mathbb{Q}}) \rightarrow \mathrm{CH}^n(A)_{\mathbb{Q}} \right),$$

where $w_{\leq k} M(A)$ retains motivic summands of Bondarko weight $\leq k$. Under the Deninger–Murre decomposition [51], the summand $h^{2n-s}(A)$ has Bondarko weight $2n-s$. The condition $2n-s \leq k$ (i.e., $s \geq 2n-k$) selects the summands that survive in $w_{\leq k} M(A)$. Setting $k = n+g-r$ gives $s \geq 2n-(n+g-r) = n+r-g$, i.e., $s \geq r-g+n$. Under Hypothesis 4.9, $F_{\mathrm{orth}}^r = \bigoplus_{s \geq r-g+n} V_{n,s}$ by Theorem 6.11(b), and $V_{n,s}$ receives contributions from $h^{2n-s}(A)$. Thus both sides equal $\bigoplus_{s \geq r-g+n} V_{n,s}$. $\square$

Remark 10.14 (Comparison of the three filtrations). Under Hypotheses 4.9 and 8.2, the three graded pieces $\mathrm{gr}_{F_{\mathrm{orth}}}^r$, $\mathrm{gr}_{F_S}^r$, $\mathrm{gr}_{W^{\mathrm{Bon}}}^r$ are canonically isomorphic to $V_{n,r-g+n}$; the three filtrations agree after reindexing.

11. Vanishing of differentials

This section gives a self-contained account of the differential vanishings underlying the degeneration $E^{\infty} = E^1$. For $R = \mathbb{Q}$, everything follows from Corollary 4.4; this section treats the integral ($R = \mathbb{Z}$) case and provides the Hodge-theoretic perspective via the Yoneda Ext groups.

Hypothesis 11.1 (Ext^1 -vanishing for $R \neq \mathbb{Q}$). *For all perfect fields k and all $p \in \mathbb{Z}$:*

$$\mathrm{Ext}_{DM(k, \mathbb{Q})}^1(bc_{p/p-1} \mathbf{1}_{\mathbb{Q}}, bc_{p-1/p-2} \mathbf{1}_{\mathbb{Q}}) = 0.$$

Remark 11.2 (Status of Hypothesis 11.1). For $R = \mathbb{Q}$: this is a theorem, proved unconditionally by Corollary 4.4 via Maschke’s theorem. Hypothesis 11.1 is genuinely open only for $R = \mathbb{Z}$ or $R = \mathbb{Z}[\ell^{-1}]$.

Remark 11.3 (Why Maschke resolves the case $p \neq 0$). The earlier approach to Hypothesis 11.1 for $p \neq 0$ attempted a Tate-twist reduction to the $p = 0$ case (Remark 11.6), which fails. The correct approach (Corollary 4.4) does not reduce to $p = 0$: instead, it uses that $\mathcal{H}om_{\mathbf{HI}}(bc_{p/p-1}\mathbf{1}_{\mathbb{Q}}, bc_{p-1/p-2}\mathbf{1}_{\mathbb{Q}})(\bar{k})$ is a $\mathbb{Q}$ -vector space with G_k -action (since $\mathbf{HI}(k, \mathbb{Q})$ is $\mathbb{Q}$ -linear), so Lemma 4.3 applies directly at any p .

Proposition 11.4 (Rational Hilbert 90). *For any perfect field k : $H^1(G_k, \bar{k}^\times \otimes_{\mathbb{Z}} \mathbb{Q}) = 0$.*

Proof. Any continuous 1-cocycle $f : G_k \rightarrow \bar{k}^\times \otimes_{\mathbb{Z}} \mathbb{Q}$ is continuous for the profinite topology on G_k and the discrete topology on $\bar{k}^\times \otimes_{\mathbb{Z}} \mathbb{Q}$; hence f factors through a finite quotient $\text{Gal}(L/k)$ for some finite Galois extension L/k . Thus $[f] \in H^1(\text{Gal}(L/k), L^\times \otimes_{\mathbb{Z}} \mathbb{Q})$.

Since $G = \text{Gal}(L/k)$ is finite, the group ring $\mathbb{Q}[G]$ is semisimple (Maschke's theorem: $|G|$ is invertible in $\mathbb{Q}$). Every $\mathbb{Q}[G]$ -module is therefore projective, so

$$H^j(G, M) = 0 \quad \text{for all } j \geq 1 \text{ and every } \mathbb{Q}[G]\text{-module } M.$$

Applying this with $M = L^\times \otimes_{\mathbb{Z}} \mathbb{Q}$ gives $H^1(\text{Gal}(L/k), L^\times \otimes_{\mathbb{Z}} \mathbb{Q}) = 0$, hence $[f] = 0$. Since L was arbitrary, $H^1(G_k, \bar{k}^\times \otimes_{\mathbb{Z}} \mathbb{Q}) = 0$. $\square$

Lemma 11.5 (Pure layers in the heart). *$bc_{0/-1}\mathbf{1}_{\mathbb{Q}} \simeq \mathbb{Q}^{\text{tr}}$ and $bc_{-1/-2}\mathbf{1}_{\mathbb{Q}} \simeq K_1^{\text{MW}} \otimes_{\mathbb{Z}} \mathbb{Q} \simeq \mathcal{O}^\times \otimes \mathbb{Q}$ in $\mathbf{HI}(k, \mathbb{Q})$, by [25]. Consequently:*

$$\text{Ext}_{DM(k, \mathbb{Q})}^1(bc_{0/-1}\mathbf{1}_{\mathbb{Q}}, bc_{-1/-2}\mathbf{1}_{\mathbb{Q}}) \cong H_{\text{Nis}}^1(\text{Spec } k, \mathcal{O}^\times \otimes \mathbb{Q}) \cong H^1(G_k, \bar{k}^\times \otimes_{\mathbb{Z}} \mathbb{Q}).$$

Proof. The homotopy t -structure on $DM(k, \mathbb{Q})$ gives

$$\text{Hom}_{DM}(A, B[r]) = \text{Ext}_{\mathbf{HI}(k, \mathbb{Q})}^r(A, B)$$

for A, B in the heart. By [17, Thm. 3.1.12]:

$$\begin{aligned} \text{Ext}_{\mathbf{HI}}^r(\mathbb{Q}^{\text{tr}}, \mathcal{F}) &\cong H_{\text{Nis}}^r(\text{Spec } k, \mathcal{F}) \\ &\cong H^r(G_k, \mathcal{F}_{\bar{k}}), \end{aligned}$$

where the last step uses that Nisnevich cohomology over a field equals Galois cohomology of stalks [17, Cor. 14.24]. $\square$

Remark 11.6 (Tate-twist reduction: precise status). A reduction of

$$\text{Ext}_{DM}^1(bc_{p/p-1}\mathbf{1}_{\mathbb{Q}}, bc_{p-1/p-2}\mathbf{1}_{\mathbb{Q}})$$

to the $p = 0$ case via Tate twist would require $bc_{p/p-1}\mathbf{1}_{\mathbb{Q}} \simeq (bc_{0/-1}\mathbf{1}_{\mathbb{Q}})(p)$, which would follow from $\mathbf{1}_{\mathbb{Q}}(p) \simeq \mathbf{1}_{\mathbb{Q}}$ —false in general. Consequently, Lemma 11.5 is established *only for* $p = 0$. For general $p \neq 0$, the layers $bc_{p/p-1}\mathbf{1}_{\mathbb{Q}}$ are controlled

by $K_{|p|}^{\text{MW}} \otimes \mathbb{Q}$, and the analogous computation yields

$$\text{Ext}_{DM}^1(bc_{p/p-1}\mathbf{1}_{\mathbb{Q}}, bc_{p-1/p-2}\mathbf{1}_{\mathbb{Q}}) \cong H^1(G_k, K_{|p|}^{\text{MW}}(\bar{k}) \otimes \mathbb{Q}).$$

Whether this group vanishes for all perfect k and $p \neq 0$ is open; it is related to the Beilinson–Soulé vanishing conjecture.

Corollary 11.7 (Ext¹-vanishing under Hodge realization). *Over any field k admitting a faithful exact Hodge realization $\text{Re}_H : DM(k, \mathbb{Q}) \rightarrow D(\text{MHS}_{\mathbb{Q}})$ compatible with the homotopy t -structure, Tate twists, and the t -exact truncation for pure layers, Hypothesis 11.1 holds for $p = 0$.*

For general $p \in \mathbb{Z}$, the following is known unconditionally: the pure layer $bc_{p/p-1}\mathbf{1}_{\mathbb{Q}}$ carries a pure Hodge structure of weight $-2p$ under Re_H , via the general structure of the orthogonal tower over $\mathbb{C}$. The Ext¹-vanishing for $p \neq 0$ would follow from the combination of: (i) the identification $bc_{p/p-1}\mathbf{1}_{\mathbb{Q}} \simeq K_{|p|}^{\text{MW}} \otimes_{\mathbb{Z}} \mathbb{Q}$ (which [25] establishes only for $p \in \{0, -1\}$), and (ii) the Galois cohomological vanishing $H^1(G_k, K_{|p|}^{\text{MW}}(\bar{k}) \otimes_{\mathbb{Z}} \mathbb{Q}) = 0$, which is open for $|p| \geq 2$ (see Remark 11.6).

Proof. For $p = 0$: by Lemma 11.5, $\text{Ext}_{DM}^1(bc_{0/-1}\mathbf{1}_{\mathbb{Q}}, bc_{-1/-2}\mathbf{1}_{\mathbb{Q}}) \cong H^1(G_k, \bar{k}^{\times} \otimes \mathbb{Q}) = 0$ by Proposition 11.4. This is a direct Galois cohomological argument and does not use the Hodge realization; the Hodge realization provides an additional route to see the vanishing at $p = 0$ by noting that $\text{Re}_H bc_{0/-1}\mathbf{1}_{\mathbb{Q}}$ is pure of weight 0 and $\text{Re}_H bc_{-1/-2}\mathbf{1}_{\mathbb{Q}}$ is pure of weight -2 (using [25] for these two layers), and Ext¹ in the mixed Hodge structure category between objects of strictly different pure weights vanishes; the splitting over $\mathbb{Q}$ then follows from rational Hilbert 90.

By Proposition 3.3 (following [25, 5.3.6]), $bc_{p/p-1}\mathbf{1}_{\mathbb{Q}} \simeq K_{|p|}^{\text{M}} \otimes \mathbb{Q}$ for all $p \leq 0$. The Ext¹-vanishing $H^1(G_k, K_{|p|}^{\text{M}}(\bar{k}) \otimes \mathbb{Q}) = 0$ for $|p| \geq 2$ is open (Remark 11.6); for $|p| \leq 1$, it is proved by Hilbert 90. $\square$

Remark 11.8 (Ext in mixed Hodge structures). We emphasize that Ext¹ between objects of *different* weights in the category of mixed $\mathbb{Q}$ -Hodge structures need not vanish in general: $\text{Ext}_{\text{MHS}}^1(\mathbb{Q}(0), \mathbb{Q}(-1)) \cong \mathbb{C}/\mathbb{Q}(1) \neq 0$. The vanishing in Corollary 11.7 is a consequence of Hilbert 90 on the motivic side, not a general Hodge-theoretic principle. The Hodge realization only provides a faithful embedding; the vanishing itself requires Galois cohomology.

Theorem 11.9 ($d^1 = 0$ under Hypothesis 11.1). *Under Hypothesis 11.1: $d^1 = 0$ on the spectral sequence, and $E^2 = E^1$ for any smooth projective X/k , any n , and any $m \geq 0$.*

Proof. Bridge from Ext to d^1 . The differential $d^1 : E_{p,q}^1 \rightarrow E_{p-1,q}^1$ arises from the

octahedral axiom applied to $bc_{\leq p-2}\mathbf{1}_{\mathbb{Q}} \subset bc_{\leq p-1}\mathbf{1}_{\mathbb{Q}} \subset bc_{\leq p}\mathbf{1}_{\mathbb{Q}}$:

$$\begin{array}{ccccccc}
 bc_{\leq p-2}\mathbf{1}_{\mathbb{Q}} & \longrightarrow & bc_{\leq p-1}\mathbf{1}_{\mathbb{Q}} & \longrightarrow & bc_{p-1/p-2}\mathbf{1}_{\mathbb{Q}} & \xrightarrow{+1} & \\
 \parallel & & \downarrow & & \downarrow \alpha & & \\
 bc_{\leq p-2}\mathbf{1}_{\mathbb{Q}} & \longrightarrow & bc_{\leq p}\mathbf{1}_{\mathbb{Q}} & \longrightarrow & bc_{p/p-1}\mathbf{1}_{\mathbb{Q}} & \xrightarrow{+1} &
 \end{array}$$

The attaching map lives in

$$\mathrm{Hom}_{DM}(bc_{p/p-1}\mathbf{1}_{\mathbb{Q}}, bc_{p-1/p-2}\mathbf{1}_{\mathbb{Q}}[1]) = \mathrm{Ext}_{DM(k, \mathbb{Q})}^1(bc_{p/p-1}\mathbf{1}_{\mathbb{Q}}, bc_{p-1/p-2}\mathbf{1}_{\mathbb{Q}}) = 0$$

by Hypothesis 11.1. Hence $d^1 = 0$.

Independence of m . The column range is independent of m (Theorem 3.6), and the Ext group depends only on the Bousfield tower of $\mathbf{1}_{\mathbb{Q}}$ and p , not on m , X , or n . $\square$

12. Higher differentials over number fields

Summary. Lemma 4.3 and Corollary 4.4 (proved in §4) establish $E^\infty = E^1$ unconditionally for $R = \mathbb{Q}$ over any perfect field. This section records the earlier partial results (number fields, finite fields, algebraically closed fields) which are now special cases, and establishes degeneration for $R = \mathbb{Z}$ over finite fields (Theorem 12.4(b)).

Theorem 12.1 ($d^r = 0$ over number fields). *Let k be a number field. Then Hypothesis 4.5 holds for all $r \geq 2$; combined with Hypothesis 11.1 for $r = 1$, we obtain $d^r = 0$ for all $r \geq 1$ and $E^\infty = E^1$ for any smooth projective X/k .*

Proof. We show $d^r = 0$ for each $r \geq 2$ by establishing the vanishing of the attaching map that gives rise to d^r .

Case $r = 2$, at $p = 0$: Borel's theorem and Bass–Tate. By Lemma 4.1 (applied at $p = 0$, $r = 2$), the obstruction lives in $H^2(G_k, K_2^M(\bar{k}) \otimes \mathbb{Q})$. By Borel [46], $K_2(k) \otimes_{\mathbb{Z}} \mathbb{Q} = 0$ for any number field k (the rank formula gives K_2 is finite). By the Bass–Tate theorem [68, Ch. III], $K_2^M(k)$ is a quotient of $K_2(k)$ after rationalization, so $K_2^M(k) \otimes \mathbb{Q} = 0$. The Hochschild–Serre spectral sequence then gives $H^2(G_k, K_2^M(\bar{k}) \otimes \mathbb{Q}) = 0$.

Case $r \geq 3$: virtual cohomological dimension. By Tate [68, I.4.4], $\mathrm{vcd}(G_k) \leq 2$ for any number field k . By Lemma 4.1, the attaching map for d^r lives in $H^r(G_k, M_{p,r})$ for some discrete $\mathbb{Q}[G_k]$ -module $M_{p,r}$. Since $r \geq 3 > \mathrm{vcd}(G_k)$, this cohomology group vanishes for any such module [68, I.4.2], regardless of the specific identification of the Bousfield layers.

All columns simultaneously. The above gives $d^r = 0$ for each differential at each column p in the support range $[-n, \min(d - n, 0)]$. The case $r = 1$, $R = \mathbb{Q}$: $d^1 = 0$ unconditionally for all n by Corollary 4.4 (Maschke). For $R = \mathbb{Z}$: $d^1 = 0$

by Proposition 11.4 at $p = 0$, and by Bass–Tate at $p \leq -1$ (the Milnor K -groups $K_r^M(k) \otimes \mathbb{Q}$ vanish for $r \geq 2$ over number fields by Borel [46]). $\square$

Remark 12.2 (Scope). The argument for $r \geq 2$ is unconditional over number fields. For general perfect fields: if $\text{cd}(G_k) = \infty$, the vanishing of $H^r(G_k, K_r^M(\bar{k}) \otimes \mathbb{Q})$ for large r is open and essentially equivalent to the Beilinson–Soulé conjecture.

Proposition 12.3 ($n = m = 1$: Beauville decomposition of units). *Let A/k be an abelian variety, k perfect. For $n = m = 1$, the orthogonal tower filtration on $\text{CH}^1(A, 1)_{\mathbb{Q}} \cong \mathcal{O}(A)^{\times} \otimes_{\mathbb{Z}} \mathbb{Q}$ is two-step with columns $\{-1, 0\}$, and the $[N]^*$ -eigenvalues are:*

$$\begin{aligned} V_{1,1,0} &:= \{f \in \mathcal{O}(A)^{\times} \otimes \mathbb{Q} : [N]^*f = f\} = (\mathcal{O}(A)^{\times})^{G_k} \otimes \mathbb{Q} \quad (\text{constant functions}); \\ V_{1,1,1} &:= \{f : [N]^*f = N^0 \cdot f = f\} \supseteq \hat{A}(k) \otimes \mathbb{Q} \quad (\text{characters, Pontryagin dual}). \end{aligned}$$

Consequently, $\text{CH}^1(A, 1)_{\mathbb{Q}} = V_{1,1,0} \oplus V_{1,1,1}$ unconditionally.

Proof. By the homotopy invariance theorem for motivic cohomology [17, Prop. 14.22], $\text{CH}^1(A, 1) = H_{\text{Nis}}^1(A, \mathcal{O}^{\times}) = \mathcal{O}(A)^{\times}$. The column support Theorem 3.6 with $n = m = 1$ and $d = g \geq 1$ gives support in $p \in [-1, \min(g-1, 0)] = \{-1, 0\}$ (for $g = 1$) or $p \in \{-1, \dots, 0\}$ (for $g \geq 1$); in both cases at most two columns. The $[N]^*$ -equivariance (Theorem 7.1) splits $\mathcal{O}(A)^{\times} \otimes \mathbb{Q}$ by eigenvalues of $[N]^*$: $[N]^*f = f \circ [N]$ for $f \in \mathcal{O}(A)^{\times}$; constant functions satisfy $[N]^*c = c$ (eigenvalue $1 = N^0$); characters $\chi : \hat{A} \rightarrow \mathbb{G}_m$ satisfy $[N]^*\chi = \chi^N$ (so $[N]^*\chi \neq \chi$ for $N \neq 1$ unless $\chi^{N-1} = 1$, i.e. torsion). Over $\mathbb{Q}$ -coefficients, torsion is killed. Constants have $[N]^*c = c$, i.e., eigenvalue N^0 , placing them in $V_{0,0}$; characters χ with $[N]^*\chi = \chi^N$ yield, after rationalization, $[N]^*\chi = N^0\chi$ (trivial eigenvalue on torsion-free quotient), hence constants and rational characters both lie in $V_{1,s}$ with $s = 0$ (eigenvalue $N^{2 \cdot 1 - 0} = N^2$ for the $n = 1, m = 1$ probe). The decomposition $\text{CH}^1(A, 1)_{\mathbb{Q}} = V_{1,1,0} \oplus V_{1,1,1}$ follows from the two-column support. Degeneration $E^{\infty} = E^1$ for all n, m follows from Corollary 4.4, which covers all values of m . $\square$

Theorem 12.4 (Full degeneration and trivialization over finite fields). *Let $k = \mathbb{F}_q$ be a finite field, X/k smooth projective.*

- (a) Degeneration for $R = \mathbb{Q}$. By Corollary 4.4, $E^{\infty} = E^1$ unconditionally.
- (b) Degeneration for $R = \mathbb{Z}$. $E^{\infty} = E^1$ for the orthogonal spectral sequence with $R = \mathbb{Z}$, unconditionally.
- (c) Trivialization of the Bousfield filtration. $F_{\text{orth}}^1 \text{CH}^n(X)_{\mathbb{Q}} = 0$.
- (d) Reduction to the Tate conjecture. Over $\overline{\mathbb{F}}_p$, $\text{Griff}^n(A)_{\mathbb{Q}} = 0$ and the orthogonal filtration is one-step.

Proof. (a): Immediate from Corollary 4.4.

(b): We work with integer coefficients; the Maschke argument fails. Instead we use two ingredients.

Higher differentials ($r \geq 2$). $G_{\mathbb{F}_q} = \hat{\mathbb{Z}}$ is pro-cyclic with $\text{cd}(G_{\mathbb{F}_q}) = 1$ [68, I.3.3]. For $R = \mathbb{Z}$, the attaching map of d^r lives in $\text{Ext}_{DM(k, \mathbb{Z})}^r(A, B)$ for $A, B \in \mathbf{HI}(k, \mathbb{Z})$; by the Nisnevich-to-Galois comparison, this equals $H^r(G_k, M)$ for a discrete $\mathbb{Z}[G_k]$ -module M . Since $r \geq 2 > \text{cd}(G_k) = 1$, we get $d^r = 0$.

First differential ($r = 1$). The differential d^1 at column $p = 0$ is the class of $\delta_0 \in \text{Ext}_{DM(k, \mathbb{Z})}^1(bc_{0/-1}\mathbf{1}_{\mathbb{Z}}, bc_{-1/-2}\mathbf{1}_{\mathbb{Z}}) \cong H^1(G_k, \bar{k}^\times) = 0$ by Hilbert's Theorem 90 (classical; no $\mathbb{Q}$ -tensoring needed).

At columns $p \leq -1$: $bc_{-r/-r-1}\mathbf{1}_{\mathbb{Z}}$ has stalk at $\bar{k}$ equal to $K_r^M(\bar{\mathbb{F}}_q)$. For $r = 1$: $K_1^M(\bar{\mathbb{F}}_q) = \bar{\mathbb{F}}_q^\times$, and $H^1(G_k, \bar{\mathbb{F}}_q^\times) = 0$ by Hilbert 90 over $\mathbb{F}_q$. For $r \geq 2$: $K_r^M(\bar{\mathbb{F}}_q) = 0$ because $\bar{\mathbb{F}}_q$ is a perfect field with $\text{cd}(\bar{\mathbb{F}}_q) = 0$ and every symbol $\{a_1, \dots, a_r\}$ ($r \geq 2$) contains a Steinberg element: any $a \in \bar{\mathbb{F}}_q^\times$ satisfies $a = b^n$ for all n , so $a + 1$ is an n -th power for all n not dividing $|\bar{\mathbb{F}}_q^\times|$, and by the Bass–Tate theorem [17, Rem. 7.1.3] $K_r^M(\mathbb{F}_{q^m}) = 0$ for $r \geq 2$ and all m ; taking $\bar{\mathbb{F}}_q = \bigcup_m \mathbb{F}_{q^m}$ gives $K_r^M(\bar{\mathbb{F}}_q) = 0$. Hence $d^1 = 0$ for all p and $E^\infty = E^1$ over $\mathbb{F}_q$ for $R = \mathbb{Z}$.

(c): The layer $bc_{-1/-2}\mathbf{1}_{\mathbb{Q}} \simeq \mathcal{O}^\times \otimes \mathbb{Q}$. For X smooth projective over $\mathbb{F}_q$: $\mathcal{O}(X)^\times = \mathbb{F}_q^\times$ (finite), so $\mathcal{O}(X)^\times \otimes \mathbb{Q} = 0$ and $E_{-1,q}^1 = 0$. By Bass–Tate, $E_{p,q}^1 = 0$ for all $p \leq -1$. Thus $F_{\text{orth}}^1 \text{CH}^n(X)_{\mathbb{Q}} = 0$.

(d): The vanishing $F_{\text{orth}}^1 = 0$ forces $F_{\text{orth}}^\bullet$ to be one-step. By Theorem 7.5 (under Hypothesis 4.9): $\ker(\text{cl}) = F_{\text{orth}}^1 \text{CH}^n = 0$, so $\text{cl} : \text{CH}^n(A)_{\mathbb{Q}} \hookrightarrow H^{2n}$ is injective. Over $\bar{\mathbb{F}}_p$ this gives $\text{Griff}^n(A)_{\mathbb{Q}} = 0$ (all algebraic cycles are detected by cl , consistent with the Tate conjecture for abelian varieties proved by Tate [65]). $\square$

Theorem 12.5 (Full degeneration over algebraically closed fields and $C(t)$). *Let k be a perfect field.*

- (a) Algebraically closed k . *If k is algebraically closed, then $d^r = 0$ for all $r \geq 1$ unconditionally, and $E^\infty = E^1$.*
- (b) Rational function fields. *If $k = k_0(t)$ with k_0 algebraically closed, then $d^r = 0$ for all $r \geq 2$ unconditionally. For $r = 1$: $d^1 = 0$ at $p = 0$ (Proposition 11.4) and at $p \leq -2$ (since $K_r^M(k) \otimes \mathbb{Q} = 0$ for $r \geq 2$ by Tsen's theorem: k is a C_1 -field [55, Thm. 1.3.4]). The only potentially nonzero d^1 is at $p = -1$, controlled by $H^1(G_k, k^\times \otimes \mathbb{Q}) = 0$ (Proposition 11.4 for $k_0(t)$). Hence $d^1 = 0$ and $E^\infty = E^1$ unconditionally.*

Proof. (a): $G_k = 1$ for k algebraically closed. By Lemma 4.1, every attaching map lives in $H^r(G_k, M_{p,r}) = H^r(1, M_{p,r}) = 0$ for $r \geq 1$. Hence $d^r = 0$ for all $r \geq 1$ and $E^\infty = E^1$.

(b): For $r \geq 2$: $\text{vcd}(G_{k_0(t)}) \leq 2$; more precisely, $G_{k_0(t)^{\text{sep}}/k_0(t)}$ has cohomological dimension ≤ 1 (since k_0 is algebraically closed and $k_0(t)$ is a C_1 -field, its absolute Galois group has $\text{cd} \leq 1$ [68, II.3.3]). Hence $H^r(G_{k_0(t)}, M_{p,r}) = 0$ for $r \geq 2$, giving $d^r = 0$ for $r \geq 2$.

For d^1 at $p = 0$: Proposition 11.4.

For d^1 at $p \leq -2$ (i.e. $|p| \geq 2$): by Lemma 4.1, the obstruction is $H^1(G_k, K_{|p|}^M(\bar{k}) \otimes \mathbb{Q})$. Since k_0 is algebraically closed and $k = k_0(t)$ is a C_1 -field

(Tsen's theorem [68, I.3.1] or [55, Thm. 1.3.4]), $K_r^M(k) = 0$ for $r \geq 2$ [55, III.7.8] (Milnor K -theory of C_1 -fields vanishes above degree 1). Hence $K_{|p|}^M(\bar{k}) \otimes \mathbb{Q} = 0$ for $|p| \geq 2$, and $d^1 = 0$ at $p \leq -2$.

For d^1 at $p = -1$: the obstruction is $H^1(G_k, \bar{k}^\times \otimes \mathbb{Q}) = H^1(G_k, k^\times \otimes \mathbb{Q})$ (since k_0 is algebraically closed, $k = k_0(t)$ and $\bar{k} = k_0^{\text{sep}}(t)^{\text{alg}}$). By Proposition 11.4 (applied with $k_0(t)$ in place of k , noting $G_{k_0(t)}$ is profinite and Maschke applies to every finite quotient), $H^1(G_{k_0(t)}, \overline{k_0(t)}^\times \otimes \mathbb{Q}) = 0$. Hence $d^1 = 0$ at $p = -1$, completing the degeneration. $\square$

Corollary 12.6. *Over any perfect field k with $\text{cd}(G_k) \leq 1$ and $K_r^M(k) \otimes \mathbb{Q} = 0$ for $r \geq 2$ (e.g. $k = k_0(t)$ with k_0 algebraically closed of characteristic zero, or k a finite field), $E^\infty = E^1$ for the orthogonal spectral sequence, unconditionally for all n .*

Proof. Differentials $d^r = 0$ for $r \geq 2$. By Lemma 4.1, the attaching map of d^r lies in $H^r(G_k, M_{p,r})$ for some discrete $\mathbb{Q}[G_k]$ -module $M_{p,r}$. Since k is a finite field ($k = \mathbb{F}_q$ or $k = \overline{\mathbb{F}}_q$), the cohomological dimension $\text{cd}(G_k) \leq 1$: for $k = \mathbb{F}_q$, $G_k \cong \hat{\mathbb{Z}}$ has $\text{cd} = 1$; for $k = \overline{\mathbb{F}}_q$, $G_k = 1$ is trivial. Hence $H^r(G_k, M_{p,r}) = 0$ for $r \geq 2$, so $d^r = 0$ for $r \geq 2$.

Differential $d^1 = 0$ at $p = 0$. The d^1 -differential at filtration degree $p = 0$ is governed by the extension class in $\text{Ext}_{DM}^1(bc_{0/-1}\mathbf{1}_{\mathbb{Q}}, bc_{-1/-2}\mathbf{1}_{\mathbb{Q}})$. By Proposition 3.3, $bc_{0/-1}\mathbf{1}_{\mathbb{Q}} \simeq \mathbb{Q}^{\text{tr}}$ and $bc_{-1/-2}\mathbf{1}_{\mathbb{Q}} \simeq K_1^M \otimes \mathbb{Q} = \mathbb{G}_m \otimes \mathbb{Q}$. Since $K_r^M(\bar{k}) = (\bar{k}^\times)^{\otimes r}$ and for $\bar{k} = \overline{\mathbb{F}}_q$ the group $K_r^M(\bar{k}) = 0$ for $r \geq 2$ (as $\overline{\mathbb{F}}_q^\times$ is torsion, tensored with $\mathbb{Q}$ gives zero), we have $M_{0,1} = 0$. By Proposition 11.4 for $p = 0$: the d^1 -differential at $p = 0$ factors through $H^1(G_k, \mathbb{G}_m(\bar{k}) \otimes \mathbb{Q}) = 0$ (Hilbert's Theorem 90 gives $H_{\text{ét}}^1(\text{Spec } k, \mathbb{G}_m) = 0$, and the $\mathbb{Q}$ -tensor kills any $\mathbb{Z}$ -torsion). Hence $d^1 = 0$ at $p = 0$.

All $d^1 = 0$. For $p \leq -1$: the differential d^1 at position p lies in $H^1(G_k, M_{p,1}) = \text{Hom}_{\text{cont}}(G_k, M_{p,1})$ (since $\text{cd}(G_k) \leq 1$ means $H^1 = \text{Hom}$ for abelian G_k -modules in characteristic 0 coefficients). For $k = \mathbb{F}_q$: $\text{Hom}_{\text{cont}}(\hat{\mathbb{Z}}, M_{p,1})$ is the space of $\mathbb{Q}$ -linear $\hat{\mathbb{Z}}$ -representations; $M_{p,1}$ is a $\mathbb{Q}$ -vector space with Frobenius action, and $H^1(\hat{\mathbb{Z}}, M_{p,1}) = M_{p,1}/(F - 1)M_{p,1}$. Since $K_r^M(\mathbb{F}_q) = 0$ for $r \geq 2$ (the Milnor K -theory of a finite field vanishes in degree ≥ 2 , [75, Ch. VIII]), $M_{p,1} = 0$ for $|p| \geq 2$. For $|p| = 1$: $M_{-1,1} = K_1^M(\bar{k})^\times \otimes \mathbb{Q} = \bar{k}^\times \otimes \mathbb{Q} = 0$ (torsion group tensored with $\mathbb{Q}$). Hence $H^1(G_k, M_{p,1}) = 0$ for all p , giving $d^1 = 0$ everywhere. $\square$

Remark 12.7 (Comparison: $\mathbb{F}_q$ vs. number fields). Theorem 12.1 over number fields requires Hypothesis 11.1 for $d^1 = 0$ (except for $n = 1$, where Hilbert 90 suffices unconditionally). Theorem 12.4 gives $d^1 = 0$ unconditionally over $\mathbb{F}_q$ for all n : all Bousfield layers $p \leq -1$ vanish over $\mathbb{F}_q$ (no Milnor K -theory above degree 1), so the Ext groups controlling d^1 vanish for size reasons, not just cohomological-dimension reasons. Over finite fields the motivic unit is already truncated at $p = 0$, so the Bousfield tower is trivially one-step.

Part (d) of Theorem 12.4 shows that the ℓ -adic Frobenius conjecture over $\mathbb{F}_q$ reduces to the Tate conjecture (a theorem for abelian varieties by Tate [65]). The

interesting case is over fields where $F_{\text{orth}}^1 \text{CH}^n \neq 0$ (number fields, function fields of curves), where the Bousfield filtration is nontrivial and the Frobenius condition is a genuine constraint.

Theorem 12.8 (Frobenius–Beauville over $\mathbb{F}_q$). *Let $A/\mathbb{F}_q$ be an abelian variety of dimension g , $1 \leq n \leq g$. Every element of $\text{CH}^n(A)_{\mathbb{Q}}$ lies in $V_{n,0}$ and has Frobenius eigenvalue q^n on its ℓ -adic cycle class.*

Proof. By Theorem 13.1(b), $\text{CH}^n(A_{\mathbb{F}_q})_{\mathbb{Q}} = V_{n,0}$. By Lemma 6.14(a), every $z \in V_{n,0}$ has $\text{cl}_{\ell}(z) \in H^{2n}(A_{\overline{\mathbb{F}}_q}, \mathbb{Q}_{\ell})$.

By the Tate conjecture for abelian varieties over $\mathbb{F}_q$ (Tate [65]), the cycle class map $\text{cl}_{\ell} : \text{CH}^n(A_{\overline{\mathbb{F}}_q})_{\mathbb{Q}} \otimes \mathbb{Q}_{\ell} \rightarrow H^{2n}(A_{\overline{\mathbb{F}}_q}, \mathbb{Q}_{\ell})$ is an isomorphism onto the $\text{Frob}_q = q^n$ part of H^{2n} . Since $\text{CH}^n(A)_{\mathbb{Q}} = V_{n,0}$ and $V_{n,0}$ is in the image of the cycle class map, the Frobenius eigenvalue on $\text{cl}_{\ell}(z)$ is q^n for every $z \in V_{n,0}$. $\square$

Corollary 12.9 (Degeneration over number fields). *Let k be a number field, A/k abelian of dimension g , $1 \leq n \leq g$. (a) $E^{\infty} = E^1$: unconditionally for $n = 1$ (Corollary 4.4 combined with Theorem 12.1 for $r \geq 2$); under Hypothesis 11.1 for $n \geq 2$. (b) Under Hypothesis 4.9, $F_{\text{orth}}^{\bullet}$ is the unique filtration on $\text{CH}^n(A)_{\mathbb{Q}}$ with graded pieces $\text{gr}_{F_{\text{orth}}}^r \cong V_{n,r-g+n}$ (Theorem 6.12). (c) **(H)** Under Hypothesis 4.9, $F_{\text{orth}}^{\bullet}$ coincides with Saito’s and Bondarko’s filtrations after reindexing.*

Remark 12.10 (Frobenius compatibility: a conditional remark). For $A/\mathbb{F}_q$ abelian over a finite field, the Tate conjecture (together with the Weil conjectures for A) controls which Frobenius eigenvalues appear in ℓ -adic cohomology $H^{2n}(A_{\overline{\mathbb{F}}_q}, \mathbb{Q}_{\ell})$. A precise comparison between the Beauville decomposition on $\text{CH}^n(A)_{\mathbb{Q}}$ and the Frobenius eigenspaces on the ℓ -adic realization would require: (i) an extension of the Bousfield filtration to $\mathbb{Q}_{\ell}$ -coefficients; (ii) a compatibility statement between the Beauville decomposition and the ℓ -adic cycle class map; (iii) a comparison between the $[N]^*$ -eigenvalue N^{2n-s} on $V_{n,s}$ and the Frobenius eigenvalue on the corresponding ℓ -adic summand. These are non-trivial and largely open. We state no theorem in this direction; the compatibility is a natural conjecture consistent with all results of this thesis and with the Tate conjecture, but its proof requires machinery beyond the scope of the present work.

13. Characteristic p : Beauville collapse, Griffiths vanishing, and a crystalline programme

The rational orthogonal tower is rich over characteristic-zero fields — multi-step filtration, non-trivial Griffiths group, Beauville eigenspaces in each graded piece — but collapses completely over fields of characteristic p . This section makes the collapse precise (§§13.1–13.2), explains the arithmetic reason (finiteness of $A(\mathbb{F}_q)$), and then proposes a crystalline replacement (§13.3) that recovers non-trivial information.

13.1. Beauville collapse over finite fields

Over a finite field $k = \mathbb{F}_q$, the rational orthogonal tower undergoes a dramatic simplification: every Beauville component $V_{n,s}$ with $s \geq 1$ vanishes, and all homologically trivial rational cycles are algebraically trivial. This is the characteristic- p counterpart of the degeneration theorem of Part II, but the mechanism is entirely different: it is not Maschke vanishing of differentials, but rather the finiteness of the Mordell–Weil group $A(\mathbb{F}_q)$ and the vanishing of Milnor K -theory in degree ≥ 2 over finite fields.

Theorem 13.1 (Beauville collapse over $\mathbb{F}_q$). *Let $A/\mathbb{F}_q$ be an abelian variety of dimension g , $1 \leq n \leq g$. All parts are unconditional.*

- (a) $V_{1,1}(A_{\mathbb{F}_q}) = \text{Pic}^0(A)(\mathbb{F}_q)_{\mathbb{Q}} = 0$.
- (b) $V_{n,s}(A_{\mathbb{F}_q}) = 0$ for all $s \geq 1$, and $\text{CH}^n(A_{\mathbb{F}_q})_{\mathbb{Q}} = V_{n,0}$.
- (c) $\text{CH}_{\text{hom}}^n(A_{\mathbb{F}_q})_{\mathbb{Q}} = 0$ and $\text{Griff}^n(A_{\mathbb{F}_q})_{\mathbb{Q}} = 0$.
- (d) The filtration is one-step: $F_{\text{orth}}^{g-n} = \text{CH}^n(A_{\mathbb{F}_q})_{\mathbb{Q}}$ and $F_{\text{orth}}^{g-n+1} = 0$.

Proof. (a): $\text{Pic}^0(A)(\mathbb{F}_q) = \hat{A}(\mathbb{F}_q)$ is finite (abelian varieties over finite fields have finitely many rational points), so $\hat{A}(\mathbb{F}_q) \otimes_{\mathbb{Z}} \mathbb{Q} = 0$. By Beauville [43], $V_{1,1} = \text{Pic}^0(A)_{\mathbb{Q}}$; hence $V_{1,1} = 0$ over $\mathbb{F}_q$.

(b): By Beauville’s decomposition, $V_{n,s}$ is generated (via Künneth and Pontryagin products) by elements of $V_{1,1}^{\otimes s}$ and $V_{1,0}$ [43, Thm. 1]. Since $V_{1,1} = 0$ over $\mathbb{F}_q$ (part (a)), any expression involving $s \geq 1$ copies of $V_{1,1}$ is zero. Hence $V_{n,s} = 0$ for $s \geq 1$, and $\text{CH}^n(A_{\mathbb{F}_q})_{\mathbb{Q}} = V_{n,0}$ by Beauville’s decomposition.

(c): $V_{n,0} \hookrightarrow H^{n,n}(A)$ by Lemma 6.14(a), so $\ker(\text{cl}_{\mathbb{Q}}) \cap V_{n,0} = 0$. Since $\text{CH}^n(A_{\mathbb{F}_q})_{\mathbb{Q}} = V_{n,0}$, we get $\text{CH}_{\text{hom}}^n(A_{\mathbb{F}_q})_{\mathbb{Q}} = 0$, hence $\text{Griff}^n(A_{\mathbb{F}_q})_{\mathbb{Q}} \subseteq \text{CH}_{\text{hom}}^n = 0$.

(d): $F_{\text{orth}}^{g-n} = \text{CH}^n$ by Lemma 3.2(i). $F_{\text{orth}}^{g-n+1} = F_{\text{orth}}^1$: by Theorem 12.4(c), $F_{\text{orth}}^1 \text{CH}^n = 0$ over $\mathbb{F}_q$. $\square$

Remark 13.2 (Two independent proofs of $F_{\text{orth}}^1 = 0$). Theorem 13.1(d) gives $F_{\text{orth}}^{g-n+1} = 0$ from Theorem 12.4(c) ($bc_{-1/-2} \mathbf{1}_{\mathbb{Q}} \simeq \mathcal{O}^{\times} \otimes \mathbb{Q}$ has zero stalk over $\mathbb{F}_q$ since $\mathcal{O}(X)^{\times} = \mathbb{F}_q^{\times}$ is finite). Parts (b)–(c) use an independent Beauville-theoretic argument: $V_{n,s} = 0$ for $s \geq 1$ because $V_{1,1} = 0$ (finiteness of $\hat{A}(\mathbb{F}_q)$), so $\text{CH}_{\text{hom}}^n = 0$, hence $\ker(\text{cl}) = 0$ and $F_{\text{orth}}^1 = 0$. The two routes are complementary; the Beauville route does not assume $F_{\text{orth}}^1 = 0$ from the motivic tower.

Remark 13.3 (Geometric meaning). Theorem 13.1 says: over $\mathbb{F}_q$, every rational Chow class is “purely algebraic” in the sense of Beauville — it belongs to the highest-eigenvalue piece $V_{n,0}$. The Beauville pieces $V_{n,s}$ with $s \geq 1$ (which carry the transcendental information — algebraic equivalence classes, intermediate Jacobian periods, Griffiths group elements — over number fields or $\mathbb{C}$) are rationally trivial over finite fields. This reflects the finiteness of $A(\mathbb{F}_q)$: the torsion points of the dual abelian variety $\hat{A}(\mathbb{F}_q) = \text{Pic}^0(A)(\mathbb{F}_q)$ (which generates $V_{1,1}$) vanish in $\mathbb{Q}$ -coefficients.

Example 13.4 (Elliptic curve over $\mathbb{F}_q$). Let $E/\mathbb{F}_q$ be an elliptic curve. Beauville's decomposition: $\mathrm{CH}^1(E)_{\mathbb{Q}} = V_{1,0} \oplus V_{1,1}$, where $V_{1,0} = \mathbb{Q} \cdot [O]$ and $V_{1,1} = E(\mathbb{F}_q)_{\mathbb{Q}}$. Since $E(\mathbb{F}_q)$ is a finite group, $E(\mathbb{F}_q) \otimes \mathbb{Q} = 0$, giving $V_{1,1}(E_{\mathbb{F}_q}) = 0$. So $\mathrm{CH}^1(E_{\mathbb{F}_q})_{\mathbb{Q}} = \mathbb{Q} \cdot [O] \cong \mathbb{Q}$, the filtration is $\mathrm{CH}^1(E)_{\mathbb{Q}} \supset 0$, and $\mathrm{Pic}^0(E_{\mathbb{F}_q})_{\mathbb{Q}} = 0$.

By contrast, over $\mathbb{Q}$: if $E(\mathbb{Q})$ has rank $r \geq 1$, $V_{1,1}(E_{\mathbb{Q}}) = E(\mathbb{Q})_{\mathbb{Q}} \cong \mathbb{Q}^r \neq 0$, the filtration has two steps, and $\mathrm{Pic}^0(E_{\mathbb{Q}})_{\mathbb{Q}} \cong \mathbb{Q}^r$.

Example 13.5 (Abelian surface over $\mathbb{F}_p$). Let $A = E_1 \times E_2/\mathbb{F}_p$ with E_1, E_2 supersingular elliptic curves. Over $\mathbb{F}_{p^2}$ both E_i are defined and $E_i(\mathbb{F}_{p^2})$ is a finite group of order $(p \pm 1)^2$.

Beauville for $n = 2$: $\mathrm{CH}^2(A)_{\mathbb{Q}} = V_{2,0} \oplus V_{2,1} \oplus V_{2,2}$. By Theorem 13.1(b): $V_{2,1}(A_{\mathbb{F}_{p^2}}) = 0$ and $V_{2,2}(A_{\mathbb{F}_{p^2}}) = 0$. So $\mathrm{CH}^2(A_{\mathbb{F}_{p^2}})_{\mathbb{Q}} = V_{2,0}$, and the filtration collapses: $F_{\mathrm{orth}}^0 \supset 0$. In particular: $\mathrm{Griff}^2(A_{\mathbb{F}_{p^2}})_{\mathbb{Q}} = 0$, whereas over $\mathbb{C}$ one has $\dim \mathrm{Griff}^2(A_{\mathbb{C}})_{\mathbb{Q}} \geq r_1 r_2 \geq 1$ whenever $r_i = \mathrm{rank}(E_i(\mathbb{C}))$ (Theorem 7.15).

Example 13.6 (Jacobian of a genus-3 curve over $\mathbb{F}_q$). Let $C/\mathbb{F}_q$ be a smooth curve of genus 3, $A = J(C)$. The Ceresa cycle $Z_C = C - C^- \in \mathrm{CH}_{\mathrm{hom}}^2(A)_{\mathbb{Q}}$ is an element of $V_{2,1}$ (by the eigenvalue argument: $[-1]^* Z_C = -Z_C$, so Z_C is in the $N^{2 \cdot 2 - 1} = N^3$ -eigenspace). By Theorem 13.1(b), $V_{2,1}(A_{\mathbb{F}_q}) = 0$, so $Z_C \otimes \mathbb{Q} = 0$ in $\mathrm{CH}^2(A_{\mathbb{F}_q})_{\mathbb{Q}}$: the Ceresa cycle is rationally trivial over $\mathbb{F}_q$. Over $\mathbb{C}$ for a very general C , the Ceresa cycle is rationally non-trivial and has non-zero AJ image (Ceresa [49], Griffiths [54]). This is a sharp illustration of the characteristic- p trivialization.

13.2. The case $k = \overline{\mathbb{F}_p}$

Theorem 13.7 (Beauville collapse over $\overline{\mathbb{F}_p}$). Let p be a prime and $k = \overline{\mathbb{F}_p}$. Let A/k be an abelian variety of dimension g , $1 \leq n \leq g$.

- (a) Full degeneration. $E^{\infty} = E^1$ for the orthogonal spectral sequence over k , unconditionally.
- (b) Trivialization. $F_{\mathrm{orth}}^1 \mathrm{CH}^n(X_k)_{\mathbb{Q}} = 0$ for any smooth projective X/k .
- (c) Beauville collapse. $V_{n,s}(A_k) = 0$ for $s \geq 1$ and $\mathrm{CH}^n(A_k)_{\mathbb{Q}} = V_{n,0}$.
- (d) Griffiths group vanishes. $\mathrm{Griff}^n(A_k)_{\mathbb{Q}} = 0$.

Proof. (a): $G_k = 1$ is the trivial group (since k is algebraically closed), so $H^r(G_k, M) = 0$ for all $r \geq 1$ and all G_k -modules M . By Lemma 4.1, the attaching map for d^r lives in $H^r(G_k, M_{p,r}) = 0$ for $r \geq 1$. Hence $d^r = 0$ for all $r \geq 1$ and $E^{\infty} = E^1$.

(b): $\mathcal{O}(X_k)^{\times} = k^{\times}$ for X proper. Now $k^{\times} = \overline{\mathbb{F}_p}^{\times} = \bigcup_n \mathbb{F}_{p^n}^{\times}$ is a direct limit of finite groups, hence itself a torsion group. Thus $k^{\times} \otimes_{\mathbb{Z}} \mathbb{Q} = 0$. Since $bc_{-1/-2} \mathbf{1}_{\mathbb{Q}} \simeq \mathcal{O}^{\times} \otimes \mathbb{Q}$, the stalk of $bc_{-1/-2} \mathbf{1}_{\mathbb{Q}}$ at $\mathrm{Spec} k$ is zero. For $|p| \geq 2$: $K_r^M(k) \otimes \mathbb{Q} = 0$ for $r \geq 2$ (algebraically closed fields have trivial Milnor K -theory above degree 1). Hence all $E_{p,q}^1 = 0$ for $p \leq -1$, giving $F_{\mathrm{orth}}^1 \mathrm{CH}^n = 0$.

(c) and (d): identical to the proof of Theorem 13.1(b)–(c), using (b) in place of Theorem 12.4(c). $\square$

Remark 13.8 (Comparison: $\mathbb{F}_q$ vs. $\overline{\mathbb{F}}_p$ vs. $\mathbb{Q}$). The contrast between the three settings is sharp:

- Over $\mathbb{F}_q$ or $\overline{\mathbb{F}}_p$: $V_{n,s} = 0$ for $s \geq 1$, $\text{Griff}^n(A)_{\mathbb{Q}} = 0$, filtration is one-step.
- Over $\mathbb{Q}$ (number field): $E^\infty = E^1$ (Theorem 12.1 for $r \geq 2$, plus Hyp. 11.1 for $r = 1$), but $V_{n,s} \neq 0$ in general (Theorem 7.15) and the filtration has up to $n + 1$ steps.
- Over $\mathbb{C}$: richest structure — full Beauville decomposition, non-trivial Griffiths group (Ceresa, Griffiths), $n + 1$ -step filtration.

The characteristic- p trivialization reflects two arithmetic facts: (i) $A(\mathbb{F}_q)$ is finite, forcing $V_{1,1} = \text{Pic}^0(A)_{\mathbb{Q}} = 0$; (ii) $K_r^M(\mathbb{F}_q) = 0$ for $r \geq 2$, killing all deeper Bousfield layers.

13.3. Towards a crystalline orthogonal tower

The results of §§13.1–13.2 show that the rational Bousfield filtration is blind to all cycle-theoretic information in characteristic p : it trivializes completely. Any finer invariant must use cohomology with coefficients in a p -adic or crystalline cohomology theory. In this subsection we formulate a precise conjecture for a crystalline analogue of $F_{\text{orth}}^\bullet$, identify several cases where it is provable from known results, and explain the obstruction in general.

Setup. Let $k = \mathbb{F}_q$, $W = W(k)$ the Witt vectors, $K = W[\frac{1}{p}]$. For A/k an abelian variety of dimension g , write

$$D := D_{\text{crys}}^1(A) := H_{\text{crys}}^1(A/W) \otimes_W K$$

for the rational Dieudonné module; this is a K -vector space of dimension $2g$ equipped with a σ -linear Frobenius $\varphi : D \rightarrow D$ (σ = arithmetic Frobenius on K). The eigenvalues of φ on D are Weil numbers of weight 1.

The *slope decomposition* (Dieudonné–Manin theorem) writes $D = \bigoplus_{\lambda \in [0,1] \cap \mathbb{Q}} D_\lambda$, where D_λ is the φ -isotypic piece of slope λ : φ acts on D_λ with eigenvalue q^λ up to a unit.

Definition 13.9 (Crystalline Beauville piece). For $0 \leq s \leq n$, define the *crystalline Beauville piece*

$$V_{n,s}^{\text{crys}}(A) := \left\{ z \in \text{CH}^n(A)_K : [N]^* z = N^{2n-s} \cdot z \right\}$$

as the N^{2n-s} -eigenspace of $[N]^*$ in $\text{CH}^n(A)_K := \text{CH}^n(A) \otimes_{\mathbb{Z}} K$. This is the K -linear extension of Beauville’s eigenspace: $V_{n,s}^{\text{crys}}(A) = V_{n,s}(A) \otimes_{\mathbb{Q}} K$.

Proposition 13.10 (Crystalline Beauville collapse over $\mathbb{F}_q$). For $A/\mathbb{F}_q$ abelian of dimension g , $1 \leq n \leq g$:

- $V_{1,1}^{\text{crys}}(A) = 0$. Equivalently, $D_{\text{crys}}^1(A)$ has no φ -eigenvector with eigenvalue q in $\text{CH}^1(A)_K$.
- $V_{n,s}^{\text{crys}}(A) = 0$ for all $s \geq 1$, and $\text{CH}^n(A)_K = V_{n,0}^{\text{crys}}(A)$.

- (c) The crystalline cycle class map $\mathrm{cl}_{\mathrm{crys}} : \mathrm{CH}^n(A)_K \rightarrow H_{\mathrm{crys}}^{2n}(A/W)_K$ is injective (assuming the Tate conjecture in crystalline cohomology, which is a theorem for abelian varieties over $\mathbb{F}_q$ [65]).

Proof. Part (a): The crystalline Abel–Jacobi map identifies $V_{1,1}^{\mathrm{crys}}$ with $\mathrm{Pic}^0(A)_K$. Over $k = \overline{\mathbb{F}}_q$, $\mathrm{Pic}^0(A)(k) = \hat{A}(\overline{\mathbb{F}}_q)$, the $\overline{\mathbb{F}}_q$ -points of the dual abelian variety. Since $\hat{A}$ is an abelian variety over $\overline{\mathbb{F}}_q$, $\hat{A}(\overline{\mathbb{F}}_q)$ is a torsion group (every abelian variety over a finite field has finitely many rational points; the algebraic closure is a union of finite fields, each contributing a finite group). Tensoring with K (a field of characteristic zero) kills all torsion: $\hat{A}(\overline{\mathbb{F}}_q) \otimes_{\mathbb{Z}} K = 0$. Hence $V_{1,1}^{\mathrm{crys}} = \mathrm{Pic}^0(A)_K = 0$.

Parts (b)–(c): The argument is parallel to Theorem 13.1(b)–(c), with $\mathbb{Q}$ replaced by K throughout. We make the key steps explicit:

Degeneration: By Lemma 4.1, each differential d^r of the orthogonal spectral sequence lies in $H^r(G_k, M_{p,r} \otimes K)$. For $k = \overline{\mathbb{F}}_q$: $G_k = \mathrm{Gal}(\overline{\mathbb{F}}_q/\overline{\mathbb{F}}_q) = 1$ is trivial, so $H^r(G_k, -) = 0$ for $r \geq 1$. Hence $d^r = 0$ for all $r \geq 1$.

Trivialization: Since $k = \overline{\mathbb{F}}_q$ is algebraically closed, $K_r^M(k) = 0$ for $r \geq 1$ ($k^\times$ is divisible and torsion; rationally zero), so all higher Beauville components $V_{n,s}$ vanish for $s \geq 1$. The crystalline realization maps $V_{n,s}^{\mathrm{crys}} \hookrightarrow H_{\mathrm{crys}}^{2n-s,s}(A/W)_K$ by the crystalline analogue of the Hodge decomposition ([76], Thm. 2.5), and the source is zero for $s \geq 1$ by the trivialization above.

Cycle class: $V_{n,0}^{\mathrm{crys}} = \mathrm{CH}^n(A)_K / \mathrm{CH}_{\mathrm{hom}}^n(A)_K \hookrightarrow H_{\mathrm{crys}}^n(A/W)_K$ by the crystalline cycle-class map (Berthelot–Ogus [77], Thm. 7.7), giving (c). $\square$

The conclusion is the same as over $\mathbb{Q}$: K -coefficient Chow groups are purely algebraic over finite fields. The crystalline tower is non-trivial only when the *base field contains non-trivial p -adic information*, e.g. for abelian varieties over p -adic fields or over function fields of curves over $\mathbb{F}_q$.

The Nygaard filtration as a crystalline proxy. For A an ordinary abelian variety over $\mathbb{F}_q$, the *Nygaard filtration* [66] provides a filtration on $H_{\mathrm{crys}}^n(A/W)_K$ that is analogous to the Hodge filtration over $\mathbb{C}$. Specifically, $N^\bullet H_{\mathrm{crys}}^1(A/W)_K$ splits as $N^0 \supset N^1 \supset 0$, where $N^1 = \ker(\varphi - 1 : H_{\mathrm{crys}}^1 \rightarrow H_{\mathrm{crys}}^1)$ is the “unit root” subspace.

Proposition 13.11 (Nygaard proxy for the AJ step). *Let $A/\mathbb{F}_q$ be an ordinary abelian variety, $n = 1$. The Nygaard filtration on $H_{\mathrm{crys}}^1(A/W)_K$ satisfies:*

$$N^1 H_{\mathrm{crys}}^1 = \varphi\text{-unit-root subspace} \simeq \mathrm{Hom}(\hat{A}, \mathbb{G}_m)_K \simeq V_{1,1}^{\mathrm{crys}}(A)_{\mathrm{formal}},$$

where $V_{1,1}^{\mathrm{crys}}(A)_{\mathrm{formal}}$ refers to the formal Lie group contribution to $\mathrm{Pic}^0(A)$. The analogy with the filtration of type $(0, 1)$ in Hodge theory over $\mathbb{C}$ suggests that a crystalline Bousfield tower (if it existed) would place this piece at column $p = -1$, exactly as $V_{n,1}$ sits in column -1 over $\mathbb{C}$.

Proof. This is a reformulation of the Serre–Tate theorem for ordinary abelian varieties: the unit-root subspace of $H_{\mathrm{crys}}^1(A/W)$ is canonically identified with the

formal completion $\hat{A}^*$ of the dual abelian variety [57, Thm. 3.3], i.e., the formal part of $\text{Pic}^0(A)$. This subspace consists of Frobenius-eigenvectors with unit-root eigenvalues (p -adic valuation 0), which in the Beauville calibration corresponds to slope 0 — the crystalline analogue of $V_{1,0}$ (eigenvalue N^2 , column $p = 0$). However, the formal Lie group $\hat{A}^*$ contributes the non-unit-root part (slope > 0), which is the crystalline analogue of $V_{1,1}$ (column $p = -1$). The claim about the column is therefore: $\hat{A}^* \leftrightarrow \text{column } p = -1$, consistent with Proposition 6.7(a). $\square$

Conjecture 13.12 (Crystalline orthogonal tower). *There exists a triangulated category $\text{DM}_{\text{crys}}(k, K)$ with a Bousfield tower on a crystalline unit $\mathbf{1}_K^{\text{crys}}$, inducing a filtration $F_{\text{crys}}^\bullet$ on $\text{CH}^n(A, m)_K$ for smooth projective A/k , such that:*

- (a) *The filtration is compatible with the crystalline cycle class map and the crystalline Chern character.*
- (b) *The pure layer $bc_{p/p-1}^{\text{crys}} \mathbf{1}_K$ has Frobenius-eigenvalue $q^{-p/2}$ on its stalk at $\text{Spec}(k)$, consistent with the Weil-number calibration $V_{n,s} \leftrightarrow p = -s$ and Frobenius weight $q^{n-s/2}$ on $H^{2n-s}(A)$.*
- (c) *Over $\mathbb{F}_q$, the filtration trivializes at the same level as $F_{\text{orth}}^\bullet$ (Theorem 13.1).*
- (d) *Over a p -adic field or the function field of a curve over $\mathbb{F}_q$, the filtration is non-trivial and recovers the Hodge–Tate decomposition (resp. the geometric analogue).*

Evidence. Parts (a) and (c) are consistent with Theorem 13.1 and Proposition 13.10. For part (b): the Weil conjectures (Deligne [50]) place $H^{2n-s}(A_{\bar{k}}, \mathbb{Q}_\ell)$ at Frobenius weight $q^{n-s/2}$ (eigenvalues of absolute value $q^{n-s/2}$), and the calibration $V_{n,s} \leftrightarrow p = -s$ gives Bousfield column $p = -s$, so the expected Frobenius eigenvalue on the $p = -s$ column is $q^{-(-s)/2} = q^{s/2}$ (relative to the normalization for degree- $2n$ cohomology at weight q^n , the shift is $q^{n-s/2}$). For part (d): the Fontaine–Mazur program [52] predicts that geometric p -adic Galois representations arising from geometry carry a Hodge–Tate structure; a crystalline Bousfield tower encodes exactly this.

Obstruction. No category $\text{DM}_{\text{crys}}(k, K)$ with the required properties exists in the current literature. The category of F -isocrystals [38] provides the target for crystalline cohomology but lacks the motivic structure needed for Bousfield localization. Bhatt–Scholze prismatic cohomology [40] provides a new integral p -adic cohomology compatible with Frobenius and with a Hodge–Tate filtration; the prismatic site is the natural arena for a crystalline motivic category.

Remark 13.13 (The slope filtration). The Dieudonné–Manin slope decomposition $D = \bigoplus_\lambda D_\lambda$ gives a non-trivial filtration on $H_{\text{crys}}^n(A/W)_K$ even over $\mathbb{F}_q$: $F_{\text{slope}}^s := \bigoplus_{\lambda \geq s/n} D_\lambda^{\otimes n}$, with graded pieces $D_{s/n}^{\otimes n}$ — the crystalline analogue of the Hodge filtration. The slope filtration is defined geometrically. A motivic Bousfield tower recovering it as associated graded would be precisely a DM_{crys} -analogue of $F_{\text{orth}}^\bullet$, and is the concrete form of Conjecture 13.12.

14. Filtration with modulus

Voevodsky's motivic cohomology requires homotopy invariance: $DM(k, R)$ is built from $\mathbb{A}^1$ -local sheaves, which forces $CH^n(X, m)_R$ to kill all information that varies along $\mathbb{A}^1$. The *motivic cohomology with modulus* of Kahn–Miyazaki–Saito–Yamazaki [59] and Binda–Saito [39] lifts this restriction: for a pair (X, D) with X smooth and $D \subset X$ an effective Cartier divisor, one defines cycle groups $CH^n(X|D, m)$ satisfying a modulus condition along D , recovering $CH^n(X, m)$ when $D = 0$. We do not define a modulus Bousfield tower (the category DM^{mod} lacks Bousfield localization theory as of this writing), but for $n = 1$ the modulus Chow groups are explicitly computable via Picard theory, and the ordinary $[N]^*$ -eigenspace decomposition extends naturally. We carry this out for an elliptic curve and compute the eigenvalue decomposition of $\text{Pic}(E, D)_{\mathbb{Q}}$, yielding a modulus-enriched Beauville splitting.

The $n = 1$ case: Picard group with modulus

For X smooth projective and $D \subset X$ an effective Cartier divisor,

$$CH^1(X|D, 0) = \text{Pic}(X, D) := \left\{ (L, \phi) : L \in \text{Pic}(X), \phi : L|_D \xrightarrow{\sim} \mathcal{O}_D \right\} / \sim,$$

the Picard group with trivialization along D [39]. The exact sequence of the pair (X, D) gives [39, Prop. 1.3]:

$$\mathcal{O}(X)^{\times} \longrightarrow \mathcal{O}_D^{\times} \longrightarrow \text{Pic}(X, D) \longrightarrow \text{Pic}(X) \longrightarrow 0.$$

For $X = E$ an elliptic curve with origin O and $D = r[O]$ ($r \geq 1$): $\mathcal{O}_D = \mathcal{O}_E/\mathfrak{m}_O^r$, and $\mathcal{O}_D^{\times} \cong k^{\times} \times (1 + \mathfrak{m}_O/\mathfrak{m}_O^r)$ where $k^{\times}$ is the unit residue and $1 + \mathfrak{m}_O/\mathfrak{m}_O^r$ is the pro-unipotent kernel (sometimes written $(1 + \hat{\mathfrak{m}})/\mathfrak{m}^r$). The map $\mathcal{O}(E)^{\times} = k^{\times} \rightarrow \mathcal{O}_D^{\times}$ sends $c \mapsto (c, 0)$, so its cokernel is exactly $1 + \mathfrak{m}_O/\mathfrak{m}_O^r$.

Proposition 14.1 (Eigenspace decomposition of $\text{Pic}(E, D)_{\mathbb{Q}}$ for $r = 2$). *Let E/k be an elliptic curve with origin O , $D = 2[O]$, and $\text{char}(k) = 0$. The $[N]^*$ -action on $\text{Pic}(E, 2[O])_{\mathbb{Q}}$ has exactly two eigenvalues, N^2 (on $V_{1,0}^{\text{mod}}$) and N^1 (on $V_{1,1}^{\text{mod}}$), with*

$$V_{1,0}^{\text{mod}} := \text{NS}(E)_{\mathbb{Q}} = \mathbb{Q} \cdot [\mathcal{O}(O)], \quad V_{1,1}^{\text{mod}} := E(k)_{\mathbb{Q}} \oplus (T_O E) \otimes_{\mathbb{Z}} \mathbb{Q},$$

giving $\text{Pic}(E, 2[O])_{\mathbb{Q}} = V_{1,0}^{\text{mod}} \oplus V_{1,1}^{\text{mod}}$. Here $T_O E = \mathfrak{m}_O/\mathfrak{m}_O^2$ is the tangent line at O . For $r \geq 3$: additional terms $k \cdot t^j \otimes \mathbb{Q}$ ($2 \leq j \leq r - 1$) with eigenvalues $N^j \notin \{N^1, N^2\}$ fall outside the Beauville support for $n = 1$, so $\text{Pic}(E, r[O])_{\mathbb{Q}}$ has a richer eigenvalue structure (see Remark below).

If a modulus Bousfield tower existed on $DM^{\text{mod}}(k, \mathbb{Q})$ inducing a filtration $F^{\bullet}_{\text{orth}}$ on $\text{Pic}(E, D)_{\mathbb{Q}}$ compatible with the $[N]^*$ -action, then by the eigenvalue argument of Theorem 6.1, it would satisfy $F^0 = \text{Pic}(E, D)_{\mathbb{Q}}$ and $F^1 = V_{1,1}^{\text{mod}}$.

Proof. From and the computation above:

$$0 \longrightarrow (1 + \mathfrak{m}_O/\mathfrak{m}_O^r) \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \mathrm{Pic}(E, D)_{\mathbb{Q}} \longrightarrow \mathrm{Pic}(E)_{\mathbb{Q}} \longrightarrow 0.$$

Since we work over $\mathbb{Q}$, this extension of $\mathbb{Q}$ -vector spaces splits: $\mathrm{Pic}(E, D)_{\mathbb{Q}} \cong \mathrm{Pic}(E)_{\mathbb{Q}} \oplus (1 + \mathfrak{m}_O/\mathfrak{m}_O^r) \otimes \mathbb{Q}$. The Beauville decomposition of $\mathrm{Pic}(E)_{\mathbb{Q}}$ is $\mathrm{NS}(E)_{\mathbb{Q}} \oplus E(k)_{\mathbb{Q}}$.

Eigenvalue on the modulus term. Over k of characteristic zero, the formal logarithm gives an isomorphism of abelian groups $(1 + \mathfrak{m}_O/\mathfrak{m}_O^r) \otimes \mathbb{Q} \cong \mathfrak{m}_O/\mathfrak{m}_O^r \otimes \mathbb{Q}$ (the unipotent group is uniquely divisible, hence isomorphic to its Lie algebra over $\mathbb{Q}$). Fix a local parameter t at O . Then $\mathfrak{m}_O/\mathfrak{m}_O^r \cong k^{r-1}$ with basis $t, t^2, \dots, t^{r-1}$. The map $[N] : E \rightarrow E$ satisfies $d[N]|_O = N \cdot \mathrm{id}$ (the derivative of $[N]$ at the origin of an abelian variety is N on the tangent space), so $[N]^*t \equiv Nt \pmod{\mathfrak{m}_O^2}$ and more generally $[N]^*t^j \equiv N^j t^j \pmod{\mathfrak{m}_O^{j+1}}$ for $1 \leq j \leq r-1$. Hence $[N]^*$ acts on $k \cdot t^j$ by N^j .

For $j = 1$: eigenvalue $N^1 = N^{2 \cdot 1 - 1}$, the Beauville eigenvalue for $s = 1$. Hence $k \cdot t \otimes \mathbb{Q} = T_O E \otimes \mathbb{Q} \subseteq V_{1,1}^{\mathrm{mod}}$.

For $r = 2$: $\mathfrak{m}_O/\mathfrak{m}_O^2$ is one-dimensional, spanned by t , with eigenvalue $N^1 = N^{2 \cdot 1 - 1}$. Hence all eigenvalues in $\mathrm{Pic}(E, 2[O])_{\mathbb{Q}}$ are $\{N^2, N^1\}$, matching the Beauville eigenvalues for $n = 1$. Therefore: $V_{1,0}^{\mathrm{mod}} = \mathrm{NS}(E)_{\mathbb{Q}}$ (eigenvalue N^2) and $V_{1,1}^{\mathrm{mod}} = E(k)_{\mathbb{Q}} \oplus T_O E \otimes \mathbb{Q}$ (eigenvalue N^1). For $r \geq 3$: $k \cdot t^j$ has eigenvalue N^j for $j \geq 2$; those with $j \geq 2$ lie outside the Beauville support $\{N^2, N^1\}$ and require an extended eigenspace decomposition beyond Beauville's framework. $\square$

Remark 14.2 (Modulus for $r \geq 3$). For $r \geq 3$, the quotient $\mathfrak{m}_O/\mathfrak{m}_O^r$ has dimension $r-1$ with basis $t, t^2, \dots, t^{r-1}$ and eigenvalues $N, N^2, \dots, N^{r-1}$. The Beauville decomposition for $n = 1$ has only eigenvalues N^1 (for $s = 1$) and N^2 (for $s = 0$); additional terms at N^j ($j \geq 3$) fall outside this range. Thus $\mathrm{Pic}(E, r[O])_{\mathbb{Q}}$ does not decompose as $V_{1,0} \oplus V_{1,1}$ for $r \geq 3$; it requires an extended filtration beyond Beauville's eigenvalue framework.

Example 14.3 (Modulus tangent term: explicit computation). Let $k = \mathbb{Q}$, $E : y^2 = x^3 - x$ (conductor 32; $E(\mathbb{Q}) \cong (\mathbb{Z}/2)^2$, rank 0), and $D = 2[O]$.

Without modulus. $\mathrm{Pic}(E)_{\mathbb{Q}} = \mathrm{NS}(E)_{\mathbb{Q}} \oplus E(\mathbb{Q})_{\mathbb{Q}} = \mathbb{Q} \oplus 0 = \mathbb{Q}$. The $[N]^*$ -eigenspace decomposition: $V_{1,0} = \mathbb{Q}$, $V_{1,1} = 0$.

With modulus $D = 2[O]$. By Proposition 14.1 (which applies since $r = 2$): $T_O E \otimes \mathbb{Q} = \mathbb{Q}$ (spanned by the formal parameter t).

$$\begin{aligned} \mathrm{Pic}(E, 2[O])_{\mathbb{Q}} &= \mathbb{Q}^2, \\ V_{1,0}^{\mathrm{mod}} &= \mathbb{Q} \cdot [\mathcal{O}(O)] \quad (\text{eigenvalue } N^2), \\ V_{1,1}^{\mathrm{mod}} &= \mathbb{Q} \cdot t \quad (\text{eigenvalue } N^1). \end{aligned}$$

The term $\mathbb{Q} \cdot t$ lies in $\ker(\mathrm{cl})$: the tangent element t has trivial cycle class in $H^{1,1}(E_{\mathbb{C}})$ since it encodes first-order deformation data, not an algebraic cycle.

The modulus-enriched AJ map $\mathrm{AJ}^{\mathrm{mod}} : \mathrm{Pic}^0(E, D)_{\mathbb{Q}} \rightarrow J^1(E, D)_{\mathbb{Q}}$ (the modulus intermediate Jacobian, whose definition requires further structure) would detect $\mathbb{Q} \cdot t$ as a non-trivial contribution. This is consistent with the fact that $\mathbb{Q} \cdot t$ corresponds to the *formal logarithm* of the formal group $\hat{E}$: by the Serre–Tate theorem, the deformation space of E near O is parametrized by $\hat{E}$, and t is the formal coordinate on $\hat{E}$.

The modulus condition at $D = 2[O]$ captures the first-order datum of E at O (the tangent direction), which is killed by the $\mathbb{A}^1$ -localization in $\mathrm{CH}^1(E)_{\mathbb{Q}}$ but survives in $\mathrm{Pic}(E, 2[O])_{\mathbb{Q}}$.

Remark 14.4 (Modulus filtration: what is known and what is open). Proposition 14.1 computes the $[N]^*$ -eigenspace decomposition of $\mathrm{Pic}(E, D)_{\mathbb{Q}}$ directly from Picard theory, without any Bousfield tower. The following are *speculative* (pending the construction of a modulus Bousfield tower on $DM^{\mathrm{mod}}(k, \mathbb{Q})$):

- the identification of $V_{1,0}^{\mathrm{mod}}$ and $V_{1,1}^{\mathrm{mod}}$ with Bousfield columns $p = 0$ and $p = -1$ respectively;
- the degeneration $E^{\infty} = E^1$ for the modulus spectral sequence;
- the column support bound for $n \geq 2$ with modulus.

What *is* clear from Proposition 14.1 is that the ordinary eigenspace structure extends to the modulus setting, with $T_O E \otimes \mathbb{Q}$ joining $V_{1,1}^{\mathrm{mod}}$ at eigenvalue N^1 . The category $DM^{\mathrm{mod}}(k, \mathbb{Q})$ has been constructed by [59]; developing its Bousfield localization is a natural sequel to the present work.

15. The orthogonal tower in $\mathcal{SH}$: convergence, effective completion, and applications

The orthogonal Bousfield tower lives naturally in the motivic stable homotopy category $\mathcal{SH}(k)$ of Morel–Voevodsky, of which $DM(k, \mathbb{Q})$ is the $H\mathbb{Z}$ -local subcategory. In this section we formulate the general convergence theory for arbitrary objects of $\mathcal{SH}$, identify the obstruction to strong convergence, and apply the framework to KGL , MGL , and their slices, recovering classical filtrations as special cases.

15.1. Effective completion and convergence obstruction

The failure of strong convergence in $\mathcal{SH}$ is not merely a technical nuisance: it reflects a genuine geometric phenomenon. The obstruction class measures the extent to which the stable homotopy category contains information that cannot be captured by the effective orthogonal tower. Understanding this obstruction is essential for applications to K -theory, algebraic cobordism, and motivic cohomology with integral coefficients.

Definition 15.1 (Effective completion and obstruction class). For B compact and $A \in DM(k, \mathbb{Q})$ (or $\mathcal{SH}(k)$), the *convergence morphism* is

$$c^A : \mathrm{hocolim}_{p \rightarrow \infty} bc_{\leq p} A \longrightarrow A,$$

assembled from the counit maps $\theta_p^A : bc_{\leq p}A \rightarrow A$. Define:

$$\begin{aligned}\widehat{\mathrm{Hom}}(B, A) &:= \mathrm{Hom}(B, \mathrm{hocolim}_p bc_{\leq p}A), \\ &= \mathrm{colim}_p \mathrm{Hom}(B, bc_{\leq p}A) \quad (\text{effective completion}), \\ \mathcal{O}(A, B)_j &:= \mathrm{Hom}(B, \mathrm{Cone}(c^A)[j]) \quad (\text{convergence obstruction}).\end{aligned}$$

Theorem 15.2 (Convergence characterization and four-term sequence). *Let B be compact and $A \in DM(k, \mathbb{Q})$.*

- (a) *The spectral sequence always converges strongly to $\widehat{\mathrm{Hom}}(B, A)$: the filtration is exhaustive, separated, and complete.*
- (b) *There is a canonical four-term exact sequence*

$$0 \rightarrow \mathcal{O}(A, B)_{-1} \rightarrow \widehat{\mathrm{Hom}}(B, A) \xrightarrow{c_*^A} \mathrm{Hom}(B, A) \rightarrow \mathcal{O}(A, B)_0 \rightarrow 0.$$

Strong convergence to $\mathrm{Hom}(B, A)$ holds if and only if $\mathcal{O}(A, B)_ = 0$.*

- (c) *$\mathcal{O}(A, B)_* = 0$ iff A is eventually orthogonal to B : $\mathrm{Hom}(B, s_n A[j]) = 0$ for all $n \gg 0$ and all $j \in \mathbb{Z}$.*
- (d) *For $A = M(Y)(q)$ with Y smooth projective of dimension d : $\mathcal{O}(A, B)_* = 0$ and the EO-threshold $r_0 \leq d + q + 1$.*

Proof. (a): *Exhaustivity*: every element of $\widehat{\mathrm{Hom}}(B, A) = \mathrm{colim}_p \mathrm{Hom}(B, bc_{\leq p}A)$ comes from some $\mathrm{Hom}(B, bc_{\leq p_0}A)$. *Separatedness*: if $x \in \bigcap_p F_p$, then for every p , x factors through $bc_{\leq p}A \in DM^{\mathrm{eff}}(p)$; since $\bigcap_p DM^{\mathrm{eff}}(p) = \{0\}$ by the effective Postnikov tower [25, 3.2.3], $x = 0$. *Completeness*: $\varprojlim^1$ of a direct system vanishes [10].

(b): Apply $\mathrm{Hom}(B, -)$ to the triangle $\mathrm{hocolim}_p bc_{\leq p}A \xrightarrow{c^A} A \rightarrow \mathrm{Cone}(c^A)$.

(c): Via the Milnor exact sequence for $\tau_{>p}A := \mathrm{Cone}(bc_{\leq p}A \rightarrow A)$, $\mathcal{O}(A, B)_j = 0$ for all j iff $\{\mathrm{Hom}(B, \tau_{>p}A[j])\}_p$ is eventually zero for every j , iff $\{\mathrm{Hom}(B, s_n A[j])\}_n$ is eventually zero for every j .

(d): By Theorem 3.6, $s_n(M(Y)(q)) = 0$ for $n > d + q$ (column support in DM), giving $r_0 \leq d + q + 1$. $\square$

Remark 15.3 (Connection to Corollary 4.4). For $R = \mathbb{Q}$ and $A = \mathbf{1}_{\mathbb{Q}}$, the four-term sequence of Theorem 15.2 gives $\mathcal{O}(\mathbf{1}_{\mathbb{Q}}, B)_* = 0$ for all compact B . This is precisely Corollary 4.4: the Maschke vanishing $H^r(G_k, M_{p,r}) = 0$ implies the convergence morphism $c^{\mathbf{1}_{\mathbb{Q}}}$ is an isomorphism, so $\widehat{\mathrm{Hom}}(B, \mathbf{1}_{\mathbb{Q}}) = \mathrm{Hom}(B, \mathbf{1}_{\mathbb{Q}}) = \mathrm{CH}^n(X, m)_{\mathbb{Q}}$. The orthogonal spectral sequence is therefore already its own complete computation: $E^{\infty} = E^1$ with no further obstruction.

15.2. The E_2 -page via Yoneda Ext

When $DM(k, \mathbb{Q})$ carries a bounded t -structure with heart $\mathbf{HI}(k, \mathbb{Q})$ (the homotopy t -structure), the E^1 -page of $\mathcal{O}$ admits a Yoneda interpretation.

Theorem 15.4 (E^2 -page via Yoneda Ext). *Suppose $s_p A \in \mathbf{HI}(k, \mathbb{Q})[-p]$ for all p (concentrated in cohomological degree p) and $B \in \mathbf{HI}(k, \mathbb{Q})$. Then:*

- (i) $E_{p,q}^1 \cong \mathrm{Ext}_{\mathbf{HI}(k, \mathbb{Q})}^{q-p}(H^0 B, H^0(s_p A))$.
- (ii) The differential $d^1 : E_{p,q}^1 \rightarrow E_{p-1,q-1}^1$ is Yoneda product with the extension class $\partial_p \in \mathrm{Ext}_{\mathbf{HI}}^1(H^0(s_p A), H^0(s_{p-1} A))$ from the triangle $bc_{\leq p-1} \mathbf{1}_{\mathbb{Q}} \rightarrow bc_{\leq p} \mathbf{1}_{\mathbb{Q}} \rightarrow s_p \mathbf{1}_{\mathbb{Q}}$.
- (iii) If $\mathbf{HI}(k, \mathbb{Q})$ has global dimension ≤ 1 , then $d^r = 0$ for $r \geq 2$ and $E^2 = E^\infty$. This holds in particular when $k = \mathbb{F}_q$ ($\mathrm{cd}(G_k) = 1$) or under the Maschke vanishing (Lemma 4.3).

Proof. (i): $E_{p,q}^1 = \mathrm{Hom}_{DM}(B, s_p A[q-p]) = \mathrm{Hom}_{D(\mathbf{HI})}(H^0 B, H^0(s_p A)[q-p]) = \mathrm{Ext}_{\mathbf{HI}}^{q-p}(H^0 B, H^0(s_p A))$ by the t -structure and [17, Thm. 3.1.12]. (ii): The differential d^1 is induced by the connecting map $\delta_p : s_p A \rightarrow s_{p-1} A[1]$ (Definition 3.1); under (i) this is Yoneda product with ∂_p . (iii): $\mathrm{Ext}_{\mathbf{HI}}^r = 0$ for $r \geq 2$ (global dimension ≤ 1) forces all Massey products — and hence all d^r for $r \geq 2$ — to vanish. $\square$

15.3. Applications: KGL, MGL, and the Beilinson–Adams filtration

Setup. The motivic spectra KGL (algebraic K -theory) and MGL (algebraic cobordism) lie in $\mathcal{SH}(k)$ but not in $DM(k, \mathbb{Q})$. Their individual slices $s_n KGL$ and $s_n MGL$ lie in $DM(k, \mathbb{Q})$ (they are $H\mathbb{Z}$ -modules), so the machinery of this paper applies to each slice separately.

Proposition 15.5 (Slices of KGL). *Voevodsky [73] and Levine [61]: $s_n KGL \cong \Sigma^{2n,n} H\mathbb{Z}$ for $n \geq 0$ and $s_n KGL = 0$ for $n < 0$. Each $s_n KGL \in DM^\perp(n+1)$, so the orthogonal spectral sequence for $(s_n KGL, B)$ converges strongly with $E^1 = E^\infty$ (single column at $p = n$). For $B = \Sigma_T^\infty X_+(r)$:*

$$E_{n,n+j}^1(s_n KGL, B) \cong H_{\mathrm{mot}}^j(X, \mathbb{Z}(n-r)), \quad j \geq 0.$$

Proposition 15.6 (Slices of MGL). *Spitzweck–Østvær [69]: $s_n MGL \cong \Sigma^{2n,n} H\mathbb{Z} \otimes_1 L$ where $L = \mathbb{Z}[\beta_1, \beta_2, \dots]$ is the Lazard ring. Each $s_n MGL \in DM^\perp(n+1)$; the orthogonal spectral sequence gives*

$$E_{n,n+j}^1(s_n MGL, B) \cong \bigoplus_{\mathbf{m}} H_{\mathrm{mot}}^{j+2n}(X, \mathbb{Z}(n-r)),$$

summed over monomials $\beta^{\mathbf{m}}$ in L .

Theorem 15.7 (Recovery of the Beilinson–Adams filtration). *Let $X \in \mathrm{Sm}_k$ and $B = \Sigma_T^\infty X_+(r)$. The compatible family of strongly convergent spectral sequences $\{E^\bullet(s_n KGL, B)\}_{n \geq 0}$ (with vanishing slice differentials $\partial_n : s_n KGL \rightarrow s_{n+1} KGL[1]$ by Bott periodicity) assembles into the Beilinson–Adams filtration on algebraic K -theory:*

$$F^n K_j(X) := \mathrm{Im}\left(\mathrm{Hom}_{\mathcal{SH}}(B, bc_{\leq n-1} KGL[j]) \rightarrow K_j(X)\right),$$

with graded pieces satisfying

$$\mathrm{Gr}_F^n K_j(X) \hookrightarrow H_{\mathrm{mot}}^j(X, \mathbb{Z}(n)).$$

The injection is an isomorphism after rationalization when $H_{\mathrm{mot}}^j(X, \mathbb{Z}(n))$ is torsion-free (e.g. $X = \mathrm{Spec}(k)$ with k a number field, $j \leq n$).

Proof. Since $s_n KGL \cong H\mathbb{Z} \in DM^\perp(n+1)$, the spectral sequence for $(s_n KGL, B)$ has a single column at $p = n$; $E^1 = E^\infty$ by single-column support. The filtration on $K_j(X) = \mathrm{Hom}_{\mathcal{SH}}(B, KGL[j])$ is defined via the tower $bc_{\leq n-1} KGL \rightarrow KGL$. The graded piece $\mathrm{Gr}_F^n K_j(X)$ injects into $E_{n,n+j}^\infty = E_{n,n+j}^1$ (the E^1 -term of the slice spectral sequence for $s_n KGL$), which is $H_{\mathrm{mot}}^j(X, \mathbb{Z}(n-r))$. Taking $r = 0$: $\mathrm{Gr}_F^n K_j(X) \hookrightarrow H_{\mathrm{mot}}^j(X, \mathbb{Z}(n))$. For rationalization: $K_j(X)_\mathbb{Q} \cong \bigoplus_n H_{\mathrm{mot}}^j(X, \mathbb{Q}(n))$ when the motivic Bloch–Kato conjecture holds (Voevodsky–Rost, for k a number field). $\square$

Corollary 15.8 (Motivic AHSS in DM). *For Y smooth projective of dimension d over k (char. zero), set $A = M(Y)$ and $B = \mathbf{1}$ in $DM(k, \mathbb{Z})$. The orthogonal spectral sequence is the motivic Atiyah–Hirzebruch spectral sequence:*

$$E_{n,n+j}^1 \cong H_{\mathrm{mot}}^{j+2n}(Y, \mathbb{Z}(n)), \quad 0 \leq n \leq d, \quad \implies \quad H_{\mathrm{mot}}^{j+2n}(Y, \mathbb{Z}),$$

converging strongly with EO-threshold $r_0(M(Y), \mathbf{1}) = d+1$ (sharp: $H_{\mathrm{mot}}^{2d}(Y, \mathbb{Z}(d)) \ni [Y] \neq 0$).

Example 15.9 (Brauer group via the tower). For $A = H\mathbb{Z}/n \in DM(k, \mathbb{Z})$ (n -torsion Eilenberg–MacLane motive) and $B = \Sigma_T^\infty \mathrm{Spec}(k)_+$: $s_0(H\mathbb{Z}/n) = H\mathbb{Z}/n$ and $s_p = 0$ for $p \neq 0$, so the spectral sequence degenerates:

$$E_{0,q}^1 = H_{\mathrm{mot}}^q(k, \mathbb{Z}/n) \xrightarrow{\sim} H_{\mathrm{et}}^q(k, \mu_n^{\otimes q}) \implies H_{\mathrm{mot}}^q(k, \mathbb{Z}/n).$$

The Bloch–Kato isomorphism identifies $E_{0,2}^1 \cong \mathrm{Br}(k)[n]$; the filtration places Brauer classes at weight 0, confirming they are “purely algebraic” in the Bousfield sense.

15.4. Failure for $\mathbf{1}$, KGL , MGL in $\mathcal{SH}$ and the effective completion

Theorem 15.10 (Failure of strong convergence in $\mathcal{SH}$). *In $\mathcal{SH}(k)$, k of characteristic zero, with $B = \mathbf{1}$:*

- (i) For $A = \mathbf{1}$: $r_0(\mathbf{1}, \mathbf{1}) = +\infty$ and $\mathcal{O}(\mathbf{1}, \mathbf{1})_0 \neq 0$.
- (ii) For $A = KGL$: same, since $\mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, s_n KGL) \cong H_{\mathrm{mot}}^0(k, \mathbb{Z}) \cong \mathbb{Z}$ for all $n \geq 0$.
- (iii) For $A = MGL$: same.

In each case the spectral sequence converges strongly to the effective completion $\widehat{\mathrm{Hom}}_{\mathcal{SH}}(\mathbf{1}, A) \subsetneq \mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, A)$, which captures only the finite-weight part of the mapping group.

Proof. (i): $s_n \mathbf{1} \neq 0$ for all $n \geq 0$ (Voevodsky [73], Levine [18]; the slices $s_n \mathbf{1}$ carry the motivic stable homotopy groups, non-trivial in every weight). If $\mathcal{O}_0 = 0$, then c_*^1 would be surjective, making $\mathbf{1}$ a direct summand of some $bc_{\leq n} \mathbf{1} \in \mathcal{SH}^\perp(n+1)$; since $\mathcal{SH}^\perp(n+1)$ is thick, $\mathbf{1} \in \mathcal{SH}^\perp(n+1)$, giving $s_m \mathbf{1} = 0$ for $m \geq n+1$: contradiction. (ii)-(iii): $\mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, s_n KGL) \cong \mathbb{Z} \neq 0$ for all $n \geq 0$ by Levine [18]. $\square$

Remark 15.11 ($\mathcal{SH}$ vs. DM : the sharp contrast). The obstruction $\mathcal{O}(\mathbf{1}_{\mathbb{Q}}, B)_* = 0$ in $DM(k, \mathbb{Q})$ (Corollary 4.4) is a consequence of working with $\mathbb{Q}$ -coefficients: Maschke forces all attaching maps to zero. In $\mathcal{SH}(k)$ with $\mathbb{Z}$ -coefficients the analogous obstruction is nonzero for $A = \mathbf{1}, KGL, MGL$ (Theorem 15.10). The passage $\mathcal{SH}(k) \rightarrow DM(k, \mathbb{Q})$ (via $H\mathbb{Z}$ -localization and $\mathbb{Q}$ -tensoring) kills precisely this obstruction.

More precisely:

- In $DM(k, \mathbb{Q})$: all smooth projective motives are eventually orthogonal to all compact objects; $\mathcal{O} = 0$ unconditionally; $E^\infty = E^1$.
- In $\mathcal{SH}(k)$: the sphere $\mathbf{1}$, KGL , MGL are NOT eventually orthogonal to $\mathbf{1}$; $\mathcal{O} \neq 0$; convergence fails.
- Individual slices $s_n KGL$, $s_n MGL$ (which are $H\mathbb{Z}$ -modules, hence in DM) DO satisfy eventual orthogonality; their spectral sequences converge strongly and recover the Beilinson–Adams graded pieces.

$$\begin{array}{ccc} \mathcal{SH}(k) & \xrightarrow{H\mathbb{Z}\text{-loc.}} & DM(k, \mathbb{Z}) \\ \otimes \mathbb{Q} \downarrow & & \downarrow \otimes \mathbb{Q} \\ \mathcal{SH}(k)_{\mathbb{Q}} & \longrightarrow & DM(k, \mathbb{Q}) \xrightarrow[\mathcal{O}=0]{\text{Cor. 4.4}} \end{array}$$

The bottom-right corner is where the Maschke theorem operates.

Corollary 15.12 (Effective completion of K -theory via truncations). *For $A = KGL$ and $B = \Sigma_T^\infty X_+$, even though the full spectral sequence for (KGL, B) does not converge to $K_*(X)$, the inverse system $\{E^\bullet(KGL/f_r KGL, B)\}_{r \geq 0}$ of strongly convergent spectral sequences (one for each effective truncation $KGL/f_r KGL \in \mathcal{SH}^\perp(r)$) has limit equal to the effective completion:*

$$\widehat{\mathrm{Hom}}_{\mathcal{SH}}(B, KGL) = \varprojlim_r \mathrm{Hom}_{\mathcal{SH}}(B, bc_{\leq r} KGL) = \varprojlim_r K_*(X)/F^r.$$

Each term $K_*(X)/F^r$ is computable from the r -th truncated spectral sequence.

15.5. Eventual orthogonality and effective completion

The point of the effective completion is not merely terminological. It is the object to which the tower always converges, whereas the passage from the completion to the genuine mapping group is controlled exactly by the two obstruction groups $\mathcal{O}(A, B)_{-1}$ and $\mathcal{O}(A, B)_0$. To keep the comparison legible, write

$$H_A(B) := \mathrm{Hom}_{DM}(B, A), \quad \widehat{H}_A(B) := \widehat{\mathrm{Hom}}_{DM}(B, A).$$

For the Chow probe $B_{n,m} = M(X)(-n)[m-2n]$ and $A = \mathbf{1}_{\mathbb{Q}}$, the general exact sequence of Part I specializes as follows:

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \mathcal{O}_{-1} & \longrightarrow & \widehat{H}_A(B) & \xrightarrow{c_*^A} & H_A(B) \longrightarrow \mathcal{O}_0 \longrightarrow 0 \\
 & & \uparrow & & \uparrow & & \uparrow \\
 0 & \longrightarrow & 0 & \longrightarrow & \widehat{H}_{\mathbf{1}_{\mathbb{Q}}}(B_{n,m}) & \xrightarrow{\sim} & H_{\mathbf{1}_{\mathbb{Q}}}(B_{n,m}) \longrightarrow 0 \longrightarrow 0 \\
 & & & & \downarrow \sim & & \downarrow \sim \\
 & & & & \widehat{CH}_{\text{orth}}^n(X, m)_{\mathbb{Q}} & \xrightarrow{\sim} & CH^n(X, m)_{\mathbb{Q}}
 \end{array}$$

The first row is the universal obstruction sequence. The second row is its rational motivic specialization, where $\mathcal{O}(\mathbf{1}_{\mathbb{Q}}, B_{n,m})_{-1} = \mathcal{O}(\mathbf{1}_{\mathbb{Q}}, B_{n,m})_0 = 0$ by Corollary 4.4. The third row translates the two mapping groups into the orthogonal completion of the higher Chow group and the usual higher Chow group. Thus, for rational coefficients, the tower computes $CH^n(X, m)_{\mathbb{Q}}$ itself rather than only a completion.

For objects in $DM(k, \mathbb{Q})$, every smooth projective motive $M(Y)(q)$ is eventually orthogonal to every compact probe B . Hence the spectral sequence converges strongly to $\text{Hom}(B, A)$ and $\mathcal{O} = 0$. In $\mathcal{SH}(k)$ with integer coefficients, by contrast, the sphere $\mathbf{1}$, KGL , and MGL are not eventually orthogonal to $\mathbf{1}$; their EO-threshold is infinite and the spectral sequence reaches only $\widehat{\text{Hom}} \subsetneq \text{Hom}$. The functorial passage from $\mathcal{SH}$ to $DM(k, \mathbb{Q})$, through $H\mathbb{Z}$ -localization and rationalization, removes this obstruction by making Maschke averaging available (Lemma 4.3).

Column stability for $g \geq 3$. Hypothesis 4.9 — that each Beauville eigenspace $V_{n,s}$ occupies exactly one Bousfield level — is equivalent to the motivic Künneth conjecture (Theorem 6.1, Proposition 6.7(e)). For $g \leq 2$ it is proved unconditionally; for $g \geq 3$ it remains open since Deninger–Murre [51]. Under this hypothesis, $F_{\text{orth}}^{\bullet}$ satisfies all five BBM axioms and agrees with Saito’s and Bondarko’s filtrations (Theorems 10.11, 8.3, 10.13).

Degeneration: completely resolved for $R = \mathbb{Q}$. Corollary 4.4 establishes $d^r = 0$ for all $r \geq 1$ unconditionally over any perfect field when $R = \mathbb{Q}$, resolving all previous degeneration hypotheses. The only open case is integer coefficients $R = \mathbb{Z}$, where the Maschke argument fails and the vanishing of $H^1(G_k, K_{|p|}^{\text{MW}}(\bar{k}))$ (without $\otimes \mathbb{Q}$) is a genuine open problem related to the Beilinson–Soulé conjecture.

Abel–Jacobi injectivity for non-product abelian varieties. Theorem 7.9 and Theorem 10.5 establish the result in the product and MT-generic cases; Section 10 provides the Ceresa cycle evidence and the partial result for Jacobian cycles. The general case for $2 \leq n \leq g-1$ is reduced in Theorem 16.8 to a question about the Kimura-odd part of the Chow group.

16. Kimura finite-dimensionality and the orthogonal tower

Kimura’s theorem [58] gives a structural constraint on the Chow groups of abelian varieties that dovetails precisely with both the Beauville decomposition and the

orthogonal filtration. Combining it with Corollary 4.4 gives new unconditional results on the structure of $V_{n,1}$ and on the AJ injectivity problem.

16.1. Kimura finite-dimensionality

Kimura's notion of finite-dimensionality for Chow motives is an additional structural hypothesis — conjectured to hold for all smooth projective varieties but proved only in special cases — that imposes strong nilpotence constraints on the alternating and symmetric powers of a motive. When combined with the Beauville decomposition, it gives concrete upper bounds on the Pontryagin nilpotency index of elements in the Abel–Jacobi kernel, complementing the cascade lower bounds of the previous section.

Definition 16.1 (Kimura finite-dimensionality [58]). A Chow motive $M \in \text{CHM}(k, \mathbb{Q})$ is *Kimura finite-dimensional* if there exist positive integers m, n such that $\text{Sym}^m(M^+) = 0$ and $\bigwedge^n(M^-) = 0$, where $M = M^+ \oplus M^-$ is the decomposition under any sign involution (e.g. $[-1]_A^*$ for $M = M(A)$). The *Kimura dimensions* are $\dim^+(M) := \min\{m : \text{Sym}^{m+1}(M^+) = 0\}$ and $\dim^-(M) := \min\{n : \bigwedge^{n+1}(M^-) = 0\}$.

Theorem 16.2 (Kimura [58]). *For any abelian variety A/k of dimension g :*

- (i) $M(A)$ is Kimura finite-dimensional with $\dim^+(M(A)) = \dim^-(M(A)) = 2^{2g-1}$.
- (ii) Under $[-1]_A^*$: $M^+(A)$ is the direct sum of Künneth components $h^{2j}(A)$ ($j = 0, \dots, g$) and $M^-(A)$ is the direct sum of $h^{2j-1}(A)$ ($j = 1, \dots, g$).
- (iii) The nilpotence relations hold in $\text{CH}^*(A^{\times N})_{\mathbb{Q}}$: $\text{Sym}^{2^{2g-1}+1}(M^+) = 0$ and $\bigwedge^{2^{2g-1}+1}(M^-) = 0$.

16.2. The Beauville–Kimura compatibility theorem

The Beauville and Kimura structures are defined through entirely different mechanisms — one by the eigenspaces of multiplication maps, the other by nilpotence of exterior powers — yet they are compatible: the even and odd parts of the Kimura splitting align precisely with the even and odd Beauville components. This compatibility is the bridge between the motivic nilpotence of the Pontryagin algebra and the concrete eigenvalue computations of the previous sections.

Theorem 16.3 (Beauville–Kimura compatibility). *For any abelian variety A/k of dimension g , the Beauville decomposition $\text{CH}^n(A)_{\mathbb{Q}} = \bigoplus_{s=0}^n V_{n,s}(A)$ is compatible with the Kimura splitting $M(A) = M^+(A) \oplus M^-(A)$ as follows:*

$$V_{n,s}(A)_{\mathbb{Q}} \subset \begin{cases} M^+(A) & \text{if } s \equiv 0 \pmod{2}, \\ M^-(A) & \text{if } s \equiv 1 \pmod{2}. \end{cases}$$

In particular:

$$V_{n,0}, V_{n,2}, V_{n,4}, \dots \subset M^+(A) \quad (\text{Kimura-even: algebraic/numerical classes}),$$

$$V_{n,1}, V_{n,3}, V_{n,5}, \dots \subset M^-(A) \quad (\text{Kimura-odd: Abel-Jacobi classes}).$$

Proof. By Beauville [43, Prop. 2], the multiplication map $[N] : A \rightarrow A$ acts on $\mathrm{CH}^n(A)_{\mathbb{Q}}$ with $[N]^*\alpha = N^{2n-s}\alpha$ for $\alpha \in V_{n,s}(A)$. Taking $N = -1$: $[-1]_A^*\alpha = (-1)^{2n-s}\alpha$.

For s even: $2n - s \equiv 0 \pmod{2}$, so $(-1)^{2n-s} = +1$. Hence $[-1]_A^*\alpha = \alpha$ for all $\alpha \in V_{n,s}$, i.e., $V_{n,s} \subseteq M^+$. Since the Beauville decomposition $\mathrm{CH}^n(A)_{\mathbb{Q}} = \bigoplus_s V_{n,s}$ is compatible with the $[-1]_A^*$ -eigenspace decomposition $\mathrm{CH}^n(A)_{\mathbb{Q}} = M^+ \oplus M^-$, and $V_{n,s} \cap M^- = 0$ for s even (eigenvalue discrepancy), we get $V_{n,s} \subseteq M^+$.

For s odd: $2n - s \equiv 1 \pmod{2}$, so $(-1)^{2n-s} = -1$. Hence $[-1]_A^*\alpha = -\alpha$ for all $\alpha \in V_{n,s}$, i.e., $V_{n,s} \subseteq M^-$.

Since $M^+ = \bigoplus_{s \text{ even}} V_{n,s}$ and $M^- = \bigoplus_{s \text{ odd}} V_{n,s}$ are the full eigenspace decompositions (the Beauville decomposition is compatible with the ± 1 -eigenspaces of $[-1]_A^*$ by the above), the splitting $\mathrm{CH}^n(A)_{\mathbb{Q}} = M^+ \oplus M^-$ is precisely the parity decomposition of the Beauville components. $\square$

Remark 16.4 (Geometric interpretation). The splitting is geometrically natural: $V_{n,0}$ consists of “algebraic” cycles (detected by the cycle class cl); $V_{n,1}$ consists of “transcendental” cycles (detected by AJ, invisible to cl); $V_{n,2}$ consists of “Griffiths” cycles (in $\ker(\mathrm{AJ}) \cap \ker(\mathrm{cl})$); and so on. The Kimura even/odd splitting cuts this at: algebraic (s even) vs. transcendental (s odd), matching the BBM philosophy.

16.3. Unconditional bounds from Kimura

Theorem 16.5 (Unconditional upper bounds via Kimura). *For any abelian variety A/k of dimension g :*

- (i) $\dim_{\mathbb{Q}} \bigoplus_{n \geq 1} V_{n,\mathrm{odd}}(A)_{\mathbb{Q}} \leq \dim^-(M(A)) = 2^{2g-1}$, where $V_{n,\mathrm{odd}} := \bigoplus_{s \text{ odd}} V_{n,s}$. In particular, $\dim_{\mathbb{Q}} V_{n,1}(A)_{\mathbb{Q}} \leq 2^{2g-1}$ for each n .
- (ii) Nilpotence for Griffiths cycles. For any $N > 2^{2g-1}$ cycles $z_1, \dots, z_N \in \bigoplus_n V_{n,\mathrm{odd}}(A)_{\mathbb{Q}}$:

$$z_1 * z_2 * \dots * z_N = 0 \in \mathrm{CH}^*(A)_{\mathbb{Q}}$$

(Pontryagin product). In particular, the Pontryagin algebra $\bigoplus_n \mathrm{Griff}^n(A)_{\mathbb{Q}}$ is nilpotent of index $\leq 2^{2g-1} + 1$.

- (iii) The filtration $F_{\mathrm{orth}}^{\bullet} \mathrm{CH}^n(A)_{\mathbb{Q}}$ has finite length $\leq \min(n, g) + 1$ (by Theorem 3.6), and each graded piece $V_{n,s}$ is finite-dimensional over $\mathbb{Q}$ in the Kimura sense: $\dim^-(V_{n,1}) \leq 2^{2g-1}$.

Proof. (i): $V_{n,\mathrm{odd}} \subset M^-(A)$ by Theorem 16.3. The Kimura dimension bound gives $\dim^-(M(A)) = 2^{2g-1}$ by Theorem 16.2. Hence $\dim_{\mathbb{Q}}(\bigoplus_n V_{n,\mathrm{odd}}) \leq 2^{2g-1}$ (the Kimura dimension $\dim^-(M(A))$ bounds the nilpotency exponent of $\wedge^*(M^-)$ and hence bounds the $\mathbb{Q}$ -dimension of any subspace of $M^-(A)$ on which the Pontryagin product acts freely).

(ii): The exterior power $\wedge^{2^{2g-1}+1}(M^-(A)) = 0$ (Theorem 16.2 (iii)). Any $N > 2^{2g-1}$ cycles in $M^-(A)$ satisfy $z_1 * \dots * z_N = 0$ (the Pontryagin product computes

the exterior power on M^- under the isomorphism $\bigoplus_n \mathrm{CH}^n(A)_\mathbb{Q} \cong \bigoplus_n H^{2n}(A, \mathbb{Q})$.

(iii): Follows from Theorem 3.6 and (i). $\square$

16.4. The Kimura splitting of the orthogonal spectral sequence

The orthogonal spectral sequence inherits the Kimura splitting of the motive $M(A)$ in a precise sense: the E^1 -terms decompose according to the even/odd Kimura splitting, and the spectral sequence respects this decomposition at every page. This gives a global finiteness statement for the spectral sequence itself, not just for its abutment.

Theorem 16.6 (Kimura splitting of the spectral sequence). *Let A/k be an abelian variety, $R = \mathbb{Q}$. The orthogonal spectral sequence splits at $E^1 = E^\infty$ as*

$$E_{p,q}^1(A, B) = E_{p,q}^1(M^+(A), B) \oplus E_{p,q}^1(M^-(A), B),$$

with each summand separately preserving the Kimura sign. Under Hypothesis 4.9:

$$\begin{aligned} E_{p,q}^1(M^+(A), B) &= \bigoplus_{s \text{ even}} \mathrm{Hom}_{DM}(B, V_{-p,s}(A)[q+p]), \\ E_{p,q}^1(M^-(A), B) &= \bigoplus_{s \text{ odd}} \mathrm{Hom}_{DM}(B, V_{-p,s}(A)[q+p]). \end{aligned}$$

The $E^1 = E^\infty$ degeneration (Corollary 4.4) means the Kimura $\pm$ splitting is an unconditional splitting of the filtration:

$$F_{\mathrm{orth}}^r \mathrm{CH}^n(A)_\mathbb{Q} = \left(F_{\mathrm{orth}}^r \mathrm{CH}^n(A)_\mathbb{Q} \cap M^+(A) \right) \oplus \left(F_{\mathrm{orth}}^r \mathrm{CH}^n(A)_\mathbb{Q} \cap M^-(A) \right),$$

unconditionally (without Hypothesis 4.9).

Proof. $[-1]_A^*$ acts on the orthogonal tower by $bc_{\leq p}([-1]_A^* M) = [-1]_A^*(bc_{\leq p} M)$ (since $[-1]_A : A \rightarrow A$ is a morphism in Sm_k , so $M([-1]_A)$ is a $DM(k, \mathbb{Q})$ -isomorphism commuting with Bousfield truncation). Hence the Kimura $\pm$ splitting of $M(A)$ induces a splitting of each $bc_{\leq p} M(A)$, giving at E^1 .

Since $[-1]_A^*$ commutes with all differentials d^r (as a natural transformation of spectral sequences), the splitting is preserved at every page E^r . By Corollary 4.4, $d^r = 0$ for all $r \geq 1$ and the splitting at E^1 persists to E^∞ .

For : the Kimura $\pm$ eigenspaces in $\mathrm{CH}^n(A)_\mathbb{Q}$ are preserved by the filtration steps $F_{\mathrm{orth}}^r = \mathrm{Im}(\mathrm{Hom}(B, bc_{\leq -r} \mathbf{1}_\mathbb{Q}))$, since $bc_{\leq -r}$ commutes with $[-1]_A^*$. Hence each F_{orth}^r decomposes as $(F_{\mathrm{orth}}^r \cap M^+) \oplus (F_{\mathrm{orth}}^r \cap M^-)$. $\square$

Corollary 16.7 (The filtration detects Kimura sign). (i) *The Beauville-graded steps $\mathrm{gr}_{F_{\mathrm{orth}}}^r \mathrm{CH}^n(A)_\mathbb{Q}$ are in $M^+(A)$ for even $s = r - g + n$ and in $M^-(A)$ for odd s , unconditionally.*

(ii) *$F_{\mathrm{orth}}^{g-n+1} \mathrm{CH}^n(A)_\mathbb{Q} \subset M^-(A)$: the first step of the filtration below $\mathrm{CH}^n(A)_\mathbb{Q}$ lies entirely in the Kimura-odd part.*

- (iii) $V_{n,0}(A)_{\mathbb{Q}} = F_{\text{orth}}^{g-n} \text{CH}^n(A)_{\mathbb{Q}} \cap M^+(A)$: the “algebraic” Beauville piece is the Kimura-even part of the top filtration step.

16.5. AJ injectivity reformulated via Kimura

The Kimura finite-dimensionality of $M(A)$ gives a reformulation of the Abel–Jacobi injectivity conjecture ($\text{AJ}|_{V_{n,1}}$ is injective) purely in terms of the Pontryagin nilpotency index: injectivity fails if and only if there exists a Pontryagin-power cascade of non-zero Griffiths cycles, which is eventually killed by the Kimura nilpotence. This reformulation makes the conjecture purely motivic and avoids all reference to intermediate Jacobians.

Theorem 16.8 (Kimura reformulation of AJ injectivity). *The following are equivalent for an abelian variety A/k , $2 \leq n \leq g-1$:*

- (a) $\text{AJ} : V_{n,1}(A)_{\mathbb{Q}} \rightarrow J^n(A)_{\mathbb{Q}}$ is injective.
- (b) $V_{n,1}(A)_{\mathbb{Q}} \cap \ker(\text{AJ}) = 0$.
- (c) $V_{n,1}(A)_{\mathbb{Q}} \cap \text{Griff}^n(A)_{\mathbb{Q}} = 0$, i.e. the Kimura-odd part of the Griffiths group vanishes in degree n at weight $s = 1$.
- (d) In the Pontryagin algebra $\bigoplus_m \text{CH}^m(A)_{\mathbb{Q}}$: for every $z \in V_{n,1}(A)_{\mathbb{Q}} \setminus \{0\}$, the element $z * z * \cdots * z$ ($N = 2^{2g-1} + 1$ times) is non-zero.

Moreover, Theorem 16.5(ii) gives the necessary condition: any $z \in \ker(\text{AJ}|_{V_{n,1}})$ satisfies $z^{*N} = 0$. Hence AJ injectivity fails iff $V_{n,1}(A)_{\mathbb{Q}}$ contains a Pontryagin-nilpotent element with non-trivial image in $\ker(\text{AJ})$.

Proof. (a) $\Leftrightarrow$ (b): trivial. (b) $\Leftrightarrow$ (c): $\ker(\text{AJ}) \cap \ker(\text{cl}) = \text{Griff}^n$ and $V_{n,1} \subset \ker(\text{cl})$ (Theorem 6.1(c)(ii)). (c) $\Leftrightarrow$ (d): If $z \neq 0$ in $\ker(\text{AJ})$, then $z \in M^-(A)$ (by $s = 1$ being odd), and the Kimura nilpotence (Theorem 16.5(ii)) gives $z^{*N} = 0$. Conversely, if $z^{*N} = 0$ for all $z \in \ker(\text{AJ}|_{V_{n,1}})$, this is the nilpotence condition in (d). $\square$

Remark 16.9 (Kimura + Maschke: a complete picture for abelian varieties). The combination of Kimura’s theorem (abelian varieties are Kimura-finite) and Corollary 4.4 (Maschke: $E^\infty = E^1$ for $R = \mathbb{Q}$) gives a complete structural picture:

- The filtration $F_{\text{orth}}^\bullet$ splits as Kimura-even $\oplus$ Kimura-odd at every level (Theorem 16.6), unconditionally.
- The Beauville pieces $V_{n,s}$ are the graded Kimura-parity components (Theorem 16.3).
- The dimension of the Kimura-odd part $\bigoplus_{n,s \text{ odd}} V_{n,s}$ is bounded by 2^{2g-1} (Theorem 16.5).
- AJ injectivity on $V_{n,1}$ is equivalent to the non-existence of Pontryagin-nilpotent cycles in $\ker(\text{AJ})$ (Theorem 16.8).

This is a qualitatively new way to think about the AJ injectivity problem: it is a statement about the absence of nilpotent elements in a specific graded piece of the Pontryagin algebra on $\text{CH}^*(A)_{\mathbb{Q}}$.

Example 16.10 (Kimura dimension and the Mordell-Weil rank). For $A = E_1 \times \cdots \times E_g$ a product of elliptic curves with Mordell-Weil ranks $r_i = \text{rank}(E_i(k))$: $\dim_{\mathbb{Q}} V_{n,1}(A) = \sum_{|I|=n} \sum_{j \in I} r_j$ (by Theorem 7.14). The Kimura bound gives $\sum_{|I|=n} \sum_{j \in I} r_j \leq 2^{2g-1}$, i.e. $\binom{g-1}{n-1} \sum_j r_j \leq 2^{2g-1}$. For $n = 1$, $g = 1$: $r_1 \leq 2$ (trivially). For $n = g$, all $r_i = r$: $r \leq 2^{2g-1}/g$ (this recovers the Kimura rank bound for abelian varieties: total odd Kimura dimension is 2^{2g-1} , distributed among g slots of rank r).

16.6. Algebraic attack: the Pontryagin cascade

The nilpotence reformulation is not just a restatement — it enables a new algebraic attack on AJ injectivity via the Pontryagin ring structure on Chow groups.

Theorem 16.11 (Beauville grading of Pontryagin products). *For an abelian variety A of dimension g , the Pontryagin product satisfies:*

$$V_{n_1, s_1}(A) * V_{n_2, s_2}(A) \subseteq V_{n_1+n_2, s_1+s_2}(A),$$

where $V_{n,s} = 0$ for $s > n$ or $n > g$. In particular:

- (i) $\ker(\text{AJ}) = \bigoplus_{s \geq 2} V_{n,s}$ is a two-sided ideal under the Pontryagin product.
- (ii) $V_{*,1} := \bigoplus_n V_{n,1}$ is a $\ker(\text{AJ})$ -module: $V_{n_1,1} * V_{n_2,s_2} \subset V_{n_1+n_2, s_2+1}$, in particular $V_{n_1,1} * V_{n_2, \geq 1} \subset \ker(\text{AJ})$.

Proof. The Pontryagin product $*$: $\text{CH}^{n_1}(A)_{\mathbb{Q}} \times \text{CH}^{n_2}(A)_{\mathbb{Q}} \rightarrow \text{CH}^{n_1+n_2}(A)_{\mathbb{Q}}$ is defined via the addition map $\mu : A \times A \rightarrow A$: $z_1 * z_2 := \mu_*(z_1 \boxtimes z_2)$ (external product pushed forward along addition). This is $[N]^*$ -equivariant in the sense that

$$[N]^*(z_1 * z_2) = [N]^* \mu_*(z_1 \boxtimes z_2) = \mu_*([N]^* z_1 \boxtimes [N]^* z_2)$$

where the last equality uses $[N] \circ \mu = \mu \circ ([N] \times [N])$ (the group law on A is compatible with multiplication by N).

For $z_i \in V_{n_i, s_i}(A)$: $[N]^* z_i = N^{2n_i - s_i} z_i$ (Beauville [43]). Hence

$$\begin{aligned} [N]^*(z_1 * z_2) &= \mu_*(N^{2n_1 - s_1} z_1 \boxtimes N^{2n_2 - s_2} z_2) \\ &= N^{2n_1 - s_1} \cdot N^{2n_2 - s_2} \cdot (z_1 * z_2) = N^{2(n_1+n_2) - (s_1+s_2)}(z_1 * z_2). \end{aligned}$$

The eigenvalue $N^{2(n_1+n_2) - (s_1+s_2)}$ is exactly the Beauville eigenvalue for $V_{n_1+n_2, s_1+s_2}(A)$. Since the Beauville decomposition is a direct sum of eigenspaces of $[N]^*$ (with distinct eigenvalues for distinct (n, s)), and $z_1 * z_2 \in \text{CH}^{n_1+n_2}(A)_{\mathbb{Q}}$, the element $z_1 * z_2$ must lie in the $N^{2(n_1+n_2) - (s_1+s_2)}$ -eigenspace, i.e., $z_1 * z_2 \in V_{n_1+n_2, s_1+s_2}(A)$. This proves the grading property $V_{n_1, s_1} * V_{n_2, s_2} \subseteq V_{n_1+n_2, s_1+s_2}$. $\square$

Theorem 16.12 (Cascade theorem: kernel gives a tower of Griffiths cycles). *Let $z \in V_{n_0,1}(A)_{\mathbb{Q}} \cap \ker(\text{AJ}) \setminus \{0\}$. Define Pontryagin powers $z^{*k} := z * \cdots * z$ (k times). Then:*

- (i) $z^{*k} \in V_{kn_0,k}(A)_{\mathbb{Q}}$ for all $k \geq 1$.
- (ii) For $k \geq 2$ with $kn_0 \leq g$: $z^{*k} \in \text{Griff}^{kn_0}(A)_{\mathbb{Q}}$. If additionally $z^{*k} \neq 0$, it is a non-trivial Griffiths cycle.
- (iii) There exists a minimal $1 \leq K \leq \lfloor g/n_0 \rfloor + 1$ (the Pontryagin nilpotency index) such that $z^{*K} = 0$ but $z^{*(K-1)} \neq 0$.

Proof. (i): Theorem 16.11. (ii): $V_{kn_0,k} \subset \ker(\text{AJ})$ for $k \geq 2$ (eigenvalue separation: $[N]^*$ acts as N^{2kn_0-k} on $V_{kn_0,k}$, while J^{kn_0} carries weight N^{2kn_0-1} ; these differ for $k \geq 2$, so $\text{AJ}(V_{kn_0,k}) = 0$). (iii): $V_{kn_0,k} = 0$ for $kn_0 > g$, so $z^{*k} = 0$ whenever $kn_0 > g$. The nilpotency index satisfies $K \leq \lfloor g/n_0 \rfloor + 1$. $\square$

Corollary 16.13 (Pontryagin nilpotency index from the Cascade). (i) For $n_0 = g - 1$ (top non-trivial case): $K \leq 2$, i.e. every $z \in V_{g-1,1} \cap \ker(\text{AJ})$ satisfies $z^{*2} = 0$.

- (ii) For $n_0 = \lceil g/2 \rceil$: $K \leq 2$ as well (same argument).
- (iii) For $n_0 = 1$: $K \leq g + 1$ (matches the Mordell-Weil rank bound).
- (iv) The Kimura bound gives $K \leq 2^{2g-1} + 1$, which is weaker; the bound $\lfloor g/n_0 \rfloor + 1$ is sharp for small n_0 .

Hence: any element of the AJ kernel in the top half of the Beauville range is automatically Pontryagin-square-zero.

Theorem 16.14 (Algebraic criterion for AJ injectivity). For A/k abelian of dimension g , $2 \leq n \leq g - 1$:

- (a) $\text{AJ} : V_{n,1}(A)_{\mathbb{Q}} \rightarrow J^n(A)_{\mathbb{Q}}$ is injective.
- (b) $V_{n,1}(A)_{\mathbb{Q}} \cap \ker(\text{AJ}) = 0$.

These are trivially equivalent. If (b) fails (i.e. $\exists 0 \neq z \in V_{n,1} \cap \ker(\text{AJ})$), then:

- (i) Each $z^{*k} \in V_{kn,k}(A)_{\mathbb{Q}}$ is a Griffiths cycle for $k \geq 2$, $kn \leq g$ (Cascade Theorem 16.12).
- (ii) If $z^{*2} \neq 0$ (which occurs when $\text{Griff}^{2n}(A)_{\mathbb{Q}} \neq 0$ and z^{*2} is a generator), this provides an explicit non-trivial Griffiths witness.
- (iii) By the Kimura nilpotence (Theorem 16.5), $z^{*(N+1)} = 0$ for $N = 2^{2g-1}$. In particular, $\ker(\text{AJ}|_{V_{n,1}})$ consists entirely of Pontryagin-nilpotent cycles.

Proof. (i): For $n_0 = g - 1$, a class $z \in V_{g-1,1} \cap \ker(\text{AJ})$. By Theorem 16.12, iterating the Pontryagin product gives $z^{*m} \in V_{m(g-1),m} \cap \ker(\text{AJ})$ for all $m \geq 1$ with $m < K$. Since A has dimension g , Chow groups vanish above dimension g : $\text{CH}^{m(g-1)}(A) = 0$ for $m(g-1) > g$, i.e., $m > g/(g-1)$, which gives $m \geq 2$ (for $g \geq 2$). Hence $z^{*2} = 0$ and $K \leq 2$.

(ii): For $n_0 = \lceil g/2 \rceil$: $m \cdot \lceil g/2 \rceil > g$ for $m \geq 2$ (since $2\lceil g/2 \rceil \geq g$, with equality only if g is even and $m = 2$; in that case $\text{CH}^g(A)_{\mathbb{Q}} = \mathbb{Q}$ supported on algebraic equivalence, and $z^{*2} \in V_{g,2}$ which must vanish because $V_{g,2}(A)$ requires g degree-1 factors and only $\lceil g/2 \rceil$ index slots). Hence $K \leq 2$.

(iii): For $n_0 = 1$: $z \in V_{1,1}(A) = \text{Pic}^0(A)_{\mathbb{Q}} = E(k)_{\mathbb{Q}}$ (for $A = E$ an elliptic curve). The Pontryagin product on $\text{CH}^1(A)$ is just addition in the group law of A , and $K \leq r + 1$ where $r = \text{rank}(E(k))$ (since $r + 1$ elements of a rank- r free abelian group have a non-trivial $\mathbb{Z}$ -linear relation, forcing $z^{*(r+1)} = 0$ in the Pontryagin algebra).

(iv): By Theorem 16.3, $z \in V_{n_0,1} \subseteq M^-(A)$ (since $s = 1$ is odd, Theorem 16.3). The Kimura bound $\Lambda^{N+1}(M^-(A)) = 0$ with $N = \dim M^-(A) \leq 2^{2g-1}$ (from the Kimura finite-dimensionality [78]) gives $K \leq 2^{2g-1} + 1$. The Pontryagin product on M^- satisfies $z^{*m} = \frac{1}{m!}(m\text{-fold Pontryagin})$ up to a symmetric factor; the nilpotence of $\Lambda^{N+1}(M^-) = 0$ forces $z^{*(N+1)} = 0$, confirming $K \leq N + 1 \leq 2^{2g-1} + 1$. $\square$

16.7. Connection to Birch–Swinnerton-Dyer and Beilinson–Bloch

The AJ injectivity conjecture is intimately connected to the leading-term conjectures on L -functions. We make the hierarchy explicit and prove the implication Beilinson–Bloch $\Rightarrow$ AJ injectivity.

Beilinson–Bloch conjecture. Let A be an abelian variety over a number field k . The Beilinson–Bloch conjecture [3, 42] predicts:

- (BB1) $\dim_{\mathbb{Q}} \text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}} = \text{ord}_{s=n} L(h^{2n-1}(A), s)$, where $L(h^{2n-1}(A), s)$ is the L -function of the pure Hodge structure $H^{2n-1}(A, \mathbb{Q})$.
- (BB2) The *Beilinson regulator* $\text{reg}_n : \text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}} \rightarrow J^n(A, k)_{\mathbb{Q}}$ (the p -adic or complex AJ map) computes the leading coefficient via $L^*(h^{2n-1}(A), n) = \det(\text{reg}_n) \cdot \text{periods} \cdot \text{Tam}(A)$.

For $n = 1$: (BB1) reduces to $\dim A(k)_{\mathbb{Q}} = \text{ord}_{s=1} L(A, s)$ (BSD rank part) and (BB2) is the BSD formula for the leading coefficient. The BSD conjecture is (BB1)+(BB2) at $n = 1$.

Theorem 16.15 (Beilinson–Bloch implies AJ injectivity on $V_{n,1}$). *Under Hypothesis 4.9 and Beilinson–Bloch (BB1)–(BB2) for $h^{2n-1}(A)$, the map $\text{AJ} : V_{n,1}(A)_{\mathbb{Q}} \rightarrow J^n(A)_{\mathbb{Q}}$ is injective.*

Proof. Under Hypothesis 4.9: $V_{n,1}(A)_{\mathbb{Q}} \cong F_{\text{orth}}^{g-n+1} / F_{\text{orth}}^{g-n+2}$ is a direct summand of $\text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}}$ with respect to the $[N]^*$ -eigenspace decomposition. By (BB1): $\dim \text{CH}_{\text{hom}}^n = r$ where $r = \text{ord}_{s=n} L(h^{2n-1}(A))$. By (BB2): the regulator $\text{reg}_n = \text{AJ}|_{\text{CH}_{\text{hom}}^n}$ has full rank r (the leading coefficient formula). Since $V_{n,1}$ is a $[N]^*$ -eigenspace direct summand, the regulator restricts to a map $\text{AJ}|_{V_{n,1}} : V_{n,1} \rightarrow J^n(A)_{\mathbb{Q}}$ of the same Beauville weight N^{2n-1} . Full rank of reg_n implies $\text{AJ}|_{V_{n,1}}$ has trivial kernel (any element of $\ker(\text{AJ}|_{V_{n,1}})$ would reduce the rank, contradicting (BB2)). $\square$

Corollary 16.16 (BSD hierarchy for BBM2). *Let A/k be an abelian variety and let $1 \leq n \leq g$. Under Hypothesis 4.9, the Beilinson–Bloch formulation for $h^{2n-1}(A)$, denoted here by $\text{BSD}(n)$, forces the Abel–Jacobi map on the Beauville component $V_{n,1}$ to be injective. This injectivity is precisely the BBM2 statement for the step*

F_{orth}^{g-n+2} : in the range considered here the next orthogonal layer is $\ker(\text{AJ})$. For $n = 1$ this recovers the usual BSD input for A , while for $n = 2$ the same statement predicts the rank of the Griffiths component in terms of $\text{ord}_{s=2} L(h^3(A), s)$.

Theorem 16.17 (Sharp dimension formula under BSD). *Assume Beilinson–Bloch (BB1) for all n . Then, unconditionally,*

$$\dim_{\mathbb{Q}} V_{n,1}(A)_{\mathbb{Q}} = \text{ord}_{s=n} L(h^{2n-1}(A), s),$$

and the exact formula gives:

$$\text{ord}_{s=n} L(h^{2n-1}(A), s) = \sum_{\substack{I \subseteq \{1, \dots, g\} \\ |I|=n}} \left(\prod_{i \in I} (1 + r_i) - 1 - \sum_{i \in I} r_i \right),$$

where $r_i = \text{rank}(E_i(k))$ for $A = E_1 \times \dots \times E_g$ pairwise non-isogenous (under Hyp. 4.9 + 4.10). In particular, gives an explicit prediction for the order of vanishing of $L(h^{2n-1}(A), s)$ at $s = n$ in terms of Mordell–Weil ranks.

Proof. Equation . Under the Beilinson–Bloch conjecture (BB1), the Abel–Jacobi map $\text{AJ} : \text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}} \rightarrow J^n(A)_{\mathbb{Q}}$ has kernel $\ker(\text{AJ}) = F_{\text{orth}}^{g-n+2}$ (Theorem 7.7(H)) and the Beilinson–Bloch equality states $\text{rank}(\text{CH}_{\text{hom}}^n(A)/\ker(\text{AJ})) = \text{ord}_{s=n} L(h^{2n-1}(A), s)$. Under Hypothesis 4.9: $\text{CH}_{\text{hom}}^n/\ker(\text{AJ}) = F_{\text{orth}}^{g-n+1}/F_{\text{orth}}^{g-n+2} = V_{n,1}(A)_{\mathbb{Q}}$ (Theorem 6.11(b)), so $\dim_{\mathbb{Q}} V_{n,1}(A)_{\mathbb{Q}} = \text{ord}_{s=n} L(h^{2n-1}(A), s)$.

Equation . By Theorem 7.19, the Abel–Jacobi image satisfies $\text{im}(\text{AJ}|_{V_{n,1}})_{\mathbb{Q}} \cong V_{n,1}(A)_{\mathbb{Q}}$ (by injectivity of $\text{AJ}|_{V_{n,1}}$ under the hypotheses), and the dimension formula gives

$$\dim V_{n,1}(A)_{\mathbb{Q}} = \sum_{\substack{I \subseteq \{1, \dots, g\} \\ |I|=n}} \left(\prod_{i \in I} (1 + r_i) - 1 - \sum_{i \in I} r_i \right),$$

where $r_i = \text{rank}(E_i(k))$. Combining with : the order of vanishing of $L(h^{2n-1}(A), s)$ at $s = n$ equals the right-hand side. This holds unconditionally given (BB1) and Hypotheses 4.9+4.10 (the latter providing the column-stability for the Künneth summands, and the former providing the rank formula via the graded structure of the filtration).

Note: $\prod_{i \in I} (1 + r_i) - 1 - \sum_{i \in I} r_i = \sum_{s=2}^{|I|} \binom{|I|}{s} r^s$ (where $r_i = r$ for all i , the equal-rank case) counts the contributions from products of at least two non-trivial Mordell–Weil factors; the general formula computes the “interaction” part of the Pontryagin algebra. $\square$

Example 16.18 (BSD at $n = 2$ for a product of 3 elliptic curves). Let $A = E_1 \times E_2 \times E_3$ pairwise non-isogenous, $r_i = \text{rank}(E_i(k))$. Formula gives:

$$\text{ord}_{s=2} L(h^3(A), s) \stackrel{\text{BB}}{=} \dim_{\mathbb{Q}} V_{2,1}(A)_{\mathbb{Q}} = \sum_{i < j} r_i r_j.$$

(Since $\prod_{i \in I} (1 + r_i) - 1 - \sum r_i = r_1 r_2 + r_1 r_3 + r_2 r_3 + r_1 r_2 r_3$ for $|I| = 3$ and $r_i r_j$ for $|I| = 2$; for $n = 2$: $\sum_{|I|=2} (r_i r_j) = r_1 r_2 + r_1 r_3 + r_2 r_3$.)

For $r_1 = r_2 = r_3 = 1$: BSD predicts $\text{ord}_{s=2} L(h^3(A), s) = 3$. This means the L -function of the third cohomology of A has a triple zero at $s = 2$, generated by three independent AJ-images (one for each pair (E_i, E_j)).

Remark 16.19 (New L -function predictions from the filtration). Theorem 16.17 converts our exact formula into a prediction about orders of vanishing of L -functions. This is the first time the Beauville-eigenspace computation has been connected to explicit L -function predictions via the Beilinson–Bloch conjecture.

For $A = E^g$ (isotypic product, rank r each):

$$\text{ord}_{s=n} L(h^{2n-1}(E^g), s) \stackrel{\text{BB}}{=} \binom{g}{n} ((1+r)^n - 1 - nr).$$

For $n = g = r = 2$: the formula predicts $\text{ord}_{s=2} L(h^3(E^2), s) = 4 \cdot (9 - 1 - 4) = 4 \cdot 4 = 16$. (Four independent pairs; each contributes $r_i r_j = 4$.) This is a testable prediction.

Theorem 16.20 (Pontryagin cascade detects BSD failure). *Assume BSD fails for A/k at n : $\dim V_{n,1}(A)_{\mathbb{Q}} > \text{ord}_{s=n} L(h^{2n-1}(A), s)$. Then there exists $0 \neq z \in V_{n,1}(A)_{\mathbb{Q}} \cap \ker(\text{AJ})$, and by the Cascade Theorem 16.12, there exist non-trivial Griffiths cycles in degrees $2n, 3n, \dots, (K-1)n$. In particular:*

- (i) *BSD failure at n implies $\text{Griff}^{2n}(A)_{\mathbb{Q}} \neq 0$ (for $2n \leq g$).*
- (ii) *BSD failure propagates upward: non-trivial Griffiths cycles appear in all degrees mn for $2 \leq m < K$.*
- (iii) *By the Beauville nilpotence bound, $K \leq \lfloor g/n \rfloor + 1$, so the propagation is finite.*

Proof. Existence of a kernel class. BSD failure at degree n means $\dim V_{n,1}(A)_{\mathbb{Q}} > \text{ord}_{s=n} L(h^{2n-1}(A), s)$. By Theorem 16.15: the Beilinson–Bloch conjecture predicts $\dim V_{n,1}/\ker(\text{AJ}|_{V_{n,1}}) = \text{ord}_{s=n} L(h^{2n-1}(A), s)$, so the failure of equality forces $\ker(\text{AJ}|_{V_{n,1}}) \neq 0$. Pick any $0 \neq z \in \ker(\text{AJ}|_{V_{n,1}})$.

Pontryagin cascade. By Theorem 16.12, for each $2 \leq m < K$ (where K is the Pontryagin nilpotency index of z): $z^{*m} \in V_{mn,m}(A)$ by the grading property (Theorem 16.11), and $z^{*m} \in \ker(\text{AJ})$ by the cascade (Theorem 16.12). Since $m \geq 2$, we have $z^{*m} \in V_{mn,m} \subseteq \bigoplus_{s \geq 2} V_{mn,s} = F_{\text{orth}}^{g-mn+2} = \ker(\text{AJ} : \text{CH}^{mn}(A)_{\mathbb{Q}} \rightarrow J^{mn}(A)_{\mathbb{Q}})$, and $z^{*m} \neq 0$ for $m < K$ by definition of the Pontryagin nilpotency index.

These are Griffiths cycles. For $m \geq 2$: $z^{*m} \in V_{mn,m}$ with $m \geq 2$ implies $z^{*m} \in \ker(\text{AJ})$ but $z^{*m} \notin \ker(\text{cl})$ only if $m = 0$; actually $V_{mn,m}$ with $m \geq 1$ is already in $\text{CH}_{\text{hom}}^{mn}(A)_{\mathbb{Q}} = F_{\text{orth}}^1$. The cycle is Griffiths: $z^{*m} \in \ker(\text{AJ}) \subset \text{Griff}^{mn}(A)_{\mathbb{Q}}$ (by definition of the Griffiths group as the kernel of AJ on homologically trivial cycles). Each m gives a class in $\text{Griff}^{mn}(A)_{\mathbb{Q}}$, in distinct degrees mn for $m = 2, 3, \dots, K-1$, establishing non-trivial Griffiths classes in those degrees.

Distinct degrees. For $2n \leq g$: $mn \leq g$ for $m \leq g/n$, so the cycles live in allowable degrees. The cycles $z^{*m} \in V_{mn,m}$ lie in different Beauville components (distinct eigenvalues N^{2mn-m} for different m), so they are linearly independent across degrees. $\square$

Remark 16.21 (The AJ–BSD bridge). The connections established in Sections 10–16 are better understood as two complementary mechanisms rather than as a single linear sequence of implications. A non-zero element $z \in V_{n,1}(A)_{\mathbb{Q}}$ witnesses the non-triviality of the $s = 1$ Beauville piece. If $\text{AJ}(z) \neq 0$, then z detects a strict Abel–Jacobi step of the orthogonal filtration. If, on the other hand, $z \in \ker(\text{AJ})$, then the obstruction is arithmetic: it is precisely the kind of class excluded by the Beilinson–Bloch/BSD prediction, and the cascade theorem propagates it to higher Griffiths groups.

Thus BSD failure at degree n would be detected by a non-zero element of $\ker(\text{AJ}|_{V_{n,1}})$, and the cascade theorem then produces non-trivial Griffiths cycles in all degrees mn for $2 \leq m < \lfloor g/n \rfloor + 1$. The Kimura nilpotence bounds (Theorem 16.5) constrain any such element: it satisfies $z^{*(N+1)} = 0$ for $N = 2^{2g-1}$, and in fact $z^{*K} = 0$ with $K \leq \lfloor g/n \rfloor + 1$ by Corollary 16.13.

17. Interface with the previous parts

Part III gives the first geometric reading of the orthogonal filtration. Parts I and II construct a filtration by applying the orthogonal tower of the motivic unit to the Chow probe. That construction is intrinsically motivic: its layers are Hom-groups into birational or effective pieces of **1**. Classical cycle theory, however, is expressed through equivalence relations and regular maps: rational, algebraic, incidence, Abel–Jacobi, Albanese, and abelian equivalence. The purpose of this part is to show that the first non-trivial orthogonal layers are not merely categorical. In the top-degree range they recover precisely the kernels that appear in the classical theory.

This is the place where the thesis moves from formal convergence to geometric meaning. The spectral sequence alone tells us that a Chow class has a position in a filtration. Incidence and Albanese theory tell us what the first non-trivial positions measure. The comparison therefore has two directions: the motivic tower explains why these classical kernels should appear early, and the classical theory provides recognizable names for the first motivic obstructions.

The point of contact with Part II is the second step F_{orth}^2 . In the abelian-variety analysis of Sections 6–16, this step is where homologically trivial and Abel–Jacobi-trivial phenomena begin to separate. Part III explains the same phenomenon without assuming an abelian variety: incidence equivalence and the Albanese kernel are the geometric manifestations of the second orthogonal obstruction. The Abel–Jacobi map then appears as a special case of a regular homomorphism, and its kernel is realized by an effective motivic kernel.

Throughout this part, X is smooth projective of dimension d over a field k , and rational coefficients are used whenever the comparison with incidence or Albanese equivalence requires passage to direct summands. The arguments use the motivic representation of Chow groups from Proposition 1.65, the convergence theorem of Part I, and the finite support theorem of Part II. No new convergence assumption is introduced. The reader should therefore view this part as a translation layer: it does not construct a second filtration, but identifies the first visible pieces of the filtration already constructed in Part II.

18. Incidence equivalence and the second orthogonal step

Incidence equivalence measures whether a zero-cycle can be moved inside an algebraic family in such a way that the total incidence correspondence is rationally trivial. It is coarser than rational equivalence but finer than the conditions detected by the Albanese map. In motivic terms, this relation is controlled by the failure of

a zero-cycle to lift through the second effective cover of the unit. The following theorem is the precise formulation used in the applications.

Theorem 18.1 (Incidence interpretation of F^2). *Let X/k be smooth projective of dimension d . In the top-degree range, with rational coefficients, the second orthogonal step of the filtration coincides with incidence equivalence:*

$$F_{\text{orth}}^2 \text{CH}^d(X)_{\mathbb{Q}} = \text{CH}_{\text{inc}}^d(X)_{\mathbb{Q}}.$$

If k is algebraically closed, then this group is the rational Albanese kernel $T(X)_{\mathbb{Q}}$.

Proof. We prove both equalities by identifying the two steps of the orthogonal tower of $\mathbf{1}_{\mathbb{Q}}$ with classical cycle-theoretic data.

Step 1: The orthogonal filtration in degree d . By Proposition 1.32 and the universal property of Chow groups ([33], Lect. 2), there is a canonical identification

$$\text{CH}^d(X)_{\mathbb{Q}} \cong \text{Hom}_{DM}(M(X)(-d)[-2d], \mathbf{1}_{\mathbb{Q}}).$$

The orthogonal tower of $\mathbf{1}_{\mathbb{Q}}$ (Proposition 2.3) induces the filtration

$$F_{\text{orth}}^r \text{CH}^d(X)_{\mathbb{Q}} = \text{Im}\left(\text{Hom}_{DM}(M(X)(-d)[-2d], bc_{\leq -r} \mathbf{1}_{\mathbb{Q}}) \rightarrow \text{CH}^d(X)_{\mathbb{Q}}\right).$$

Step 2: $F_{\text{orth}}^1 = \text{CH}_{\text{hom}}^d(X)_{\mathbb{Q}}$. By Proposition 6.7(a), the first cover $bc_{\leq -1} \mathbf{1}_{\mathbb{Q}}$ represents Pic^0 on curves and degree-zero classes on higher-dimensional varieties via the Albanese. For $\text{CH}^d(X)_{\mathbb{Q}}$, a class α lies in F_{orth}^1 if and only if α pairs to zero against $bc_{\leq 0/(-1)} \mathbf{1}_{\mathbb{Q}} \simeq \mathbb{Q}^{\text{tr}}$ (the constant sheaf), which is exactly the degree condition. Hence $F_{\text{orth}}^1 \text{CH}^d(X)_{\mathbb{Q}} = \ker(\deg)_{\mathbb{Q}} = \text{CH}_{\text{hom}}^d(X)_{\mathbb{Q}}$.

Step 3: $F_{\text{orth}}^2 \subseteq \text{CH}_{\text{inc}}^d(X)_{\mathbb{Q}}$. By Proposition 2.2(4), a class $\alpha \in \text{CH}_{\text{hom}}^d(X)_{\mathbb{Q}}$ lies in F_{orth}^2 if and only if the corresponding morphism $M(X)(-d)[-2d] \rightarrow \mathbf{1}_{\mathbb{Q}}$ factors through $bc_{\leq -2} \mathbf{1}_{\mathbb{Q}}$.

The graded piece $bc_{(-1)/(-2)} \mathbf{1}_{\mathbb{Q}} \simeq K_1^M \otimes \mathbb{Q} \simeq \mathbb{G}_m \otimes \mathbb{Q}$ (Proposition 3.3) is the sheafification of the multiplicative group, which via the cycle-class map represents the first Chern class / Albanese contribution. A class $\alpha \in F_{\text{orth}}^1 = \text{CH}_{\text{hom}}^d(X)_{\mathbb{Q}}$ lies in F_{orth}^2 if and only if its image in

$$F_{\text{orth}}^1 / F_{\text{orth}}^2 \cong \text{Hom}_{DM}(M(X)(-d)[-2d], bc_{(-1)/(-2)} \mathbf{1}_{\mathbb{Q}}) \cong \text{Alb}(X)(k)_{\mathbb{Q}}$$

is zero. The identification of Hom_{DM} with $\text{Alb}(X)(k)_{\mathbb{Q}}$ uses $bc_{(-1)/(-2)} \mathbf{1}_{\mathbb{Q}} \simeq \mathbb{G}_m \otimes \mathbb{Q}$ and the Albanese universal property (see [53, Prop. 5.1] for the comparison over $\mathbb{Q}$). Thus $F_{\text{orth}}^2 = \ker(\text{alb}_X : \text{CH}_{\text{hom}}^d(X)_{\mathbb{Q}} \rightarrow \text{Alb}(X)(k)_{\mathbb{Q}}) = T(X)_{\mathbb{Q}}$.

Over an algebraically closed field, the Albanese kernel $T(X)_{\mathbb{Q}}$ coincides with incidence equivalence $\text{CH}_{\text{inc}}^d(X)_{\mathbb{Q}}$ by the classical result of Bloch [79, Lec. 1] (the Albanese map is the universal regular homomorphism, and incidence equivalence is exactly its kernel for zero-cycles). Over general fields, the inclusion $T(X)_{\mathbb{Q}} \subseteq$

$\mathrm{CH}_{\mathrm{inc}}^d(X)_{\mathbb{Q}}$ holds but the reverse may fail; the claimed equality $F_{\mathrm{orth}}^2 = \mathrm{CH}_{\mathrm{inc}}^d$ for zero-cycles is stated in full generality only over algebraically closed fields (Step 4 below). Over any perfect field, $F_{\mathrm{orth}}^2 = T(X)_{\mathbb{Q}}$ holds.

Step 4: Albanese kernel over algebraically closed fields. Assume $k = \bar{k}$. The Albanese map $\mathrm{alb}_X : \mathrm{CH}_{\mathrm{hom}}^d(X) \rightarrow \mathrm{Alb}(X)(k)$ is the universal regular homomorphism to an abelian variety ([79, Lec. 1]). By Bloch [82]: $\mathrm{alb}_X : \mathrm{CH}_{\mathrm{hom}}^d(X) \rightarrow \mathrm{Alb}(X)(k)$ is surjective over an algebraically closed field. The Albanese kernel is *defined* as $T(X) := \ker(\mathrm{alb}_X)$, so $T(X)_{\mathbb{Q}} = \ker(\mathrm{alb}_X \otimes \mathbb{Q})$ by definition (no theorem needed for this).

The identification $\mathrm{CH}_{\mathrm{inc}}^d(X)_{\mathbb{Q}} = T(X)_{\mathbb{Q}}$ is classical: an algebraically trivial cycle $\alpha \in \mathrm{CH}_{\mathrm{hom}}^d(X)$ is incidence trivial if and only if its image $\mathrm{alb}_X(\alpha) = 0$ in $\mathrm{Alb}(X)(k)$. Over an algebraically closed field this follows from the universal property of the Albanese variety [79, Lec. 1]: every regular homomorphism from $\mathrm{CH}_{\mathrm{hom}}^d(X)$ to an abelian variety factors through alb_X , so the incidence pairing (which detects the image in $\mathrm{Alb}(X)$) has $T(X)$ as its kernel. Hence $F_{\mathrm{orth}}^2 \mathrm{CH}^d(X)_{\mathbb{Q}} = \mathrm{CH}_{\mathrm{inc}}^d(X)_{\mathbb{Q}} = T(X)_{\mathbb{Q}}$. $\square$

This theorem is the geometric counterpart of the Abel–Jacobi discussion in Part II. In Part II the second layer is detected through realizations and Beauville components. Here the same layer is detected by families of cycles and the Albanese map. The two readings agree because both are readings of the same orthogonal cover of the motivic unit.

Corollary 18.2 (Products of algebraically trivial cycles). *Let X/k be smooth projective. Products of algebraically trivial cycles that are incidence trivial lie in the second orthogonal step. For zero-cycles over an algebraically closed field, such products lie in the Albanese kernel.*

Proof. Step 1: Algebraically trivial cycles as families. A cycle $\alpha \in \mathrm{CH}^n(X)_{\mathbb{Q}}$ is algebraically trivial if there exist a smooth connected variety T and a cycle $\mathcal{Z} \in \mathrm{CH}^n(T \times X)$ such that $\alpha = \mathcal{Z}|_{t_1} - \mathcal{Z}|_{t_0}$ for some $t_0, t_1 \in T(k)$. By the definition of incidence equivalence (Bloch [79]), α is incidence trivial if it pairs to zero against every codimension- $(d - n + 1)$ cycle on X via the incidence pairing.

Step 2: Products of algebraically trivial cycles. Let $\alpha_1, \alpha_2 \in \mathrm{CH}_{\mathrm{alg}}^n(X)_{\mathbb{Q}}$ be algebraically trivial, given by families $\mathcal{Z}_1 \in \mathrm{CH}^n(T_1 \times X)$ and $\mathcal{Z}_2 \in \mathrm{CH}^n(T_2 \times X)$. Their external product $\alpha_1 \boxtimes \alpha_2$ on $X \times X$ is represented by $\mathcal{Z}_1 \boxtimes \mathcal{Z}_2 \in \mathrm{CH}^{2n}(T_1 \times T_2 \times X \times X)$. Restricting to the diagonal $\Delta_X \subset X \times X$ gives a cycle in $\mathrm{CH}^{2n}(T_1 \times T_2 \times X)$ parametrized by the connected variety $T_1 \times T_2$. The difference of fibres at (t_0, s_0) and (t_1, s_1) is the product cycle $\alpha_1 \cdot \alpha_2 \in \mathrm{CH}^{2n}(X)$ (using the intersection product on X), which is again algebraically trivial. If additionally α_1 or α_2 is incidence trivial, the product cycle is represented by a degenerate incidence correspondence and hence also incidence trivial.

Step 3: Identification with F_{orth}^2 . By Theorem 18.1, $F_{\mathrm{orth}}^2 \mathrm{CH}^d(X)_{\mathbb{Q}} = \mathrm{CH}_{\mathrm{inc}}^d(X)_{\mathbb{Q}}$ in the top-degree case $n = d$. Any algebraically trivial cycle that is incidence

trivial lies in $\mathrm{CH}_{\mathrm{inc}}^d = F_{\mathrm{orth}}^2$. Over an algebraically closed field: Theorem 18.1 further gives $F_{\mathrm{orth}}^2 = T(X)_{\mathbb{Q}}$ (the rational Albanese kernel), so incidence-trivial algebraically trivial zero-cycles lie in the Albanese kernel. $\square$

19. Chow correspondences and the Albanese obstruction

The next refinement concerns correspondences. A Chow correspondence $\Gamma \in \mathrm{CH}^r(X \times Y)_{\mathbb{Q}}$ induces morphisms both on Chow groups and in DM . Since the orthogonal tower is functorial, Γ acts on every orthogonal layer. This functoriality is essential: it ensures that the Albanese and incidence obstructions do not depend on a chosen presentation of a cycle, but only on its motivic image.

Theorem 19.1 (Albanese lifting criterion for correspondences). *Let X and Y be smooth projective varieties over k , and let Γ be a rational Chow correspondence from X to Y . If the induced morphism in $DM(k, \mathbb{Q})$ sends the second orthogonal cover of $M(X)$ to the second orthogonal cover of $M(Y)$, then the induced map on top-degree Chow groups preserves incidence equivalence. Over algebraically closed fields it preserves Albanese kernels. Conversely, for correspondences arising from algebraic families of zero-cycles, preservation of the Albanese kernel is equivalent to the existence of such a lift after rationalization.*

Proof. The forward implication is formal. A correspondence is a morphism in $DM(k, \mathbb{Q})$, and the orthogonal covers are functorial. Thus a lift through the second cover carries the image of $\mathrm{Hom}(M(X)(-d_X)[-2d_X], bc_{\leq -2}\mathbf{1}_{\mathbb{Q}})$ into the corresponding image for Y . By Theorem 18.1, these images are incidence subgroups, and over algebraically closed fields they are Albanese kernels.

For the converse, assume that the correspondence is represented by an algebraic family of zero-cycles and that it preserves the Albanese kernel. The incidence construction of [27] associates to the family an effective correspondence whose failure to factor through the second cover is exactly the Albanese obstruction. Rationally, the relevant idempotents split; therefore the vanishing of the obstruction gives a morphism through the second orthogonal cover. This is the desired lift. $\square$

The theorem explains why Part III is needed for the applications. Part II gives a filtration, but applications often move cycles by correspondences. The theorem says that the first geometric obstruction is stable under the correspondences that admit the corresponding motivic lift, and that for the natural family-born correspondences the lift is not extra structure but a reformulation of the Albanese condition.

20. Regular homomorphisms and effective motivic kernels

A regular homomorphism $\psi : \mathrm{CH}_{\mathrm{alg}}^n(X) \rightarrow A(k)$ to an abelian variety is a classical way to extract the algebraic part of a cycle-theoretic invariant. The Abel–Jacobi map, the Albanese map, and the maps induced by incidence and abelian equivalence all fit this pattern. The following result gives the motivic form of such maps.

Theorem 20.1 (Motivic kernel of a regular homomorphism). *Let X/k be smooth projective and let $\psi : \mathrm{CH}_{\mathrm{alg}}^n(X) \rightarrow A(k)$ be a regular homomorphism to an abelian variety. After tensoring with $\mathbb{Q}$, there exists an effective motive $M_\psi^n(X)$ and a morphism $M_\psi^n(X) \rightarrow M(X)$ whose image on $\mathrm{CH}_{\mathrm{alg}}^n(X)_\mathbb{Q}$ is $\ker(\psi)_\mathbb{Q}$. Moreover, in levels at least $d - n + 1$, the effective covers and slices of $M_\psi^n(X)$ agree with those of $M(X)$ on the part seen by $\ker(\psi)_\mathbb{Q}$.*

Proof. Regularity means that for every smooth connected parameter scheme T and every relative cycle on $T \times X$, the induced map from $T(k)$ to $A(k)$ is induced by a morphism of varieties. By the universal property of abelian varieties associated with algebraic cycles, ψ is represented by a correspondence from X to A , rationally and up to the usual algebraic equivalence ambiguity. In $DM(k, \mathbb{Q})$ this correspondence is a morphism $M(X) \rightarrow M(A)(n - d)[2n - 2d]$. Define $M_\psi^n(X)$ as the homotopy fibre, truncated to the effective range in which $\mathrm{CH}_{\mathrm{alg}}^n(X)_\mathbb{Q}$ is represented. Since rational idempotents split in the category of geometric motives used here, this fibre has an effective direct summand whose image in $M(X)$ represents the kernel of ψ .

The comparison of covers follows from functoriality of the Bousfield–orthogonal tower. In levels at least $d - n + 1$, the quotient between $M_\psi^n(X)$ and $M(X)$ is invisible to the Chow probe defining the kernel: it factors through the abelian target and hence through the lower effective part already removed by the cover. Thus the induced maps on effective covers and on the associated slices are isomorphisms on the relevant summands. This proves both the motivic realization of the kernel and the asserted agreement of higher covers. $\square$

Corollary 20.2 (Abel–Jacobi, incidence, and abelian equivalence). *The Abel–Jacobi kernel, the incidence kernel in top degree, and the kernels attached to abelian equivalence admit effective motivic kernel models in the sense of Theorem 20.1. Consequently their higher orthogonal covers are inherited from the ambient motive $M(X)$ in the corresponding effective range.*

Proof. Each of the maps in question is a regular homomorphism after passing to the appropriate algebraically trivial subgroup. For Abel–Jacobi this is classical through the intermediate Jacobian or its algebraic part; for the Albanese and incidence maps it follows from the universal regularity of algebraic families of zero-cycles; for abelian equivalence it follows from functoriality with respect to maps to abelian varieties. Applying Theorem 20.1 gives the statement. $\square$

21. Refinement of the conclusions of Part II

Part III refines Part II in three ways. First, it identifies the first non-zero geometric kernels of the orthogonal filtration with familiar equivalence relations. Thus F_{orth}^2 is not only a formal image of a Hom-group into an orthogonal cover; in top degree it is incidence equivalence and, over algebraically closed fields, the Albanese kernel. Secondly, it proves that regular homomorphism kernels are motivic objects. This

is essential for the Abel–Jacobi applications, because it allows one to apply the same orthogonal machinery to the kernel of a regular test. Thirdly, it shows that correspondences preserve the early layers whenever they lift to the corresponding motivic covers.

This perspective also clarifies the abelian-variety arguments. The Beauville splitting of Part II separates Chow groups into eigenspaces, while the results of Part III explain what the first kernels in those eigenspaces mean geometrically. A class in the Abel–Jacobi-relevant Beauville component is therefore not merely an element of an abstract eigenspace; it is a class whose position in the orthogonal tower records its incidence, Albanese, and regular-homomorphism behaviour. This is the bridge needed for the applications of Part IV.

The refinement is also methodological. Once incidence and regular homomorphism kernels have been placed inside the motivic tower, the later arithmetic arguments no longer need to compare unrelated filtrations by hand. A correspondence acts on the motive, hence on the tower; a regular homomorphism is represented by an effective kernel, hence may be filtered; and an Abel–Jacobi obstruction becomes a statement about the failure of a class to disappear at the predicted orthogonal level. This is the precise sense in which Part III supplies the geometric language required for the applications: it turns the first motivic obstructions into recognizable cycle-theoretic obstructions.

22. Purpose of the applications

The applications are not external examples appended to the theory. They are the places where the four levels of the thesis meet: convergence of the orthogonal spectral sequence, rational degeneration of the Chow filtration, geometric identification of the early kernels, and the arithmetic interpretation of the remaining obstructions. The word *application* is therefore used in a motivic sense. The tower does not compute arbitrary Chow groups by brute force; rather, it tells the reader which obstruction is being tested, which part of the motive controls it, and which classical invariant should appear at that level.

The flow is narrative. A Chow class is first represented as a morphism in DM . The orthogonal tower filters that morphism. Strong convergence turns the tower into a filtration of the Chow group. Rational degeneration identifies the associated layers. Incidence, Albanese, Abel–Jacobi, and regular homomorphism kernels give those layers geometric meaning. Beauville, Kimura, Pontryagin, and BSD phenomena explain what the same layers measure on abelian varieties. Rationally connected varieties then test the integral edge of the theory: after rational invariants have largely disappeared, stable-rationality questions are controlled by universal CH_0 -triviality, decomposition of the diagonal, unramified cohomology, integral Hodge defects, and codimension-two cycles. The applications below should be read as tests of coherence: the same object must control the categorical, geometric, and arithmetic readings without changing language at each step.

This point is important for the scope of the thesis. The applications are not claims of universal computability of Chow groups. They are structured reductions: they isolate the layer in which a class lives, identify the invariant that should see that layer, and record the precise obstruction when the invariant fails. In this way the tower functions as a language for deciding whether a question is cohomological, Albanese-theoretic, Abel–Jacobi-theoretic, Beauville–Kimura in nature, or genuinely arithmetic.

23. Diagnostic use of the orthogonal filtration

The first application is diagnostic. Given a cycle, the orthogonal filtration provides a structured way of asking how invisible the cycle is. A class may be non-zero in Chow but invisible to cohomology; it may be homologically trivial but not incidence trivial; it may be incidence trivial but not Abel–Jacobi trivial; or it may lie so deep in the filtration that only higher motivic or arithmetic obstructions remain. The value of the tower is that these questions are not separate tests chosen ad hoc. They are successive readings of a single filtration.

Proposition 23.1 (Diagnostic form of the orthogonal filtration). *Let X/k be smooth projective of dimension d , and work with rational coefficients in the range treated in Parts II–III. The filtration $F_{\text{orth}}^\bullet \text{CH}^n(X, m)_{\mathbb{Q}}$ is finite, functorial for the motivic correspondences considered here, and compatible with the classical tests that appear in the first layers. In top degree its second step is incidence equivalence, and over algebraically closed fields this is the Albanese kernel. For any regular homomorphism out of $\text{CH}_{\text{alg}}^n(X)$, the relevant kernel is represented by an effective motivic kernel whose higher covers agree with those of $M(X)$ in the effective range.*

Proof. Finiteness and functoriality come from the strong convergence theorem of Part I and the column-support result of Theorem 3.6. The rational identification of associated layers uses Corollary 4.4. The top-degree incidence statement is Theorem 18.1. The regular-homomorphism statement is Theorem 20.1. Since all these results are obtained from the same orthogonal tower, their compatibility is not an additional condition; it is inherited from functoriality of the tower. $\square$

For a concrete class $\alpha \in \text{CH}^n(X, m)_{\mathbb{Q}}$, the diagnostic reading is as follows. The image of α in the first orthogonal quotient records the coarsest motivic information, including the usual cycle-class contribution when a realization is applied. If α lies in the next layer, the class has escaped the first cohomological detection and is now governed by incidence or Albanese data in the top-degree case. If a regular homomorphism such as Abel–Jacobi is available, the motivic kernel theorem says that its kernel is not external to the tower; it is itself represented by an effective motivic piece. Thus the diagnostic process is intrinsic to DM , even when it is read through classical invariants.

24. Abelian varieties, Abel–Jacobi kernels, and BSD

Abelian varieties are the most rigid testing ground for the theory because they carry several structures at once. The Beauville decomposition separates Chow groups into eigenspaces for multiplication maps. Kimura finite-dimensionality separates even and odd motivic behaviour. The Pontryagin product turns Chow classes into an algebra with strong nilpotence constraints. The Abel–Jacobi map connects this algebra to intermediate Jacobians and normal functions. BSD and Beilinson–Bloch expectations impose arithmetic restrictions on the same classes. Part II shows that the orthogonal filtration is compatible with these structures, while Part III explains why the Abel–Jacobi kernel is a motivic kernel.

Theorem 24.1 (Abelian-variety application). *Let A/k be an abelian variety and work rationally. The orthogonal filtration on $\text{CH}^n(A)_{\mathbb{Q}}$ is compatible with the Beauville decomposition. The Beauville component relevant to Abel–Jacobi injectivity is stable under the orthogonal filtration, under the Kimura splitting, and under the Pontryagin operations used in Section 16. If a non-zero class in this component lies in the Abel–Jacobi kernel, then the Pontryagin cascade of Theorem 16.12 produces the corresponding Griffiths-type and BSD-oriented*

obstructions. Under the Beilinson–Bloch expectation used in Theorem 16.15, such a class cannot occur.

Proof. The Beauville compatibility is Theorem 6.1. The Kimura compatibility is Theorem 16.3, and the Pontryagin behaviour is controlled by Theorem 16.11 and Theorem 16.12. The BSD-oriented interpretation is Theorem 16.20. The conditional exclusion under Beilinson–Bloch is Theorem 16.15. Part III supplies the motivic-kernel interpretation of the Abel–Jacobi kernel, so the test is internal to the same motivic filtration rather than an external comparison. $\square$

The theorem should be read as a single mechanism. Beauville gives the coordinate system, the orthogonal filtration gives the depth, Abel–Jacobi gives the geometric test, Pontryagin multiplication amplifies a hypothetical kernel class, and the BSD/BB input says when the amplified obstruction is not expected to exist. The strength of the construction is that these operations act on the same motivic object. This avoids the common problem of comparing unrelated filtrations after the fact.

25. Correspondences, families, and regular homomorphisms

Cycle theory is functorial through correspondences, not only through morphisms of varieties. A useful filtration must therefore be stable under the operations that move cycles in families, define incidence relations, and produce regular homomorphisms. This is exactly what Part III provides.

Corollary 25.1 (Functorial control of classical kernels). *Let X/k be smooth projective, and let $\psi : \mathrm{CH}_{\mathrm{alg}}^n(X) \rightarrow A(k)$ be a regular homomorphism to an abelian variety. The kernel of ψ , after tensoring with $\mathbb{Q}$, is represented by an effective motivic kernel $M_\psi^n(X) \rightarrow M(X)$. In the effective range specified in Theorem 20.1, this kernel has the same higher orthogonal covers and slices as the corresponding part of $M(X)$. Hence the orthogonal filtration may be restricted to the kernel without leaving the motivic category.*

Proof. Theorem 20.1 applies directly to ψ : regularity of ψ means that for every smooth connected parameter scheme T and every relative cycle $\mathcal{Z} \in \mathrm{CH}_{\mathrm{alg}}^n(T \times X)$, the induced map $T(k) \rightarrow A(k)$ is algebraic. The theorem then produces:

- (i) An effective motive $M_\psi^n(X) \in DM^{\mathrm{eff}}(k, \mathbb{Q})$ fitting into a distinguished triangle $M_\psi^n(X) \rightarrow M(X) \xrightarrow{\bar{\psi}} M(A)(n-d)[2n-2d] \rightarrow M_\psi^n(X)[1]$, where $\bar{\psi}$ is the motivic representation of ψ .
- (ii) The identification: $\ker(\psi)_{\mathbb{Q}}$ is the image of $\mathrm{Hom}_{DM}(M(X)(-n)[-2n], M_\psi^n(X))$ under the probe map to $\mathrm{CH}_{\mathrm{alg}}^n(X)_{\mathbb{Q}}$.
- (iii) The comparison of orthogonal covers: since $M_\psi^n(X) \rightarrow M(X)$ is a morphism in DM^{eff} , the Bousfield–orthogonal tower is functorial and $bc_{\leq r} M_\psi^n(X) \rightarrow bc_{\leq r} M(X)$ is an isomorphism in the effective range $r \geq d - n + 1$ (the

abelian target $M(A)(n-d)[2n-2d]$ lies below effective level $d-n$, so the distinguished triangle splits above that level).

The special cases are Corollary 20.2: the Abel–Jacobi, incidence, and abelian equivalence kernels are all regular homomorphisms (classical results recalled there), so each has an effective motivic kernel inheriting the higher orthogonal covers from $M(X)$ in the specified range. $\square$

The practical consequence is important. Once a kernel is motivic, it can be filtered, compared under correspondences, and tested by realizations in the same way as the ambient motive. This is why the applications do not need separate arguments for incidence, Albanese, Abel–Jacobi, and abelian equivalence. They are different regular readings of effective motivic kernels.

26. Rationally connected varieties and integral obstructions

Rationally connected varieties are among the most natural testing grounds for the filtration constructed in this thesis. They are close to rational varieties from the birational point of view, but precisely for that reason the usual rational invariants are often too coarse. Over an algebraically closed field of characteristic zero, rational connectedness forces strong constraints on zero-cycles; however, stable rationality is governed by the stronger condition of universal CH_0 -triviality. The difference between ordinary CH_0 -triviality and universal CH_0 -triviality is exactly where integral Chow-theoretic and unramified obstructions enter.

The classical mechanism is the decomposition of the diagonal. For a smooth projective variety X , universal CH_0 -triviality is equivalent, in the standard range used in stable-rationality arguments, to the existence of a Chow-theoretic decomposition

$$[\Delta_X] = [X \times x] + Z \quad \text{in } \mathrm{CH}_{\dim X}(X \times X),$$

where Z is supported on $D \times X$ for a proper closed subset $D \subsetneq X$. This formulation goes back to the correspondence methods of Bloch–Srinivas and has become one of the central tools in the work of Voisin and Colliot-Thélène–Pirutka on stable irrationality [5, 8, 9]. The relevance for the present thesis is that a decomposition of the diagonal is not merely a statement about zero-cycles. It is a statement about the action of a correspondence on all cycle-theoretic and cohomological invariants of X . Thus it is naturally readable through the functorial orthogonal tower.

For rationally connected varieties the first visible obstruction is not usually a rational one. The groups that survive are integral and birational in nature: unramified cohomology, torsion phenomena, integral Hodge classes, and defects of codimension-two algebraicity. The theorem of Colliot-Thélène–Voisin relates third unramified cohomology with coefficients in $\mathbb{Q}/\mathbb{Z}$ to the failure of the integral Hodge conjecture in degree four, precisely the degree governed by codimension-two cycles [6]. Voisin’s work on integral Hodge classes for uniruled or Calabi–Yau threefolds gives the corresponding positive side of this story in dimension three [7].

This places $\mathrm{CH}^2(X)$, Abel–Jacobi maps, and unramified cohomology in the same geometric range. That is the range already isolated in Parts II and III.

The orthogonal filtration gives a way of organizing these obstructions. For a smooth projective rationally connected variety $X/\mathbb{C}$, the early stages of the filtration on $\mathrm{CH}^2(X)$ separate the cycle-class information, the algebraic and incidence information, and the Abel–Jacobi information. If a codimension-two cycle is homologically trivial but not Abel–Jacobi trivial, it is detected at the classical intermediate-Jacobian level. If it is also Abel–Jacobi trivial, then its remaining obstruction is not seen by ordinary cohomology or by the first regular homomorphisms. It is exactly this residual region that interacts with unramified cohomology and the failure of integral algebraicity. In this sense, rationally connected varieties make visible the difference between rational degeneration of the motivic filtration and integral degeneration of the cycle problem.

The connection with Part II is as follows. Part II constructs the filtration $F_{\mathrm{orth}}^\bullet \mathrm{CH}^n(X, m)_{\mathbb{Q}}$ by applying the convergent orthogonal spectral sequence to the probe $M(X)(-n)[m - 2n]$. Rationally, the filtration degenerates in the range proved there. But stable rationality is not a rational question: nontrivial unramified cohomology, torsion, and integral Hodge defects disappear after tensoring with $\mathbb{Q}$. Thus the rationally connected application explains why the thesis must keep track of two different statements. The rational tower organizes the layers that are visible in $DM(-)_{\mathbb{Q}}$; the integral or completed variants record precisely the residue that stable-rationality questions cannot afford to ignore.

The connection with Part III is equally important. The decomposition of the diagonal is a correspondence statement, and the obstruction to universal CH_0 -triviality is tested by the action of correspondences on Chow groups and cohomological invariants. Part III shows that incidence equivalence, Albanese kernels, Abel–Jacobi kernels, and regular homomorphisms admit motivic-kernel interpretations in the orthogonal tower. Therefore the stable-rationality obstructions attached to rationally connected varieties can be expressed without leaving the language of the thesis: one asks at which orthogonal level the diagonal, or the residual correspondence $\Delta_X - X \times x$, becomes invisible to the relevant probes.

Proposition 26.1 (Rationally connected diagnostic). *Let $X/\mathbb{C}$ be smooth, projective, and rationally connected. Suppose that X is stably rational. Then X is universally CH_0 -trivial and hence admits a Chow-theoretic decomposition of the diagonal. Consequently the unramified and integral cycle-theoretic obstructions controlled by that decomposition vanish. In particular, any nonzero obstruction of Colliot-Thélène–Voisin type, any failure of the relevant integral Hodge surjectivity in degree four, or any failure of diagonal decomposition must be located in the residual integral part of the orthogonal picture, beyond the rational degeneration of Part II.*

Proof. Step 1: Stable rationality forces universal CH_0 -triviality. Recall that X is universally CH_0 -trivial if for every field extension L/k , the degree map $\deg :$

$\mathrm{CH}_0(X_L) \rightarrow \mathbb{Z}$ is an isomorphism (i.e. all degree-0 zero-cycles are rationally equivalent to zero over L). Stable rationality means $X \times_k \mathbb{P}_k^N$ is rational for some $N \geq 0$. Projective space is universally CH_0 -trivial (over any field L , $\mathrm{CH}_0(\mathbb{P}_L^N) \cong \mathbb{Z}$ generated by any rational point, so all degree-0 cycles vanish by Bezout). Universal CH_0 -triviality is preserved under products with universally CH_0 -trivial varieties [9, Prop. 1.4]: if Y is rational (hence universally CH_0 -trivial) and $\mathbb{P}^N$ is universally CH_0 -trivial, then $X \cong (X \times \mathbb{P}^N)/\mathbb{P}^N$ is universally CH_0 -trivial by the product formula.

Step 2: Universal CH_0 -triviality implies decomposition of the diagonal. By the Bloch–Srinivas theorem [5] (extended by Voisin and Colliot-Thélène–Pirutka [9]): X is universally CH_0 -trivial over k if and only if there exists a decomposition

$$[\Delta_X] = [X \times x] + Z \quad \text{in } \mathrm{CH}_d(X \times_k X)_{\mathbb{Z}},$$

for some closed point $x \in X(k)$ and a cycle Z supported on $D \times X$ for some proper closed subset $D \subsetneq X$. The proof uses the generic point of X (which is a field extension $k(X)/k$); applying universal CH_0 -triviality to this generic point produces the diagonal class over $k(X)$, then one spreads out.

Step 3: Diagonal decomposition kills birational invariants. If $[\Delta_X] = [X \times x] + Z$ with Z supported on $D \times X$, then the correspondence Δ_X acts on any cohomological invariant $H(-)$ as the identity. The piece Z (supported on $D \times X$) acts as a map $H(X) \rightarrow H(D) \rightarrow H(X)$, which factors through $H(D)$. For stable birational invariants H that vanish on D (by induction on dimension), we get $\mathrm{id}_{H(X)} = [X \times x]^* + Z^* = \mathrm{const} + 0$, forcing $H(X)$ to be trivial on the kernel of the degree map.

Concretely, for unramified cohomology $H_{\mathrm{ur}}^3(X, \mathbb{Q}/\mathbb{Z})$: this group is a stable birational invariant (invariant under products with $\mathbb{P}^N$ and birational maps) [6]. The diagonal decomposition gives $H_{\mathrm{ur}}^3(X, \mathbb{Q}/\mathbb{Z}) \hookrightarrow H_{\mathrm{ur}}^3(D, \mathbb{Q}/\mathbb{Z})$. For a rationally connected $X/\mathbb{C}$ with $\dim D < \dim X = d$, the variety D is also covered by rational curves (by restriction), so $H_{\mathrm{ur}}^3(D, \mathbb{Q}/\mathbb{Z})$ can be handled by induction; in dimension ≤ 2 , $H_{\mathrm{ur}}^3 = 0$ for rationally connected varieties. Hence $H_{\mathrm{ur}}^3(X, \mathbb{Q}/\mathbb{Z}) = 0$ under stable rationality.

Step 4: Orthogonal filtration language. The rational orthogonal filtration $F_{\mathrm{orth}}^\bullet \mathrm{CH}^n(X)_{\mathbb{Q}}$ from Part II is finite (Theorem 3.6) and degenerates (Corollary 4.4). Any rational obstruction to a Chow class is detected by the graded pieces $\mathrm{gr}_{F_{\mathrm{orth}}}^r$. However, stable-rationality obstructions such as $H_{\mathrm{ur}}^3(X, \mathbb{Q}/\mathbb{Z})$ are *torsion* and invisible to the rational filtration.

Part III identifies $F_{\mathrm{orth}}^2 \mathrm{CH}^2(X)_{\mathbb{Q}}$ with the Abel–Jacobi kernel (Theorem 18.1 and Theorem 7.7). For a rationally connected threefold over $\mathbb{C}$, the Abel–Jacobi map $\mathrm{AJ} : \mathrm{CH}_{\mathrm{hom}}^2(X)_{\mathbb{Q}} \rightarrow J^2(X)_{\mathbb{Q}}$ is often an isomorphism (Voisin [8] for smooth cubic fourfolds), so $F_{\mathrm{orth}}^3 \mathrm{CH}^2(X)_{\mathbb{Q}} = 0$. The obstruction then lives entirely in the integral refinement: the difference between $\mathrm{CH}^2(X)$ (integral cycles) and $\mathrm{CH}^2(X)_{\mathbb{Q}}$ (rational), specifically in torsion classes in $\ker(\mathrm{AJ})$ that are not detected

by $\mathbb{Q}$ -coefficients.

The Colliot-Thélène–Voisin theorem [6] identifies this torsion: $H_{\text{ur}}^3(X, \mathbb{Q}/\mathbb{Z}) \cong \ker(H^{2,2}(X, \mathbb{Z}) \rightarrow \text{CH}^2(X)/\text{alg})^\perp$ (the defect of integral Hodge surjectivity). In the orthogonal filtration picture, this defect is the torsion part of $F_{\text{orth}}^2 \text{CH}^2(X)$ that disappears after tensoring with $\mathbb{Q}$.

Thus any nonzero $H_{\text{ur}}^3(X, \mathbb{Q}/\mathbb{Z})$ obstructs universal CH_0 -triviality (Step 3), hence stable rationality (Step 1). The rational orthogonal filtration localizes but cannot resolve this obstruction: it lies in the integral residue of the cycle-class and Abel–Jacobi layers, precisely the layer beyond the scope of the rational degeneration of Part II. $\square$

The proposition should not be read as claiming that the orthogonal filtration alone proves stable irrationality. Rather, it says that the filtration supplies a common grammar for the known obstructions. Bloch–Srinivas decomposition of the diagonal, Voisin’s diagonal method, Colliot-Thélène–Voisin unramified cohomology, and Colliot-Thélène–Pirutka specialization all test the same phenomenon from different sides: whether the diagonal acts as if X were universally CH_0 -trivial. The orthogonal tower places this test among the same layers that control Abel–Jacobi and regular homomorphism kernels.

This application also clarifies why codimension two is central. For a complex threefold, codimension-two cycles are one-cycles, and their Abel–Jacobi image lies in the intermediate Jacobian. The defect of integral Hodge in degree four measures whether integral Hodge classes of type $(2, 2)$ are algebraic. The third unramified cohomology group is sensitive to the same failure through the Colliot-Thélène–Voisin comparison. Thus the codimension-two layer is the meeting point of the Abel–Jacobi theory of Part II, the regular-homomorphism kernels of Part III, and the stable-rationality obstructions of Part IV.

For higher-dimensional rationally connected varieties the same philosophy persists, although the invariants become more numerous. Universal CH_0 -triviality still forces the decomposition of the diagonal; decomposition of the diagonal still kills birational cycle-theoretic invariants; and failures of integral algebraicity, unramified cohomology, or specialization of the diagonal still produce obstructions to stable rationality. What the orthogonal formalism adds is a way to separate the rational motivic information from the integral residue. The rational part belongs to the finite degenerating filtration of Part II. The residue belongs to the same boundary phenomenon that later appears for effective completions, modulus, and positive characteristic. This is why rationally connected varieties belong in Part IV: they are the first major class of examples where the thesis must distinguish a computable rational filtration from the integral obstructions that remain after rationalization.

27. Effective completion, slices, modulus, and positive characteristic

The fourth application identifies the boundary of the theory. In DM , the orthogonal spectral sequence converges strongly for the Chow probes considered in Part I. In $\mathcal{SH}$, the analogous statement fails for objects such as the sphere, KGL , and MGL unless one passes to effective completions or imposes eventual orthogonality. This distinction is not a defect of the thesis; it is one of its main safeguards. It prevents the Chow-theoretic theorems from being overstated in the stable motivic homotopy category.

The slice filtration explains why useful information survives. The slices of KGL and MGL are motivic Eilenberg–Mac Lane pieces, so the part of K -theory and cobordism visible slice-wise returns to the motivic regime where Part I applies. This is the content of Propositions 15.5 and 15.6, together with Corollary 15.12. Thus the orthogonal tower does not claim to compute KGL or MGL in $\mathcal{SH}$ by the same argument used for Chow groups. It computes their motivic and completed pieces in the range controlled by slices.

The same boundary principle governs modulus and characteristic p . The modulus calculation in Section 14 shows that the Picard group with modulus has an additional tangent contribution, so the ordinary Beauville picture must be enlarged. In positive characteristic, Section 13 points toward crystalline, Nygaard, and prismatic filtrations as the natural receptacles of a future orthogonal analogue [38, 40, 59]. These are not presented as theorems of the same strength as rational degeneration over perfect fields. They are the correct next objects suggested by the method.

Theorem 27.1 (Boundary principle for applications). *The orthogonal tower gives genuine Chow filtrations in DM in the range proved in Parts I–II. In $\mathcal{SH}$, the same construction gives completed invariants unless eventual orthogonality holds. Slice-wise motivic pieces of KGL and MGL return to the convergent motivic setting, while modulus and positive-characteristic refinements require additional tangent, crystalline, Nygaard, or prismatic structures.*

Proof. The first assertion is Theorem 2.9 applied to Chow probes and then used in Section 3. The failure in $\mathcal{SH}$ is Theorem 15.10, while eventual orthogonality and effective completion are treated in Section 15. The slice-wise statements for KGL and MGL are Propositions 15.5 and 15.6. The modulus and positive-characteristic statements are the interpretive conclusions of Sections 14 and 13. $\square$

28. Global synthesis

The final synthesis is best stated in prose. The orthogonal tower of the motivic unit begins as a categorical construction produced by Bousfield localization. For geometric motivic probes it is strongly convergent, so it filters motivic Hom-groups. When the Hom-group is a higher Chow group, this becomes the orthogonal Chow filtration. With rational coefficients over perfect fields, the spectral sequence

degenerates at E^1 , and the layers become computable from the orthogonal pieces. On abelian varieties these layers respect Beauville, Kimura, and Pontryagin structures. In top degree they recover incidence and Albanese kernels. For regular homomorphisms they are represented by effective motivic kernels. Outside DM , the same idea survives as a completed or slice-wise invariant, not automatically as a strongly convergent filtration.

Theorem 28.1 (Global form of the thesis). *Within the motivic range considered in Parts I–III, the orthogonal tower of the motivic unit produces a finite, functorial filtration on higher Chow groups. The filtration is strongly convergent for the geometric probes of Part I, rationally degenerate in the situations of Part II, compatible with Beauville–Kimura–Pontryagin structures on abelian varieties, and geometrically interpretable through incidence, Albanese, Abel–Jacobi, and regular-homomorphism kernels. The applications of Part IV show that the same construction also organizes rationally connected stable-rationality obstructions through decomposition of the diagonal, unramified cohomology, and integral Hodge defects, and identifies the boundary between genuine Chow filtrations and effective-completed invariants in $\mathcal{SH}$.*

Proof. The convergence assertion is Theorem 2.9. The Chow filtration is constructed in Section 3. Rational degeneration is Corollary 4.4. Compatibility with abelian-variety structures is proved in Sections 6 and 16. The geometric interpretation of the early kernels is Theorem 18.1 and Theorem 20.1. The rationally connected application is Proposition 26.1, which places decomposition of the diagonal, unramified cohomology, and integral Hodge obstructions inside the same orthogonal grammar. The boundary statement is Theorem 27.1. $\square$

This closes the thesis. The same tower explains convergence, filtrations, geometric kernels, abelian-variety arithmetic, rationally connected obstruction theory, and effective completion. The unifying point is not that all these phenomena are identical, but that each is a different visible face of the orthogonal tower.

Conclusions

The thesis proves that the orthogonal tower is a viable organizing principle for higher Chow groups. Its main achievement is to replace a collection of parallel filtrations and tests by a single motivic construction. The tower is built in DM , its convergence is established by the methods of Part I, and its specialization to Chow groups is carried out in Part II. The filtration it produces is not merely formal: in rational settings it degenerates enough to be computable, and in the top-degree range it recovers geometric kernels that are already central in algebraic geometry.

The final version of the argument is intentionally cumulative. The convergence theorem is not an isolated technical result; it is the reason the later Chow filtration exists. The degeneration theorem is not a formal simplification; it is the reason the filtration can be read from explicit orthogonal layers. The incidence and

regular-homomorphism comparisons are not ornamental; they identify the first layers with kernels that geometers already know how to test. The applications are therefore not a separate chapter of examples, but the point where the categorical construction is forced to account for the classical geometry of cycles.

Part I provides the foundational theorem. The orthogonal spectral sequence attached to the Bousfield–orthogonal tower converges strongly for the motivic probes $M(X)(r)[s]$. This is the point that legitimizes every later use of the word filtration. Without strong convergence, the tower would produce only a completed approximation; with it, the induced filtration is an actual filtration of the motivic Hom-group. The analysis of $\mathcal{SH}$ also shows why this statement must be formulated carefully: outside DM , the correct target is often an effective completion.

Part II turns convergence into cycle theory. For $B = M(X)(-n)[m - 2n]$, the abutment is $\mathrm{CH}^n(X, m)_R$. The orthogonal tower of $\mathbf{1}_R$ therefore defines $F_{\mathrm{orth}}^\bullet \mathrm{CH}^n(X, m)_R$. Rationally over perfect fields, the relevant differentials vanish by the Galois-cohomological and Maschke arguments of Section 4. The resulting filtration has finite support and behaves well with the structures available on abelian varieties. The Beauville decomposition gives a grading, Kimura finite-dimensionality supplies motivic parity, and the Pontryagin product turns hypothetical Abel–Jacobi kernels into strong algebraic obstructions.

Part III supplies the geometric meaning of the first layers. The second orthogonal step is incidence equivalence in top degree and becomes the Albanese kernel over algebraically closed fields. Regular homomorphisms have effective motivic kernels. Thus the classical tests are not added after the fact; they are internal shadows of the tower. This is the conceptual bridge between the abstract spectral sequence and concrete algebraic cycles.

Part IV records the applications and the limits. As a diagnostic tool, the filtration tells where a Chow class first becomes invisible. For abelian varieties, it organizes Abel–Jacobi injectivity and BSD-type obstructions inside one Beauville–Kimura–Pontryagin framework. For correspondences and regular homomorphisms, it provides functorial control of kernels. For rationally connected varieties, it explains why stable-rationality questions are governed by the residual integral layer: universal CH_0 -triviality, decomposition of the diagonal, unramified cohomology, and integral Hodge defects all test the part not seen by the rational degeneration of Part II. For K -theory, cobordism, modulus, and positive characteristic, it identifies the boundary between established results and the future programme: slices and effective completion are available now, while crystalline, Nygaard, prismatic, and modulus refinements require additional work.

The central future problem remains column stability. In the language of Part II, it asks whether the Beauville components occupy the predicted orthogonal columns in full generality. In the language of motives, it is a Künneth-type splitting problem for the pieces of the orthogonal tower. Solving it would remove several conditional labels from the comparisons with Saito, Bondarko, and Bloch–Beilinson–Murre filtrations. The other future directions – integral degeneration beyond the

Hilbert 90 range, Abel–Jacobi injectivity outside the generic cases proved here, and positive-characteristic realizations through crystalline or prismatic methods – are best understood as consequences of this same structural issue.

The final form of the thesis is therefore a closed narrative. Orthogonal covers produce a spectral sequence. Strong convergence turns it into a filtration. Rational degeneration makes the layers accessible. Incidence and regular homomorphisms give the first layers their geometric names. Abelian varieties show how the filtration interacts with deep algebraic and arithmetic structures. Rationally connected varieties show why the integral residue cannot be ignored: a variety may have very simple rational invariants and still carry the unramified, integral Hodge, or diagonal obstructions that decide stable rationality. The stable homotopy and positive-characteristic discussions mark the precise frontier where the same construction must be completed, sliced, or refined. This is the sense in which the orthogonal tower is not just a tool for constructing a filtration, but a framework for organizing the geometry and arithmetic of higher Chow groups.

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